



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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27-00-00-001

FLIGHT CONTROLS, GENERAL

Introduction

The flight controls consist of:

- Roll control
- Yaw control
- Pitch control

General Description

Refer to Figures 1 , 2 , 3 and 4.

The primary flight control surfaces are operated by cables from left or right control columns and rudder pedals, allowing the aircraft to be flown from either the pilot or copilot positions.

The positions of the:

- Rudder
- Roll spoilers
- Elevators,

are displayed on the Powered Flight Control Surfaces Indicator (PFCSI). The PFCSI display is on the pilot's Multi Functional Display. The inputs to the display are provided by the Integrated Flight Cabinet, which receives information on surface position from the Flight Control Electronic Control Unit (FCECU).

The Flight Control System has the following subsystems:

- Flight Control Electronic Control Unit (FCECU) (27-00-01)
- Aileron Control System (27-10-00)
- Rudder Control System (27-20-00)
- Elevator Control System (27-30-00)
- Wing Flaps (27-50-00)
- Spoilers Control System (27-60-00)
- Gust Lock Systems (27-70-00)

Detailed Description

Flight Control Electronic Control Unit (FCECU)

The Flight Control Electronic Control Unit (FCECU) receives electrical/electronic inputs and supplies electrical/electronic outputs, for all of the primary and secondary flight control systems.

The FCECU is a single unit divided into two channels, 1 and 2, (also referred to as the left and right channels). The left and right channels receive inputs from the pilot and/or copilot flight control commands. The two channels function separately from each other throughout the entire system, from the FCECU to the individual Line Replaceable Units (LRUs) and electrical busses.

The FCECU is part of these Flight Control System (FCS) functions:

- Rudder Hinge Moment Limiting (HML)
- Pitch Trim

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-00-00
Config 001

27-00-00

Config 001
Page 2
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

- Elevator Feel
- Spoiler Lift–Dump
- Outboard Spoiler Lockout
- Aural Take–Off Warning
- Flight Control Surface position reporting to the Flight Data Recorder (FDR) and the Powered Flight Control Surface Indicator (PFCSI)
- Flight Control System Caution and Warning Signals
- Built–In–Test
- Maintenance Interface to Central Diagnostic System (CDS).

Aileron Control System

The ailerons give roll control of the aircraft in flight. The pilot or copilot supplies roll commands to the aileron control system by turning the handwheels. Roll input commands are mechanically transmitted from the pilot's and copilot's handwheels to the aileron surfaces through the aileron control system. The aileron system transmits the roll commands from the pilot's and copilot's control handwheels to the aileron surfaces through a series quadrants, cables, levers and pushrods. Movement of the control handwheels causes the aileron surfaces to deflect asymmetrically and in proportion to control handwheel rotation.

The aileron system also has inputs from the Auto Flight Control System (AFCS) and the Aileron Trim and Centering Unit. The Automatic Flight Control System (AFCS) also gives roll commands,

through the autopilot servo connected to the aileron forward quadrant located below the flight compartment floor.

Ailerons Trim and Centering

Aileron trim is achieved by electrically trimming both aileron surfaces through the actuator located in the Aileron Trim and Centering Unit. The trim and centering unit is connected to the aileron forward quadrant by a spring torsion bar that makes sure the aileron surfaces always return to a neutral position.

Aileron Surfaces

There are two aileron control surfaces. One is installed on the outboard end of the left wing and the other is installed on the outboard end of the right wing. Each aileron surface forms part of the trailing edge of its respective wing.

Aileron Gust Lock System

The gust lock system locks the aileron control surfaces in the neutral position to protect the aileron surfaces from damage.

The aileron control system is a mechanically operated system. Gust protection for the ailerons is supplied by a gust lock mechanism operated from the flight compartment. When the CONTROL LOCK lever is moved to the ON position, the ailerons are locked in neutral.

Roll Spoilers System

The roll spoilers are a secondary flight control system. They are hydraulically powered and operate with, and in proportion to, the aileron control system to give lateral control of the aircraft in flight. The roll spoilers also decrease the lift made by the wings on landing.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

There are two hydraulically powered roll spoilers on each wing. The roll spoilers operate with the ailerons to give lateral control of the aircraft in flight. Rotary movement of either the pilot's or the copilot's control handwheels operates the ailerons and the spoilers at the same time through an interconnect rod joining the two control columns. The ailerons and the roll spoilers can be disconnected from each other if a jam occurs in either system. The aircraft can then be operated with limited roll control by using the unjammed system.

Ground Spoiler System

The spoiler panels are deployed symmetrically to decrease or “dump” the lift made by the wings. This action puts all of the weight of the aircraft on the landing gear, so that the wheel brakes can slow the aircraft more efficiently.

The flight spoilers are symmetrically deployed as ground spoilers, when signalled by the Weight-On-Wheels (WOW) and power lever sensors, if the spoiler FLIGHT/TAXI switch is in the FLIGHT position.

The roll spoiler system automatically changes from the flight mode to the ground mode after the aircraft has landed. The roll spoilers operate symmetrically on landing (overriding flight mode operation).

Spoiler Panels

Four spoiler panels are installed on the upper surface of the wing forward of the outboard flaps. The panels are identified as inboard and outboard and are operated hydromechanically using control wheel inputs in conjunction with the ailerons. The inboard and outboard spoilers are hinged to the upper surface of the wing. The Power Control Units (PCUs) attached to the inboard panels, are supplied with hydraulic pressure from No.1 hydraulic system. The

PCUs attached to the outboard panels are supplied by No.2 hydraulic system.

Roll Disconnect System

The roll disconnect system disengages the aileron and roll spoiler control circuits from each other in the event of a jam in either control circuit. The remaining operational circuit maintains roll (lateral) control of the aircraft.

The pilot's control handwheel operates the spoiler control circuit and the copilot's control handwheel operates the aileron control circuit. The control handwheels are connected to each other so that either control handwheel operates both circuits at the same time. This lets the aircraft be flown from either the pilot's or copilot's position. If either control circuit becomes jammed, the ROLL DISC handle is pulled to disconnect the aileron system from the roll spoiler system. The pilot with the unjammed control handwheel maintains roll control of the aircraft.

Rudder System

The rudder system transmits the yaw commands from the pilot's and copilot's rudder pedals to the rudder through a series of quadrants, cables, levers and pushrods. Movement of the rudder pedals is transmitted to two hydraulic Power Control Units (PCUs). The PCUs move the rudder in proportion to the amount of rudder pedal input. The rudder system also has inputs from the Automatic Flight Control System (AFCS) and the rudder trim actuator.

Rudder Feel and Trim

The rudder feel and trim unit gives artificial feel to the rudder pedals during operation. Because the rudder system is a powered flight



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

control, the unit is used to prevent the pilot from overcontrolling the aircraft in flight.

The rudder trim actuator gives the pilot trim control around the yaw axis of the aircraft in flight. Rudder trimming biases the complete rudder system, so that a newly trimmed position is sensed by the pilot as the neutral position.

Elevator System

The elevator system transmits the pitch commands from the pilot's and copilot's control columns to the elevator control surfaces. The elevator surfaces move in proportion to the amount of control column movement.

The left and right elevators are installed on the trailing edge of the left and right horizontal stabilizers, respectively. The elevators are moved down to pitch the aircraft nose down and moved up to pitch the aircraft nose up, around the lateral axis.

The elevators are positioned by commands from either the pilot control system or the Automatic Flight Control System (AFCS). Trim control and surface position indication is also supplied for each elevator.

Trimming of the elevator control system gives the aircraft longitudinal stability during flight. Trimming counters the effects of changes to the aircraft centre of gravity, airspeed and lift during flap extension or retraction.

Stall Warning/Stick Pusher

The Stall Warning / Stick Pusher system supplies the flight crew with warnings of an impending stall during flight. The system pushes the control columns forward to assist in stall recovery.

Warnings of an impending stall are supplied by column stick shakers, and an audio warning. The system has two identical independent channels, two separate Angle of Attack (AOA) vanes, and inputs from various parameters. If both stall protection systems reach the stick pusher reference, stick pusher assistance will be provided to reduce the angle of attack.

Elevator Surfaces

There are two elevator surfaces, left and right, which are installed on the trailing edge of the left and right horizontal stabilizers. Three elevator Power Control Units (PCUs) are attached to each elevator surface. A Linear Variable Differential Transformer (LVDT) is attached to the front spar of each elevator surface to supply position information to the Powered Flight Controls Surface Indicator (PFCSI).

Pitch Disconnect System

The pitch disconnect system lets the left and right elevator control circuits be disconnected from each other, in the event a jam occurs in either control circuit. The remaining operational circuit controls the pitch of the aircraft around the lateral axis.

The pilot's control column operates the left elevator circuit and the copilot's control column operates the right elevator circuit. The two control columns are connected to each other so that either column can operate both elevator circuits at the same time. This lets the



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

aircraft be flown from either the pilot's or copilot's position. If a jam occurs in either control circuit, the PITCH DISC handle in the flight compartment is pulled to disconnect the left elevator circuit from the right elevator circuit. The pilot with the free control column will have pitch control.

Flaps

The wing flaps supply increased lift during take-off and landing.

Each wing has an inboard and an outboard flap surface. The inboard flap on each wing is installed between the fuselage structure and the engine nacelle inboard structure. The outboard flap on each wing is installed between the engine nacelle outboard structure and the aileron (refer to SDS 27-54-00). Indication of flap position is given by an indicator on the copilot's Multi Function Display (refer to SDS 27-53-00).

The flap surfaces are electronically controlled and hydraulically operated. The flaps are extended when the flap lever is moved in its quadrant. The flap quadrant is located in the power quadrant on the right side of the centre console. The flap quadrant has five positions corresponding to the five possible flap positions: 0°, 5°, 10°, 15° and 35°.

Flap Control System

The flap control system uses a digital position control loop to operate the flap surfaces by modulating the flow to the hydraulic motor of the flap power unit.

The flap control system is an electronically controlled, hydro-mechanical system. It monitors flap position and selection

signals and produces the output signals to the Flap Power Unit (FPU) to control flap movement.

Flap Drive System

The flap drive system moves the flaps from the 0° position (full up) to the 35° position (full down) or to the intermediate 5°, 10° or 15° positions. It is designed to move the four flap sections together to the position set by the flap selector lever. When the flaps have moved into the selected position, the flap drive system makes sure that the flaps stay in that position.

Flap Position Indication

The flap position indication system has a Flap Position Indication Unit (FPIU) located below the cabin floor aft of the High Intensity Radio Frequency (HIFR) enclosure. It also has two flap position transducers (RVDTs) located on the outboard flaps.

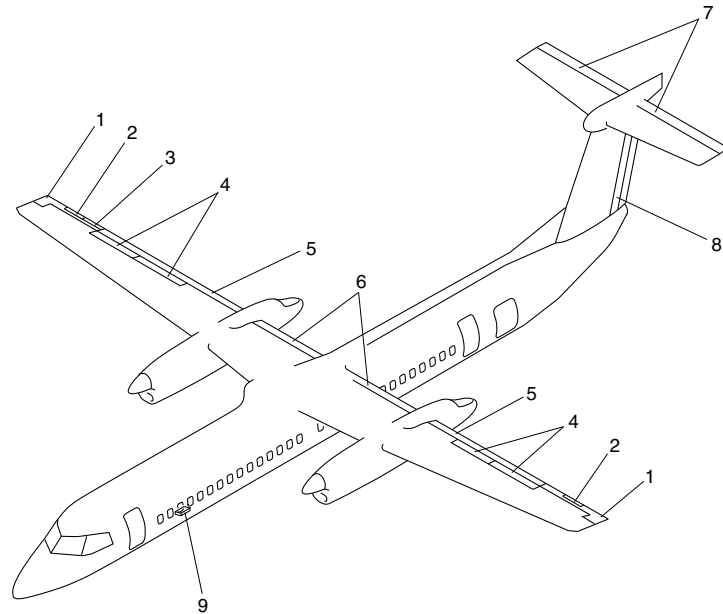
The flap position indication system has two functions:

- Supplies indication of the current flap position to the copilot's Multi-Function Display indicator
- Monitors the position of the flaps during flap deployment and sends the position data to the Flap Control Unit (FCU).



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Aileron.
2. Geared Tab.
3. Ground Adjustable Tab.
4. Spoilers.
5. Outboard Flaps.
6. Inboard Flaps.
7. Elevator.
8. Rudder.
9. Flight Control Electronic Unit.

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FLIGHT CONTROL SYSTEM
Figure 1

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EFFECTIVITY:
See first effectivity on page 2 of 27-00-00
Config 001

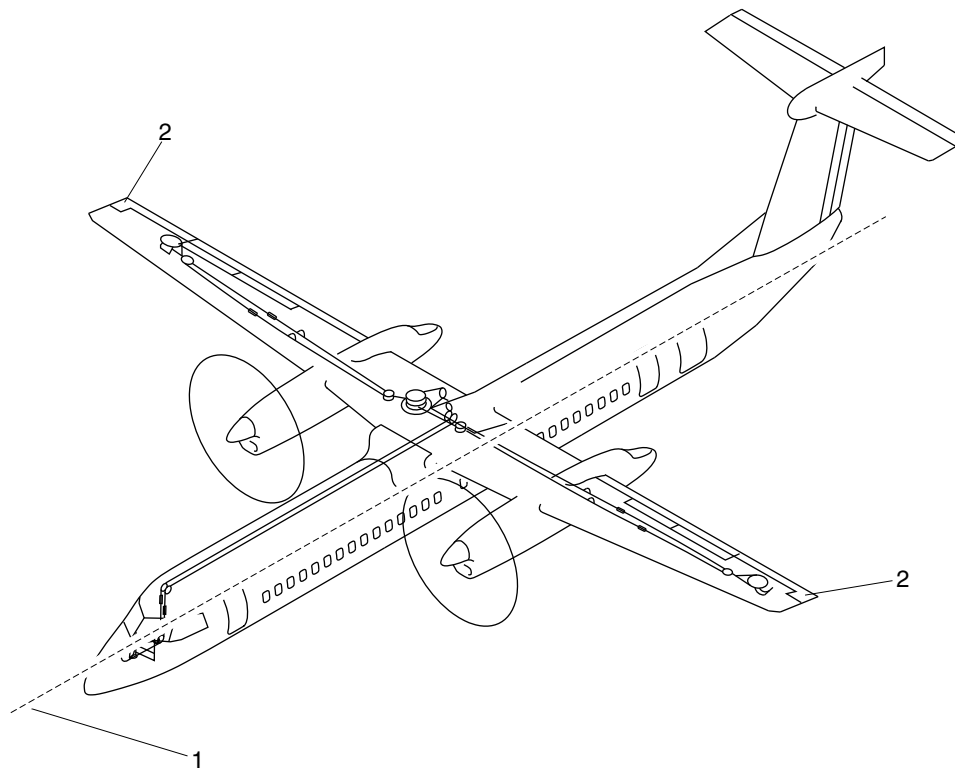
27-00-00

Config 001
Page 7
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Longitudinal Axis.
- 2. Ailerons.

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AILERON CONTROL SYSTEM
Figure 2

PSM 1-84-2A
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Config 001

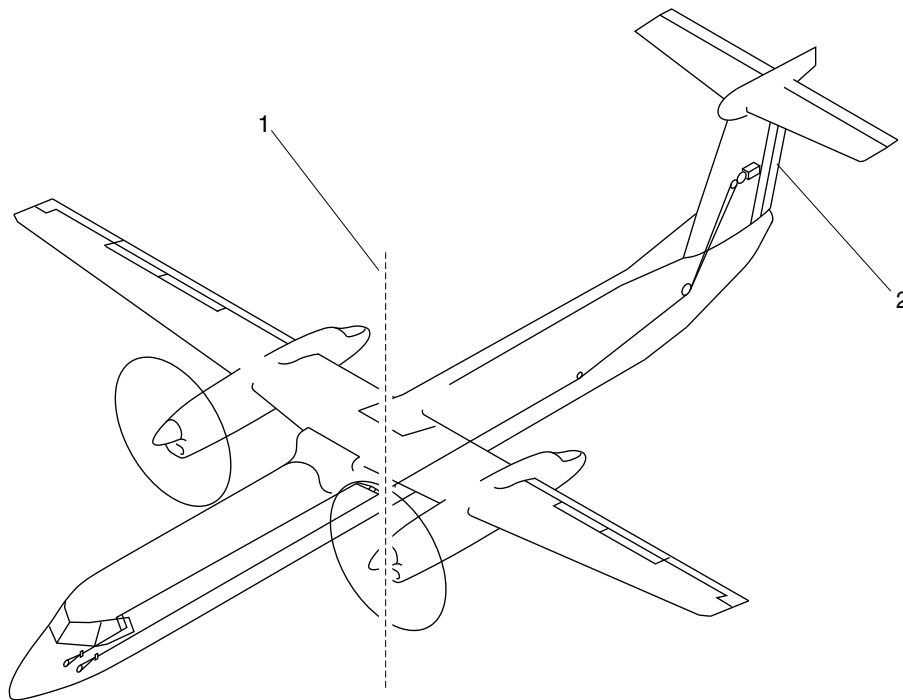
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Config 001
Page 8
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Vertical Axis.
- 2. Rudder.

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RUDDER CONTROL SYSTEM
Figure 3

PSM 1-84-2A
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See first effectivity on page 2 of 27-00-00
Config 001

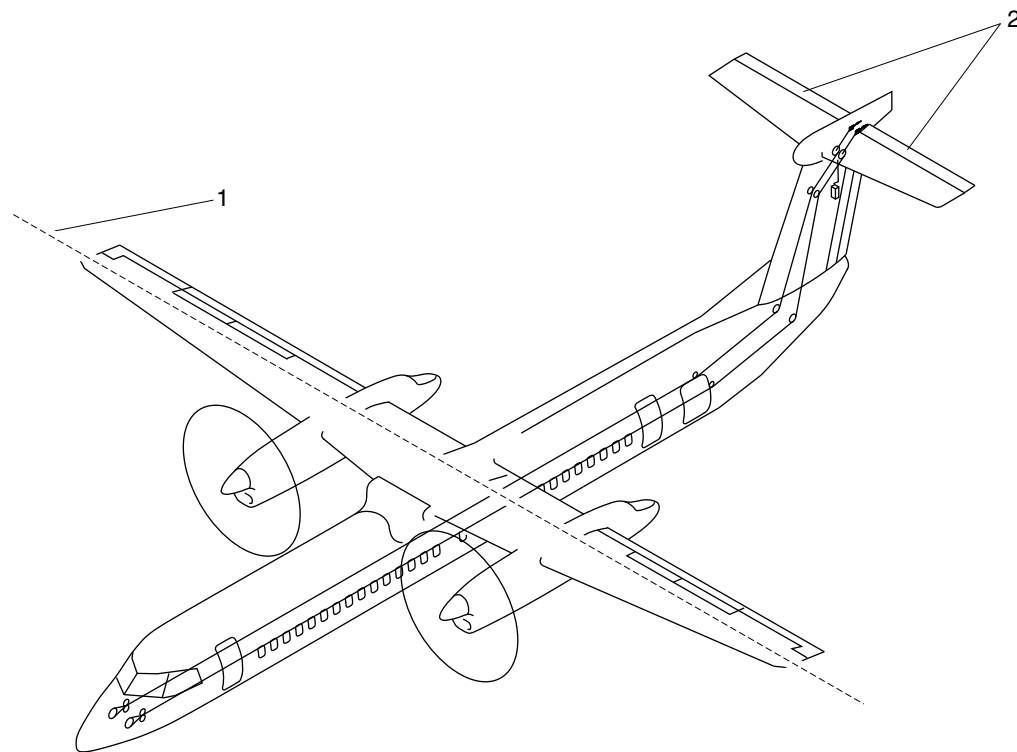
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Config 001
Page 9
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Lateral Axis.
- 2. Elevators.

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ELEVATOR CONTROL SYSTEM
Figure 4

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-00-00
Config 001

27-00-00

Config 001
Page 10
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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27-00-01-001

FLIGHT CONTROL ELECTRONIC CONTROL UNIT (FCECU)

Introduction

The Flight Control Electronic Control Unit (FCECU) receives electrical/electronic inputs and supplies electrical/electronic outputs, for all of the primary and secondary flight control systems.

General Description

The FCECU is a single unit divided into two channels, 1 and 2, (also referred to as the left and right channels). The left and right channels receive inputs from the pilot and/or copilot flight control commands. The two channels function separately from each other throughout the entire system, from the FCECU to the individual Line Replaceable Units (LRUs) and electrical busses.

The FCECU is part of these Flight Control System (FCS) functions:

- Rudder Hinge Moment Limiting (HML)
- Pitch Trim
- Elevator Feel
- Spoiler Lift-Dump
- Outboard Spoiler Lockout
- Aural Take-Off Warning

- Flight Control Surface position reporting to the Flight Data Recorder (FDR) and the Powered Flight Control Surface Indicator (PFCSI)
- Flight Control System Caution and Warning Signals
- Built-In-Test
- Maintenance Interface to Central Diagnostic System (CDS).

Detailed Description

[Refer to Figures 1 , 2 and 3.](#)

The FCECU has one motherboard with two PC cards, one for each channel. It weighs 5.5 lb and uses 16 watts per channel. Channel 1 (left) is powered by the 28 Vdc left essential bus through the FCS ECU 1 circuit breaker (M3). Channel 2 (right) is powered by the 28 Vdc right essential bus through the FCS ECU 2 circuit breaker (L6). The FCECU is located under the passenger cabin floor, forward of the rotor burst zone. It is located inside the High Intensity Radio Frequency (HIRF) enclosure for protection from electromagnetic interference. The HIRF box is not installed on aircraft with ModSum 4-153780 or 4-126136 incorporated.

The FCECU functions in each of the flight control systems that follow:

- Aileron Control System
- Elevator Control System
- Rudder Control System
- Spoiler Control System
- Flap Control System

PSM 1-84-2A

EFFECTIVITY:

See first effectivity on page 2 of 27-00-01

Config 001

27-00-01

Config 001

Page 2

Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Aileron Control System

The FCECU does the functions that follow:

- Supplies electrical excitation of the Linear Variable Differential Transducer (LVDT) attached to the right aileron
- Receives right aileron position data from the LVDT and transmits the data to the Flight Data Recorder (FDR) by way of the Integrated Flight Cabinet (IFC).

Elevator Control System

The FCECU does the functions that follow:

- Changes the elevator feel forces on the control columns as a function of airspeed and acceleration
- Changes manual pitch trim rate as a function of airspeed
- Supplies automatic pitch trim when in the autopilot mode or when the flaps are extended or retracted
- Prioritizes all valid pitch trim commands in the following order: pilot, copilot and then autopilot trim
- Monitors pitch trim actuator failures and sends a signal to the PITCH TRIM caution light and the ELEVATOR TRIM SHUTOFF switchlights
- Supplies an aural warning on takeoff if the elevator trim is outside the takeoff range
- Monitors elevator asymmetry and supplies a signal to the ELEVATOR ASYMMETRY caution light

- Monitors elevator feel actuator failures and supplies a signal to the ELEVATOR FEEL caution light
- Monitors the airspeed and acceleration signals for a mismatch condition and sets the pitch trim and elevator feel to a default mode
- Supplies electrical excitation of the left and right elevator position LVDTs
- Receives position data from the left and right elevator LVDTs and transmits the data to the FDR and EFIS by way of the IFC.

Rudder Control System

The FCECU does the functions that follow:

- Limits the rudder hinge moment by regulating the pressure in the rudder PCU according to aircraft speed.
- Monitors the airspeed signals for a mismatch condition and sets the rudder PCU regulated pressure to a default mode.
- Monitors the regulated pressure in the rudder PCU for discrepancies and depressurizes the PCU if it is faulty.
- Depressurizes the relevant rudder PCU if either of the RUD 1 or RUD 2 PUSH OFF switchlight is pushed.
- Increases the pressure to the operating rudder PCU if one of the rudder PCUs fails or is depressurized.
- Sends signals to the rudder caution lights and pushoff switchlights to indicate system malfunctions.
- Supplies electrical excitation of the upper and lower rudder position LVDTs



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

- Receives position data from the upper and lower rudder LVDTs and transmits the data to the FDR and EFIS by way of the IFC.

Spoiler Control System

The FCECU does the functions that follow:

- Locks out the outboard spoilers for airspeeds greater than 170 kt (315 km/h).
- Operates both the inboard and outboard spoilers to dump the lift from the wings when the aircraft lands.
- Sends a signal to the spoiler caution and warning lights to indicate system malfunctions
- Supplies electrical excitation of the inboard and outboard spoiler position LVDTs
- Receives position data from the inboard and outboard spoiler LVDTs and transmits the data to the FDR and EFIS by way of the IFC.

Flap Control System

The FCECU does the functions that follow:

- Receives left and right flap position data from the Flap Position Indication Unit (FPIU)
- Receives “Flaps-in-Motion” signal from the Flap Control Unit (FCU)

Interfaces

The FCECU has these interfaces:

SOURCE	PARAMETER
AIR DATA UNITS	AIRSPEED
ATTITUDE HEADING AND REFERENCE SYSTEM	ACCELERATION
FLAP POSITION INDICATION UNIT	FLAP POSITION
FLAP CONTROL UNIT	FLAPS IN MOTION SIGNAL
MANUAL PITCH TRIM SWITCHES	PILOT/COPILOT PITCH TRIM COMMAND
FLIGHT GUIDANCE MODULES	AUTOPILOT PITCH TRIM AND FLAP AUTO TRIM COMMANDS
POWER LEVER ANGLE MICROSWITCHES	FLIGHT IDLE +12° SIGNAL
FLIGHT/TAXI SWITCH	FLIGHT OR TAXI MODE SIGNALS
LVDTs	FLIGHT CONTROL SURFACE POSITION
MAINTENANCE PANEL	FLIGHT CONTROL SYSTEM FAILURES
RUDDER PCU PRESSURE SENSORS	RUDDER PCU PRESSURE



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

PITCH DISCONNECT RELAY	ENGAGED/DISENGAGED SIGNAL
PUSHOFF SWITCHLIGHTS	SWITCH ON/OFF SIGNAL

Caution Lights

The FCECU sends signals to the Caution and Warning panel to turn on the caution lights that follow: [Refer to Figure 4.](#)

- ELEVATOR FEEL
- PITCH TRIM
- ELEVATOR ASYMMETRY
- ROLL SPLR INBD GND
- ROLL SPLR OUTBD GND
- SPLR OUTBD
- ROLL SPLR INBD HYD
- ROLL SPLR OUTBD HYD
- #1 RUD HYD
- #2 RUD HYD
- RUD CNTRL.

Switchlights

The FCECU sends signals to turn on the following pushoff switchlights: [Refer to Figure 5.](#)

- RUD 1

- RUD 2
- ELEVATOR TRIM SHUTOFF (pilot/copilot).

Advisory Lights

The FCECU sends signals to turn on the following advisory lights: [Refer to Figure 6.](#)

- ROLL OUTBD (spoiler)
- ROLL INBD (spoiler).

Rudder Hinge Moment Limiting (HML)

The left and right FCECU channels supply rudder Hinge Moment Limiting (HML) commands to the lower and upper rudder PCUs. The HML commands are based on airspeed signals from the Air Data Units (ADUs) 1 and 2. As the airspeed signals change, the FCECU uses the HML commands to regulate the pressure in each rudder Power Control Unit (PCU), by controlling the pressure control valve in the PCU. As airspeed increases, the regulated PCU pressure decreases to reduce total rudder travel at higher airspeeds.

The FCECU also receives pressure signals from two pressure sensors located in each PCU. This lets the FCECU monitor the regulated PCU pressure for any failure modes. If a pressure failure occurs, the FCECU commands the rudder PCU shutoff valve to depressurize the PCU and the RUD 1 HYD or RUD 2 HYD caution lights come on. If No.1 and No.2 hydraulic systems fail the FCECU will cause the RUD CNTRL caution light to come on.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Pitch Trim Rate

The FCECU receives manual pitch trim input commands from either the pilot or copilot pitch trim switches or from the autopilot. The FCECU prioritizes the commands in the order that follows: pilot, copilot and then autopilot. The FCECU sends commands to the two pitch trim actuators located in the vertical stabilizer. The FCECU controls the pitch trim rate based on airspeed data received from the Air Data Units (ADUs). For airspeeds below 150 kt (278 km/h), the FCECU sets a pitch trim rate of 0.75 degrees/second. For airspeeds above 250 kt (463.3 km/h), the trim rate is set at 0.10 degrees/second. Between 150 and 250 kt (275 and 250 km/h), the FCECU ramps the pitch trim rate from 0.75 degrees down to 0.10 degrees/second.

Each pitch trim actuator has an LVDT which supplies feedback signals to the FCECU. The FCECU uses this feedback signal to monitor pitch trim actuator position, speed, and direction. If the FCECU senses a failure mode it sets the pitch trim rate to a default rate of either 0.10 degrees or 0.75 degrees/second and the applicable caution lights and pushoff switchlights come on.

The FCECU sends the feedback signal from the right pitch trim actuator LVDT to the Flight Data Recorder (FDR). The FCECU sends the feedback signal from the left pitch trim actuator LVDT to the FDR and to the pitch trim indicator located in the flight compartment.

Elevator Feel

The FCECU sends commands to the left and right pitch feel actuators located in the vertical stabilizer. The FCECU varies the pitch feel command signals based on ADU airspeed data and

Attitude Heading and Reference System (AHRS) acceleration data. The FCECU uses the airspeed data to control the amount of pitch feel actuator extension or retraction, and the acceleration data to control the speed of pitch feel actuator extension or retraction.

If the FCECU senses a fault in the airspeed or acceleration data, the FCECU commands the pitch feel actuators to hold in their last valid position and the ELEVATOR FEEL caution light comes on.

Spoiler Lift Dump Valves

The FCECU receives signals from the FLIGHT/TAXI switch and from the two engine power levers. If the switch is in the FLIGHT position and the position of the two power levers is less than FLIGHT IDLE +12 degrees, the FCECU sends the high side signal to the lift dump valves. When the aircraft lands, the PSEU receives the Weight On Wheels (WOW) signal and then sends the low side signal to the lift dump valves. The spoiler panels extend fully to the 75 degree position dumping the lift made by the wings. If the FLIGHT/TAXI switch is in the TAXI position or either of the two power levers is set greater than FLIGHT IDLE +12 degrees, the lift dump valves are not operational.

The FCECU sends a signal to turn on the ROLL INBD and ROLL OUTBD advisory lights if the spoiler panels extend beyond 65 degrees. If the spoiler panels fail to extend after landing, the ROLL SPLR INBD GND and ROLL SPLR OUTBD GND caution lights are turned on.

Outboard Spoiler Lockout

The FCECU uses the airspeed data from the ADUs to prevent the outboard spoilers from extending when the aircraft's airspeed is greater than 170 kt (315 km/h). When the aircraft reaches 170 kt



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

(3150km/h) airspeed, the FCECU closes an electrical path to the outboard spoiler unloading valve. The unloading valve then directs hydraulic system No.2 fluid pressure from the outboard spoiler PCUs to return, thus preventing the outboard spoiler panels from extending.

If the FCECU senses a fault in the airspeed data, it keeps the outboard spoilers operational and the SPLR OUTBD caution light turns on. The pilot is instructed to decrease the aircraft speed to no greater than 170 kt (315 km/h).

Aural Take-Off Warning

The FCECU supplies a signal to the Flight Guidance Module (FGM1) if a take-off is attempted when the aircraft is not in the proper take-off configuration. The FCECU sends a signal to the FGM1 if:

- the pointer of the pitch trim indicator is not in the take-off range.
- the left and right spoiler panels are extended.

When the FGM 1 receives the FCECU signal, it sounds the aural take-off warning.

If the FCECU fails to receive a pitch trim position signal, the aural take-off warning automatically sounds. If a spoiler LVDT fails to supply the position of a spoiler panel, the FCECU uses the LVDT data from the related panel on the opposite wing.

LVDT Excitation and Monitoring

The FCECU supplies the excitation voltage to activate the LVDTs on the aircraft. The LVDTs are installed on each flight control surface, the pitch feel actuators, the pitch trim actuators and the pressure sensors in the rudder PCU. The FCECU uses the LVDT data to

monitor the serviceability of each actuator and its position compared to the FCECU command signals. The FCECU also transmits the LVDT data to the Flight Data Recorder (FDR) and to the Powered Flight Control Surface Indicator (PFSCI).

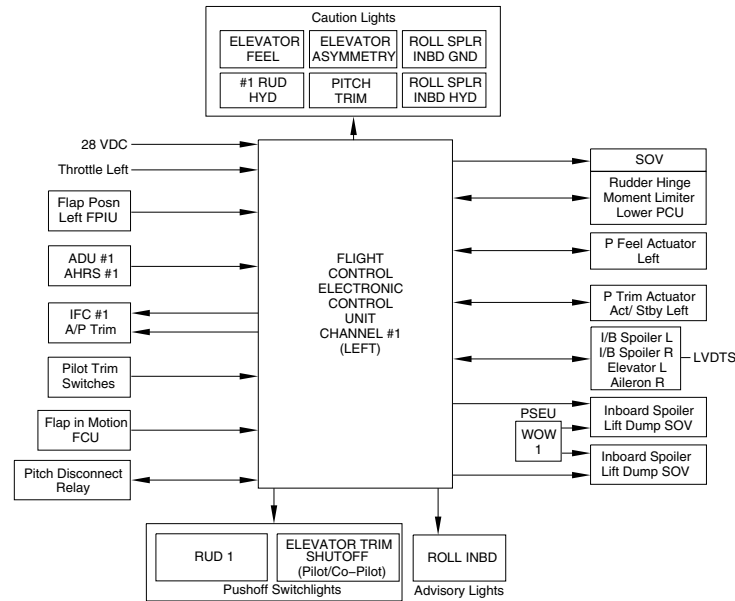
The LVDTs for the aileron and rudder trim actuators are not controlled by the FCECU.

Central Diagnostic System Interface

The FCECU supplies flight control system maintenance data to the (CDS) maintenance panel through the ARINC 429 data bus. The right channel of FCECU transmits all flight control system maintenance data to the CDS on three data pages (For software requirement data of FCECU refer to CQ406515-01 Rev.B). The first page shows the present system status. The second page shows the last ten faults detected. The third page shows the values of the FCECU inputs: the actuator positions, HML pressure and the position of the flight control surfaces.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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FCECU / LEFT CHANNEL BLOCK DIAGRAM
Figure 1

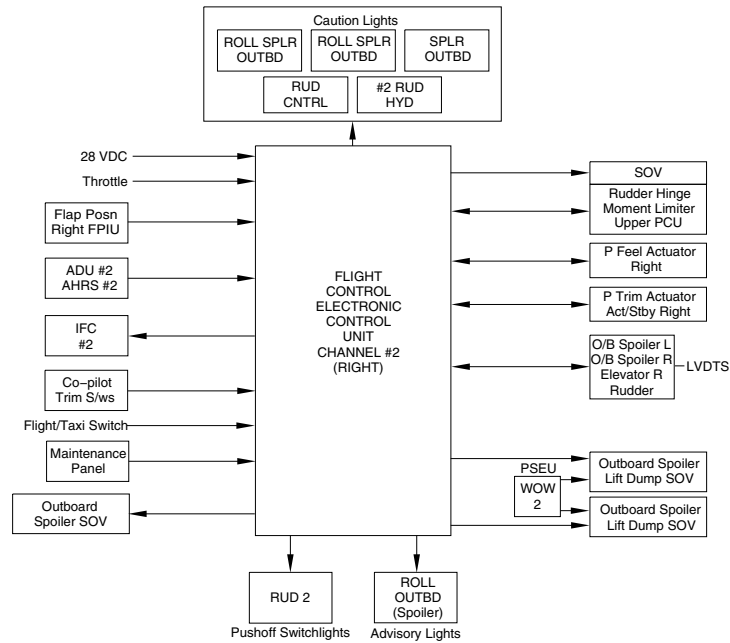
PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-00-01
Config 001

27-00-01

Config 001
Page 8
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



fs792a01.cgm

FCECU / RIGHT CHANNEL BLOCK DIAGRAM
Figure 2

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-00-01
Config 001

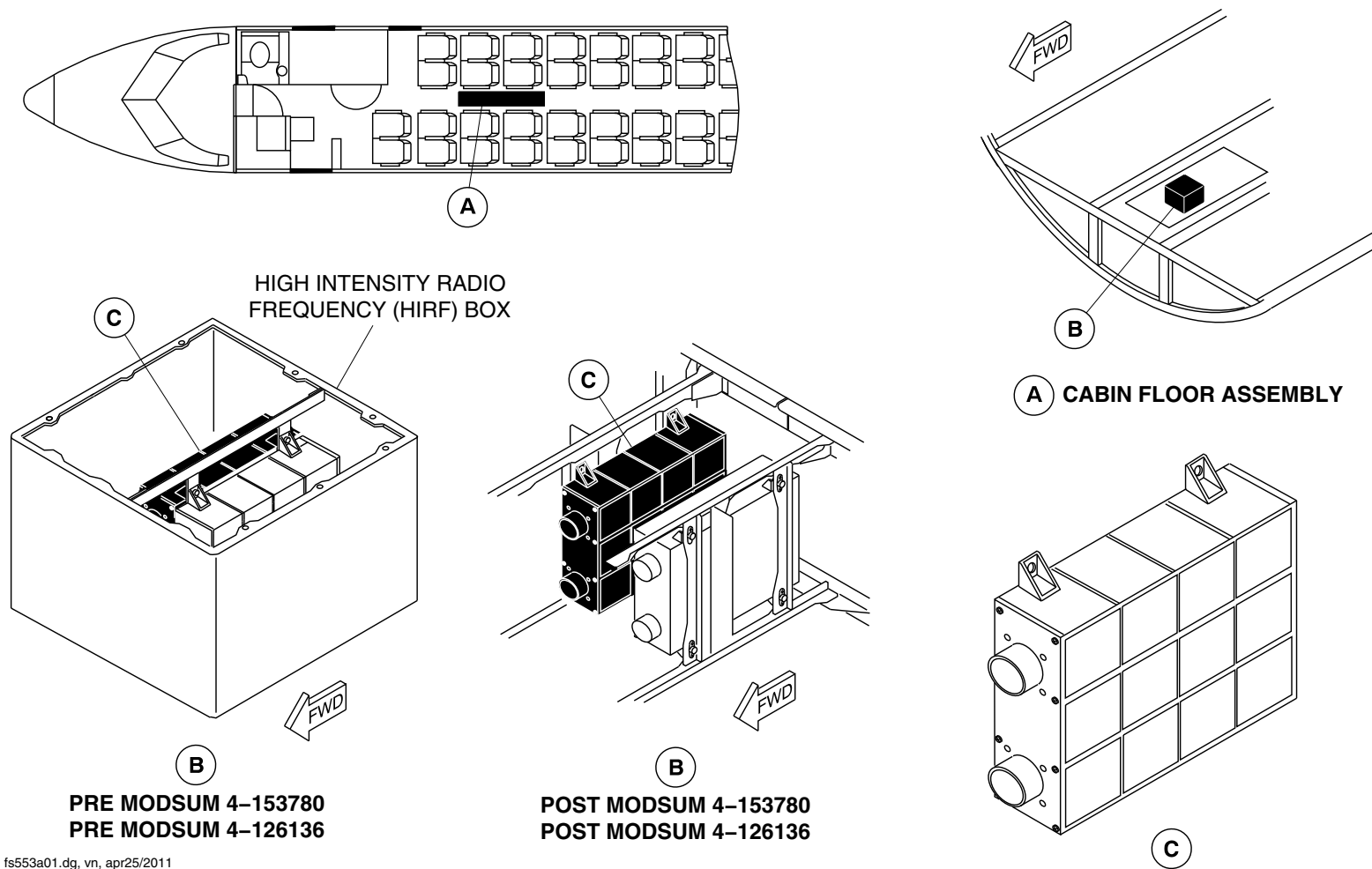
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Config 001
Page 9
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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Flight Control Electronic Control Unit (FCECU) Locator
Figure 3

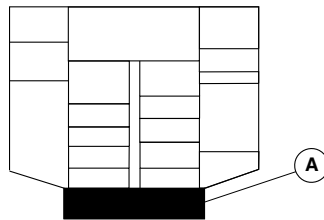
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Config 001
Page 10
Sep 05/2021



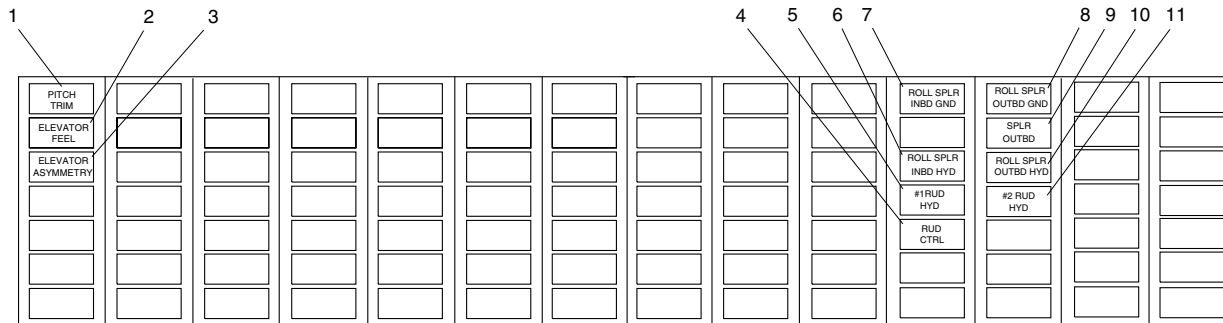
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



OVERHEAD CONSOLE

LEGEND

1. Pitch Trim (Amber).
2. Elevator Feel (Amber).
3. Elevator Asymmetry (Amber).
4. Rudder Control (Amber).
5. #1 Rudder Hydraulic Pressure (Amber).
6. Roll Spoiler Inboard Hydraulic (Amber).
7. Roll Spoiler Inboard Ground (Amber).
8. Roll Spoiler Outboard Ground (Amber).
9. Spoiler Outboard (Amber).
10. Roll Spoiler Outboard Hydraulic (Amber).
11. #2 Rudder Hydraulic Pressure (Amber).



A

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CAUTION AND WARNING PANEL
Figure 4

PSM 1-84-2A
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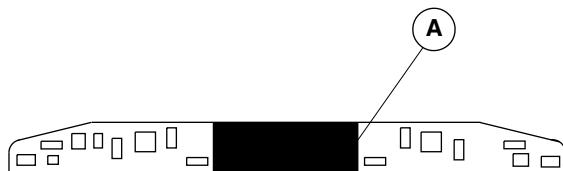
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Config 001
Page 11
Sep 05/2021



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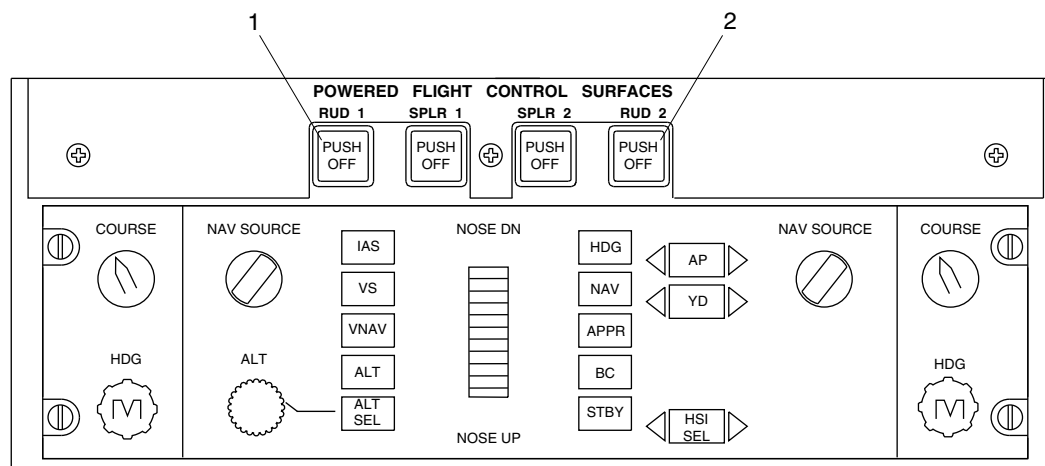
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GLARESHIELD PANEL

LEGEND

1. Rudder 1 Pushbutton Switchlight.
2. Rudder 2 Pushbutton Switchlight.



A

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RUDDER PUSHOFF SWITCHLIGHTS
Figure 5

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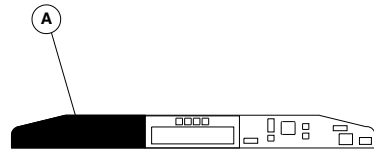
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Page 12
Sep 05/2021



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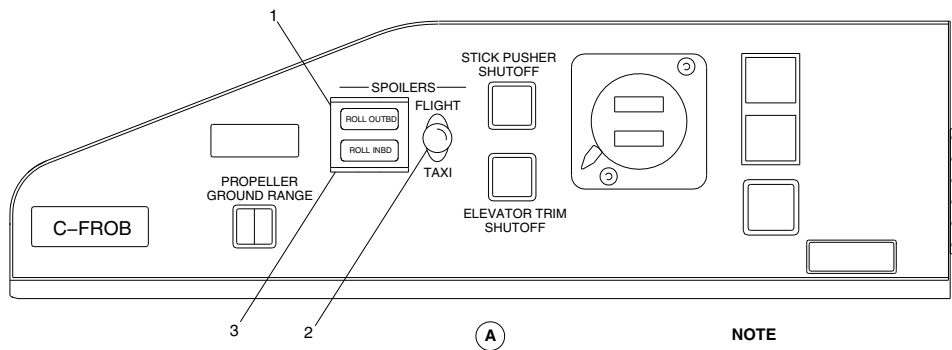
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GLARESHIELD PANEL

LEGEND

1. Roll Outboard Annunciator Light.
2. Flight/Taxi Switch.
3. Roll Inboard Annunciator Light.



NOTE

Pilot side shown.
Copilot side similar.

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ROLL SPOILER ADVISORY LIGHTS
Figure 6

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EFFECTIVITY:
See first effectivity on page 2 of 27-00-01
Config 001

27-00-01

Config 001
Page 13
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-00-01
Config 001

27-00-01

Config 001
Page 14
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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AILERON CONTROL SYSTEM

Introduction

The aileron control system is a primary flight control system. The ailerons, assisted by the spoilers, give roll control (lateral control) of the aircraft in flight.

General Description

Refer to Figures 1 and 2.

The ailerons give roll control of the aircraft in flight. The pilot or copilot supplies roll commands to the aileron control system by turning the handwheels. Roll input commands are mechanically transmitted from the pilot's and copilot's handwheels to the aileron surfaces through the aileron control system. The Automatic Flight Control System (AFCS) also gives roll commands, through the autopilot servo connected to the aileron forward quadrant located below the flight compartment floor.

The aileron control system is divided into these systems:

- Aileron System (27-12-00)
- Aileron Trim and Centering System (27-13-00)
- Aileron Surfaces (27-14-00)
- Roll Disconnect System (27-16-00).

Detailed Description

Refer to Figures 1 and 2.

The aileron system, assisted by the spoiler system, gives lateral control of the aircraft in flight. The pilot or copilot maintains lateral control of the aircraft in flight by turning the handwheels. The Automatic Flight Control System (AFCS) supplies lateral control through an autopilot servo which is connected to the aileron forward quadrant located below the flight compartment floor. The roll commands are mechanically transmitted from the pilot's and copilot's handwheels to the aileron surfaces through an aileron control circuit that includes cables, quadrants and pushrods.

Aileron System (27-12-00)

Refer to Figure 3.

The aileron surfaces are cable operated from the copilot's handwheel. The roll spoiler surfaces are cable operated from the pilot's handwheel (Refer to SDS 27-61-00 Roll Spoilers). An interconnect rod, located below the flight compartment floor, joins the aileron and roll control systems. This allows the aircraft to be flown from either the pilot's or the copilot's position. Roll commands from the pilot's and copilot's handwheels are transmitted simultaneously to the roll spoilers and ailerons through a series of cables, quadrants and pushrods. The roll disconnect system isolates the two handwheels from each other in the event of an aileron or roll spoiler jam (Refer to SDS 27-16-00 Roll Disconnect System).

The aileron autopilot servo is connected to the aileron forward quadrant by a cable and a clutched capstan. The Automatic Flight Control System (AFCS) controls the autopilot servo.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Ailerons Trim and Centering System (27–13–00)

Refer to Figures 4 and 6.

The Aileron Trim and Centering Unit (ATCU) has a torsion spring bar which is connected to the aileron trim actuator and to the aileron forward quadrant. The AILERON trim switch, located on the TRIM panel, controls the trim actuator and allows the pilot to trim the ailerons to a desired position. The aileron trim actuator moves the aileron surfaces ± 10 degrees of neutral. Input commands are transmitted from the aileron trim switch, through the trim and centering unit to the aileron forward quadrant to move both aileron control surfaces. The aileron trim position is shown on the aileron trim indicator located on the TRIM panel in the center console. The torsion spring bar keeps the handwheels centered about this trimmed position to reduce pilot workload.

The AILERON trim switch has three positions:

- LWD (left wing down)
- RWD (right wing down)
- Center–off.

The switch is spring–loaded to return to the center–off position.

The pilot controls the amount of aileron trim by operating the aileron trim switch, which supplies the electrical signal to the aileron trim actuator. Selection of the LWD position moves the left aileron up and the right aileron down. Selection of the RWD position moves the left aileron down and the right aileron up.

If the aileron trim switch fails in the closed position, a limit switch on the trim actuator opens. This shuts off the electrical power to the trim actuator causing it to stop at the maximum trim input. If the limit

switch fails to stop the trim actuator, a mechanical stop on the trim actuator stops it at the maximum trim setting.

When the autopilot is engaged and aileron trim is required, MISTRIM [TRIM L WING DN] or MISTRIM [TRIM R WING DN] is shown on the Primary Flight Display (PFD). The autopilot is disengaged and the handwheel is turned until the trim message goes out. The pilot then sets the required aileron trim with the AILERON trim switch until the control forces are removed from the handwheel. The handwheel then stays centered around the newly trimmed position.

The aileron trim indicator, located on the center console shows the aileron trim position. The aileron trim indicator receives its information from a Linear Variable Differential Transformer (LVDT) which is located inside the aileron trim actuator. The LVDT also transmits aileron trim position data to the Flight Data Recorder (FDR).

Aileron Surfaces (27–14–00)

Refer to Figures 5 , and #####.

The aileron surfaces are mass and aerodynamically balanced. They are attached to the wing rear spar by fail–safe hinges.

Each aileron has a geared tab. When the aileron control surface moves, its geared tab moves in the opposite direction. This action supplies aerodynamic assistance to aileron movement by reducing the force required to turn the pilot's and copilot's handwheels.

The right aileron has a ground adjustable trim tab to adjust for the individual flight characteristics of each aircraft. This tab is adjusted by maintenance personnel when required.

A gust lock system protects the ailerons from wind damage when the aircraft is parked or being taxied. The gust lock system locks the



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

ailerons in the neutral position (Refer to SDS 27-71-00 Aileron Gust Lock System).

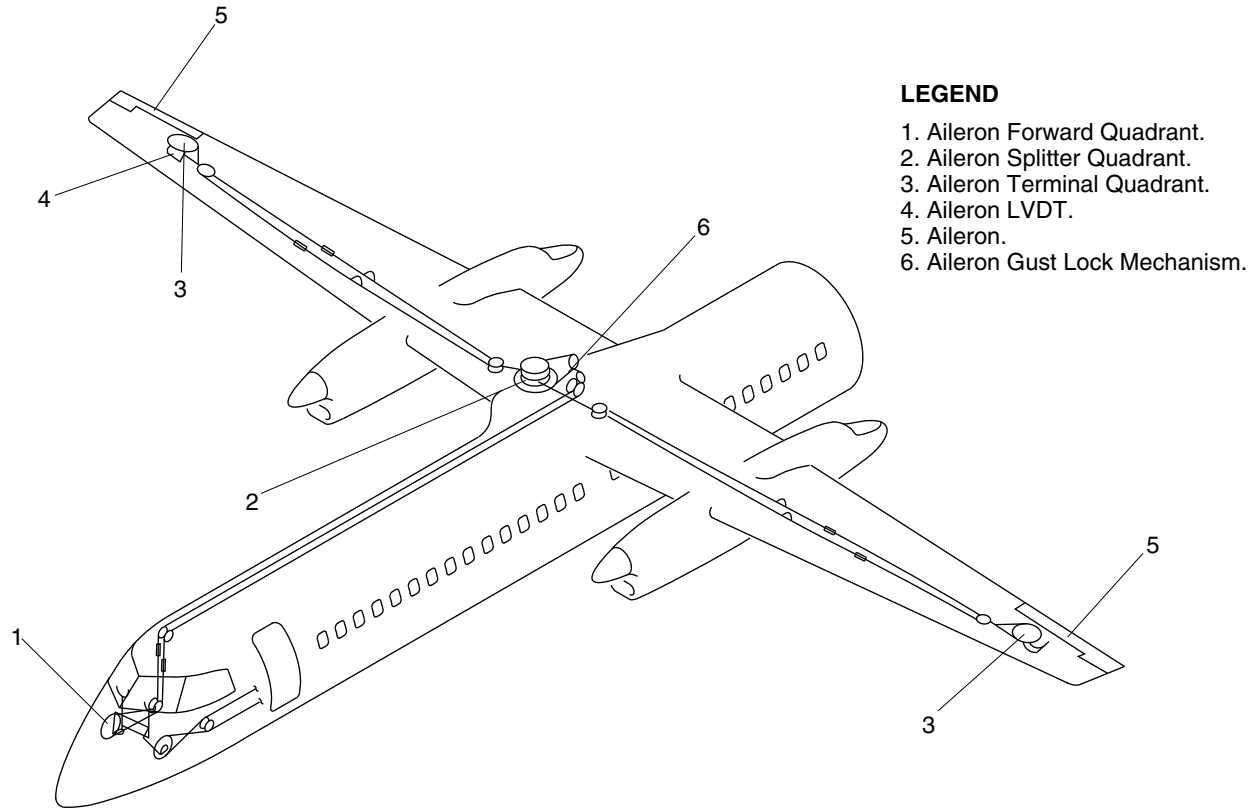
Roll Disconnect System (27-16-00)

An interconnect rod located under the flight compartment floor joins the aileron and the roll spoiler systems. This allows the ailerons and spoilers to be operated from either the pilot's or copilot's handwheel. A roll disconnect handle located on the right side of the center console connects to the roll disconnect mechanism. If a roll jam occurs in either system, the handle is pulled out to the limit and turned through 90 degrees counterclockwise. This isolates the jammed system from the operating system. The unjammed handwheel is now used for roll control of the aircraft (Refer to SDS 27-16-00 Roll Disconnect System).



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Aileron Forward Quadrant.
- 2. Aileron Splitter Quadrant.
- 3. Aileron Terminal Quadrant.
- 4. Aileron LVDT.
- 5. Aileron.
- 6. Aileron Gust Lock Mechanism.

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AILERON CONTROL SYSTEM PAGE 1
Figure 1

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-10-00
Config 001

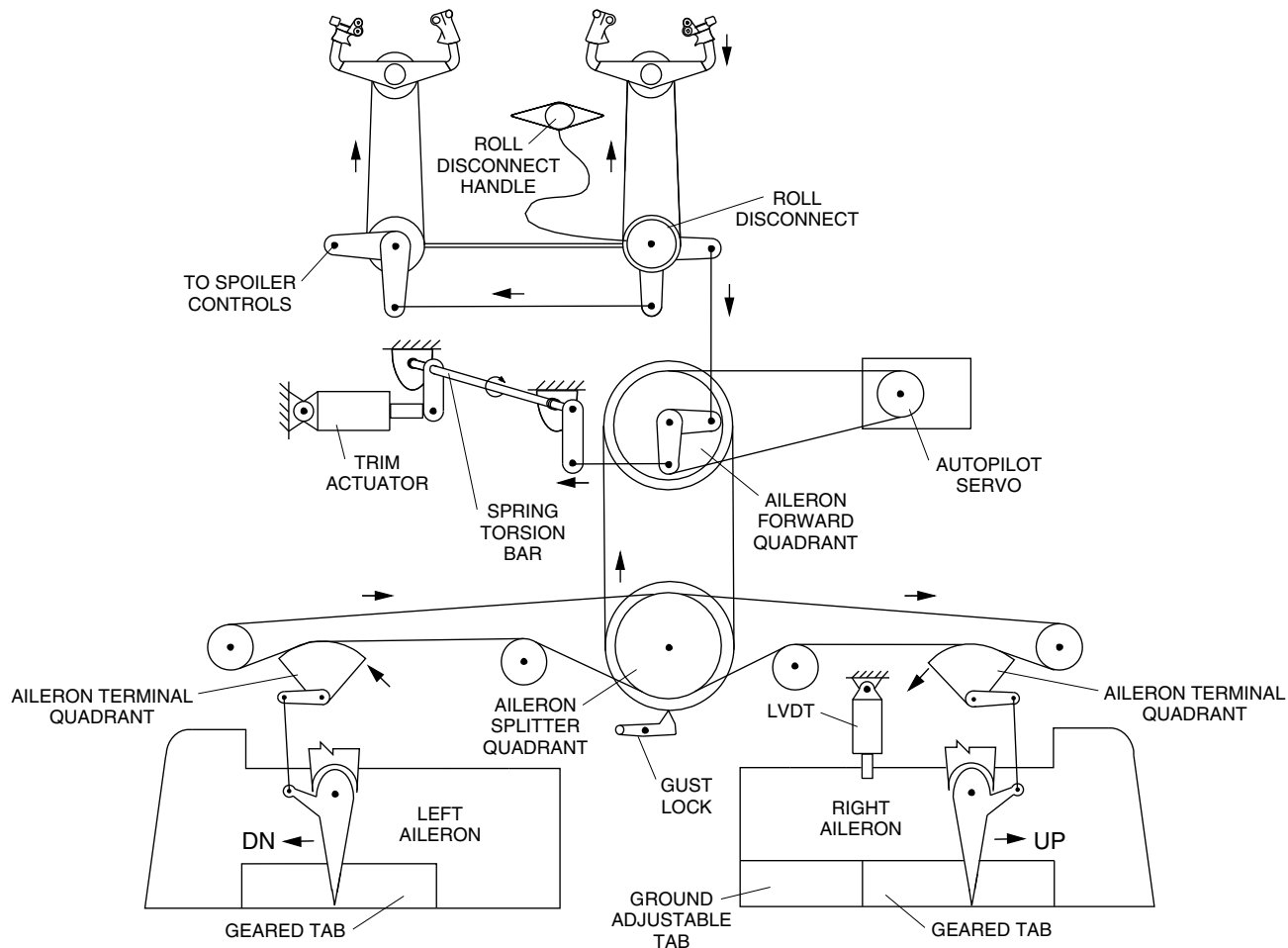
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Config 001
Page 5
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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AILERON CONTROL SYSTEM PAGE 2
Figure 2

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-10-00
Config 001

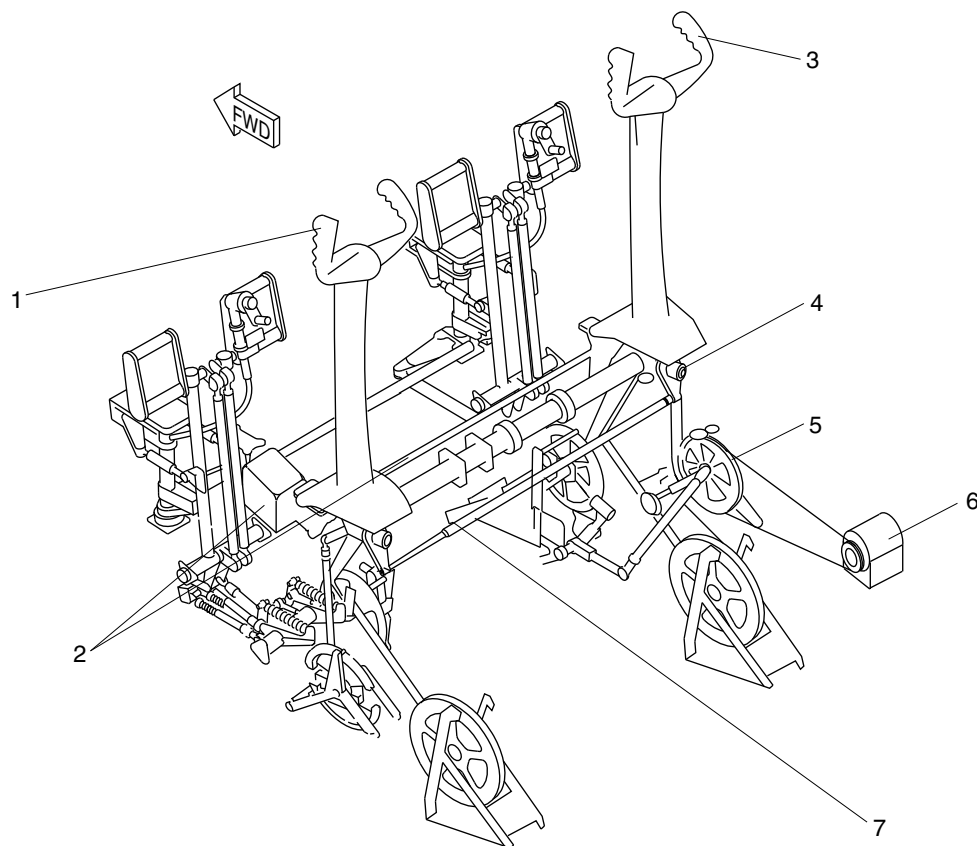
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Config 001
Page 6
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Pilot Handwheel.
2. Aileron Trim and Centering Unit.
3. Copilot Handwheel.
4. Roll Disconnect Clutch.
5. Aileron Forward Quadrant.
6. Aileron Autopilot Servo.
7. Interconnect Rod.

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FLIGHT CONTROLS / FLIGHT COMPARTMENT
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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Config 001

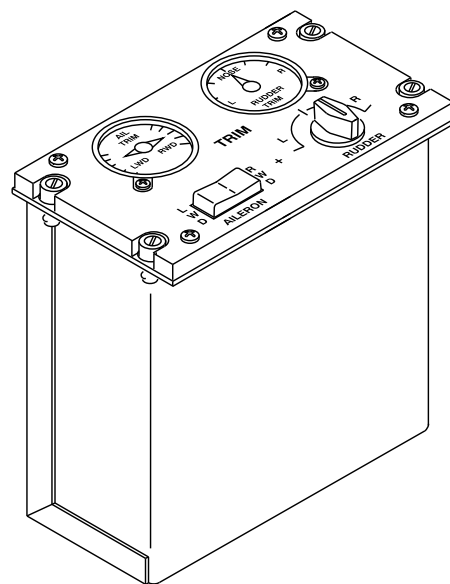
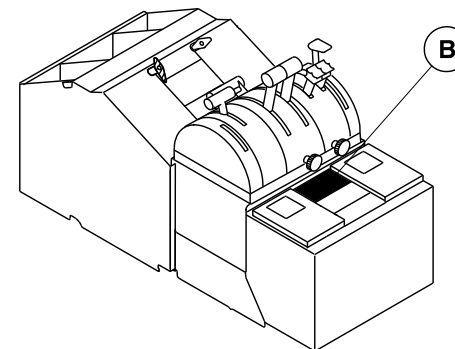
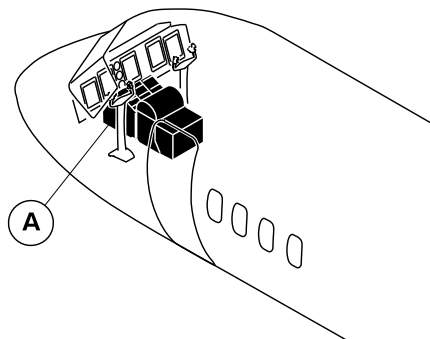
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Config 001
Page 7
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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Aileron Trim Switch and Indicator Locator
Figure 4

PSM 1-84-2A
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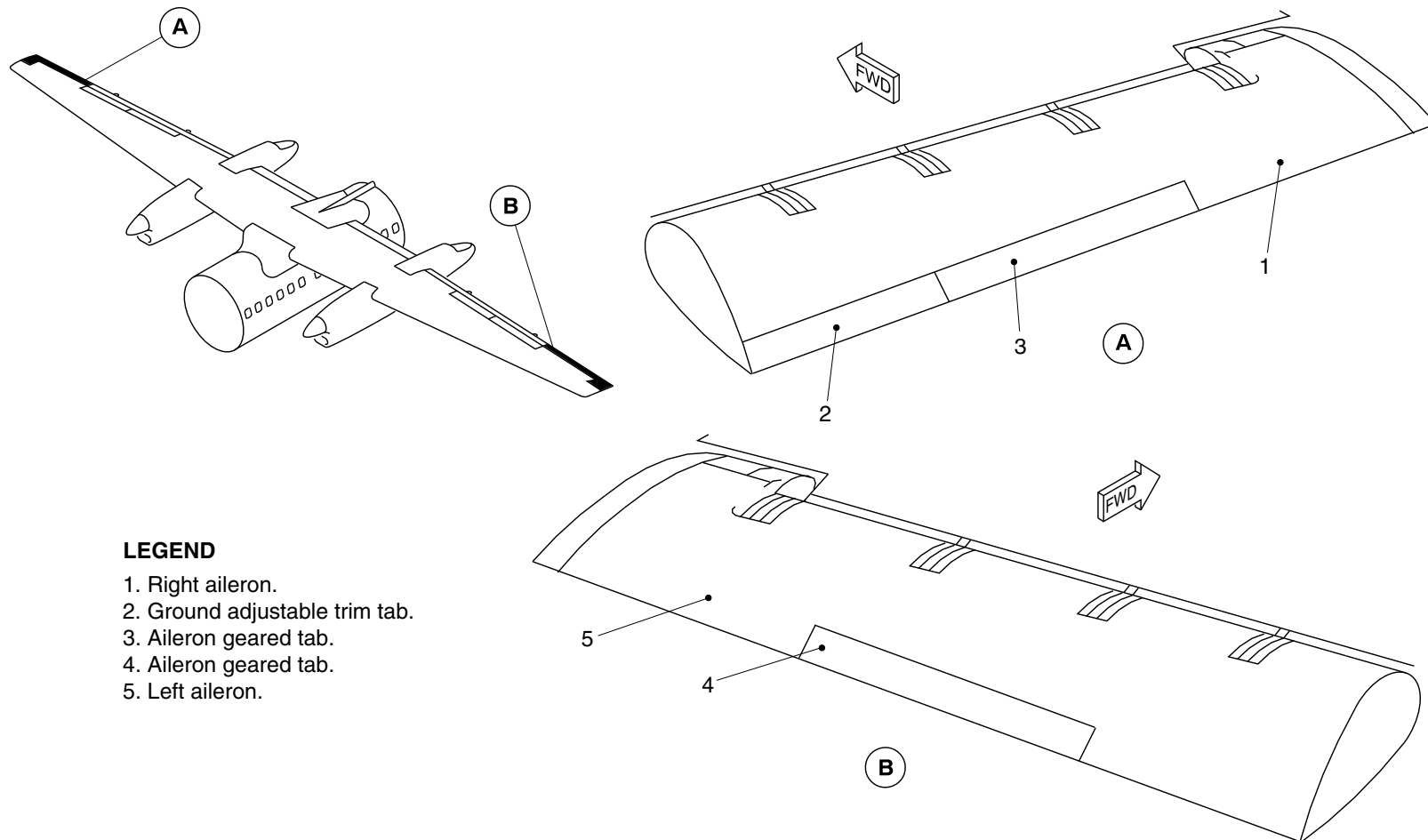
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Config 001
Page 8
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Right aileron.
- 2. Ground adjustable trim tab.
- 3. Aileron geared tab.
- 4. Aileron geared tab.
- 5. Left aileron.

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Aileron Surfaces Compartment Index
Figure 5

PSM 1-84-2A
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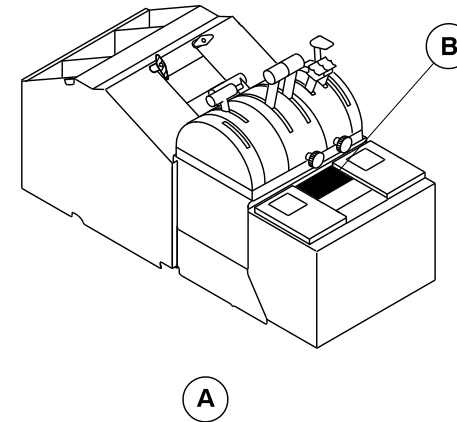
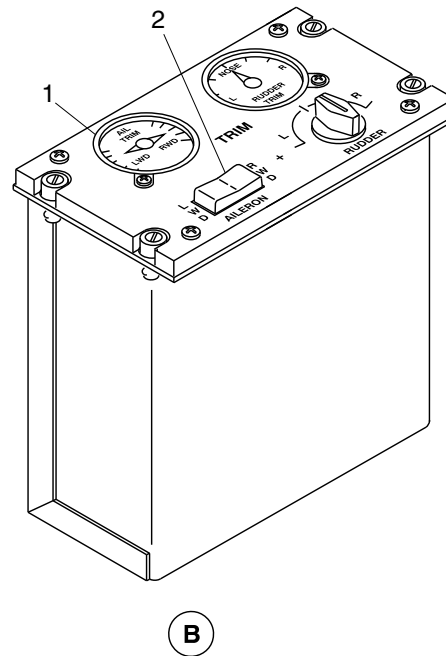
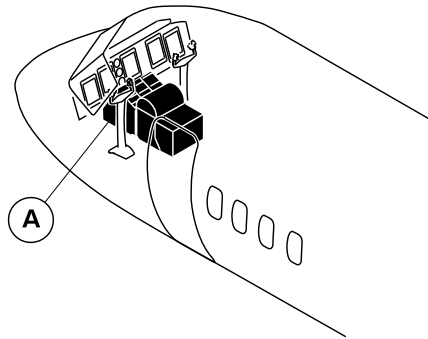
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Config 001
Page 9
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Aileron trim position indicator.
- 2. Aileron trim switch.

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Aileron Trim Switch and Indicator Detail
Figure 6

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-10-00
Config 001

27-10-00

Config 001
Page 10
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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AILERONS SYSTEM

Introduction

The aileron system transmits the roll commands from either the autopilot or the pilot's and copilot's control handwheels to the aileron control surfaces.

General Description

The aileron system transmits the roll commands from the pilot's and copilot's control handwheels to the aileron surfaces through a series of quadrants, cables, levers and pushrods. Movement of the control handwheels causes the aileron surfaces to deflect asymmetrically and in proportion to control handwheel rotation. The aileron system also has inputs from the Auto Flight Control System (AFCS) and the Aileron Trim and Centering Unit.

The aileron system has the components that follow:

- Quadrant, Aileron Forward (27-12-01)
- Quadrant, Aileron Splitter (27-12-11)
- Quadrant, Aileron Terminal (27-12-06)
- Tab, Aileron Geared (27-12-16)
- LVDT, Aileron Position (27-12-21).

Detailed Description

Rotary movement of the copilot's control handwheel is transmitted by a sprocket and chain mechanism to a sprocket and lever assembly at the base of the copilot's control column. A pushrod connects the lever to the aileron forward quadrant below the flight compartment floor.

On aircraft with ModSum 4-309210 OR SB84-31-33 OR SB84-31-39 OR SB84-31-40 incorporated ([Refer to Figure 1.](#)): The pushrod is replaced with a force sensor. The aileron force sensor is installed between the right control column and the aileron quadrant. The aileron quadrant is installed below the floor on the right side of the forward fuselage. The sensor supplies a signal in proportion to the force applied to the handwheel, which is recorded by the Flight Data Recorder.

The sprocket and lever assembly at the base of the copilot's control column has a clutch which is used for roll disconnect function. A roll disconnect (ROLL DISC) handle installed at the forward centre console operates the clutch. When the handle is in the engaged position, the clutch is engaged (refer to 27-16-00 Roll Disconnect System). This connects the aileron circuit to the spoiler circuit to give synchronized movement of the handwheels.

The Aileron Trim and Centering Unit has a trim actuator and a torsion spring bar connected to the aileron forward quadrant by levers and a pushrod. The autopilot servo is also connected to the aileron forward quadrant by a cable and a clutched capstan.

Aileron control cables are routed aft from the aileron forward quadrant, up the forward face of the flight compartment bulkhead and aft along the cabin roof. The cables then pass through pressure seals in the cabin roof to terminate at the aileron splitter quadrant on the aft face of the wing rear spar. Left and right aileron control wing

PSM 1-84-2A

EFFECTIVITY:

See first effectivity on page 2 of 27-12-00

Config 001

27-12-00

Config 001

Page 2

Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

cables are routed outboard from the aileron splitter quadrant, one cable along each wing. Each aileron control wing cable then terminates at an aileron terminal quadrant located on the aft face of the wing rear spar. Aileron terminal quadrant movement is transmitted to the aileron by a pushrod.

Quadrant, Aileron Forward

Refer to Figures 2 and 3.

The aileron forward quadrant transmits inputs from the copilot's control handwheel to the aileron splitter quadrant. The quadrant also has flight control input from the autopilot and the Aileron Trim and Centering Unit.

The aileron forward quadrant is installed in the right side of the lower forward fuselage, below the flight compartment floor. The quadrant has two pulleys with travel stops and a lever assembly all installed on a single shaft. Two mounting brackets attach the aileron forward quadrant to the lower forward fuselage. A pushrod connects the lever at the base of the copilot's control column to the lever on the aileron forward quadrant.

Quadrant, Aileron Splitter

Refer to Figures 4 and 5.

The aileron splitter quadrant transmits aileron cable inputs from the copilot's control handwheel to the aileron terminal quadrants located in each wing.

The splitter quadrant is installed on the aft face of the wing rear spar. The quadrant has three pulleys installed on a common shaft. The

center pulley has a removable fitting with two cable retaining holes. The holes for the top and bottom pulleys are inset.

Gust locking of the ailerons is supplied by an internal spring-loaded lever attached off-center to the bottom of the splitter quadrant. When the gust lock is engaged a plunger is inserted in the gust lock plate at the base of the splitter quadrant, locking the ailerons in neutral.

Quadrant, Aileron Terminal

Refer to Figures 6 and 7.

There are two aileron terminal quadrants, one in each wing. Each aileron terminal quadrant is located on the wing rear spar adjacent to its related aileron.

The aileron terminal quadrant is the last quadrant in the control linkage from the copilot's control column to the aileron. The terminal quadrant transmits aileron cable control from the aileron splitter quadrant to the aileron itself. The quadrant is connected to the aileron by a lever and a control rod. For rigging procedures, a 0.25 in. (6.35 mm) rig pin hole is used to lock the quadrant in neutral.

Tab, Aileron Geared

Refer to Figures 8 and 9.

Each aileron has a geared tab, which gives aerodynamic assistance to movement of the aileron control surface.

The aileron geared tab is attached by four hinge fittings to the trailing edge of each aileron. The tab is connected to the aileron by a geared linkage so that the geared tab moves 70% of the total aileron movement. At full aileron deflection of 17 degrees, the aileron geared tab deflects 11.9 degrees in the opposite direction. This action gives



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aerodynamic assistance and reduces the force required to turn the control hand–wheels.

LVDT, Aileron Position

[Refer to Figure 10.](#)

A Linear Variable Differential Transformer (LVDT) monitors the position of the ailerons by supplying an electrical signal equivalent to the angular position of the flight control surface.

A single LVDT is located in the right wing adjacent to the aileron. It monitors the position of the right aileron surface then transmits the aileron position data to the Flight Data Recorder (FDR).

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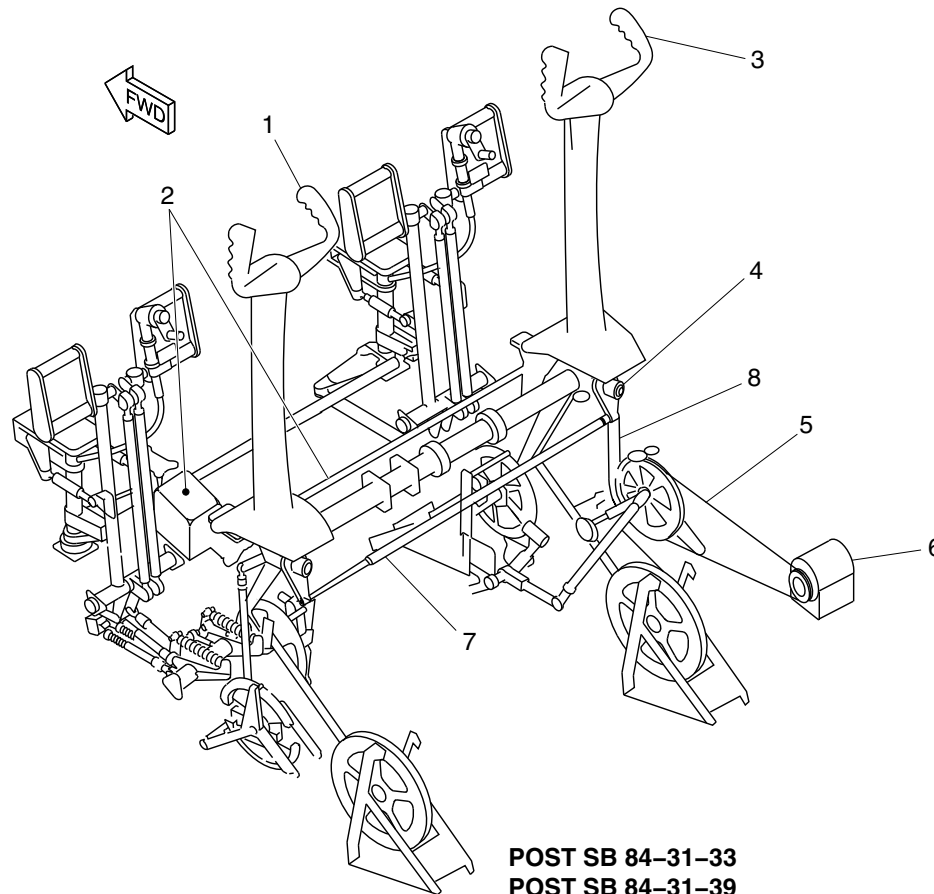
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Config 001
Page 4
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Pilot handwheel.
2. Aileron trim and centering unit.
3. Copilot handwheel.
4. Roll disconnect clutch.
5. Aileron forward quadrant.
6. Aileron autopilot servo.
7. Interconnect rod.
8. Force sensor.

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POST SB 84-31-39
POST SB 84-31-40
POST MODSUM 4-309210

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Flight Controls / Aileron System
Figure 1

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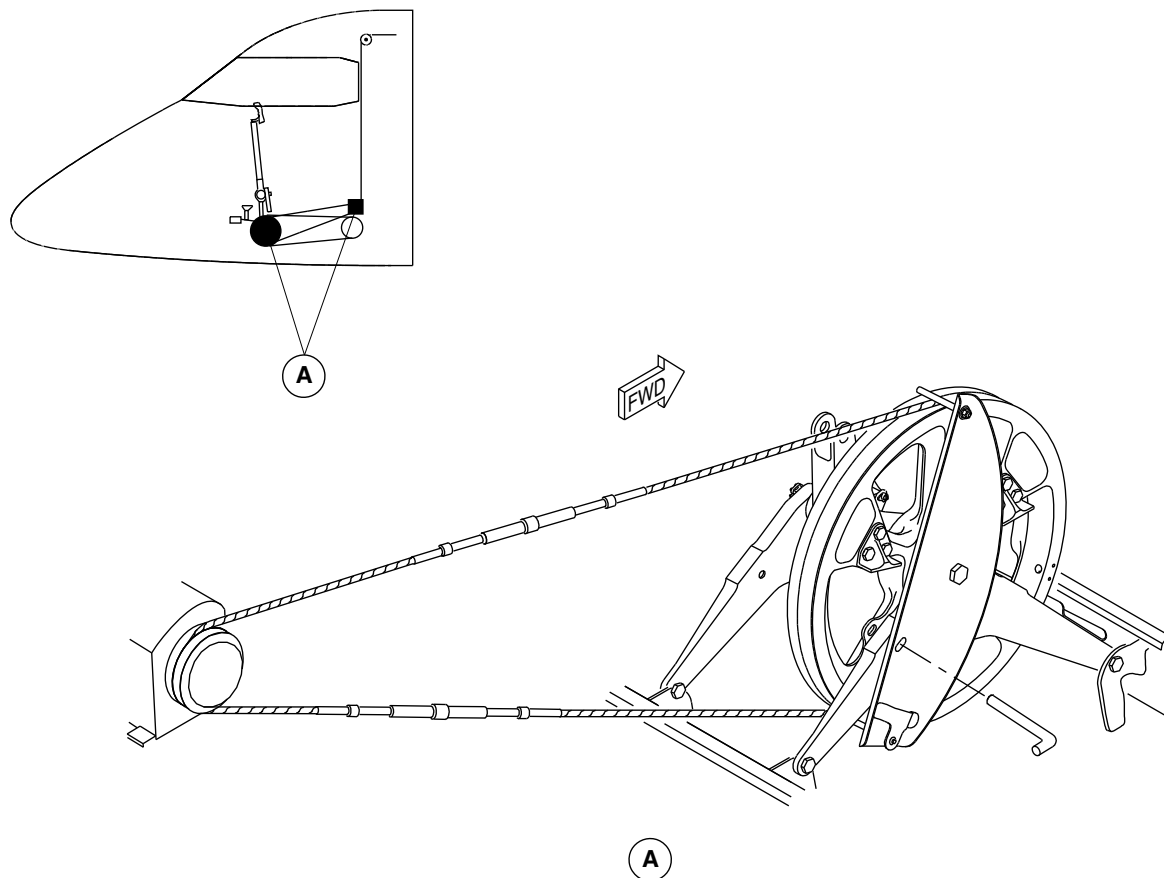
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Config 001
Page 5
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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AILERON FORWARD QUADRANT PAGE 1
Figure 2

PSM 1-84-2A
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See first effectivity on page 2 of 27-12-00
Config 001

27-12-00

Config 001
Page 6
Sep 05/2021

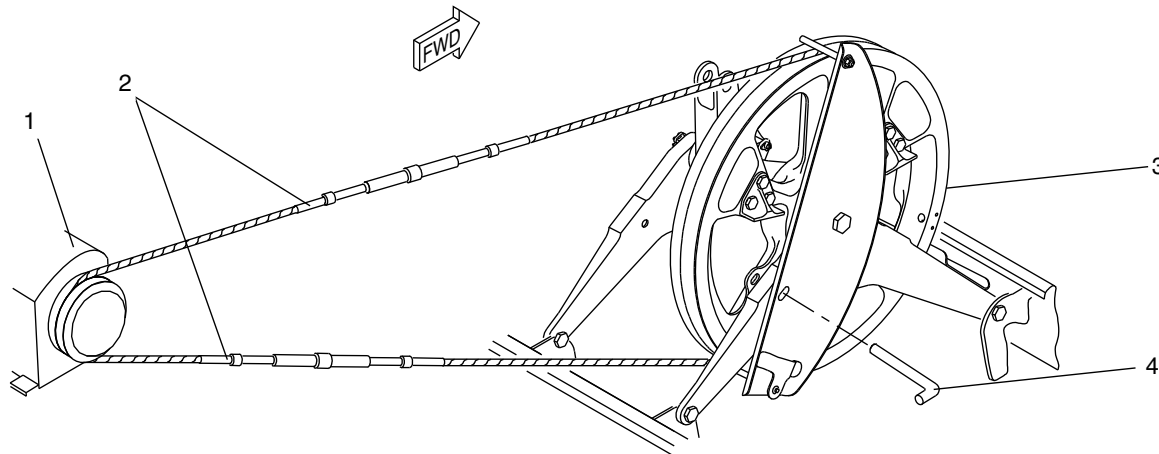


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LEGEND

1. Aileron Autopilot Servo.
2. Cables.
3. Aileron Forward Quadrant.
4. Rig Pin.



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AILERON FORWARD QUADRANT PAGE 2
Figure 3

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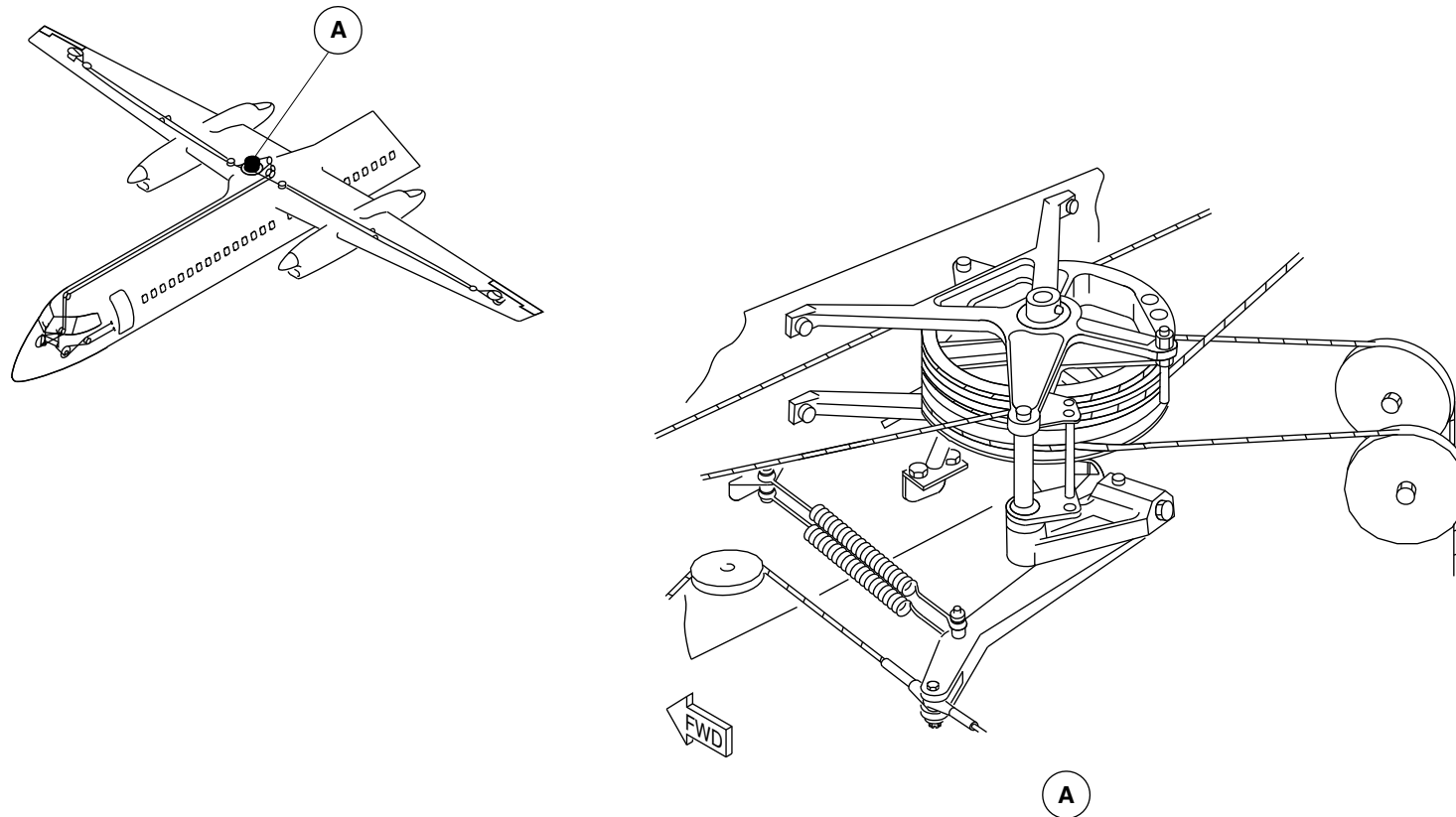
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Config 001
Page 7
Sep 05/2021



DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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AILERON SPLITTER QUADRANT PAGE 1
Figure 4

PSM 1-84-2A
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See first effectivity on page 2 of 27-12-00
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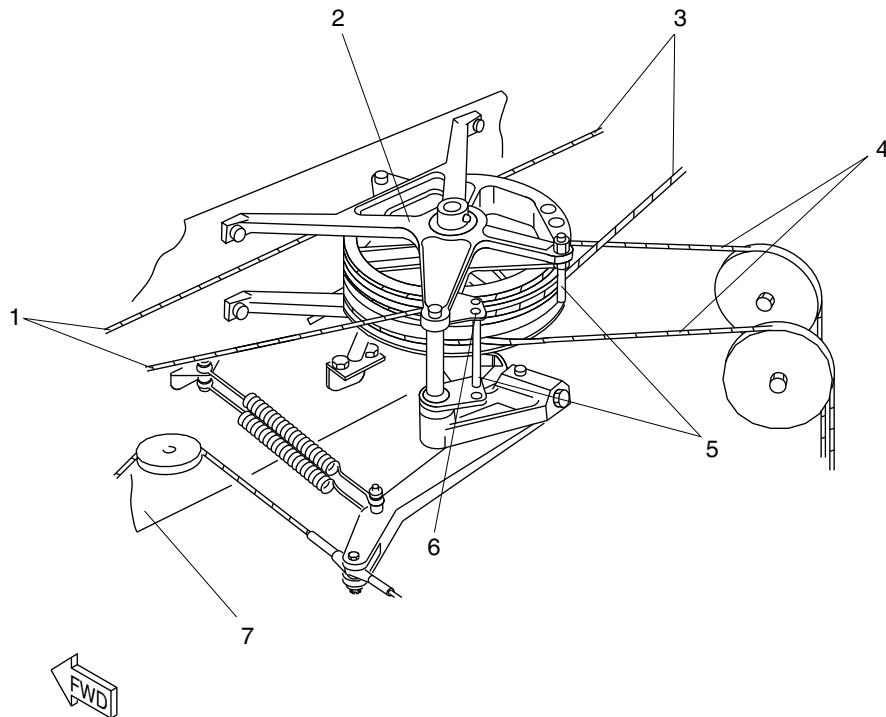
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Config 001
Page 8
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. LH Aileron Control Cables.
- 2. Aileron Splitter Quadrant.
- 3. RH Aileron Control Cables.
- 4. Quadrant Input Cables.
- 5. Cable Guards.
- 6. Rig Pin Hole.
- 7. Wing Rear Spar.

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AILERON SPLITTER QUADRANT PAGE 2
Figure 5

PSM 1-84-2A
EFFECTIVITY:
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Config 001

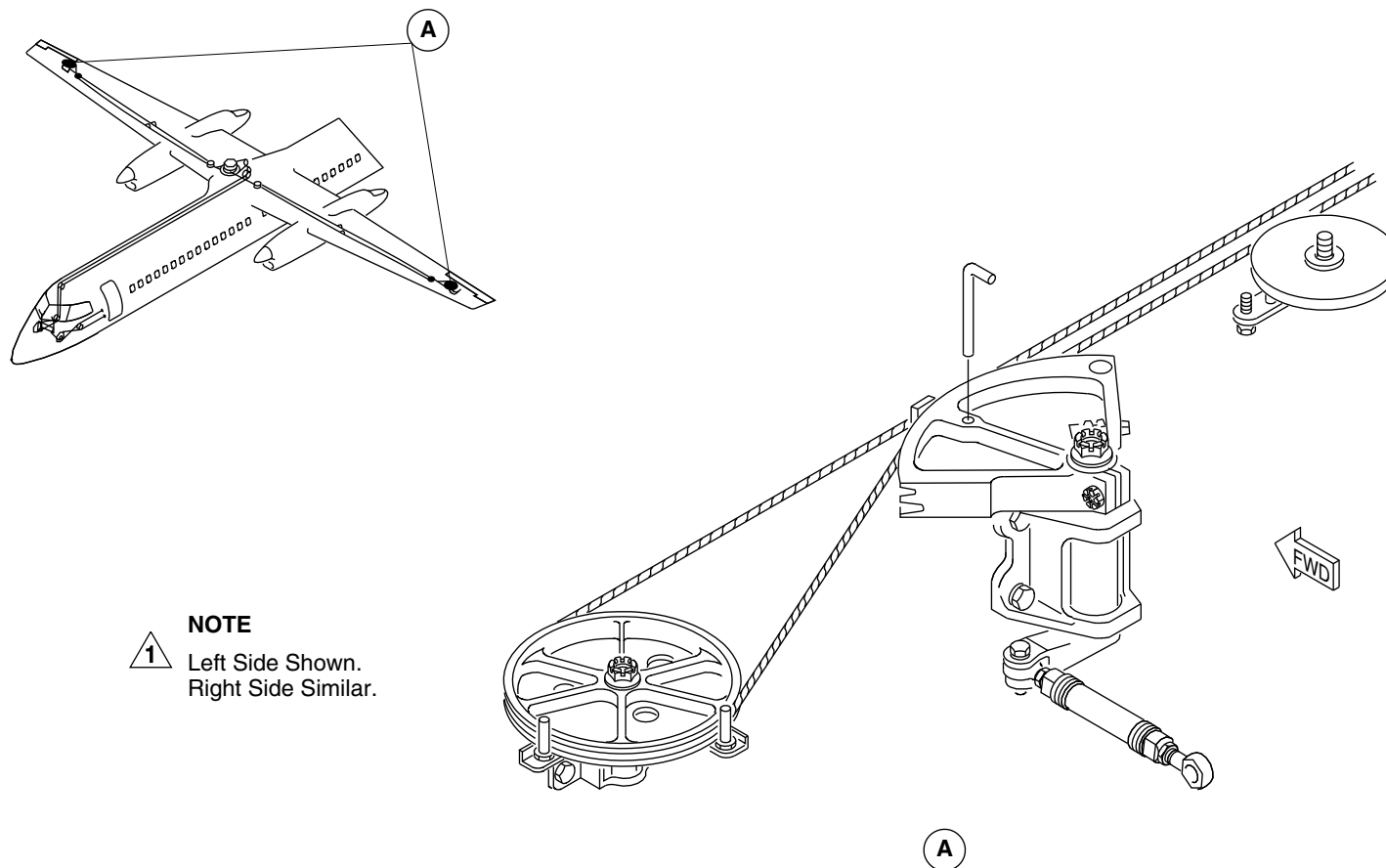
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Config 001
Page 9
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



NOTE
1 Left Side Shown.
Right Side Similar.

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AILERON TERMINAL QUADRANT PAGE 1
Figure 6

PSM 1-84-2A
EFFECTIVITY:
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Config 001

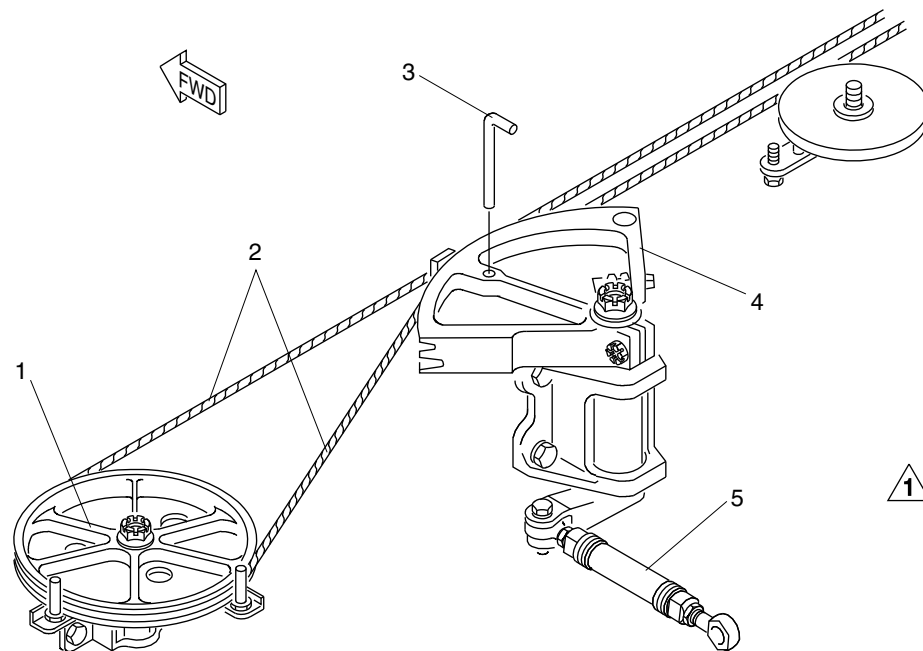
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Config 001
Page 10
Sep 05/2021



DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Pulley.
- 2. Cables.
- 3. Rig Pin.
- 4. Terminal Quadrant.
- 5. Rod.

NOTE

- 1 Left Side Shown.
Right Side Similar.

fs163a01.cgm

AILERON TERMINAL QUADRANT PAGE 2
Figure 7

PSM 1-84-2A
EFFECTIVITY:
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Config 001

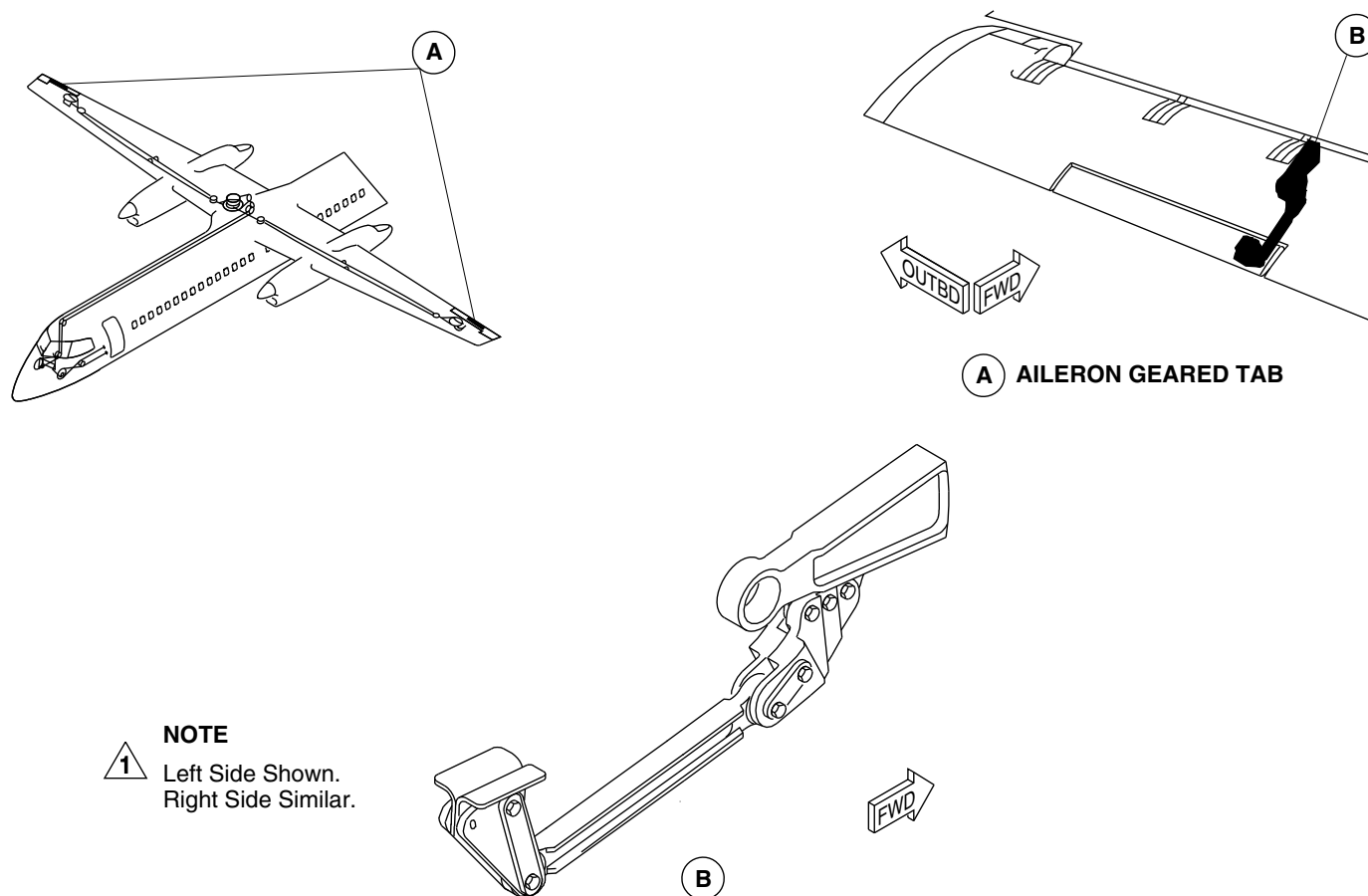
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Config 001
Page 11
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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AILERON GEARED TAB PAGE 1
Figure 8

PSM 1-84-2A
EFFECTIVITY:
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Config 001

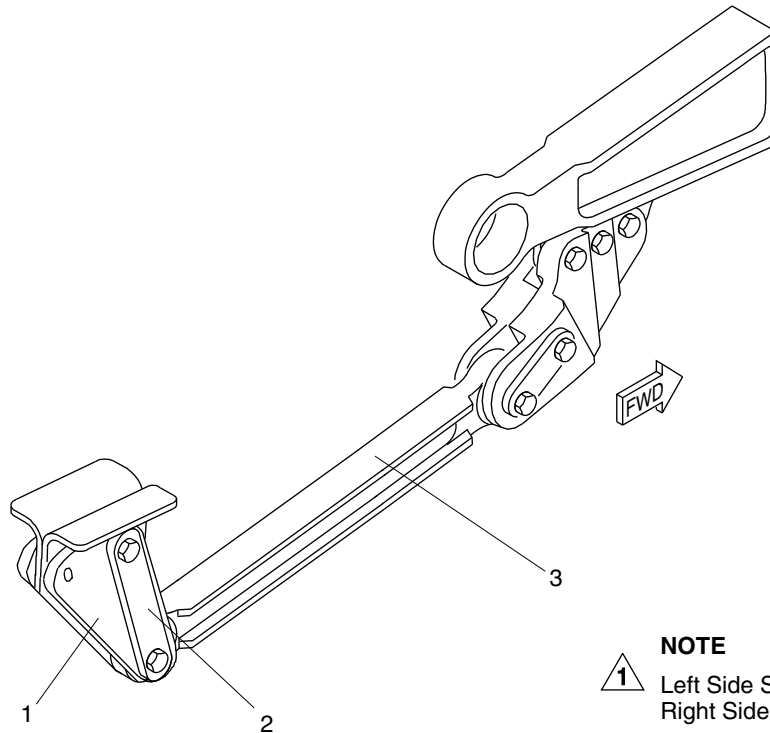
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Config 001
Page 12
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Geared Tab Bracket.
- 2. Geared Tab Sleeve Assembly.
- 3. Geared Tab Link Arm.

NOTE

- 1 Left Side Shown.
Right Side Similar.

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AILERON GEARED TAB PAGE 2
Figure 9

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-12-00
Config 001

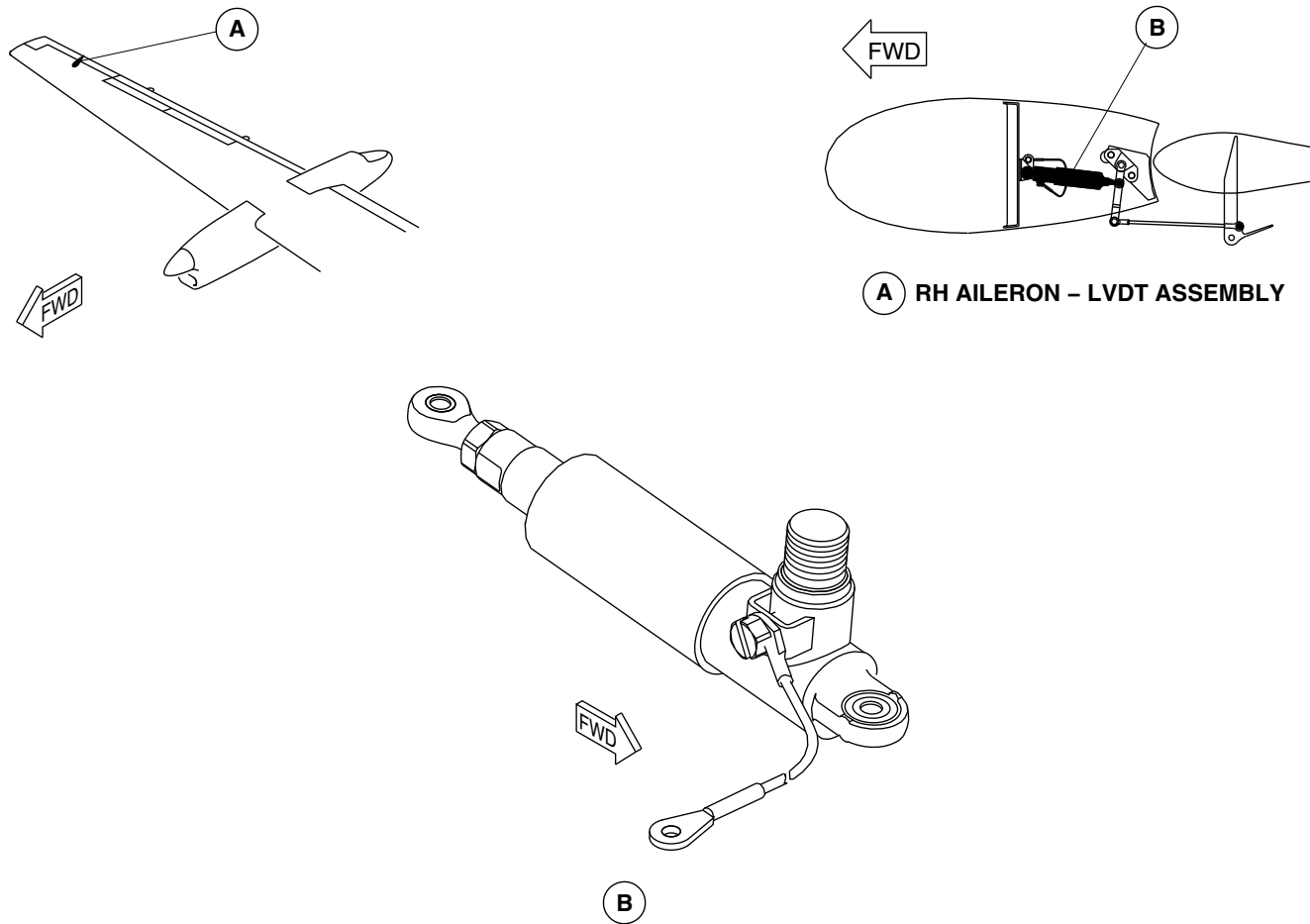
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Config 001
Page 13
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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AILERON LINEAR VARIABLE DIFFERENTIAL TRANSFORMER (LVDT)
Figure 10

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-12-00
Config 001

27-12-00

Config 001
Page 14
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

27-13-00-001

AILERONS TRIM AND CENTERING SYSTEM

Introduction

Trimming the aileron control system gives the aircraft lateral stability during flight. Aileron trim counters the effects of lateral imbalance. Lateral imbalance may be due to uneven fuel distribution, asymmetric power, or any other forces that affect the lateral axis of the aircraft.

General Description

Aileron trim is achieved by electrically trimming both aileron surfaces through the actuator located in the Aileron Trim and Centering Unit. The trim and centering unit is connected to the aileron forward quadrant by a spring torsion bar that makes sure the aileron surfaces always return to a neutral position.

The aileron trim and centering system has the components that follow:

- Actuator, Trim (27-13-01)
- Unit, Centering (27-13-06)
- Tab, Trim-Ground Adjustable (27-13-11).

Detailed Description

The aileron trim and centering system transfers input commands from the aileron trim actuator to the aileron forward quadrant. The forward quadrant rotates and transfers the trim command to the

ailerons through the aileron control circuit. Aileron trim is achieved by deflecting both aileron control surfaces. The amount of aileron trim is controlled by the AILERON trim switch on the TRIM panel located in the flight compartment on the center console. The amount of trim is shown on the AIL TRIM LWD, RWD indicator located on the aileron and rudder TRIM control panel.

The aileron trim and centering system has a spring torsion bar attached to the aileron forward quadrant by a lever and a pushrod. Movement of the CONTROL handwheels causes rotation of the aileron forward quadrant and twisting of the spring torsion bar. The twist in the spring torsion bar makes sure that the ailerons return to the neutral position after any control handwheel rotation.

Actuator, Trim

[Refer to Figure 1.](#)

The trim actuator is a part of the Aileron Trim and Centering Unit. The AILERON trim switch located on the aileron and rudder TRIM control panel gives electrical control of the aileron trim actuator.

The aileron trim switch has three positions:

- LWD (left wing down)
- RWD (right wing down)
- Center-off.

The switch is spring-loaded to return to the center-off position to reduce the possibility of accidental operation.

[Refer to Figures 2 and 3.](#)

Selection of the LWD position extends the aileron trim actuator and turns the aileron forward quadrant, causing both aileron surfaces to



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

move differentially. Selection of the RWD position reverses the process, causing the ailerons to move in the opposite direction.

The pilot controls the amount of aileron trim by operating the AILERON trim switch which supplies the electrical signal to the aileron trim actuator. If the AILERON trim switch fails to the closed position, a limit switch on the trim actuator housing opens the electrical circuit and causes the trim actuator to stop at the maximum trim input. If the limit switch fails to stop the trim actuator, a mechanical stop on the trim actuator housing stops the trim actuator at the maximum trim setting.

When the autopilot is engaged, a message MISTRIM (TRIM L WING DN) or (TRIM R WING DN) is shown on the Primary Flight Display if aileron trim is required. The autopilot is disengaged and the control handwheel is turned until the message goes out. The pilot then applies trim using the AILERON trim switch until the control forces are removed from the handwheel.

An aileron trim indicator, located on the TRIM panel in the center console shows aileron trim position. The aileron trim indicator receives its information from a Linear Variable Differential Transformer (LVDT) which is located inside the aileron trim actuator. The LVDT also transmits aileron trim position data to the Flight Data Recorder (FDR). In the event of LVDT failure or power failure to the aileron trim indicator, the pointer will indicate the last indicated trim position.

Unit, Centering

[Refer to Figures 4 and 5.](#)

The centering unit is part of the Aileron Trim and Centering System. It has a spring torsion bar that links the aileron trim actuator to the

forward quadrant. An input to the control handwheel turns the forward quadrant, which displaces the aileron surfaces. A torsional load is applied on the spring torsion bar. When the input to the handwheel is released, the load on the torsion bar causes the control handwheel to return to a neutral position. This control handwheel centering action reduces the workload of the pilot or co-pilot.

Tab, Trim–Ground Adjustable

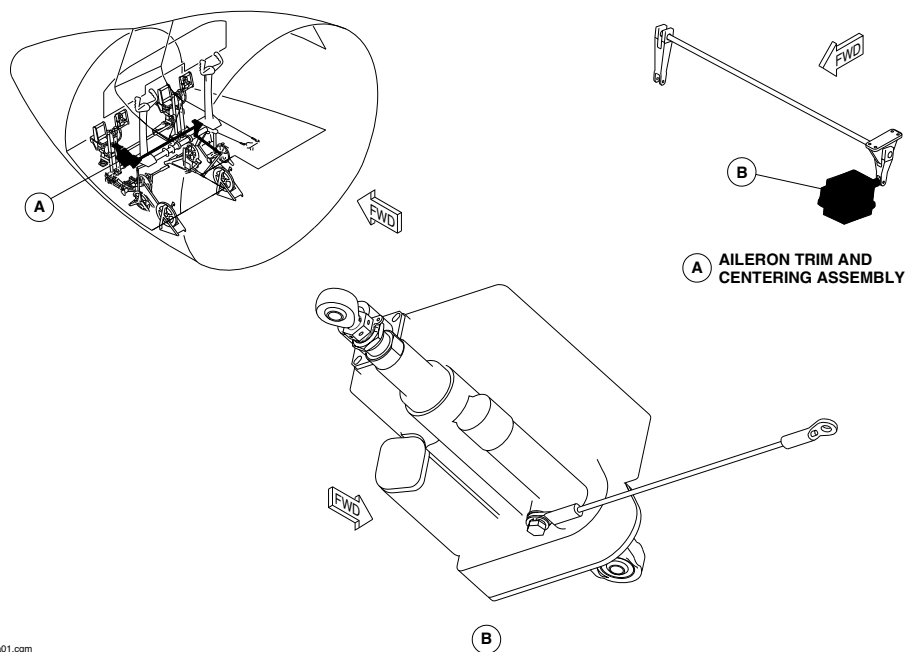
[Refer to Figures 6 and 7.](#)

A Ground Adjustable Trim tab (GAT) is located on the trailing edge of the right aileron, inboard of the geared tab. The GAT is used to adjust for any aerodynamic asymmetry between the left and right ailerons due to manufacturing tolerances. The GAT is adjusted by maintenance personnel after a test flight and only as required thereafter. The necessary adjustment is calculated from the aileron trim indication and control handwheel position.



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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AILERON TRIM ACTUATOR
Figure 1

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-13-00
Config 001

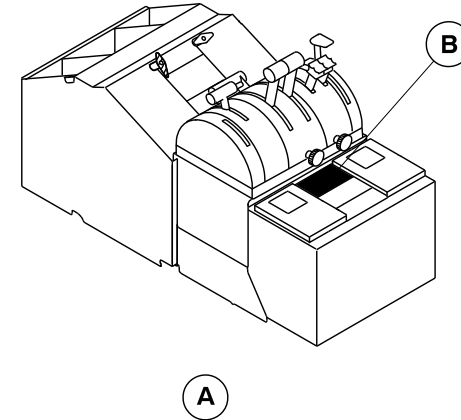
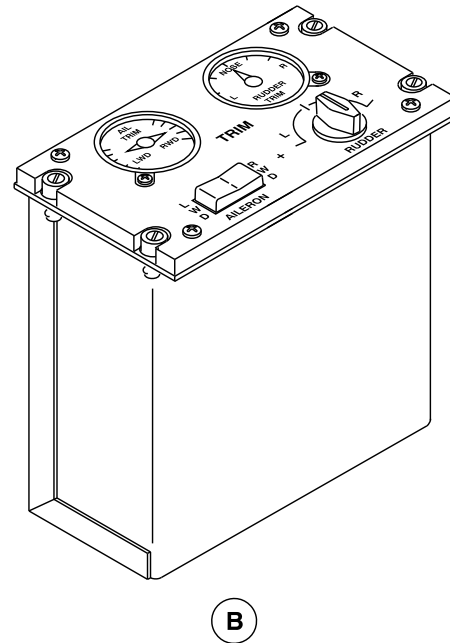
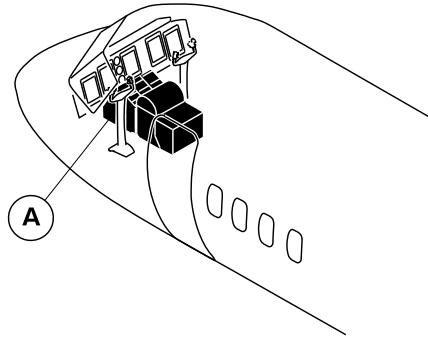
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Config 001
Page 4
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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Aileron Trim Switch and Indicator Locator
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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Config 001

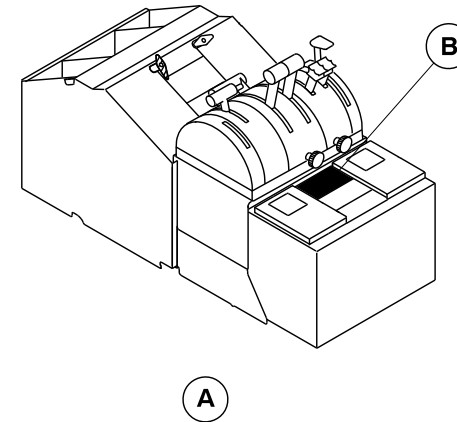
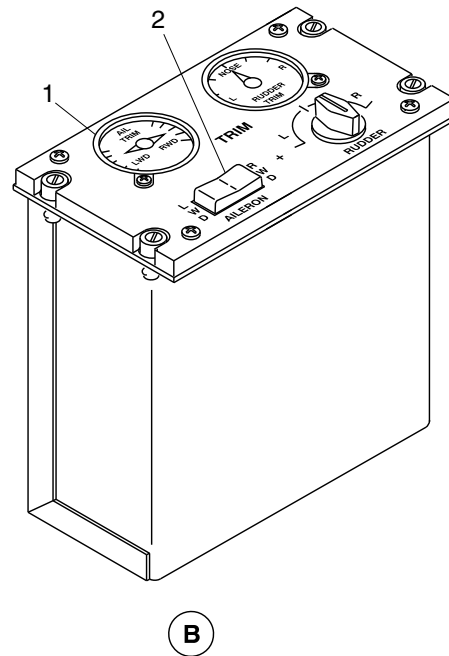
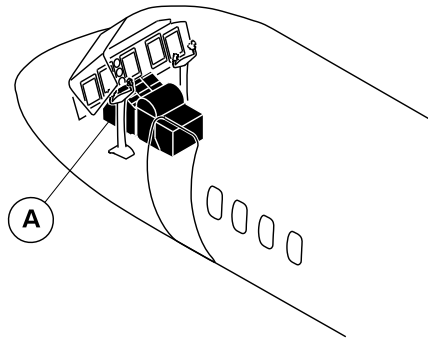
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Config 001
Page 5
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Aileron trim position indicator.
- 2. Aileron trim switch.

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Aileron Trim Switch and Indicator Detail
Figure 3

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-13-00
Config 001

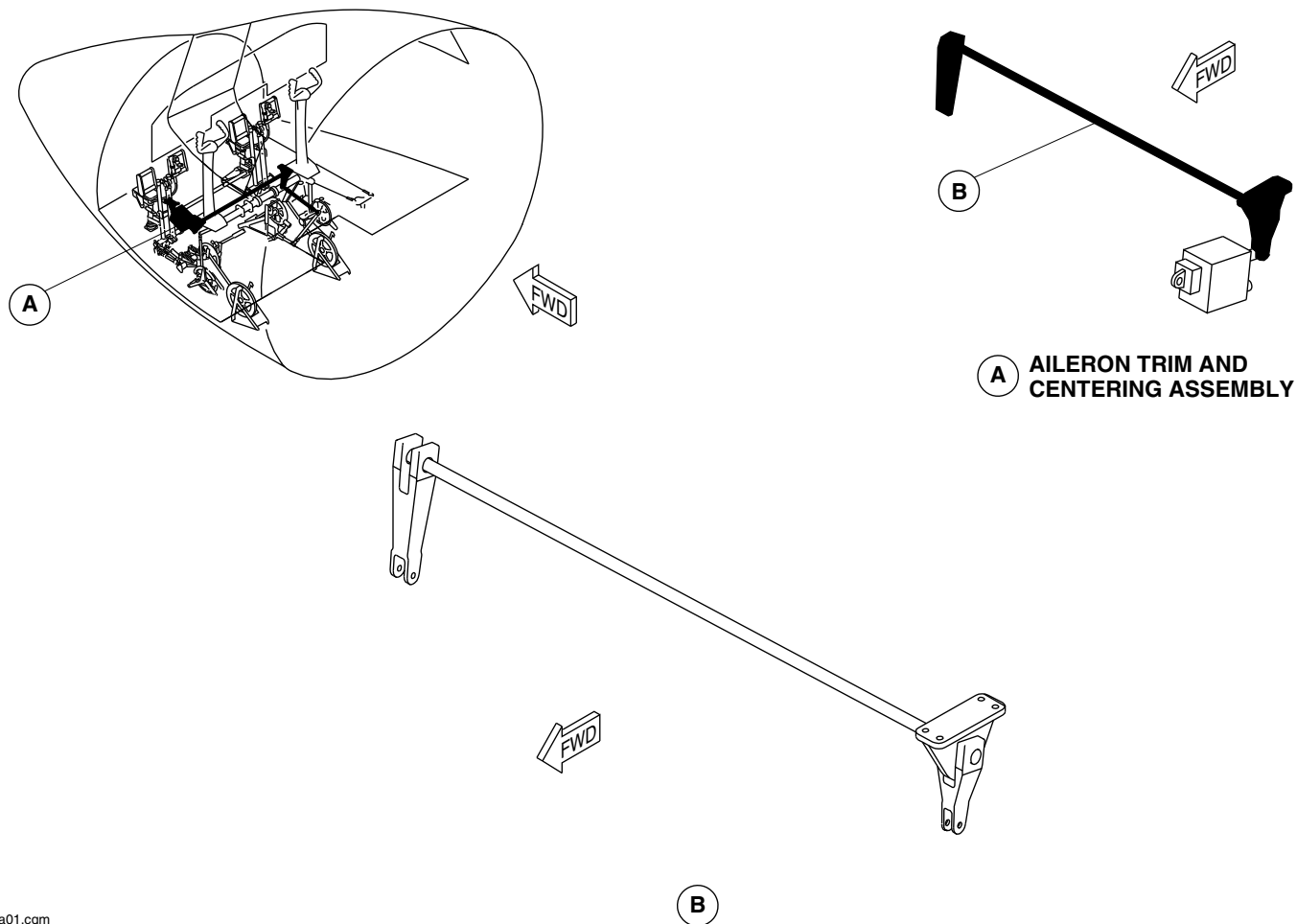
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Config 001
Page 6
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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AILERON CENTERING UNIT
Figure 4

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-13-00
Config 001

27-13-00

Config 001
Page 7
Sep 05/2021

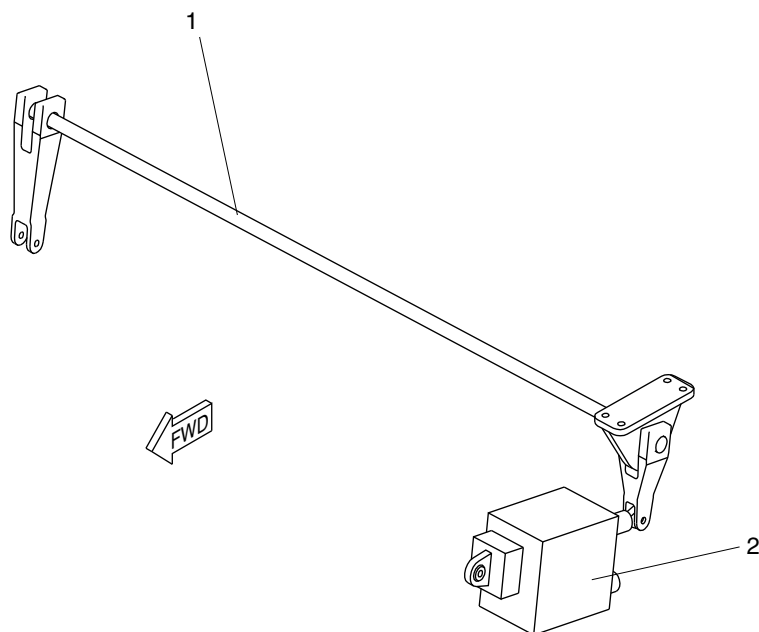


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LEGEND

- 1. Aileron Trim Actuator.
- 2. Spring Torsion Bar.



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AILERON TRIM AND CENTERING UNIT
Figure 5

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-13-00
Config 001

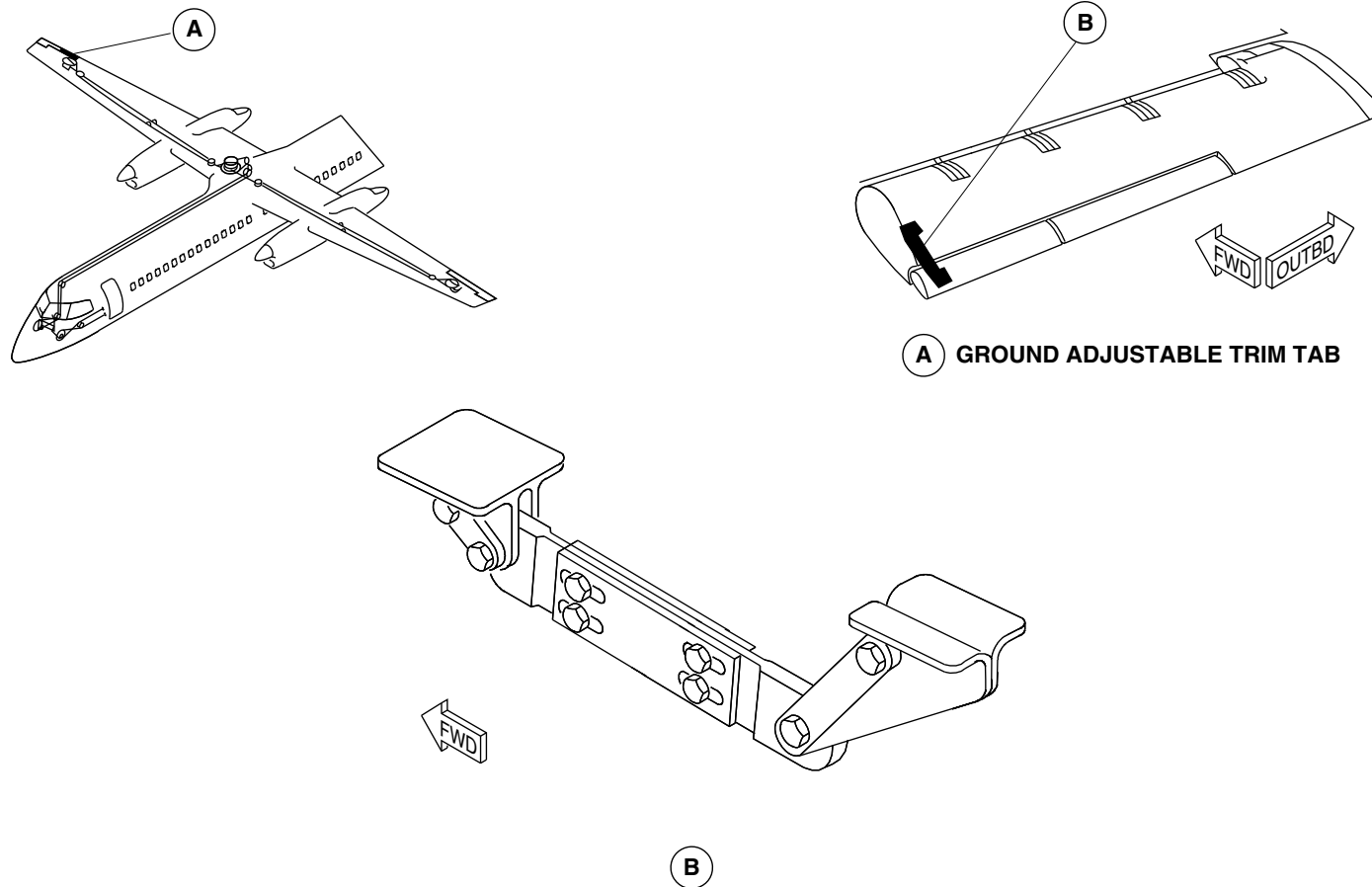
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Config 001
Page 8
Sep 05/2021



DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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GROUND ADJUSTABLE TRIM TAB PAGE 1
Figure 6

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-13-00
Config 001

27-13-00

Config 001
Page 9
Sep 05/2021

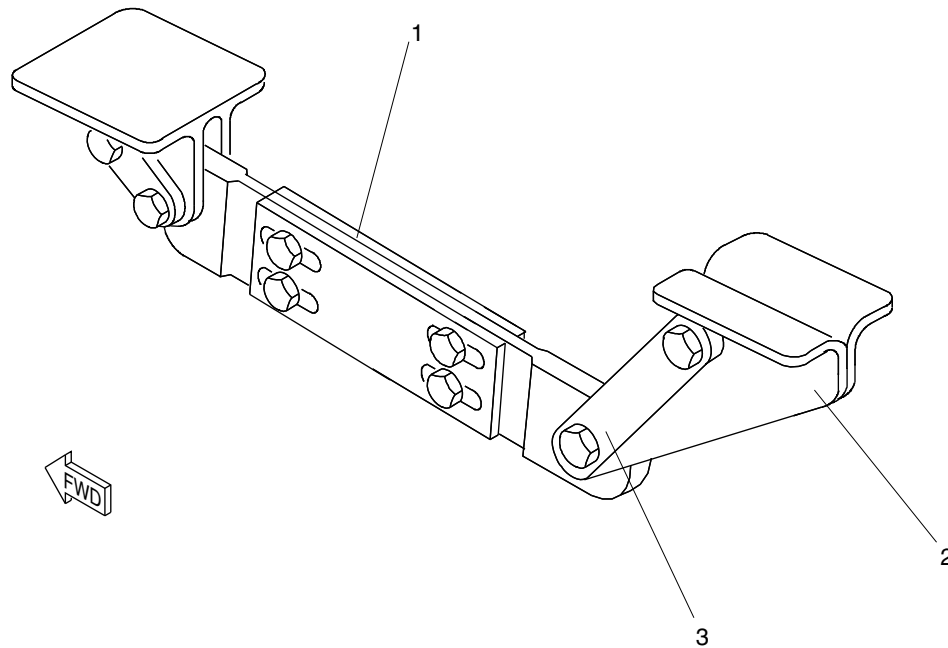


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LEGEND

- 1. Connecting Link.
- 2. Link Bracket.
- 3. Sleeve.



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GROUND ADJUSTABLE TRIM TAB PAGE 2
Figure 7

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-13-00
Config 001

27-13-00

Config 001
Page 10
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

27-14-00-001

AILERON SURFACES

Introduction

The aileron surfaces assisted by the roll spoilers give roll (lateral) control of the aircraft in flight.

General Description

Refer to Figures 1 and 2.

There are two aileron control surfaces. One is installed on the outboard end of the left wing and the other is installed on the outboard end of the right wing. Each aileron surface forms part of the trailing edge of its respective wing.

The Aileron Surfaces have the components that follow:

- Surfaces, Aileron Control (27-14-01).

Detailed Description

Each aileron control surface can travel ± 17 degrees around the neutral position. At airspeeds above 170 kt (315 km/h), the ailerons and the inboard roll spoilers give sufficient lateral control of the aircraft. At airspeeds below 170 kt (315 km/h), the ailerons are assisted by both the inboard and the outboard roll spoilers.

Each aileron has a geared tab that gives aerodynamic assistance to the movement of the aileron surfaces in flight. The geared tab also

helps to overcome the load put on the control wheels by the torsion bar in the aileron trim and centering unit.

The right hand aileron has a ground adjustable trim tab located inboard of the aileron geared tab.

Surfaces, Aileron Control

Each aileron is attached to the trailing edge of its related wing by four hinge fittings. View holes are located on the bottom of the aileron for visual inspection of the hinge pins. An access panel is located at the center hinge point. A horn with internal mass balance weights is located on the outboard end of each aileron to give aerodynamic balance.

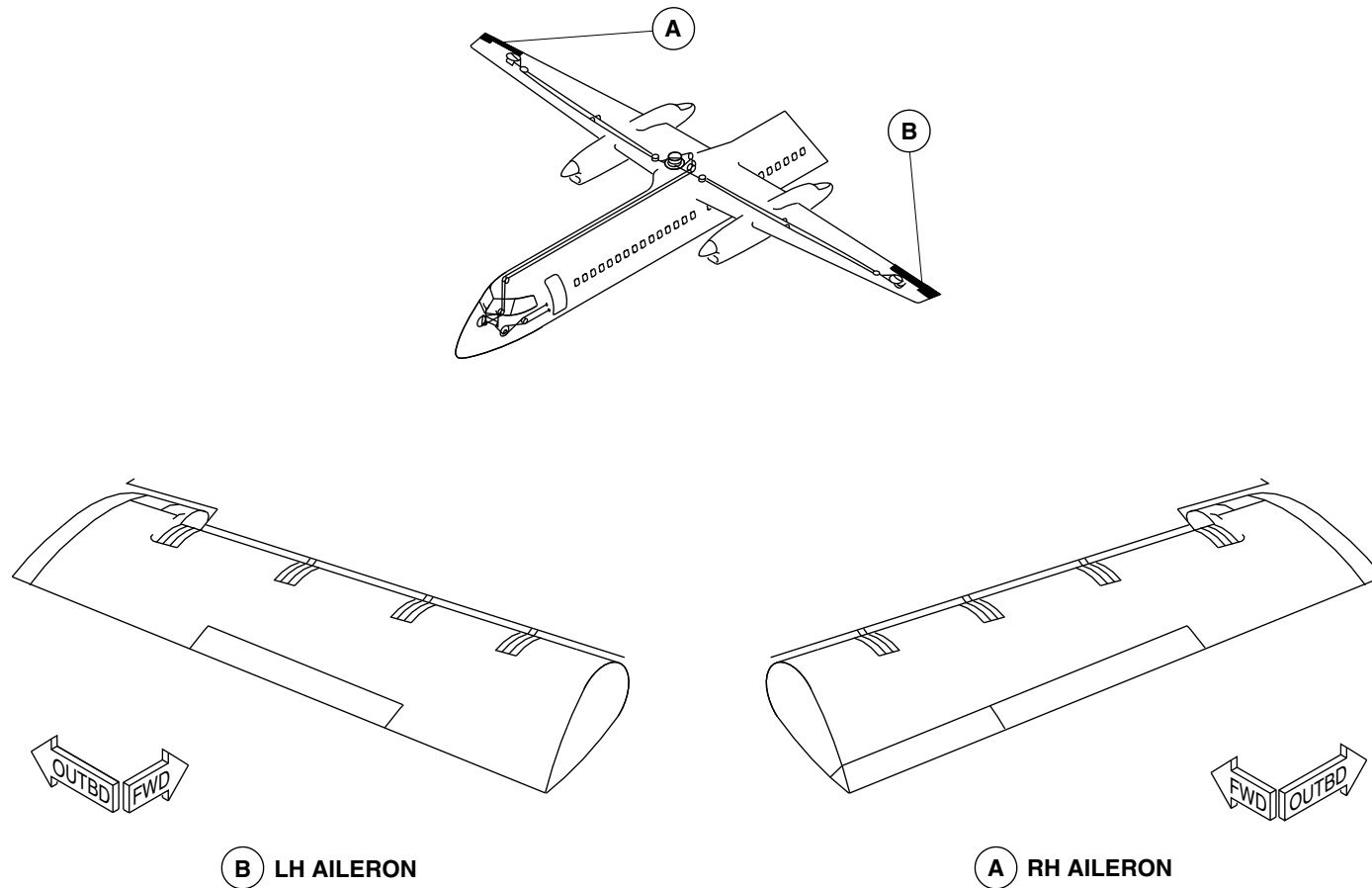
Each aileron has a geared tab, which gives aerodynamic assistance to the aileron control surface. The aileron geared tabs are attached by four hinge fittings to the trailing edge of each aileron. The tab geared linkage is connected to the aileron hinge point to give controlled tab movement, in proportion to aileron movement. The geared linkage has one link, one end connected to the geared tab hinge, the other end to the aileron hinge point. Each geared tab has adjustable bolts located on the hinge arm of the tab.

Only the right aileron has a ground adjustable trim tab located inboard of the geared tab. The adjustment is made on the serrated plate, at the tab end of the geared linkage.



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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AILERON SURFACES PAGE 1
Figure 1

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-14-00
Config 001

27-14-00

Config 001
Page 3
Sep 05/2021

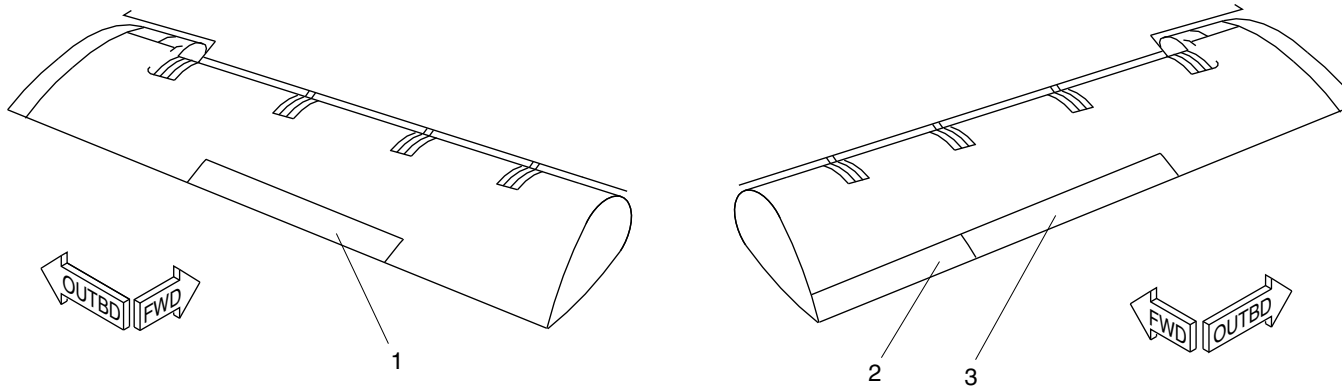


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

LEGEND

1. LH Aileron Geared Tab.
2. Ground Adjustable Trim Tab.
3. RH Aileron Geared Tab.



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AILERON SURFACES PAGE 2
Figure 2

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-14-00
Config 001

27-14-00

Config 001
Page 4
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

27-16-00-001

ROLL DISCONNECT SYSTEM

Introduction

The roll disconnect system disengages the aileron and roll spoiler control circuits from each other in the event of a jam in either control circuit. The remaining operational circuit maintains roll (lateral) control of the aircraft.

General Description

The pilot's control handwheel operates the spoiler control circuit and the copilot's control handwheel operates the aileron control circuit. The control handwheels are connected to each other so that either control handwheel operates both circuits at the same time. This lets the aircraft be flown from either the pilot's or copilot's position. If either control circuit becomes jammed, the ROLL DISC handle is pulled to disconnect the aileron system from the roll spoiler system. The pilot with the unjammed control handwheel maintains roll control of the aircraft.

The roll disconnect system is interfaced with the roll spoiler advisory switchlights (SPLR 1 and SPLR 2) on the center glareshield panel (refer to SDS 27-61-00 Roll Spoiler System).

The roll disconnect system has the components that follow:

- Roll Disconnect Handle (27-16-01)
- Lever Assembly Roll Disconnect (27-16-06)

- Clutch Mechanism Roll Disconnect (27-16-11).

Detailed Description

[Refer to Figures 1 and 2.](#)

The aileron and roll spoiler control circuits are joined to each other by an interconnect rod. The roll disconnect system has a horizontally mounted ROLL DISC handle located on the right side of the forward center console. The handle is connected to a clutch mechanism located at the base of the copilot's control column by cables, pulleys and a spring loaded lever assembly. The cables between the handle and clutch mechanism are routed under the flight compartment floor. The system is adjusted by loosening or tightening turnbuckles, one in the forward roll disconnect cable and one in the cable from the lever assembly to the clutch mechanism.

If a roll jam occurs in either system, the handle is pulled out to the limit and turned through 90 degrees. This action isolates the jammed system from the operating system. The unjammed wheel is now used to maintain roll control of the aircraft.

[Refer to Figures 3 , 4 and 5.](#)

The SPLR 1 and SPLR 2 switchlights are located on the center glareshield panel. When the roll disconnect handle is pulled, a microswitch is closed by the roll disconnect lever assembly. If the copilot control handwheel is rotated more than 50 degrees to overcome the effects of a jammed spoiler control system, the aileron quadrant also closes a microswitch. This microswitch is located on the aileron forward quadrant. When these two microswitches are closed, the SPLR 1 and SPLR 2 switchlights come on indicating a jammed spoiler. The pilot then pushes the two switchlights, which depressurizes the spoiler PCUs allowing the jammed spoiler panel(s) to retract flush with the wing surface. Pushing the SPLR 1 switchlight

PSM 1-84-2A

EFFECTIVITY:

See first effectivity on page 2 of 27-16-00

Config 001

27-16-00

Config 001

Page 2

Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

depressurizes the inboard spoiler PCUs. Pushing the SPLR 2 switchlight depressurizes the outboard spoiler PCUs.

Roll Disconnect Handle

Refer to Figure 6.

The ROLL DISC handle is located in the flight compartment, on the copilot's side of the forward center console. The handle is used to separate the controls of the aileron and spoiler systems.

The roll disconnect handle is normally in the engaged position. The handle is connected to a clutch mechanism at the base of the copilot's control column by cables, pulleys, and a spring-loaded lever assembly.

Refer to Figure 7.

The roll disconnect mechanism is disengaged when the roll disconnect handle is pulled to the stop and rotated by 90 degrees in either direction. A pin in the shaft of the roll disconnect handle rides in a slot in the shaft housing. When the handle is pulled out to the limit and rotated 90 degrees, the pin slides into the slot to keep the handle locked in the disengaged position. The roll disconnect mechanism is re-engaged when the pilot's and copilot's handwheels are moved to the neutral position, the roll disconnect handle is rotated 90 degrees and released. The spring on the clutch mechanism pulls the handle back to the engaged position.

Lever Assembly Roll Disconnect

Refer to Figures 8 and 9.

The roll disconnect lever assembly is located below the flight compartment floor near the base of the copilot's control column. The

lever assembly is attached by a cable to the roll disconnect handle. Another cable attaches the lever to the engagement spring cover of the clutch mechanism.

When the roll disconnect handle is pulled, the lever assembly rotates and pulls the engagement spring cover, which causes the clutch to disengage. At the same time, the rotating lever assembly closes a microswitch which is part of the roll spoiler advisory light circuit (refer to SDS 27–60–00). A return spring attached to the lever assembly keeps the tension off the cable joining the lever assembly to the clutch mechanism. This prevents the clutch from accidentally disengaging and helps to return the roll disconnect handle to the engaged position.

Clutch Mechanism Roll Disconnect

Refer to Figures 10 and 11.

The roll disconnect clutch mechanism is attached to the base of the copilot's control column. When the clutch mechanism is in the engaged position, it connects the chain drive of the copilot's control wheel through an interconnect rod to the pilot's control wheel. When disengaged by the roll disconnect handle in the flight compartment, the clutch mechanism separates the copilot's column from the interconnect rod.

The roll disconnect clutch mechanism consists of an engagement spring, an engagement spring cover and a clutch.

The clutch is made up of two halves:

- clutch and sprocket assembly
- clutch plate and lever assembly.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

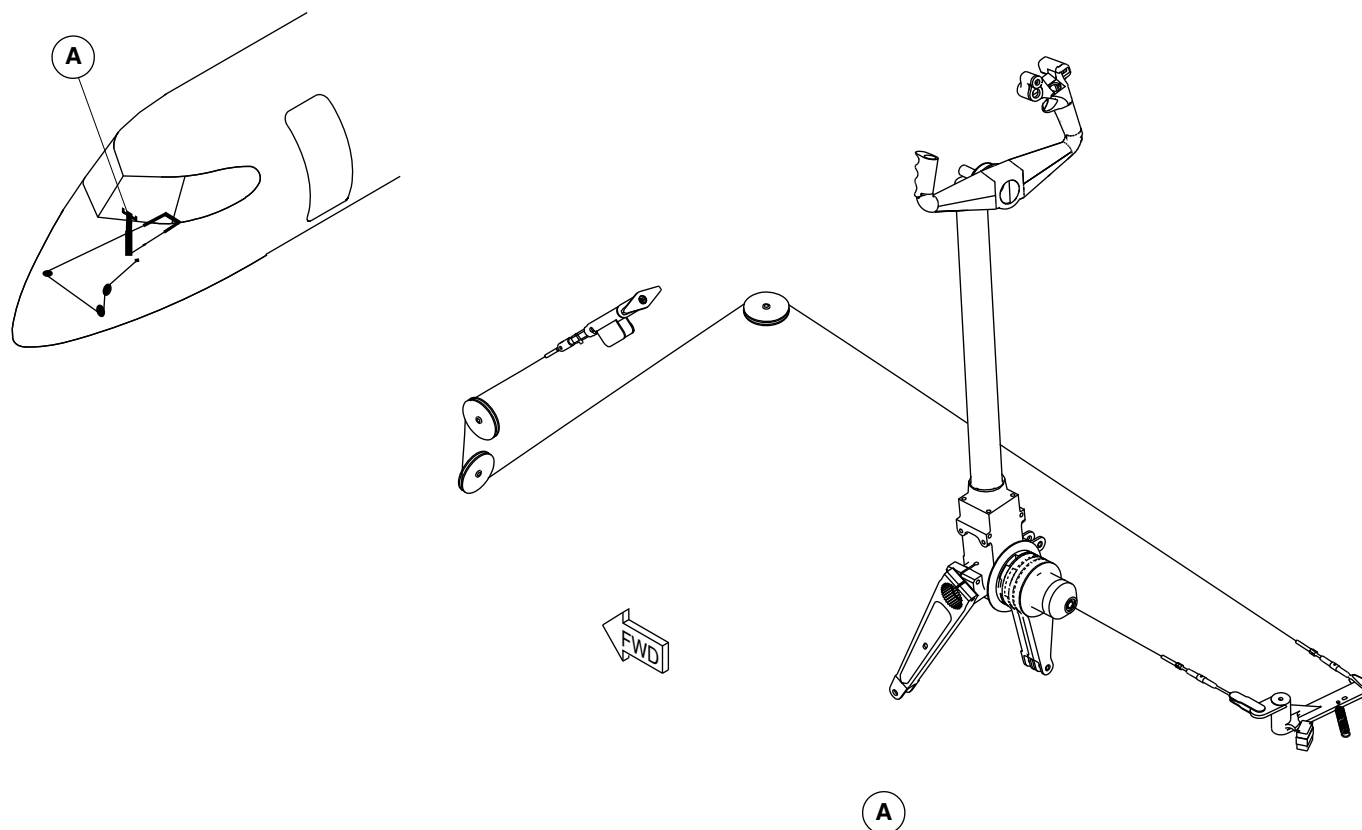
The chain in the copilot's control column rotates the sprocket and clutch assembly, which drives the clutch plate and lever assembly when in the engaged position. The lever part of the clutch plate and lever assembly is attached to the pilot's control column through the interconnect rod. The interconnect rod is the link which transfers aileron and spoiler control between the pilot's and copilot's control columns.

When the roll disconnect handle is pulled to the limit and turned 90 degrees, the clutch plate and lever assembly is separated from the clutch and sprocket assembly. This action lets the control handwheels rotate independently of each other. To re-engage the clutch, the handle must be rotated 90 degrees and released. The handle and clutch mechanism return to the engaged position by the action of the return and engagement springs. The lever and clutch plate assembly re-engages with the clutch lever and sprocket assembly when the two control wheels come into alignment. The engagement spring keeps the clutch in the engaged position.



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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ROLL DISCONNECT SYSTEM PAGE 1
Figure 1

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-16-00
Config 001

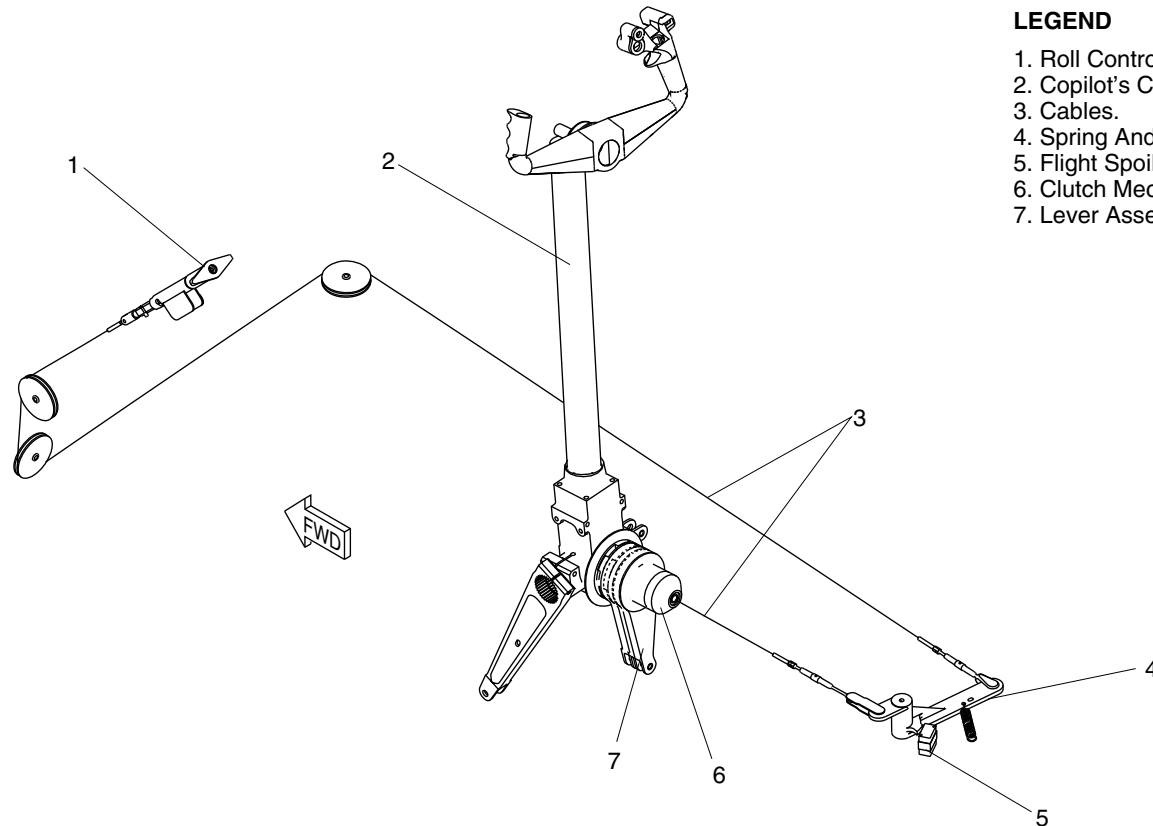
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Config 001
Page 5
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Roll Control Disconnect Handle.
2. Copilot's Control Column.
3. Cables.
4. Spring And Lever Assembly.
5. Flight Spoiler De-Clutch Micro-Switch.
6. Clutch Mechanism.
7. Lever Assembly Roll Disconnect.

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ROLL DISCONNECT SYSTEM PAGE 2
Figure 2

PSM 1-84-2A
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Config 001

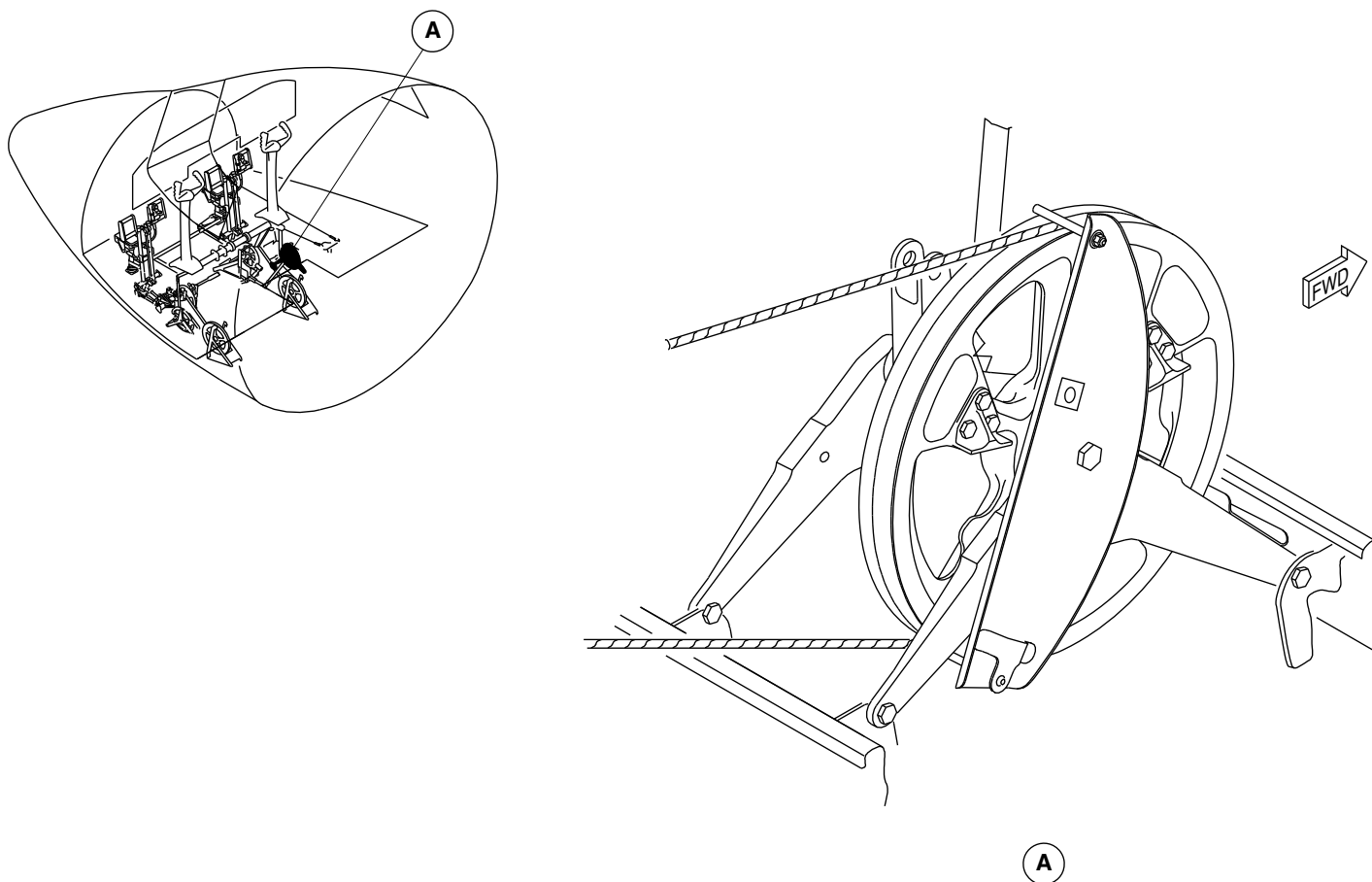
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Config 001
Page 6
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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AILERON FORWARD QUADRANT LOCATOR
Figure 3

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-16-00
Config 001

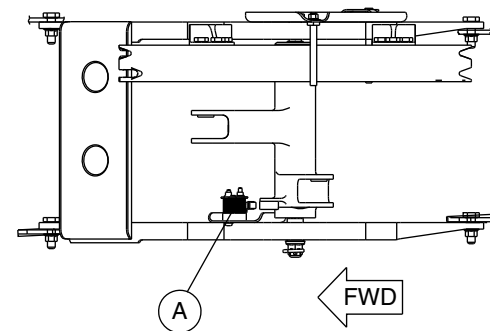
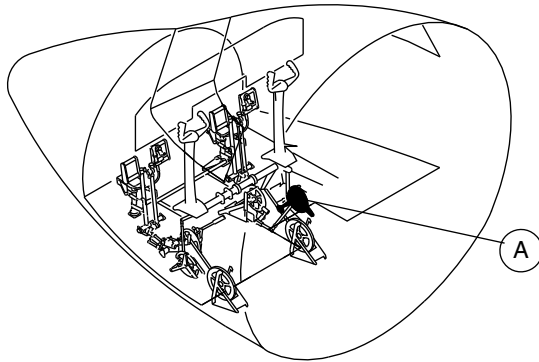
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Config 001
Page 7
Sep 05/2021

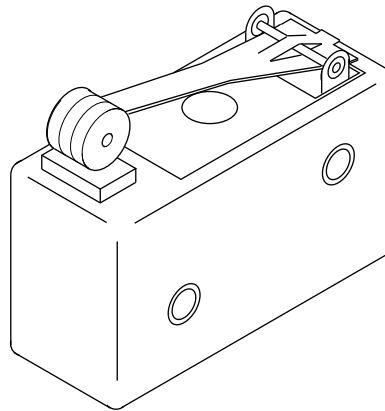


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AILERON FORWARD QUADRANT



4271a089 Roll Disconnect Sys – Aileron Fwd Quadrant Detail.iso

AILERON FORWARD QUADRANT MICROSWITCH
Figure 4

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-16-00
Config 001

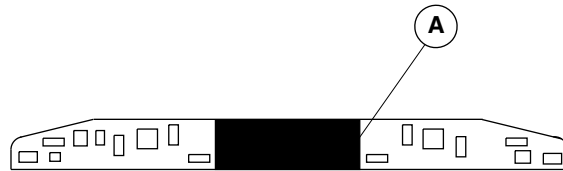
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Config 001
Page 8
Sep 05/2021



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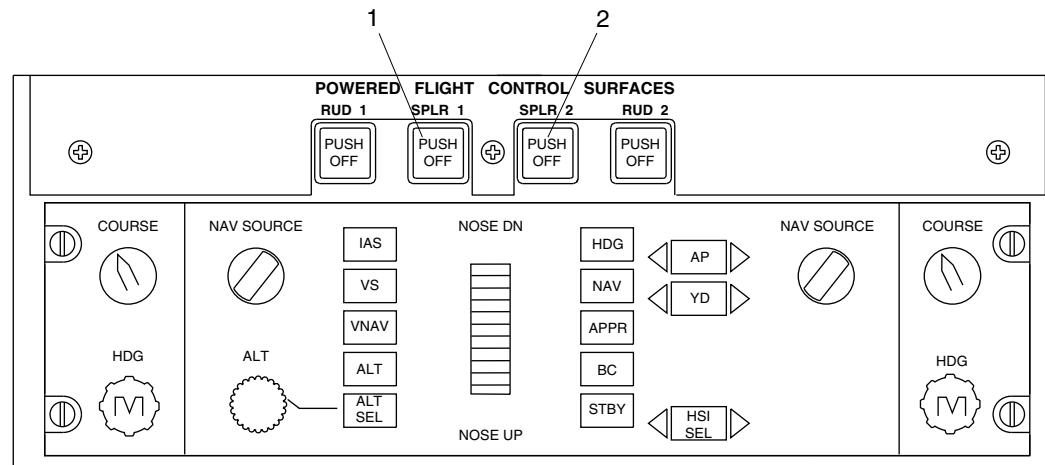
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



GLARESHIELD PANEL

LEGEND

1. Spoiler 1 Push Off Switchlight.
2. Spoiler 2 Push Off Switchlight.



A

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SPOILER PUSHOFF SWITCHLIGHTS
Figure 5

PSM 1-84-2A
EFFECTIVITY:
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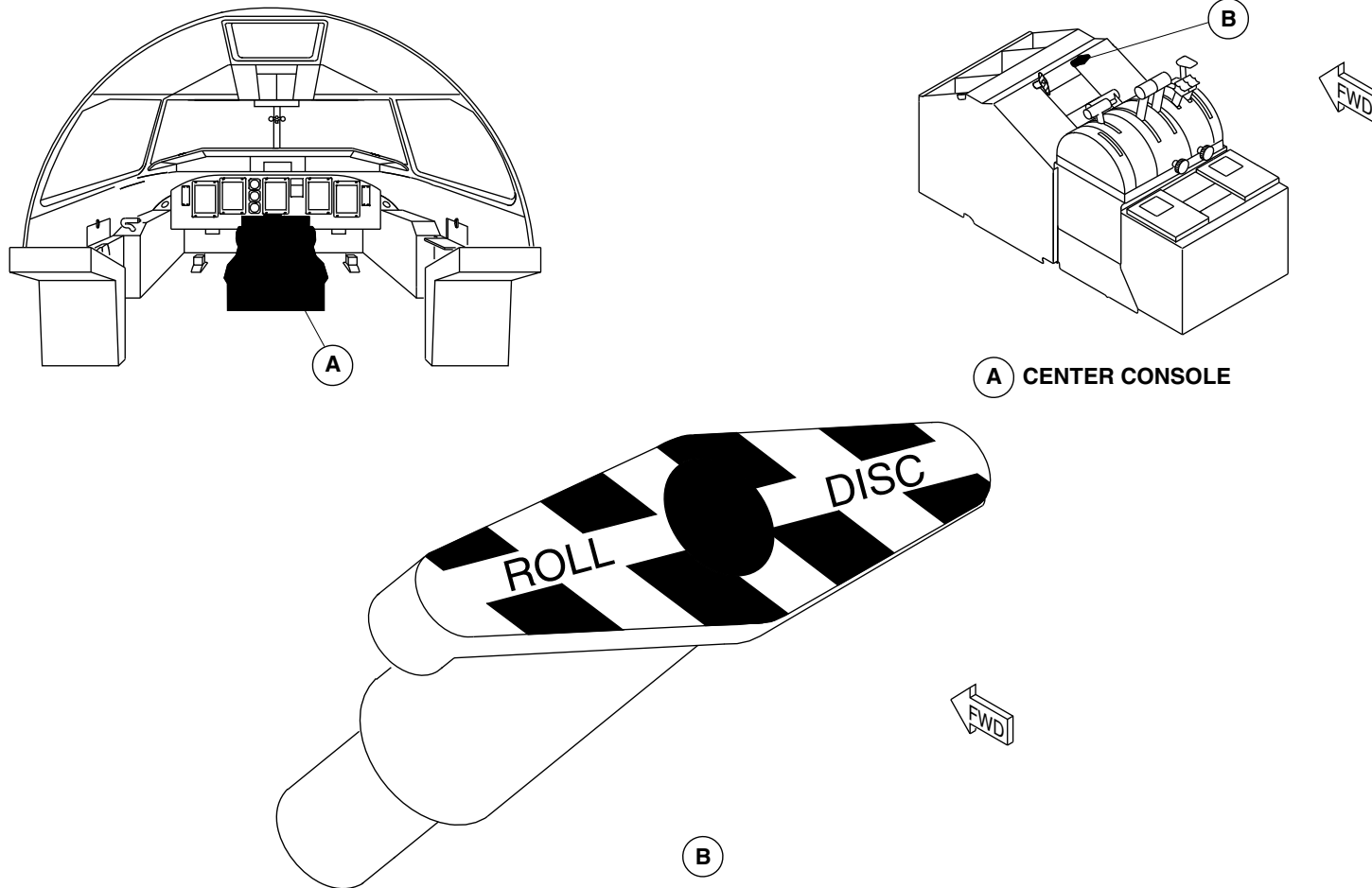
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Config 001
Page 9
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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ROLL DISCONNECT HANDLE PAGE 1
Figure 6

PSM 1-84-2A
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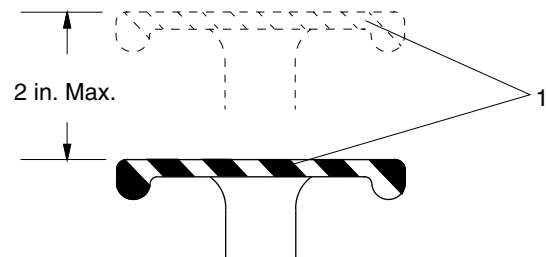
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Config 001
Page 10
Sep 05/2021



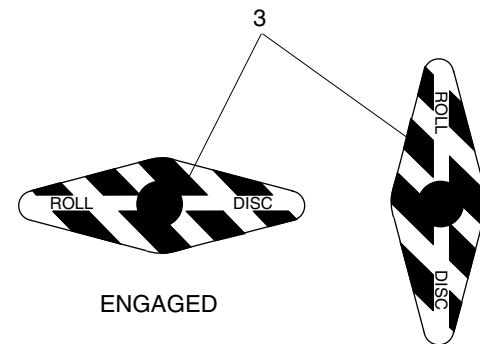
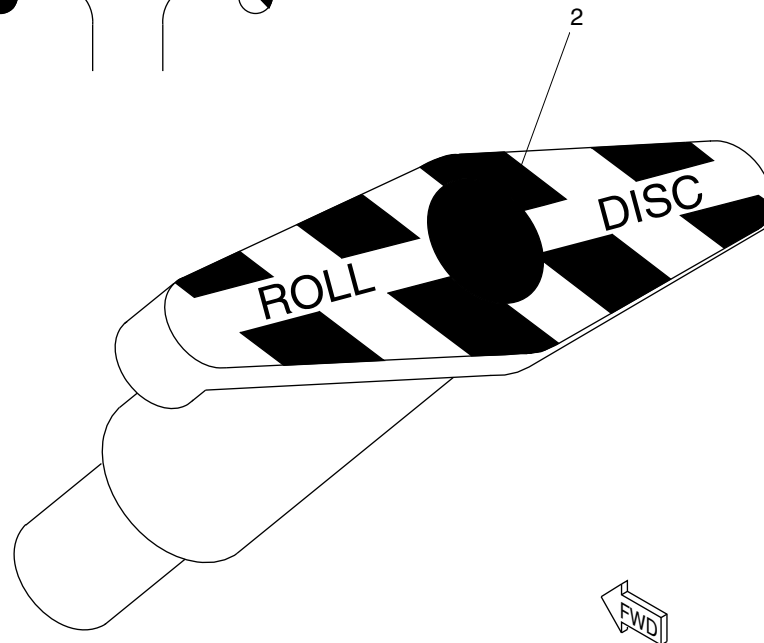
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LEGEND

1. Roll Disconnect Handle Side View Operation.
2. Roll Disconnect Handle.
3. Roll Disconnect Handle Top View Operation.



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ROLL DISCONNECT HANDLE PAGE 2
Figure 7

PSM 1-84-2A
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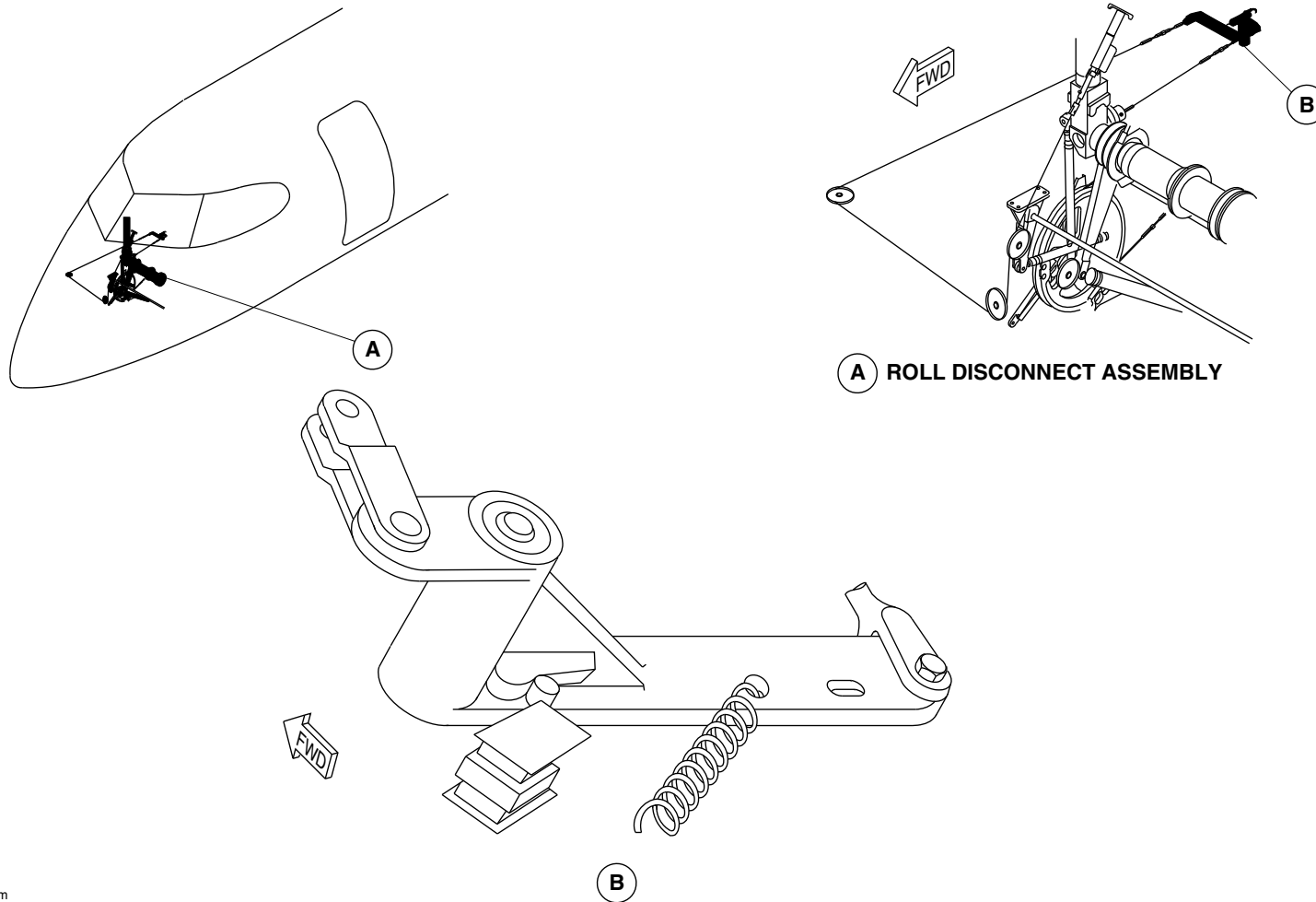
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Config 001
Page 11
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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ROLL DISCONNECT LEVER ASSEMBLY PAGE 1
Figure 8

PSM 1-84-2A
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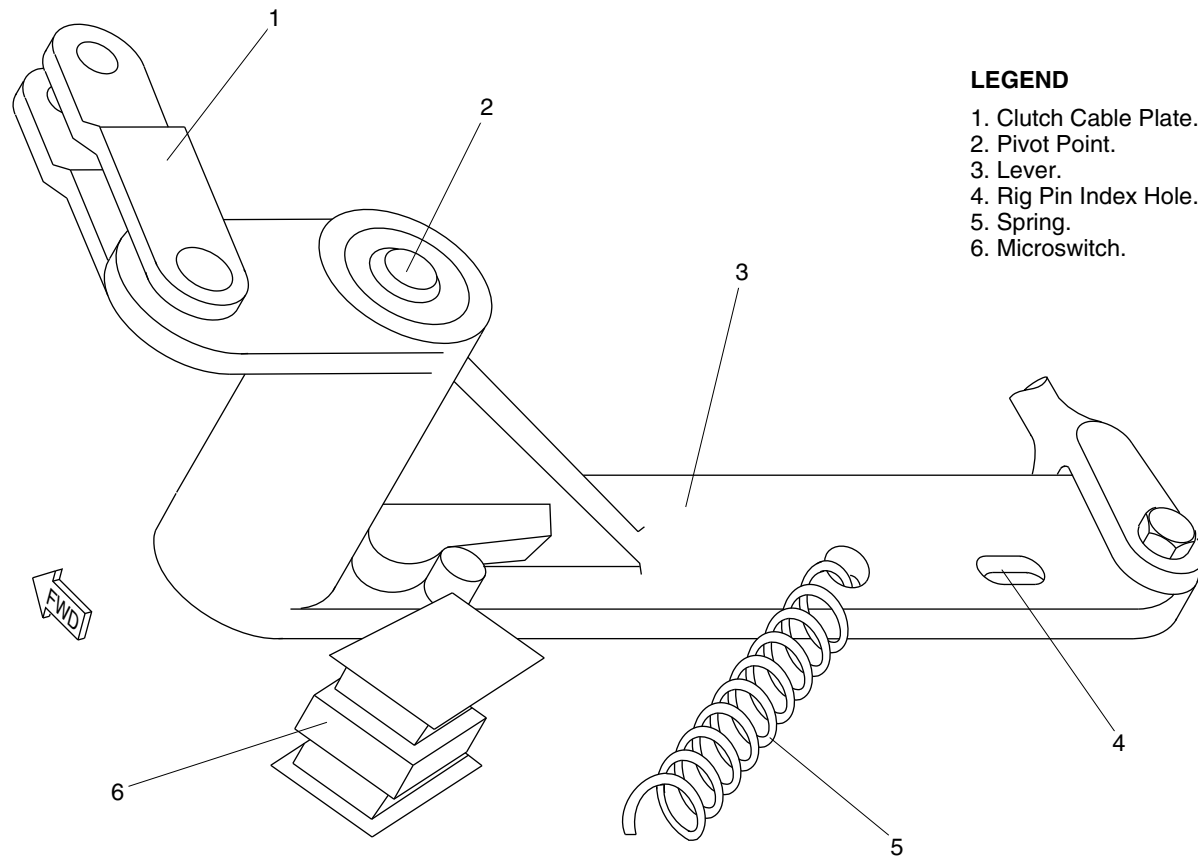
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Config 001
Page 12
Sep 05/2021



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OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Clutch Cable Plate.
- 2. Pivot Point.
- 3. Lever.
- 4. Rig Pin Index Hole.
- 5. Spring.
- 6. Microswitch.

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ROLL DISCONNECT LEVER ASSEMBLY PAGE 2
Figure 9

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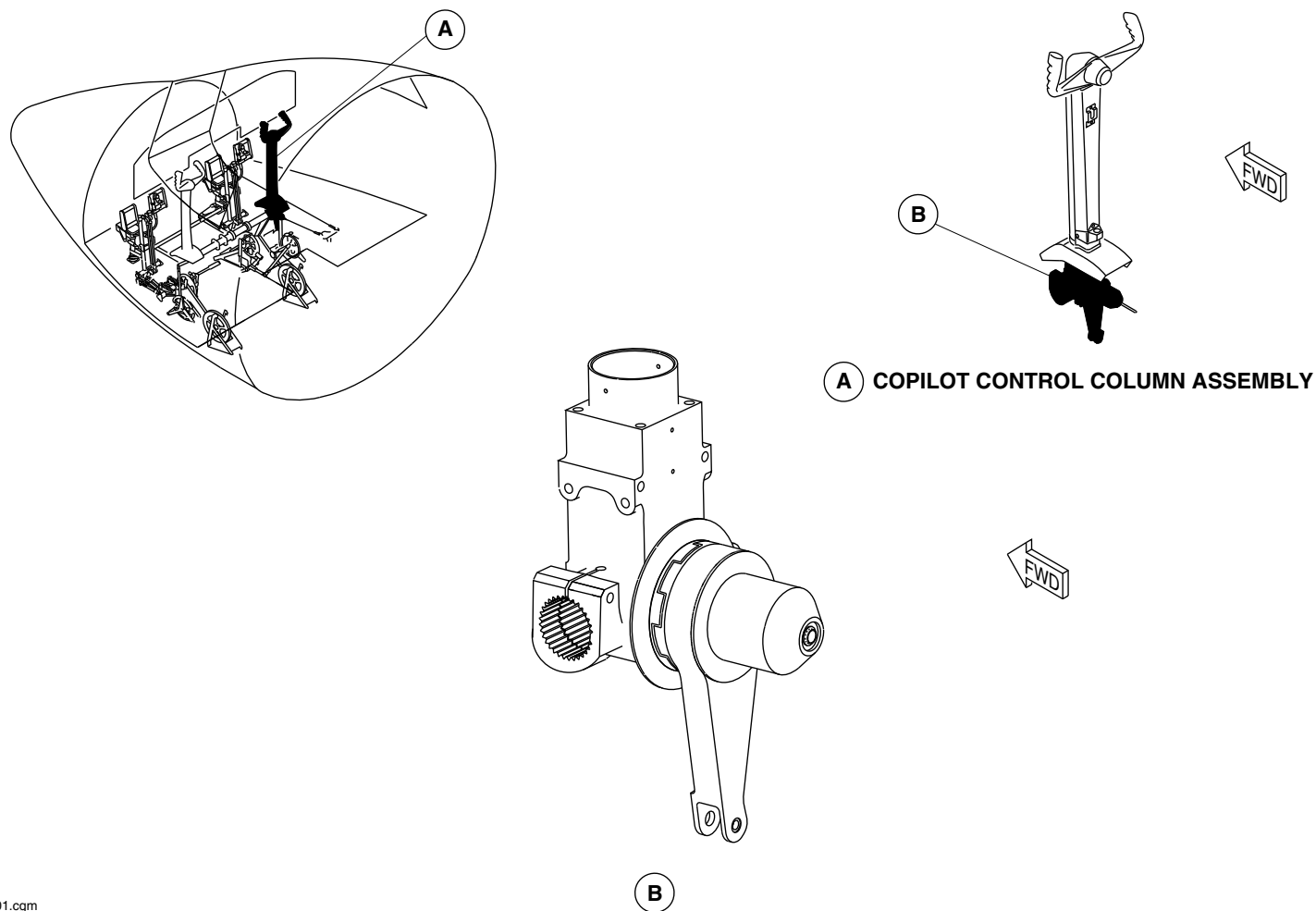
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Config 001
Page 13
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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ROLL DISCONNECT CLUTCH MECHANISM PAGE 1
Figure 10

PSM 1-84-2A
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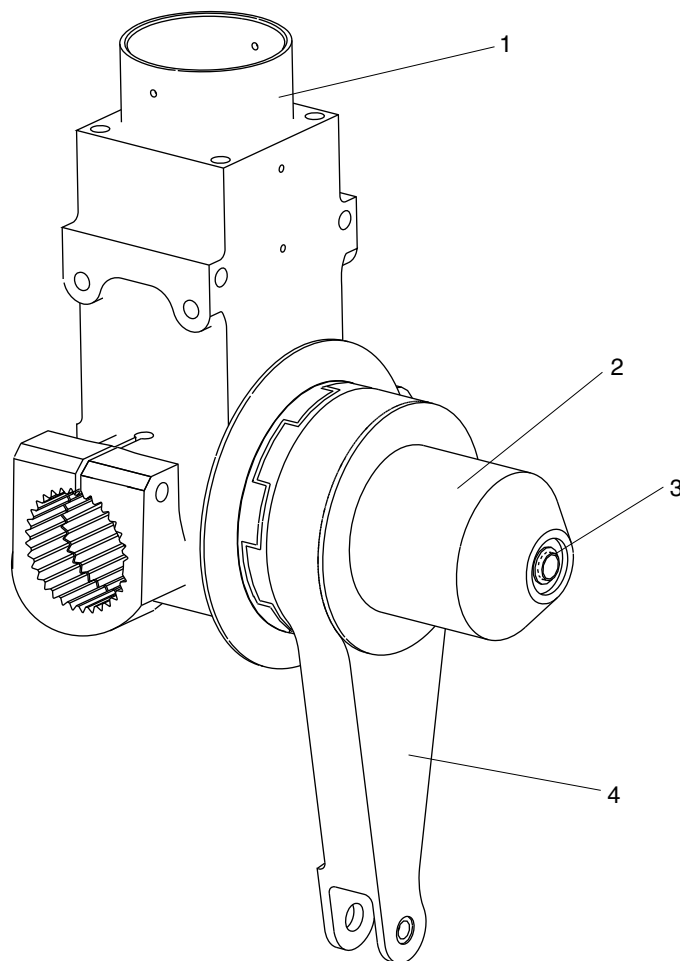
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Config 001
Page 14
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Copilot Control Column Attachment.
- 2. Roll Disconnect Clutch Housing.
- 3. Clutch Mechanism Shaft.
- 4. Roll Control Lever.

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ROLL DISCONNECT CLUTCH MECHANISM PAGE 2
Figure 11

PSM 1-84-2A
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See first effectivity on page 2 of 27-16-00
Config 001

27-16-00

Config 001
Page 15
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-16-00
Config 001

27-16-00

Config 001
Page 16
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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RUDDER CONTROL SYSTEM

Introduction

The rudder control system is a primary flight control system. It supplies directional control of the aircraft about the yaw (normal) axis during flight.

General Description

Refer to Figure 1.

The rudder control system supplies directional or yaw control of the aircraft in flight. Either the pilot, copilot or the autopilot can supply yaw commands to the rudder control system. The pilot's and copilot's rudder pedals are connected to each other through an interconnect rod.

The rudder control system has the components that follow:

- Rudder System (27-21-00)
- Rudder Feel and Trim (27-22-00)
- Rudder Surfaces (27-23-00).

Detailed Description

Refer to Figure 2.

The rudder has two sections: the fore rudder and the trailing rudder. The rudder surfaces move in proportion to the amount of rudder pedal input.

Movement of the rudder pedals supplies yaw commands to the rudder control system. The yaw commands are mechanically transmitted from the pilot's and copilot's rudder pedals to the rudder surfaces. These commands are transmitted through the Rudder Feel and Trim Unit and the rudder Power Control Units (PCUs). The PCUs move the rudder surfaces using hydraulic power.

Refer to Figure 3.

The hydraulic pressure supplied to the PCUs is varied with airspeed by the Flight Control Electronic Control unit (FCECU). The FCECU gets airspeed information from the Air Data Units. Failures in the rudder hydraulic systems are shown by three caution lights #1 RUD HYD, #2 RUD HYD and RUD CTRL.

Refer to Figure 4.

A jam in the rudder control system is shown by a rudder advisory switchlight RUD 1 or RUD 2 on the center glareshield panel.

Rudder System

The rudder has two sections designated: the fore rudder and the trailing rudder. The fore rudder is hydraulically actuated, the trailing rudder is moved mechanically by movement of the fore rudder.

There are two hydraulic Power Control Units (PCUs), one installed above the other in the vertical stabilizer, that control the fore rudder



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

movement. The pilot's and copilot's rudder pedals control both of the rudder PCUs. Hydraulic system No. 1 supplies pressure to the lower PCU and hydraulic system No. 2 supplies pressure to the upper PCU. Two pushrods transfer the directional control movements of the fore rudder to the trailing rudder. The trailing rudder moves twice as far as the fore rudder.

A rudder pedal adjustment handle gives fore and aft adjustment of either pair of rudder pedals to accommodate individual pilot preference for pedal position. Each adjustment handle is located on the pedestal between the pilot's feet. The pilot's and copilot's rudder pedals are connected to each other through an interconnect rod.

The rudder pedals rotate about the pedal support tubes to supply foot brake control. A cable connecting the pilot's and copilot's brake pedals, allows for operation of the brake system from either pilot's position.

A RUDDER trim switch in the flight compartment operates an electrical trim actuator. The actuator supplies trim signals to the rudder for yaw control. The amount of rudder trim is shown on the RUDDER trim indicator, on the center console, in the flight compartment. A Rudder Feel Trim Unit gives artificial feel to the rudder pedals during flight operation.

A rudder yaw damper actuator, supplies automatic compensation for minor rudder deflections to improve directional stability and turn coordination. The yaw damper actuator works independently or in conjunction with the Automatic Flight Control System (AFCS). The yaw damper is always engaged when the autopilot is engaged. Yaw damper authority is 4.5 degrees of rudder movement either side of center. The yaw damper gets its inputs from Flight Guidance Modules NO.1 and No. 2 and needs both inputs for operation (refer to SDS 22–11–36).

The No. 1 and No. 2 hydraulic systems supply hydraulic power to the rudder PCUs. If either hydraulic system fails, the remaining system is able to supply sufficient hydraulic power for rudder control. If either rudder hydraulic actuator system jams, a microswitch turns on a rudder advisory switchlight (RUD 1 or RUD 2) PUSH OFF on the glareshield panel. When the RUD 1 or RUD 2 switchlight is pushed, the jammed PCU depressurizes and the functioning system operates the rudder. Pushing the RUD 1 switchlight depressurizes the lower rudder PCU. Pushing the RUD 2 switchlight depressurizes the upper rudder PCU.

A rudder input restrictor mechanism, limits rudder pedal movement with flap selector operation. The flap selector lever is mechanically linked to the copilot's rudder forward quadrant. The restrictor limits rudder travel to 12 degrees left and right of center when 0 degrees flap is selected. When 5 degree flap or greater is selected, full rudder travel of 18 degrees left and 20 degrees right of center is available.

A hydraulic pressure limiting system, reduces the pressure available to the rudder PCUs as airspeed increases (refer to SDS 27–21–00 Rudder System). Therefore, rudder authority is limited as a function of airspeed in order to reduce excessive yaw rate and structural loading of the aircraft. Airspeed data is obtained from the Air Data Units (ADUs). The pressure limiting function is under the control of the Flight Control Electronic Unit (FCECU).

Rudder Feel and Trim

The Rudder Feel and Trim Unit gives artificial feel to the rudder pedals. The unit is installed just behind the rudder aft quadrant, in series with the rudder control system (refer to SDS 27–22–00).

The trim actuator is attached to the vertical stabilizer at one end and to the Rudder Feel and Trim Unit at the other end. The trim actuator



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

uses a brushless motor to extend or retract the actuator arm. An integral Linear Variable Displacement Transformer (LVDT) monitors the trim actuator position and shows it on the RUDDER trim indicator. The LVDT signal is also transmitted to the Flight Data Recorder (FDR). If the LVDT fails, the trim actuator stays functional but no trim position is shown on the RUDDER trim indicator.

The trimming system is electrically powered from the Left Essential bus through two circuit breakers; one for the trim actuator and the other for the RUDDER trim indicator.

Turning the RUDDER trim switch fully to the left (L) or right (R), gives trim at a high rate. When the switch is turned to the first graduation line, a slower trim rate is given. The trim actuator can move the rudder surface a maximum of 10 degrees to the left or right of center. As the rudder is being trimmed the position is shown on the RUDDER trim indicator (refer to SDS 27–22–00).

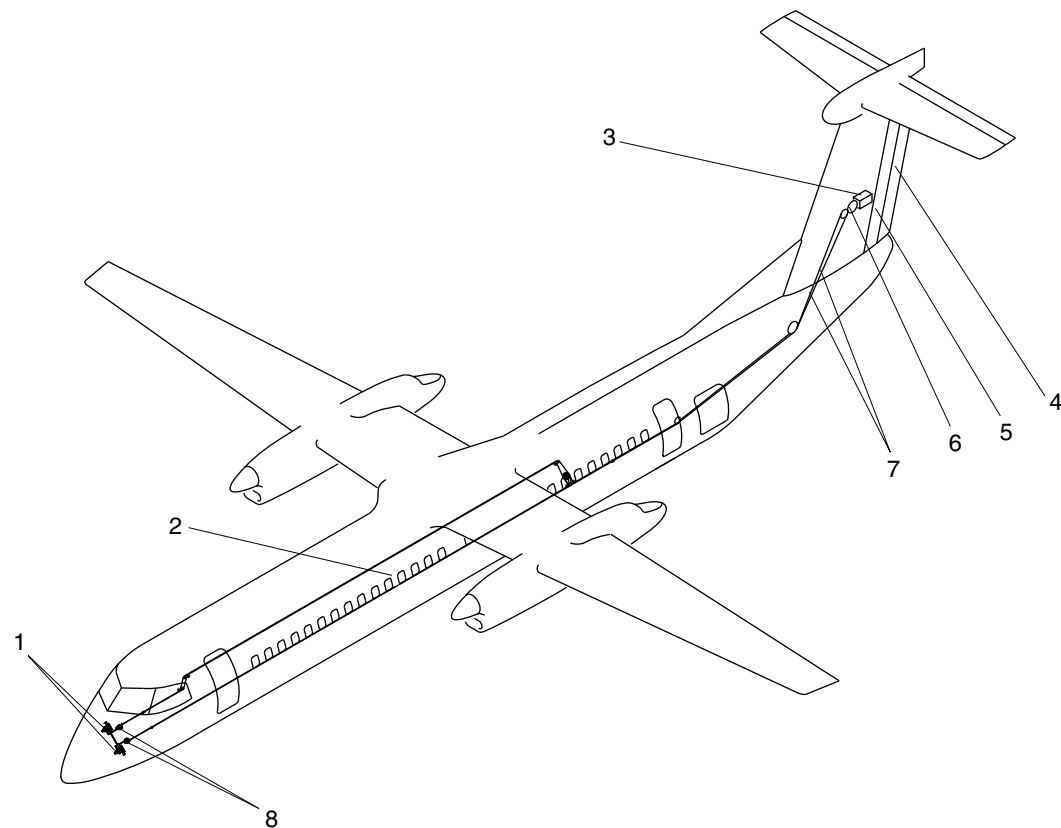
Rudder Surfaces

There are two rudder surfaces, a fore rudder and a trailing rudder. The fore rudder is attached to the rear spar of the vertical stabilizer. The trailing rudder is attached to the fore rudder by push rods so that the trailing rudder moves twice as far as the fore rudder. Each rudder is attached by hinge bolts to sealed bearings. The rudders are attached to the aircraft through a series of hinges and safety hinges (refer to SDS 27–23–00).



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LEGEND

1. Rudder Pedals.
2. Rudder Mixer Quadrant.
3. Rudder PCUs (2).
4. Trailing Rudder.
5. Fore Rudder.
6. Rudder Aft Quadrant.
7. Rudder Cables.
8. Rudder Forward Quadrants.

RUDDER CONTROL SYSTEM PAGE 1
Figure 1

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See first effectivity on page 2 of 27-20-00
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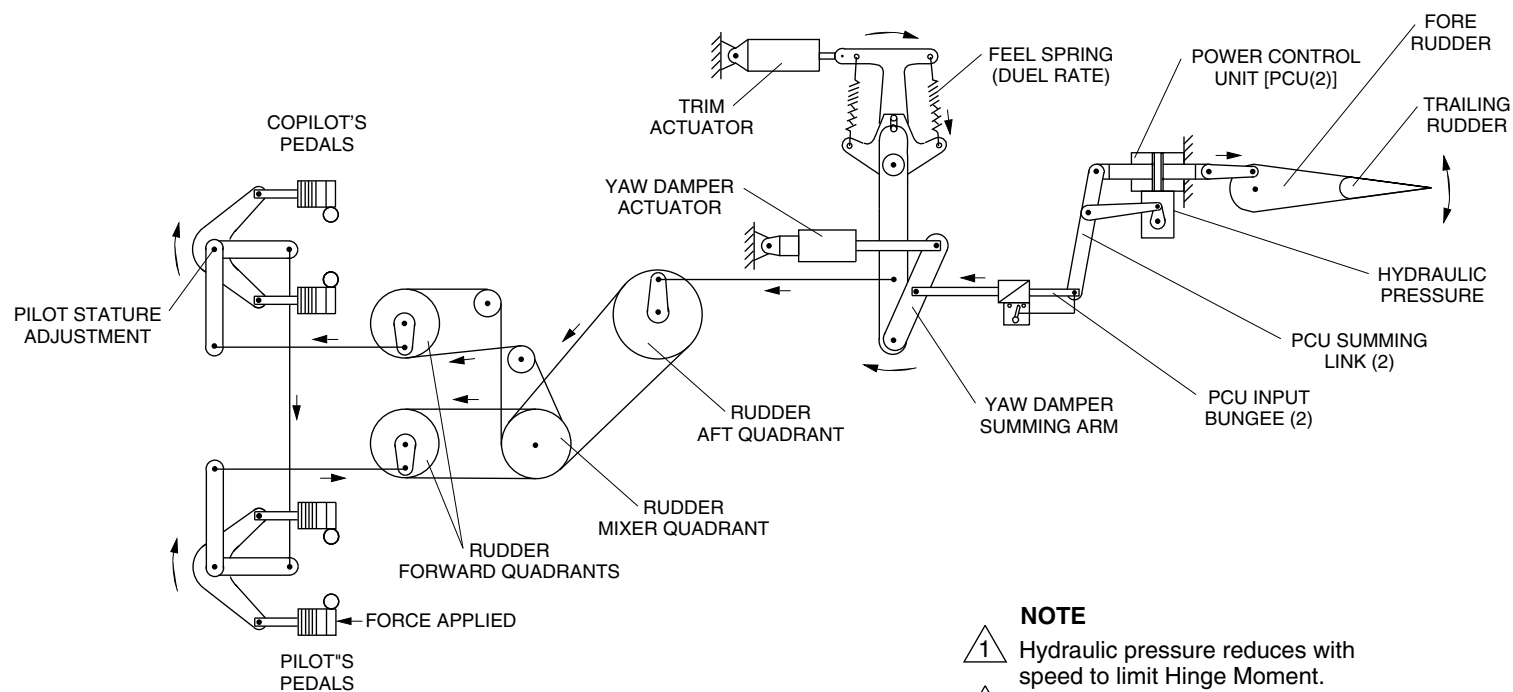
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Config 001
Page 5
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



NOTE

- 1 Hydraulic pressure reduces with speed to limit Hinge Moment.
- 2 Trailing Rudder moves twice as far as the Fore Rudder.

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RUDDER CONTROL SYSTEM PAGE 2
Figure 2

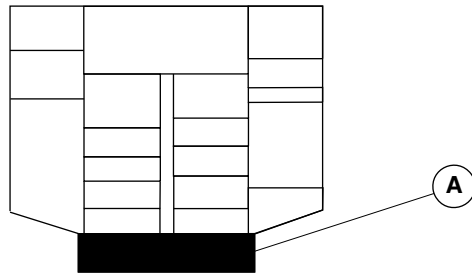
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27-20-00

Config 001
Page 6
Sep 05/2021



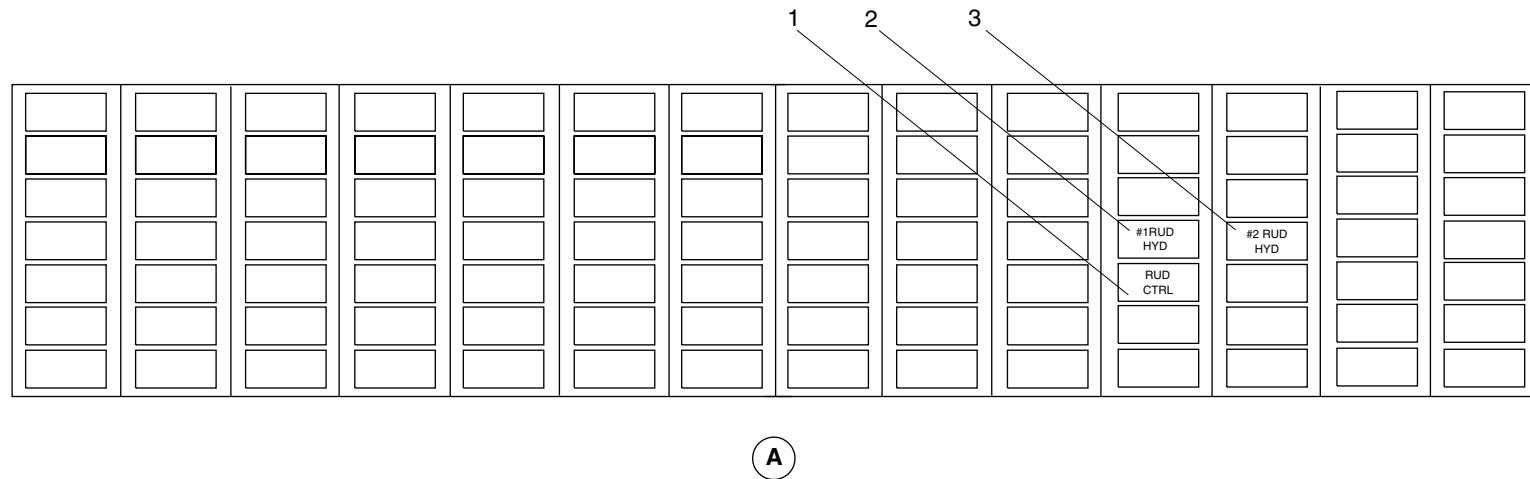
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



OVERHEAD CONSOLE

LEGEND

1. Rudder Control (Amber).
2. #1 Rudder Hydraulic Pressure (Amber).
3. #2 Rudder Hydraulic Pressure (Amber).



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RUDDER CONTROL SYSTEM CAUTION LIGHTS
Figure 3

PSM 1-84-2A
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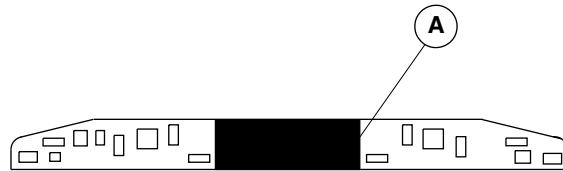
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Config 001
Page 7
Sep 05/2021



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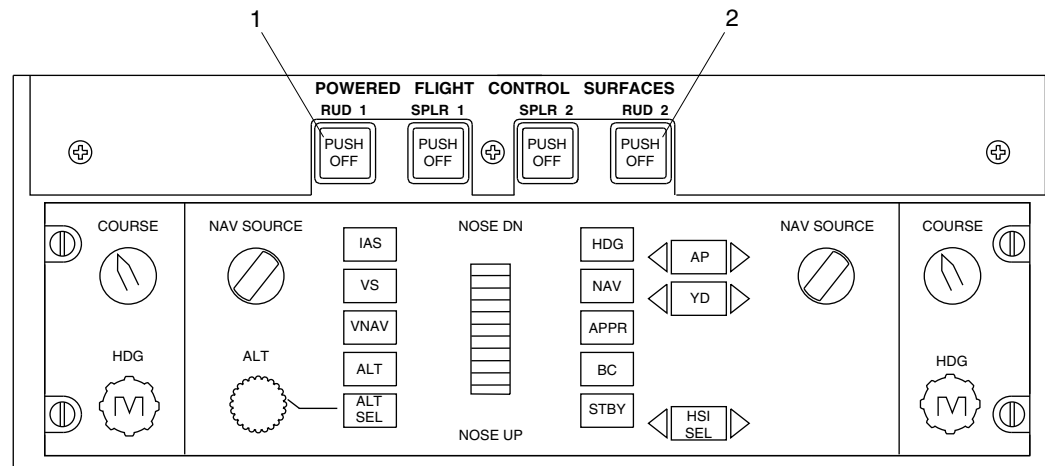
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



GLARESHIELD PANEL

LEGEND

1. Rudder 1 Pushbutton Switchlight.
2. Rudder 2 Pushbutton Switchlight.



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RUDDER PUSH OFF SWITCHLIGHTS
Figure 4

PSM 1-84-2A
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See first effectivity on page 2 of 27-20-00
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27-20-00

Config 001
Page 8
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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RUDDER SYSTEM

Introduction

The rudder system transmits the yaw commands from the pilot's and copilot's rudder pedals to the rudder control surfaces. The rudder surfaces are moved in proportion to the amount of rudder pedal input.

General Description

The rudder system transmits the yaw commands from the pilot's and copilot's rudder pedals to the rudder through a series of quadrants, cables, levers and pushrods. Movement of the rudder pedals is transmitted to two hydraulic Power Control Units (PCUs). The PCUs move the rudder in proportion to the amount of rudder pedal input. The rudder system also has inputs from the Automatic Flight Control System (AFCS) and the rudder trim actuator.

The rudder system has the components that follow:

- Pedals, Rudder (27-21-01)
- Quadrants, Rudder Forward (27-21-06)
- Quadrant, Rudder Mixer(27-21-11)
- Restrictor, Rudder (27-21-16)
- Quadrant, Rudder Aft (27-21-21)
- Unit/Actuator, Power Control (27-21-26)

- Bungee, PCU Input (27-21-31)
- LVDT, Rudder Position (27-21-36).

Detailed Description

There are two pairs of mechanically linked rudder pedals located in the flight compartment, one pair for the pilot and one pair for the copilot. Directional control movements from either the pilot's or copilot's rudder pedals are transmitted simultaneously by two independent cable circuits to the mixer quadrant. The mixer quadrant is located below the cabin floor, aft of the turbine burst zone. From the mixer quadrant, a single cable circuit is routed aft and below the left side of the cabin floor. The cables pass through the pressure seals at the aft bulkhead, then up to the aft quadrant located behind the front spar of the vertical stabilizer.

From the aft quadrant, a pushrod actuates two hydraulic Power Control Units (PCUs) through an artificial feel spring unit. The two PCUs are installed, one above the other, on the forward face of the vertical stabilizer rear spar and connected to the rudder control surface at the midpoint of the fore rudder. The No.2 system supplies hydraulic power to the upper PCU and the No.1 system supplies hydraulic power the lower PCU.

A mechanical rudder input restrictor is attached to the right rudder forward quadrant. The restrictor is also connected to the flap select lever with pushrods. The restrictor limits rudder travel when the flaps are fully retracted.

A pressure compensator in the rudder PCU keeps 50 psi (344.7 kPa) pressure in the actuator when both hydraulic systems are inoperative. This retained pressure supplies the rudder with gust dampening protection when the aircraft is parked. It also supplies flutter suppression in flight when there is no system hydraulic

PSM 1-84-2A

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See first effectivity on page 2 of 27-21-00

Config 001

27-21-00

Config 001
Page 2
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

pressure available. A leak detect valve monitors for internal leakage in the rudder PCU.

Pedals, Rudder

Refer to Figures 1 , 2 and 3.

The pilot's and copilot's rudder pedals are pivot-mounted to brackets below the flight compartment floor. A rudder pedal adjustment handle allows fore and aft adjustment of either pair of rudder pedals to accommodate individual pilot preference for pedal difference. Each adjustment handle is located on the pedestal between the pilot's feet.

Each pair of rudder pedals is attached through a series of levers to its applicable rudder forward quadrant installed below the flight compartment floor.

Refer to Figure 4.

On aircraft with ModSum 4-309210 OR SB84-31-33 OR SB84-31-39 OR SB84-31-40 incorporated, the force sensors are installed on the rudder input connecting rods. Each pair of rudder pedals is attached to its applicable rudder forward quadrant, through a series of levers. There are four force sensors, one for each rudder pedal. The sensors supply a signal in proportion to the force applied to the rudder pedals, which is recorded by the Flight Data Recorder.

Forward and aft movement of the rudder pedals turns the rudder forward quadrant giving control of the rudder. The pilot's and the copilot's rudder pedals are connected to each other through an interconnect rod. The pilot or the copilot can operate the rudder control system.

The rudder pedals rotate about the pedal support tubes to supply foot brake control. A system of rods and levers transmits movement

of the rudder pedals to the brake cable system. Two single cables from the pilot's brake pedals, operate the hydraulic brake control valves. A cable connecting the pilot's and copilot's brake pedals to each other allows for operation of the brake system from either the pilot's or the copilot's position.

Quadrant, Rudder Forward

Refer to Figures 5 and 6.

There are two rudder forward quadrants, one located below the pilot's rudder pedals and the other located below the copilot's rudder pedals. Each rudder forward quadrant is connected to its applicable pair of rudder pedals through a series of tubes and levers. The rudder forward quadrant transmits rudder cable control movement from the rudder pedals aft to the mixer quadrant.

Each rudder forward quadrant is located below the flight compartment floor and is attached to the aircraft by two mounting brackets.

The copilot's rudder forward quadrant has a restrictor mechanism lever and two travel stops. This is part of the restrictor mechanism, which limits rudder travel in conjunction with flap deployment.

Quadrant, Rudder Mixer

Refer to Figures 7 and 8.

The mixer quadrant is a three-pulley quadrant installed below the cabin floor, aft of the turbine rotor burst zone. It combines rudder control from the two rudder forward quadrants and transmits it to the rudder aft quadrant. The two pairs of control cables are routed aft from the rudder forward quadrants to the upper two pulleys of the mixer quadrant. From the lower pulley of the mixer quadrant, a single



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

pair of cables is routed aft and below the left side of the cabin floor. The cables pass through the pressurization seal at the rear pressure dome in the aft fuselage, then up to the rudder aft quadrant in the vertical stabilizer. This dual cable arrangement ensures continued rudder control if a single cable failure occurs between either the pilot or copilot's rudder pedals and the mixer quadrant.

Restrictor, Rudder

Refer to Figures 9 and 10.

The rudder restrictor is installed next to the copilot's rudder forward quadrant below the flight compartment floor. Both the forward quadrant and the rudder restrictor are secured to the same support structure.

The rudder restrictor is designed to limit the rudder to 12 degrees right or left rudder travel when 0 degrees flap is selected. If 5 degrees or more flap is selected, then full rudder travel of 18 degrees left and right of centre is restored. The flap selector lever is mechanically linked to the copilot's rudder forward quadrant through a series of levers, connecting rods and a layshaft. At 0 degrees flap lever setting, the rudder restrictor plunger is fully engaged between the two travel stops on the copilot's rudder pulley quadrant, permitting only partial rotation of the forward quadrant. At 5 degrees of flap or more, the rudder restrictor plunger is withdrawn by means of levers, pushrods and a layshaft. This allows full rotation of the copilot's rudder forward quadrant thus restoring full rudder travel.

Quadrant, Rudder Aft

Refer to Figures 11 and 12.

The rudder aft quadrant is located halfway up the vertical stabilizer next to the Power Control Unit. It is the last quadrant in the rudder control system. The rudder aft quadrant transmits rudder control inputs from the mixer quadrant to the Power Control Units. The rudder aft quadrant has the primary travel stops for the rudder cable system.

Rotation of the aft quadrant is opposed by two feel springs, which supply an artificial feel to the rudder pedals.

Unit Actuator, Power Control

Refer to Figures 13 and 14.

There are two Power Control Units (PCUs), one located above the other in the vertical stabilizer. The PCUs are attached directly to the mid-point of the fore rudder section of the rudder. The No.2 hydraulic system powers the upper PCU while the No.1 hydraulic system powers the lower PCU. Movement of the PCUs causes the rudder to move and give directional control and stability to the aircraft.

Each PCU has a single fully balanced ram and integral hydraulic manifold. A mechanically operated four-way Main Control Valve (MCV) positions the ram. Position feedback is done by a mechanical linkage, which compares the commanded ram position with the actual position. Input from the pilot or the Automatic Flight Control System causes the four way control valve to reposition the ram towards the commanded position. Movement of the ram (and the rudder) stops when the actual ram position is the same as the commanded position.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

The hydraulic pressure supplied to the rudder PCUs is limited as a function of airspeed in order to reduce excessive yaw rate and structural loading of the aircraft. This pressure limiting function is performed by the Flight Control Electronic Control Unit (FCECU) in conjunction with the Rudder Hinge Moment Limiter System (HML). Airspeed data is obtained from the Air Data Units (ADUs).

The Rudder Hinge Moment Limiter System (HML) is a part of each PCU. There are two pressure sensors and a pressure control valve for each PCU. The HML also controls an external pressure shut-off valve. The HML is interfaced with and controlled by the FCECU.

During normal operation, the No.1 and No.2 hydraulic systems supply 3000 psi (20684 kPa) of pressure. The HML and FCECU reduce this pressure to 1500 psi (10342 kPa) at each PCU. As airspeed increases, the FCECU and the HML gradually reduce the pressure at each PCU down to a minimum pressure of 600 psi (4137 kPa). Limiting the hydraulic pressure available to the rudder PCUs reduces total rudder travel at higher aircraft speeds. If either No.1 or No.2 hydraulic systems fails, the FCECU energizes the pressure shut-off valve and sets the pressure control valve of the failed system to zero pressure. At the same time, the FCECU increases the pressure to the active PCU to compensate for the loss. The rudder control system continues to operate if one PCU fails.

[Refer to Figures 15 and 16.](#)

If hydraulic failure does occur, the related No. 1 RUD HYD or No.2 RUD HYD caution light comes on. These caution lights also come on if the FCECU has shut down one of the hydraulic systems due to valve malfunction or if either RUD 1 or RUD 2 switchlight is pushed.

The RUD CNTRL caution light comes on if:

- FCECU detects a difference of 20 knots (37.06 km/h) or more between the Airspeed 1 and Airspeed 2 from the Air Data Units
- FCECU loses its ability to limit hydraulic pressure
- Both hydraulic systems fail.

Bungee, PCU Input

[Refer to Figures 17 and 18.](#)

There are two PCU input bungees, one for each rudder PCU. Each PCU input bungee has a spring strut and a motion detector. Each spring strut is installed between the feel unit assembly and its related PCU summing bar.

A spring strut motion detector is installed parallel to each spring strut and has a cam arm, link, and limit switch.

[Refer to Figure 16.](#)

During normal operation, each spring strut length stays constant. Any rotation of the rudder aft quadrant by the pilot is directly transmitted through the spring strut to its related PCU. In the event of a jam of the PCU or its linkage, any further pilot input to the rudder aft quadrant causes an inner spring in the strut to either stretch or compress. This action either increases or decreases the overall length of the strut. The change in strut length causes the motion detector cam arm to trigger the limit microswitch. When the limit microswitch has been triggered, the applicable RUD 1 or RUD 2 rudder switchlight on the glareshield panel comes on with the inscription, PUSH OFF, to warn of the jammed condition. The switchlight can then be pushed to depressurize the jammed system.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

The illuminated inscription on the switchlight then changes to OFF to show the pilot that the switch has been pushed.

LVDT, Rudder Position

Refer to Figures 19 , 20 and 21.

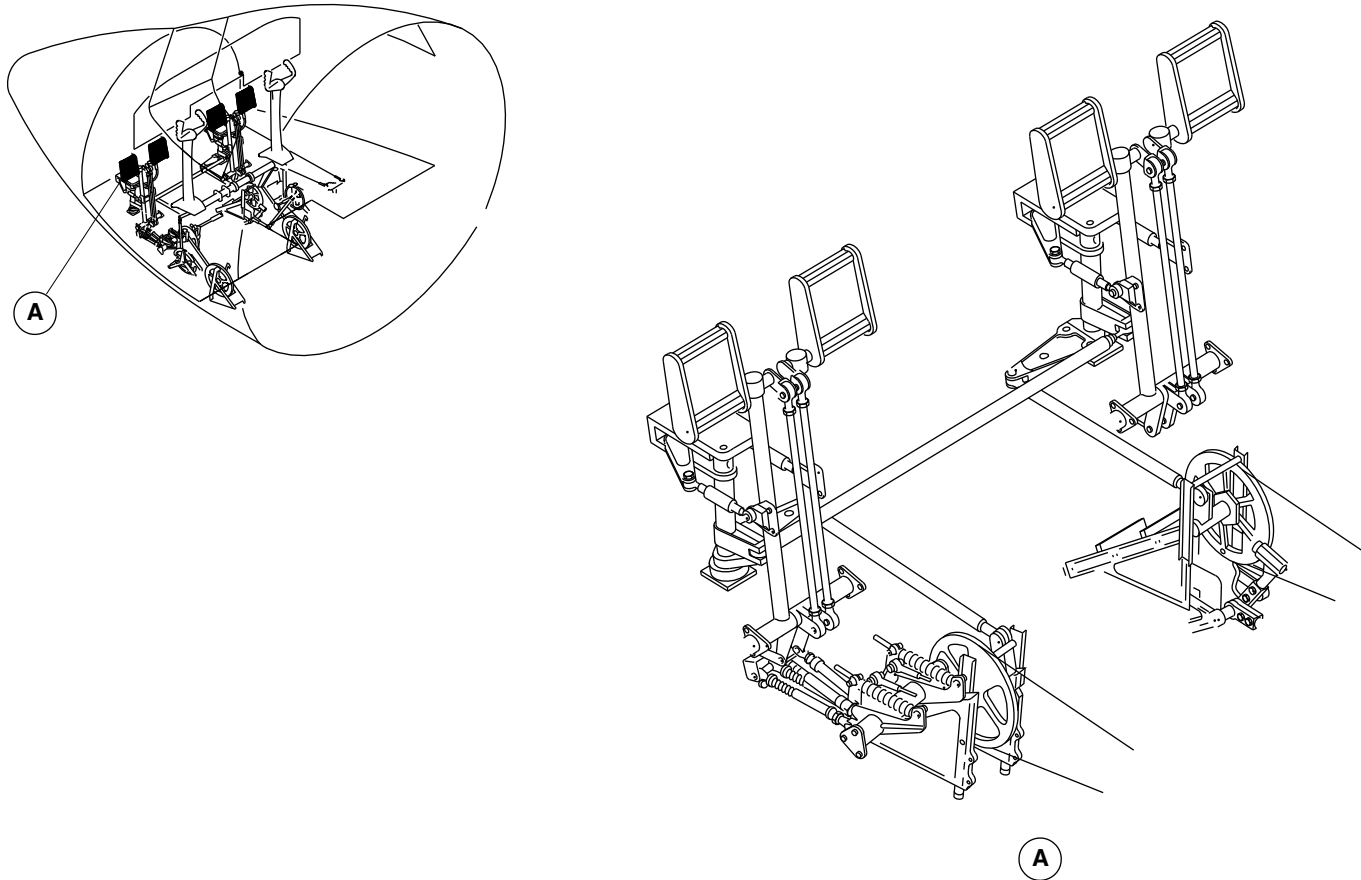
A Linear Variable Differential Transformer (LVDT) is installed to monitor the position of a flight control surface. The LVDT supplies an electrical signal, which is equivalent to the angular position of a flight control surface.

A rudder position LVDT is located in the vertical stabilizer next to the rudder. It monitors the position of the rudder then transmits the rudder position data to the Flight Data Recorder (FDR). The rudder position LVDT also transmits the rudder position data to the Powered Flight Control Surface Indicator (PFSCI) located on the pilot's Multi Functional Display (MFD) to display the position of the rudder.



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RUDDER PEDALS PAGE 1
Figure 1

PSM 1-84-2A
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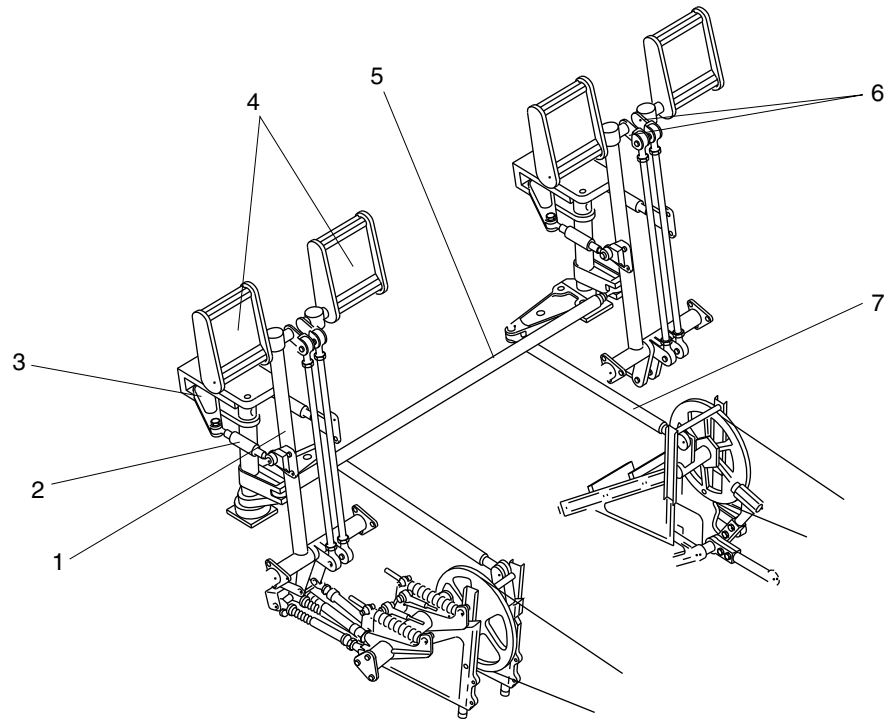
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Config 001
Page 7
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Pedal Column.
2. Rudder Input Connecting Rod.
3. Pedal Adjustment Mechanism.
4. Rudder Pedals.
5. Rudder Control Interconnect Push Rod.
6. Brake Linkages.
7. Input Quadrant Connecting Rod.

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RUDDER PEDALS PAGE 2
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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Config 001

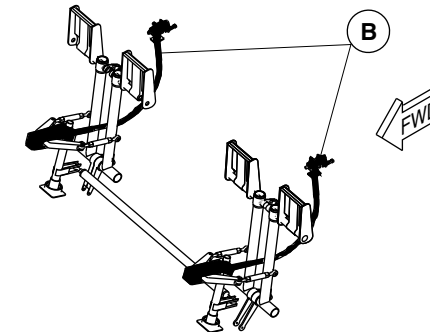
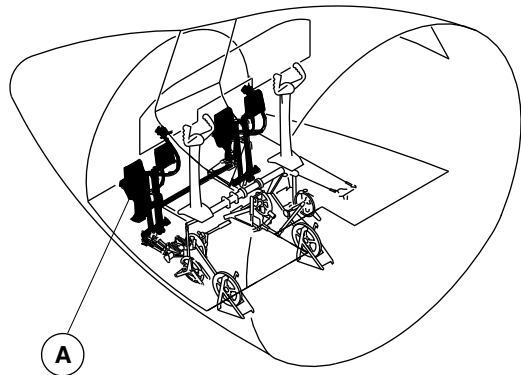
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Config 001
Page 8
Sep 05/2021



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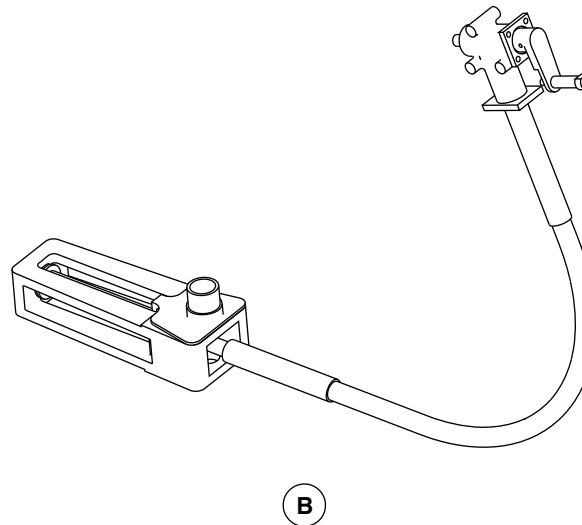
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



A RUDDER PEDAL ASSEMBLY

NOTE

Left side shown.
Right side similar.



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RUDDER PEDAL ADJUSTMENT ASSEMBLY
Figure 3

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-21-00
Config 001

27-21-00

Config 001
Page 9
Sep 05/2021

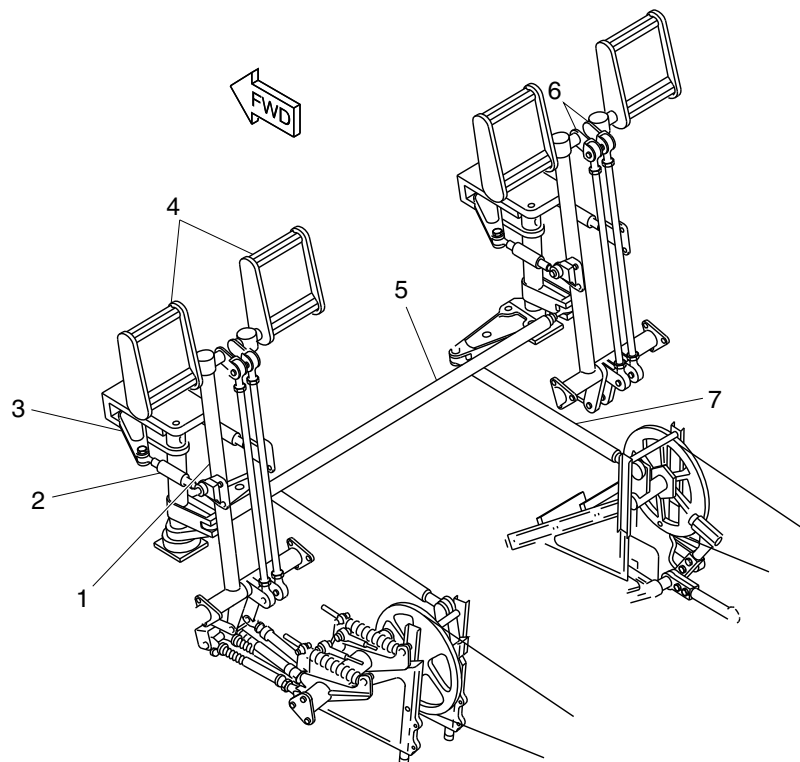


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

LEGEND

1. Pedal column.
2. Rudder input connecting rod – force sensor.
3. Pedal adjustment mechanism.
4. Rudder pedal.
5. Rudder control interconnect push rod.
6. Brake linkage.
7. Input quadrant connecting rod.



POST SB 84-31-33
POST SB 84-31-39
POST SB 84-31-40
POST MODSUM 4-309210

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Rudder Pedals
Figure 4

PSM 1-84-2A
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Config 001

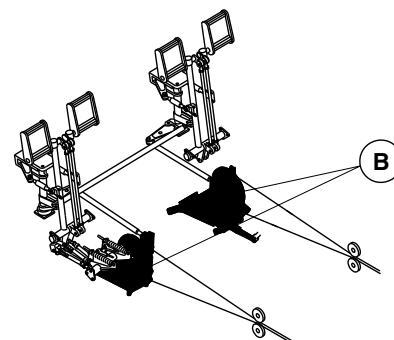
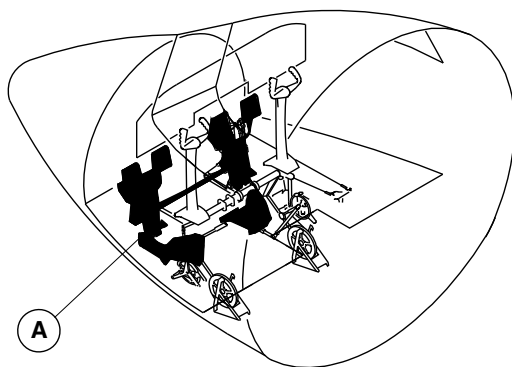
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Config 001
Page 10
Sep 05/2021

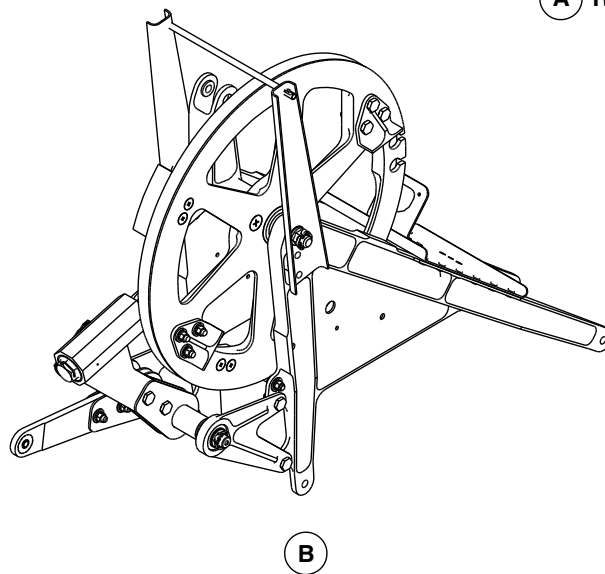


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A RUDDER FORWARD ASSEMBLY



NOTE

Right Side Shown.
Left Side Similar.

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RUDDER FORWARD QUADRANT PAGE 1
Figure 5

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-21-00
Config 001

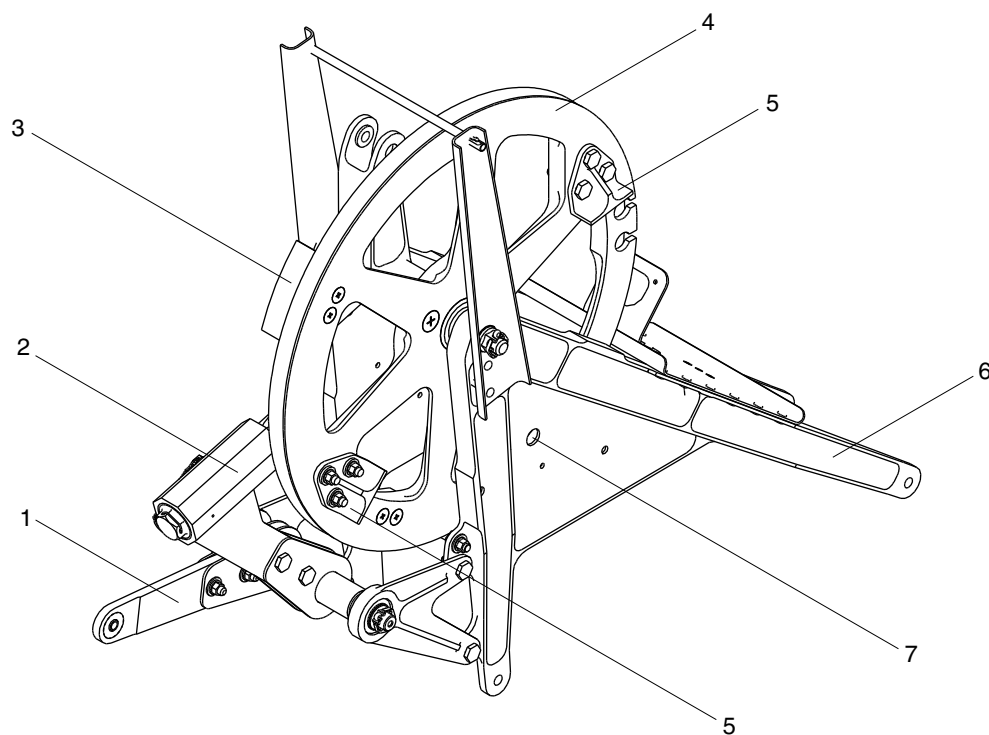
27-21-00

Config 001
Page 11
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Rudder Restrictor Input Lever.
- 2. Rudder Restrictor.
- 3. 12° Stop.
- 4. Rudder Input Quadrant.
- 5. Travel Stop.
- 6. Mounting Bracket.
- 7. Rig Pin Hole.

fs254a01.cgm

RUDDER FORWARD QUADRANT PAGE 2
Figure 6

PSM 1-84-2A
EFFECTIVITY:
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Config 001

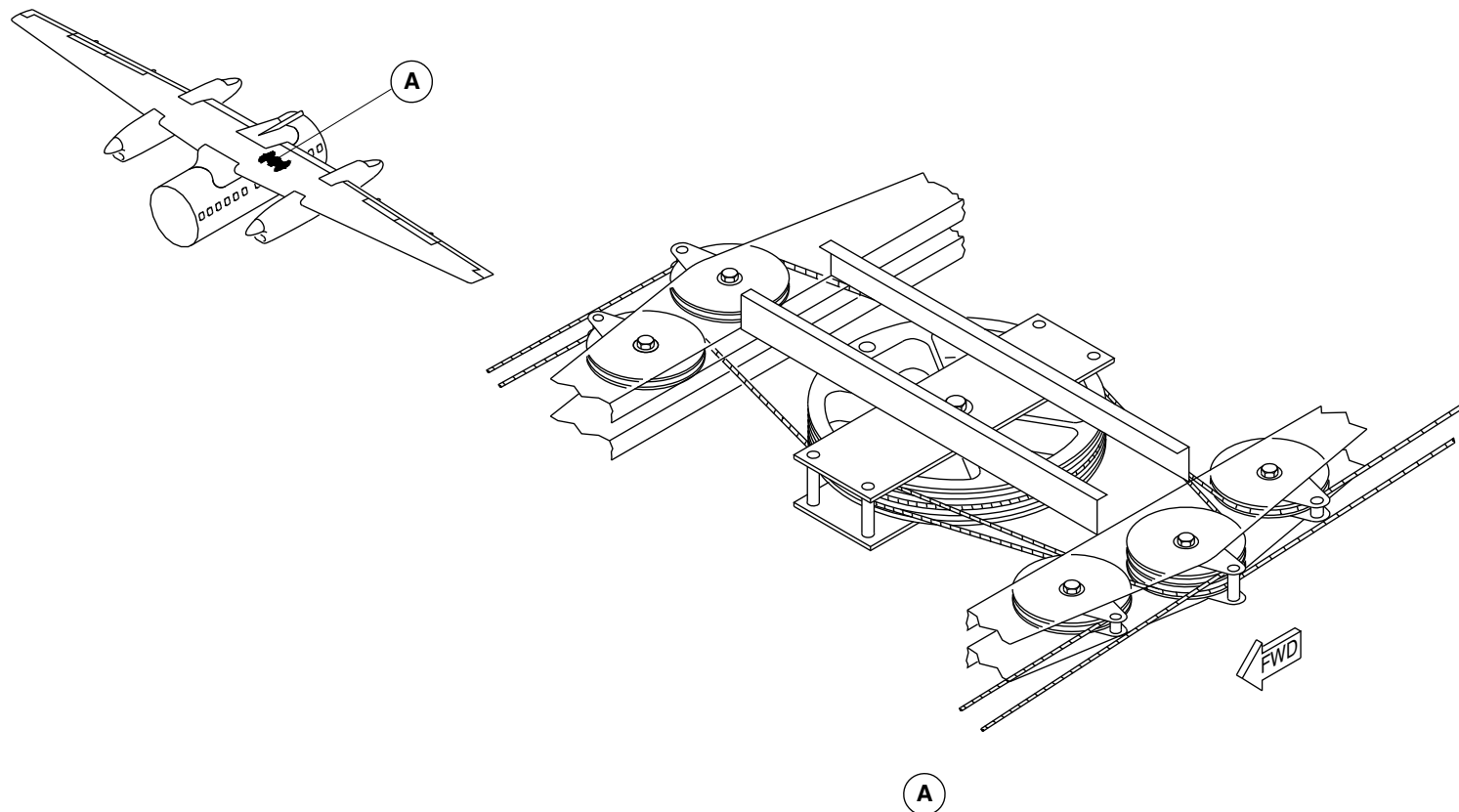
27-21-00

Config 001
Page 12
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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RUDDER MIXER QUADRANT PAGE 1
Figure 7

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-21-00
Config 001

27-21-00

Config 001
Page 13
Sep 05/2021

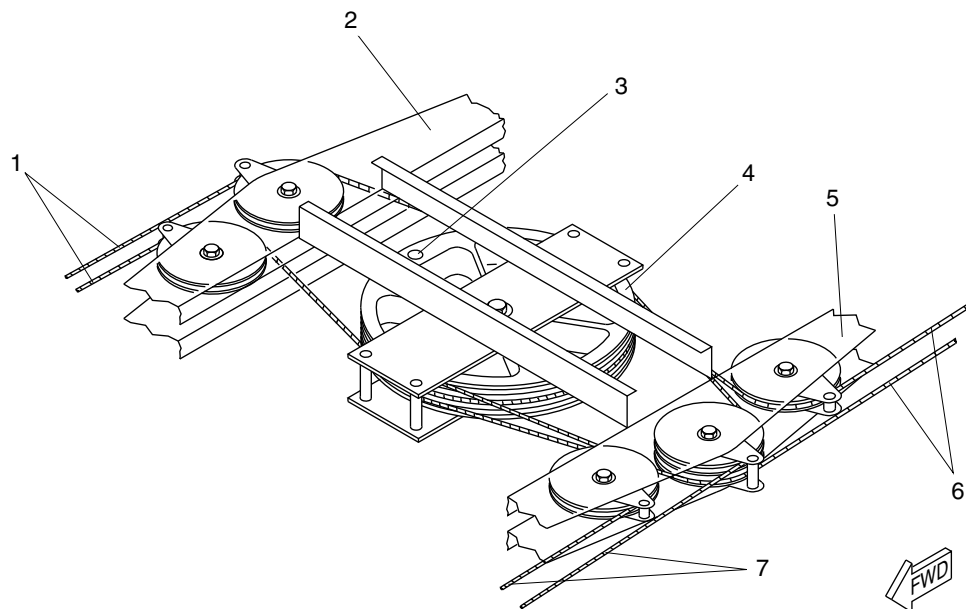


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

LEGEND

1. RH Forward Quadrant Control Cables.
2. RH Pulley Bracket.
3. Rig Pin Hole.
4. Rudder Mixer Quadrant.
5. LH Pulley Bracket.
6. Aft Rudder Quadrant Control Cables.
7. LH Forward Quadrant Control Cables.



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RUDDER MIXER QUADRANT PAGE 2
Figure 8

PSM 1-84-2A
EFFECTIVITY:
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Config 001

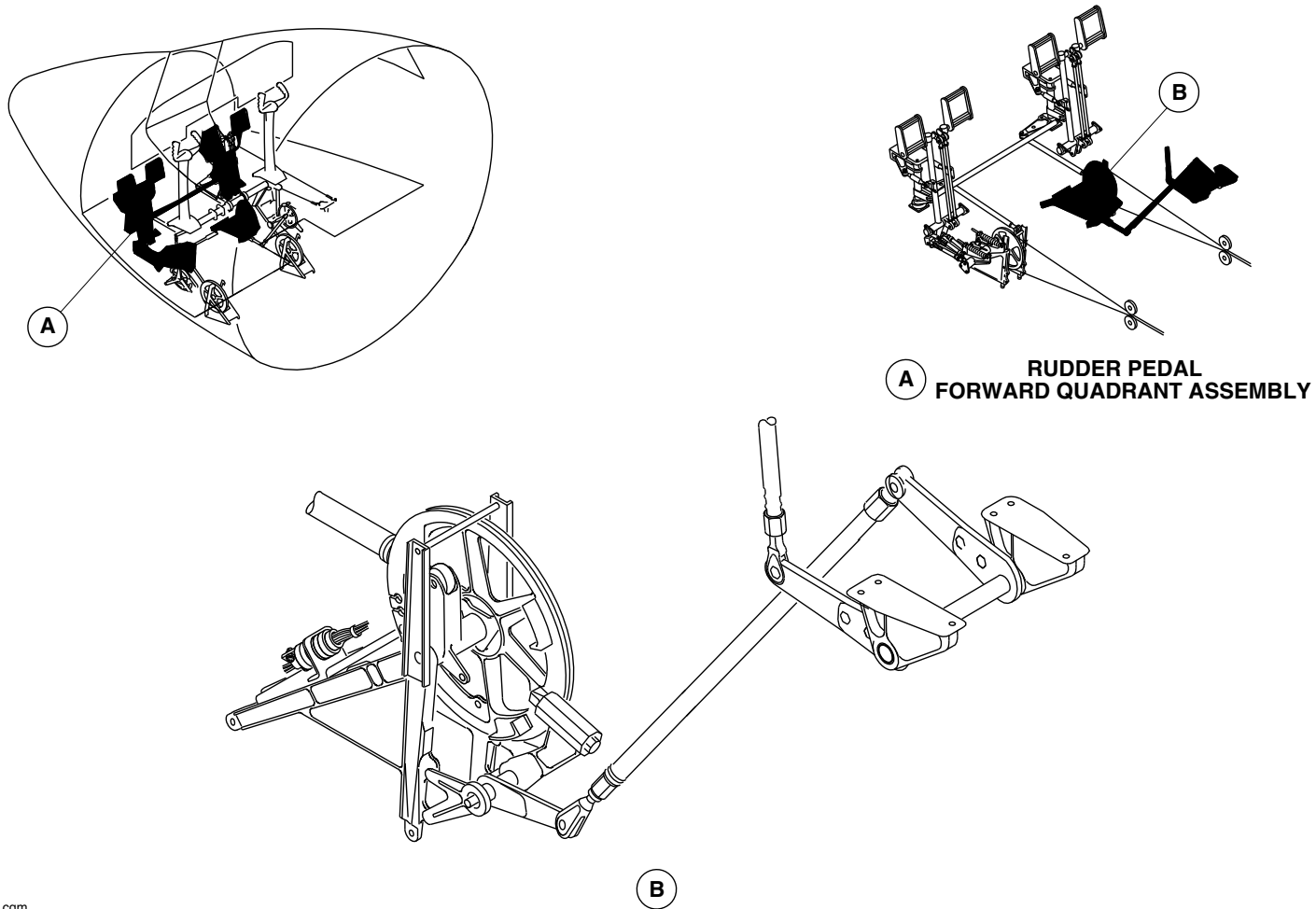
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Config 001
Page 14
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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RUDDER RESTRICTOR PAGE 1
Figure 9

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-21-00
Config 001

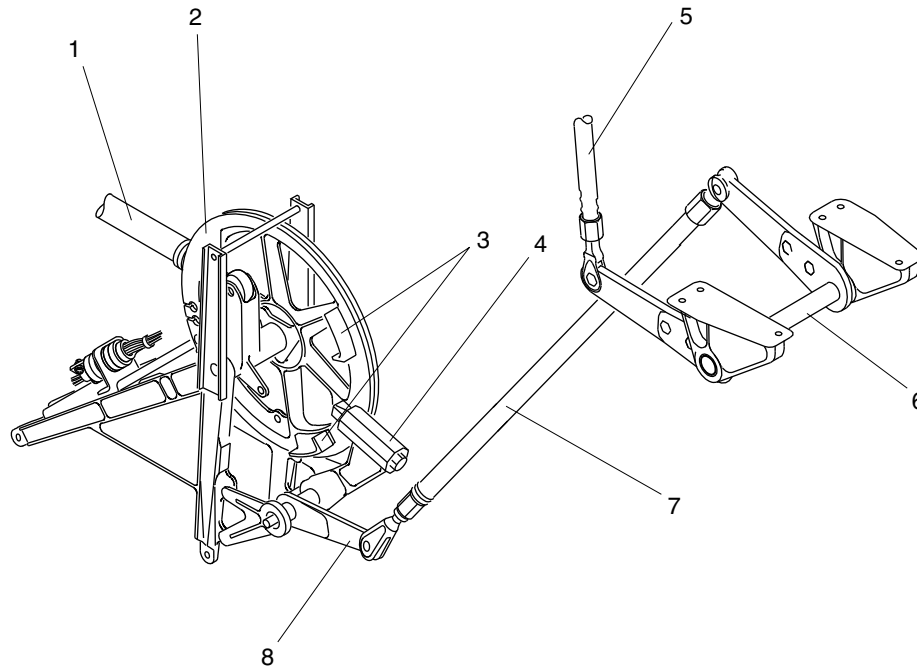
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Config 001
Page 15
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Input Quadrant Connecting Rod.
2. Rudder Forward Quadrant.
3. Rudder Restrictor Stops.
4. Rudder Restrictor Mechanism.
5. Flap Selector Lever Connecting Rod.
6. Layshaft.
7. Connecting Rod.
8. Rudder Restrictor Input Lever.

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RUDDER RESTRICTOR PAGE 2
Figure 10

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-21-00
Config 001

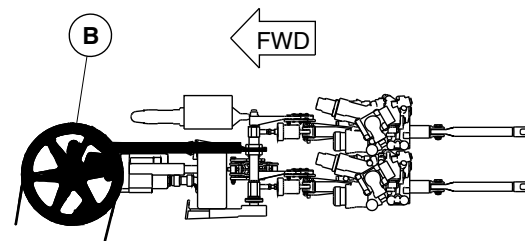
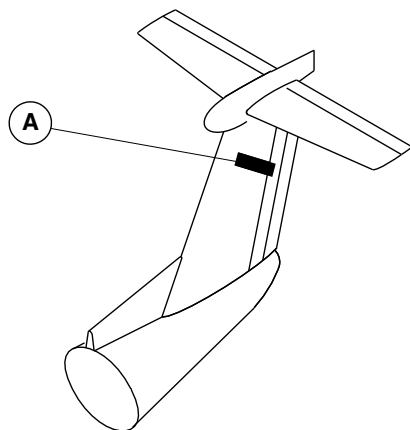
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Config 001
Page 16
Sep 05/2021

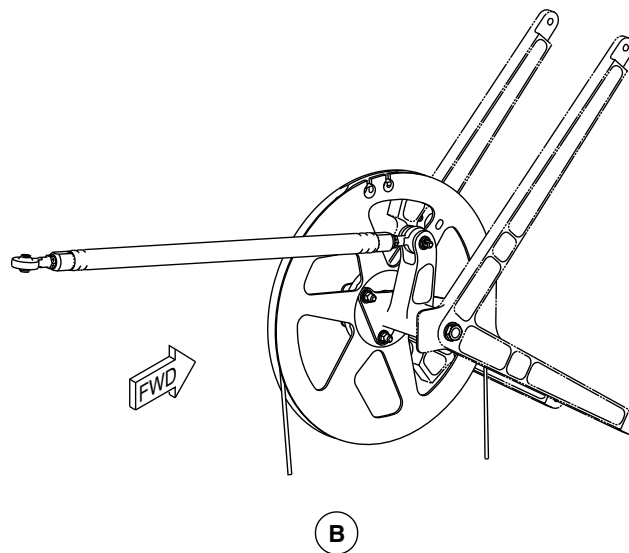


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



A RUDDER SYSTEM CONTROL
VERTICAL STABILIZER



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RUDDER AFT QUADRANT PAGE 1
Figure 11

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-21-00
Config 001

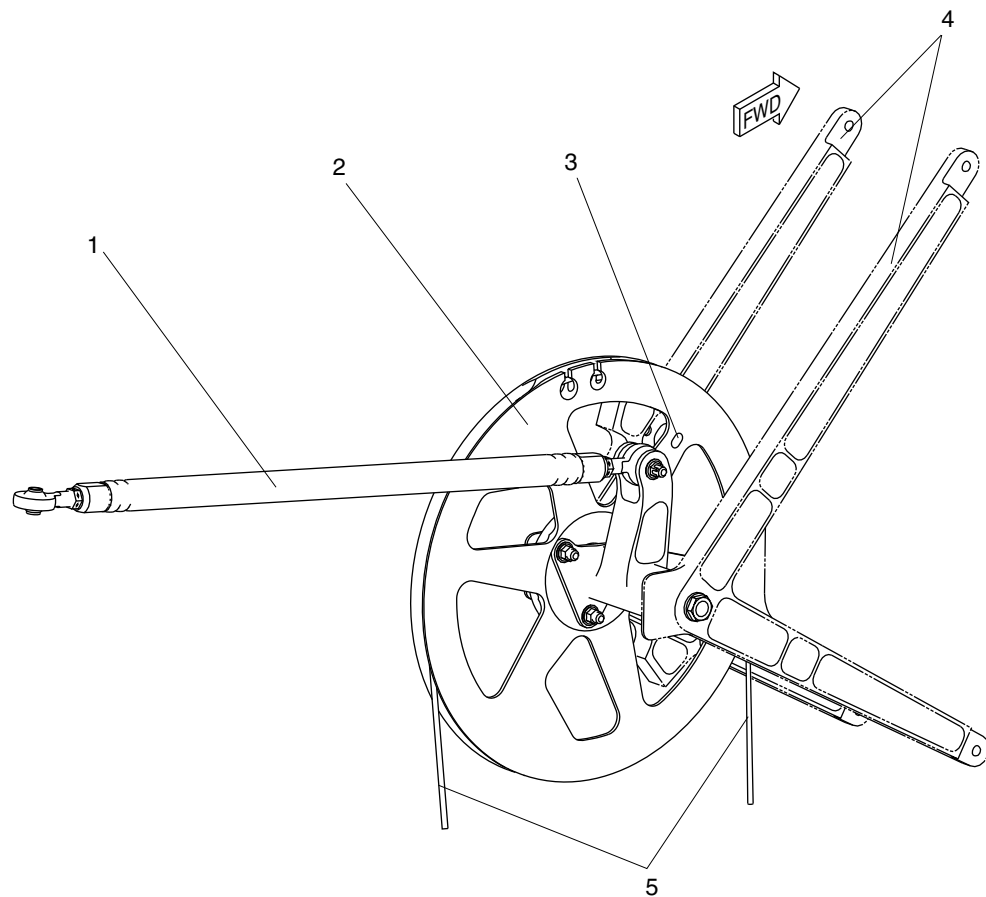
27-21-00

Config 001
Page 17
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Push Rod.
- 2. Rudder Aft Quadrant.
- 3. Rig Pin Hole.
- 4. Mounting Brackets.
- 5. Aft Quadrant Control Cables.

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RUDDER AFT QUADRANT PAGE 2
Figure 12

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-21-00
Config 001

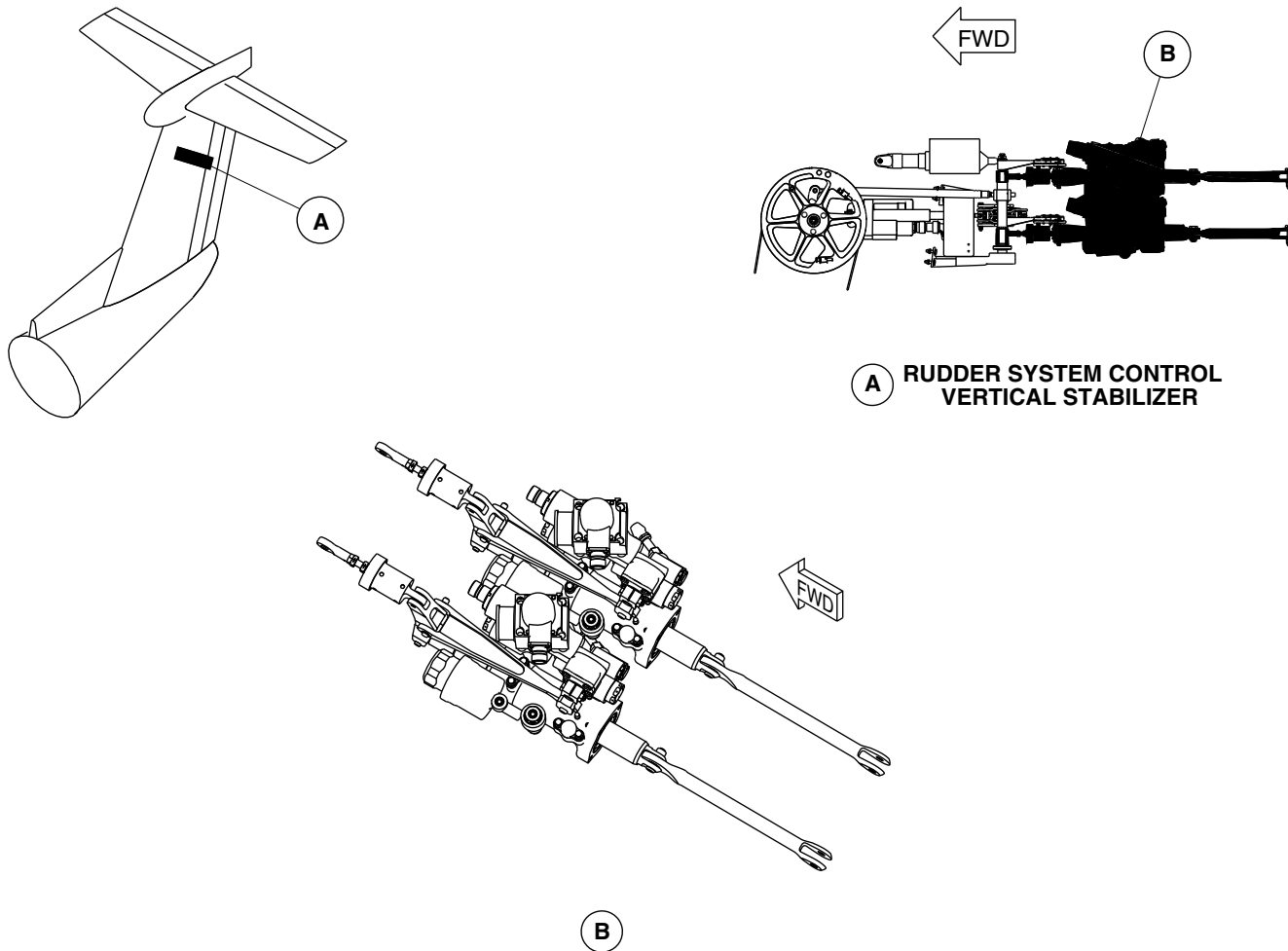
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Config 001
Page 18
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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RUDDER POWER CONTROL UNIT (PCU) PAGE 1
Figure 13

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-21-00
Config 001

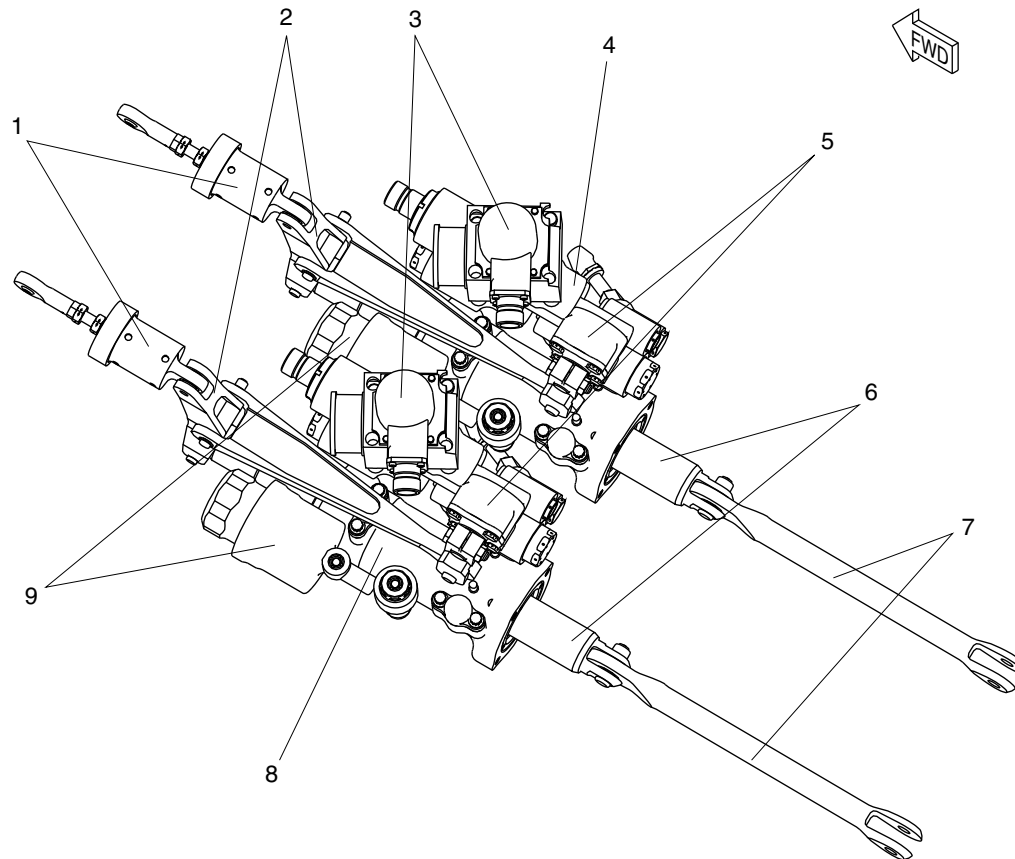
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Config 001
Page 19
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. PCU Input Bungee.
2. Summing Bar.
3. Pressure Control Valve.
4. Upper PCU.
5. Main Control Valve.
6. Actuator Output Rod.
7. Rudder Connecting Rod.
8. Lower PCU.
9. Compensator.

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RUDDER POWER CONTROL UNIT (PCU) PAGE 2
Figure 14

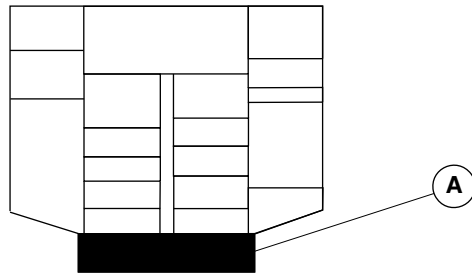
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27-21-00

Config 001
Page 20
Sep 05/2021



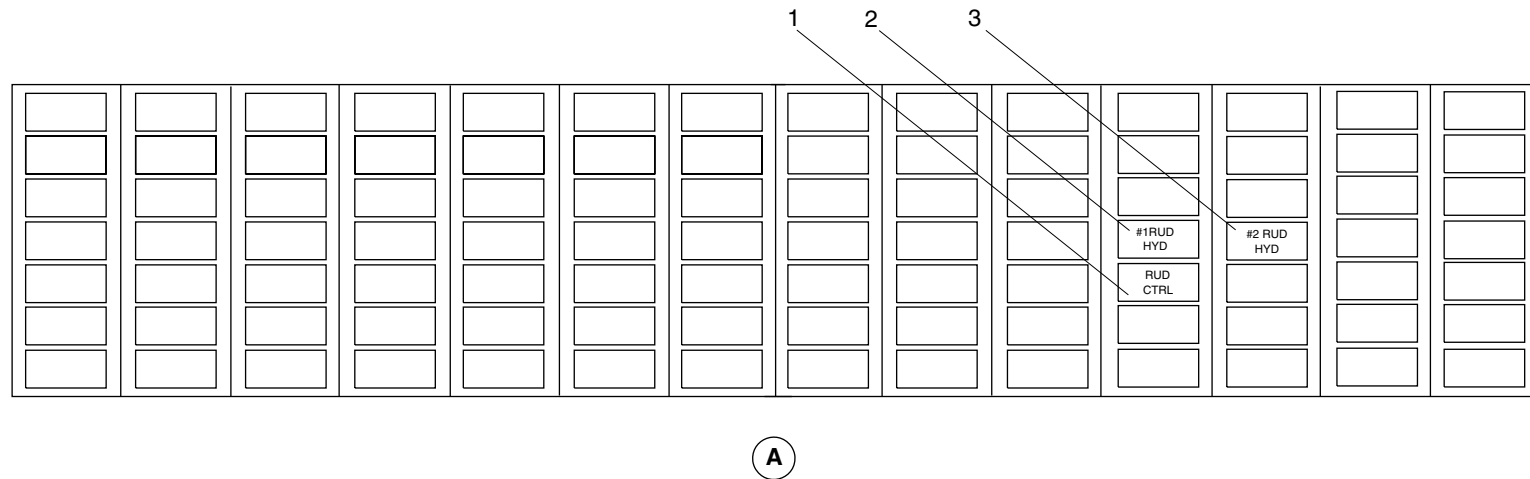
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



OVERHEAD CONSOLE

LEGEND

1. Rudder Control (Amber).
2. #1 Rudder Hydraulic Pressure (Amber).
3. #2 Rudder Hydraulic Pressure (Amber).



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RUDDER CONTROL SYSTEM CAUTION LIGHTS
Figure 15

PSM 1-84-2A
EFFECTIVITY:
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Config 001

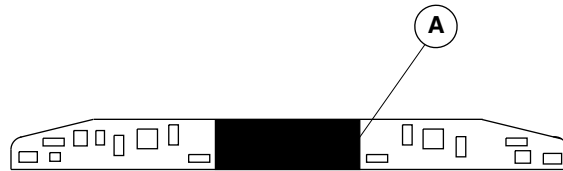
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Config 001
Page 21
Sep 05/2021



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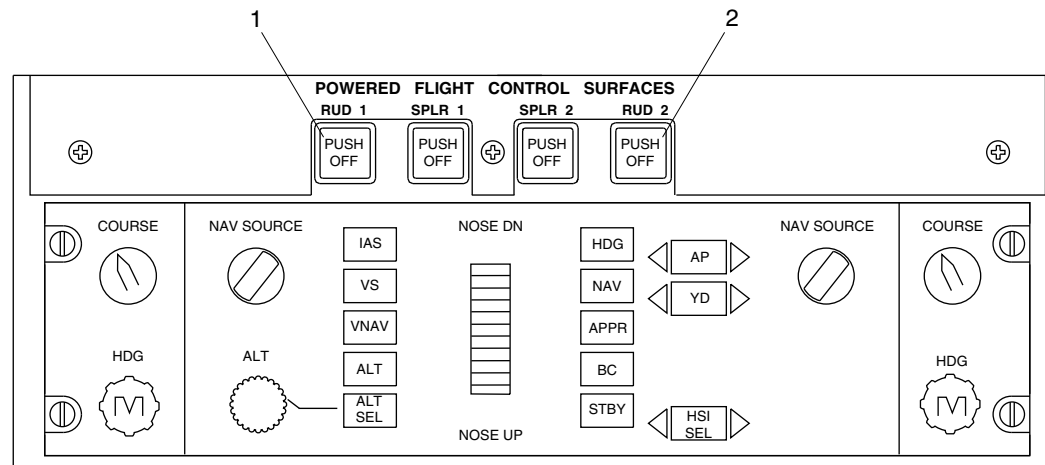
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



GLARESHIELD PANEL

LEGEND

1. Rudder 1 Pushbutton Switchlight.
2. Rudder 2 Pushbutton Switchlight.



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RUDDER PUSH OFF SWITCHLIGHTS
Figure 16

PSM 1-84-2A
EFFECTIVITY:
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Config 001

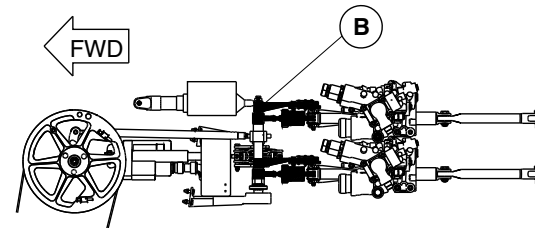
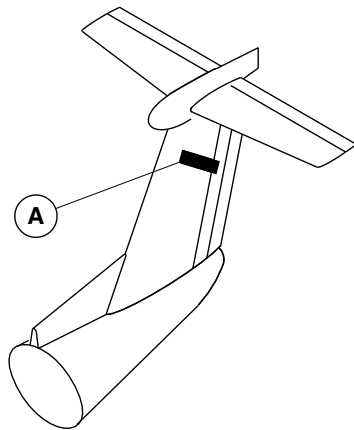
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Config 001
Page 22
Sep 05/2021

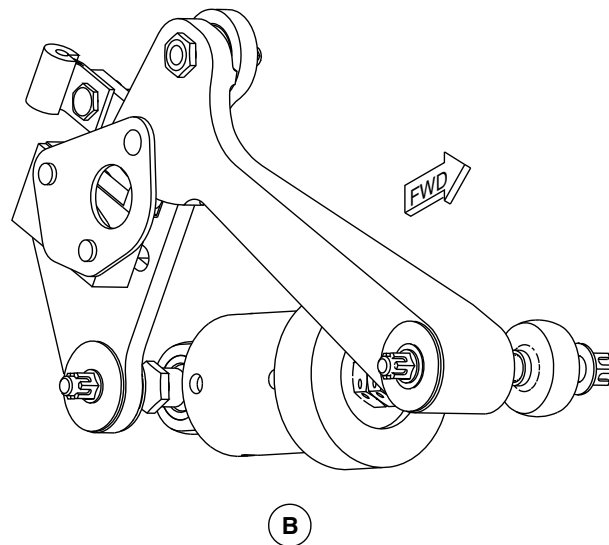


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



A RUDDER CONTROLS –
VERTICAL STABILIZER



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RUDDER PCU INPUT BUNGEE PAGE 1
Figure 17

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-21-00
Config 001

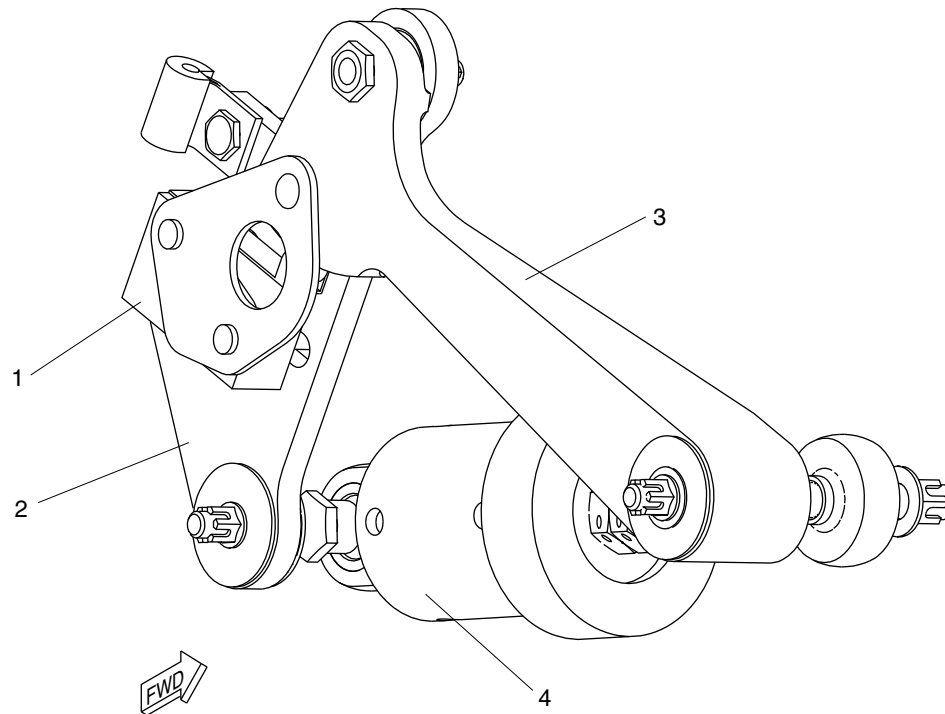
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Config 001
Page 23
Sep 05/2021



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LEGEND

1. Motion Detector Limit Switch.
2. Motion Detector Linkage.
3. Motion Detector Cam Arm.
4. Spring Strut.

fs213a01.cgm

RUDDER PCU INPUT BUNGEE PAGE 2
Figure 18

PSM 1-84-2A
EFFECTIVITY:
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Config 001

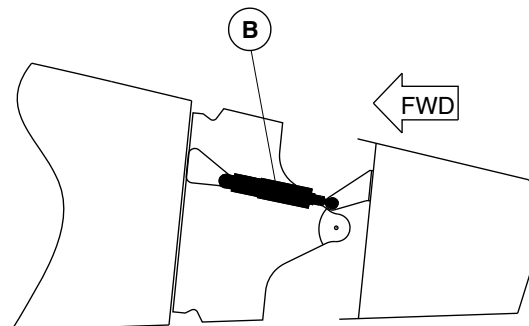
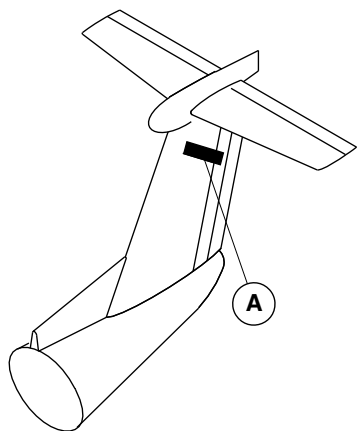
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Config 001
Page 24
Sep 05/2021

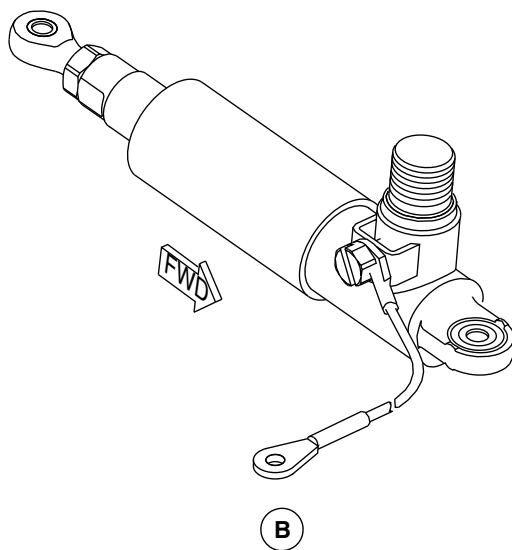


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(A) RUDDER POSITION LVDT ASSEMBLY



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RUDDER POSITION LVDT
Figure 19

PSM 1-84-2A
EFFECTIVITY:
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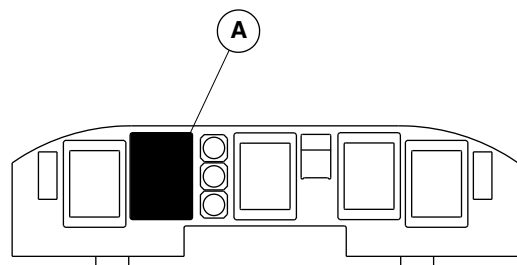
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Config 001
Page 25
Sep 05/2021



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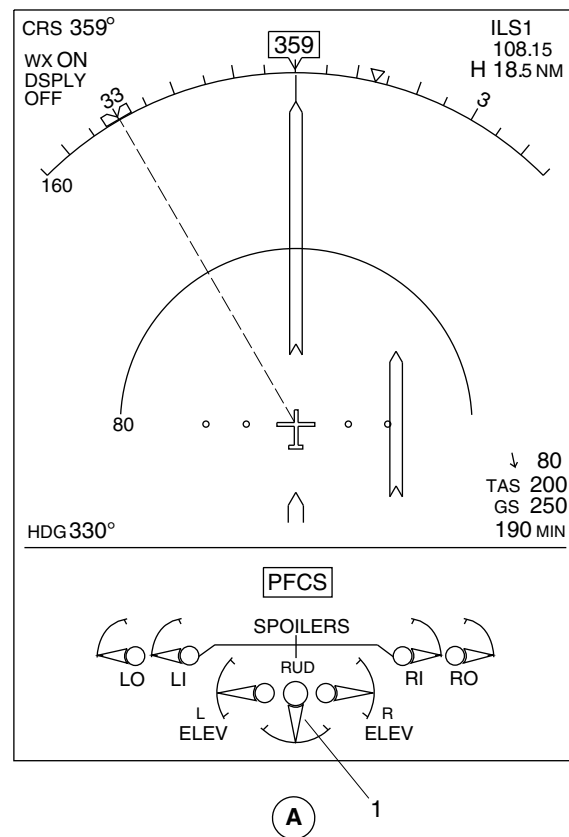
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MAIN INSTRUMENT PANEL

LEGEND

1. Rudder Position Indication.



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MULTIFUNCTION DISPLAY / RUDDER POSITION PAGE 1
Figure 20

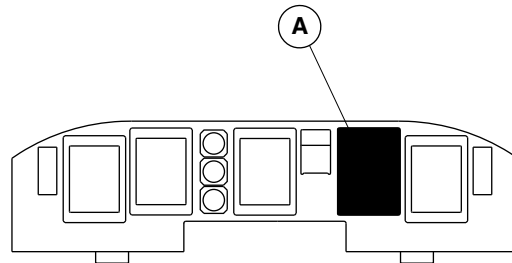
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Config 001

27-21-00

Config 001
Page 26
Sep 05/2021



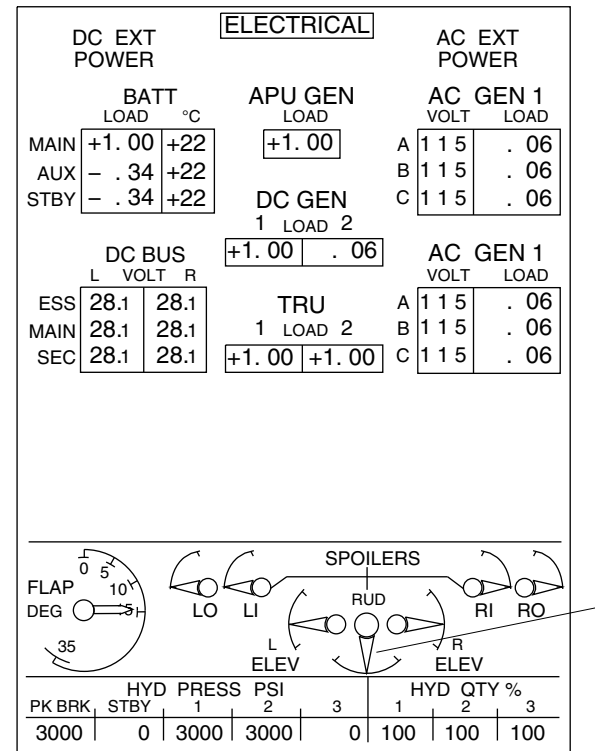
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Rudder Position Indication.



A

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MULTIFUNCTION DISPLAY / RUDDER POSITION PAGE 2
Figure 21

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-21-00
Config 001

27-21-00

Config 001
Page 27
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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See first effectivity on page 2 of 27-21-00
Config 001

27-21-00

Config 001
Page 28
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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27-22-00-001

RUDDER FEEL AND TRIM

Introduction

The rudder feel and trim unit gives artificial feel to the rudder pedals during operation. Because the rudder system is a powered flight control, the unit is used to prevent the pilot from overcontrolling the aircraft in flight.

The rudder trim actuator gives the pilot trim control around the yaw axis of the aircraft in flight. Rudder trimming biases the complete rudder system, so that a newly trimmed position is sensed by the pilot as the neutral position.

General Description

Movement of the rudder pedals extends one of the two springs in the Rudder Feel and Trim Unit. The force required to extend the springs gives artificial feel to the pilots, when operating the rudder pedals. When the force on the pedals is removed, the feel spring contracts, and the pedals return to the neutral position.

The rudder trim switch, located on the centre console, is used to extend or retract the rudder trim actuator. The actuator has an input into the Rudder Feel and Trim Unit, that causes the rudder to move left or right, to trim the aircraft in flight.

The rudder feel and trim system has the components that follow:

- Actuator, Trim (27-22-01)

- Unit, Rudder Feel and Summing (27-22-06)
- Arm, Summing-Yaw Damper (27-22-11)
- Rudder Trim Indicator.

Detailed Description

The rudder aft quadrant is mechanically linked to three distinct units:

- Actuator, Trim
- Unit, Rudder Feel and Trim (Summing)
- Arm, Summing-Yaw Damper.

Each unit input has its own effect on the rudder control system. All three units are attached to a summing bar. The bar sums the inputs and then transmits the force as a single input to the rudder Power Control Units (PCUs).

Actuator, Trim

[Refer to Figure 1.](#)

The trim actuator is attached to the front of the vertical stabilizer and connected at the rear to two trim arms, that form part of the summing unit. The trim actuator has a DC motor which extends and retracts the actuator. The position of the trim actuator is monitored by a Linear Variable Differential Transformer (LVDT) and shown on the RUDDER trim indicator located in the flight compartment on the centre console. The LVDT signal is also transmitted to the Flight Data Recorder (FDR). If the LVDT fails, the trim actuator remains functional but no trim position data will be transmitted to the RUDDER TRIM indicator.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

The trimming system is powered from the left essential bus, through two circuit breakers, RUD TRIM ACT F7 for the trim actuator and RUD TRIM IND G7 for the RUDDER trim indicator.

Refer to Figures 2 and 3.

When the RUDDER trim switch in the flight compartment is selected fully to L, power is directed through a normally closed extend limit switch for rapid left trim, causes the trim actuator to extend. If the actuator extends to its limit, the extend limit switch disconnects power to the motor. When the RUDDER trim switch is only partially rotated to the mid graduation L line, the voltage to the trim actuator is reduced and the trim rate is slowed. Extension of the rudder trim actuator resets the rudder aft quadrant biasing the complete rudder control system to the left. As the rudder is trimmed the position is shown on the RUDDER trim indicator.

When the RUDDER trim switch is rotated to R, the operation is similar but in the opposite direction. The rudder trim actuator retracts.

Unit, Rudder Feel and Summing

Refer to Figures 4 and 5.

The rudder feel and summing unit gives artificial feel to the pilots when operating the rudder pedals. The rudder feel and summing unit is installed in series with the rudder control system and is located behind the rudder aft quadrant.

The rudder feel and summing unit has two feel springs attached to a summing bar that is connected to the rudder hydraulic Power Control Units. Rotation of the feel and summing unit from neutral, in either direction, extends one or the other of the feel springs to give artificial feel during movement of the rudder pedals. The springs are attached to the feel unit arms at the actuator end, rotation of the shaft

actuating arms extends the spring in the direction of rotation while the opposite spring is unaffected. This gives the pilot artificial feel of loading on the rudder pedals, simulating aerodynamic forces on the rudder surface. During trimming, the feel springs and trim arms move together. On completion of trimming, the springs are set at the newly trimmed position and give rudder pedal feel whenever the rudder pedals are operated.

Arm, Summing–Yaw Damper

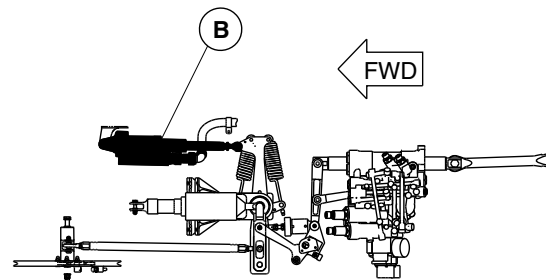
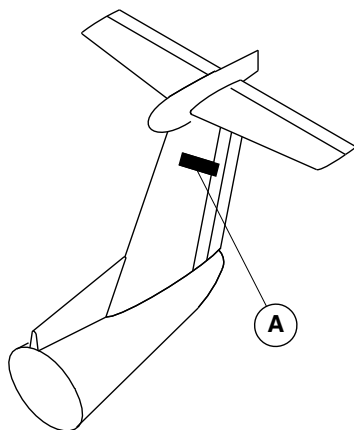
Refer to Figure 6.

The yaw damper is a linear servo actuator installed between the rudder aft quadrant and the summing arm. The yaw damper responds to signals from a dedicated accelerometer to supply small rudder deflections to compensate for yaw oscillations in flight. Yaw damper authority is 4.5 degrees either side of centre. The yaw damper has its own summing arm, which mechanically transmits any yaw damper inputs on to the rudder feel and summing unit. The yaw damper acts independently, or as an autopilot servo in conjunction with the Automatic Flight Control System (AFCS).

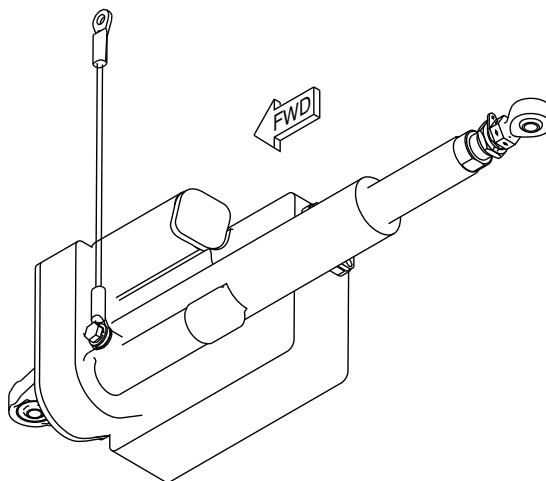


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



A RUDDER CONTROLS –
VERTICAL STABILIZER



B

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RUDDER TRIM ACTUATOR
Figure 1

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-22-00
Config 001

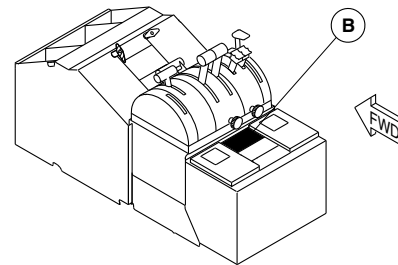
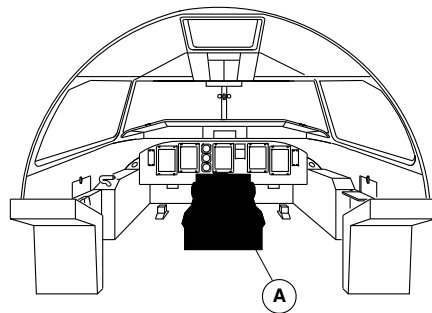
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Config 001
Page 4
Sep 05/2021

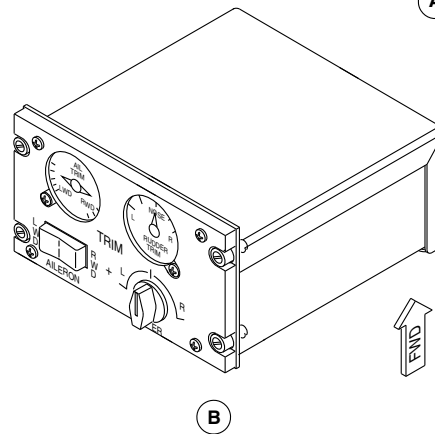


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A CENTRE CONSOLE



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RUDDER TRIM POSITION INDICATOR LOCATOR
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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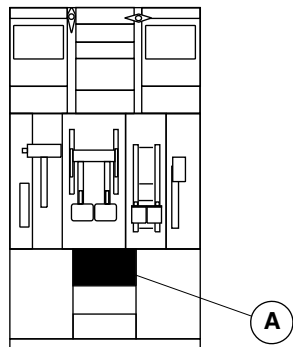
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Config 001
Page 5
Sep 05/2021



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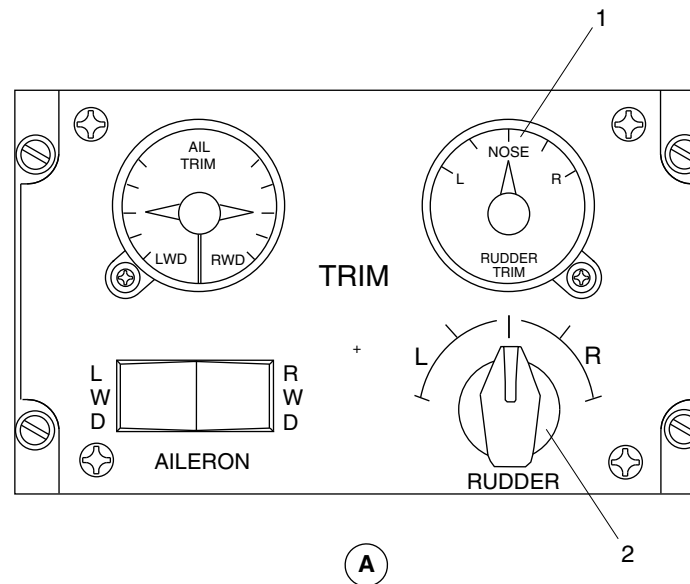
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



CENTRE CONSOLE

LEGEND

1. Rudder Trim Indicator.
2. Rudder Trim Switch.



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RUDDER TRIM POSITION INDICATOR DETAIL
Figure 3

PSM 1-84-2A
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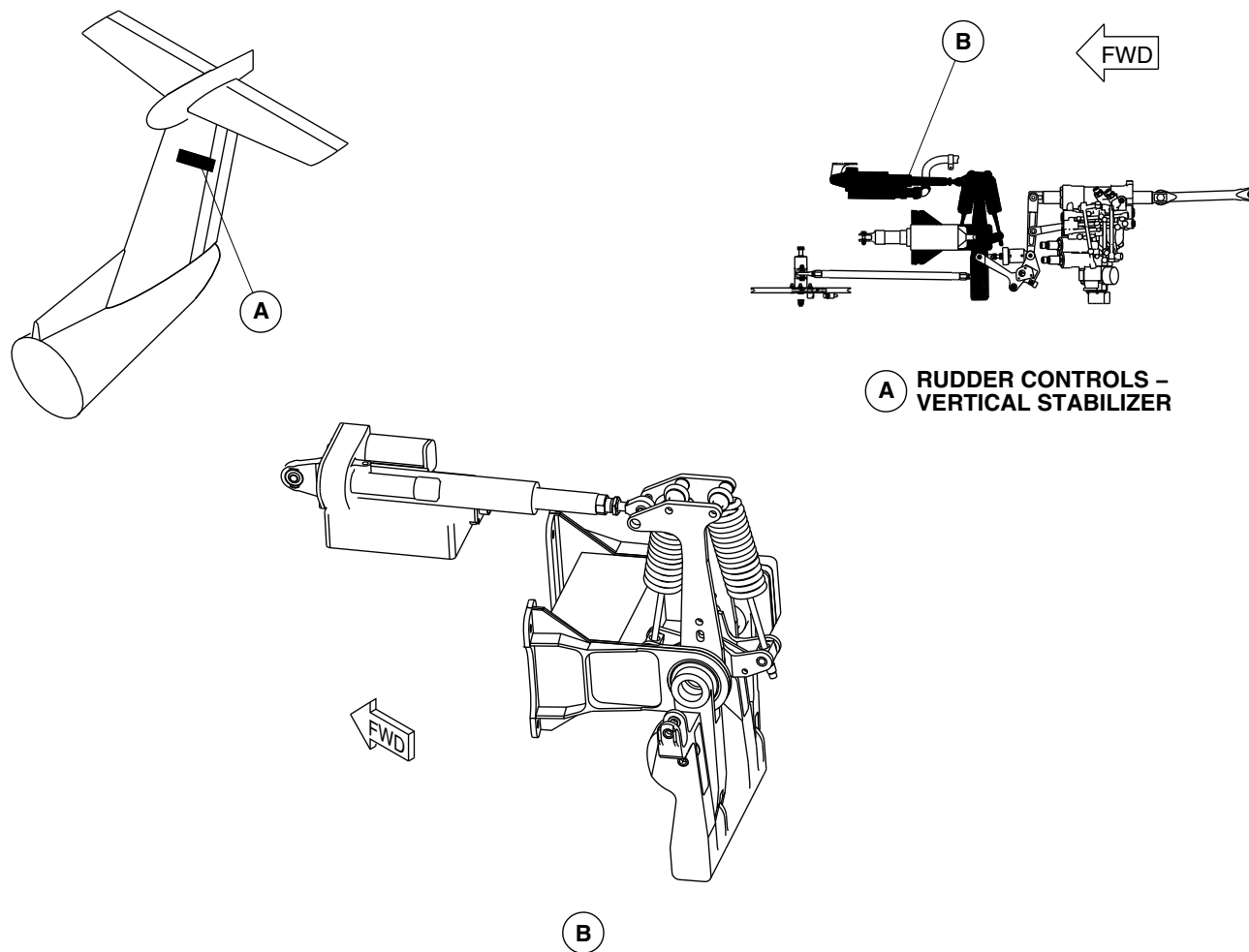
27-22-00

Config 001
Page 6
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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RUDDER FEEL AND TRIM UNIT PAGE 1
Figure 4

PSM 1-84-2A
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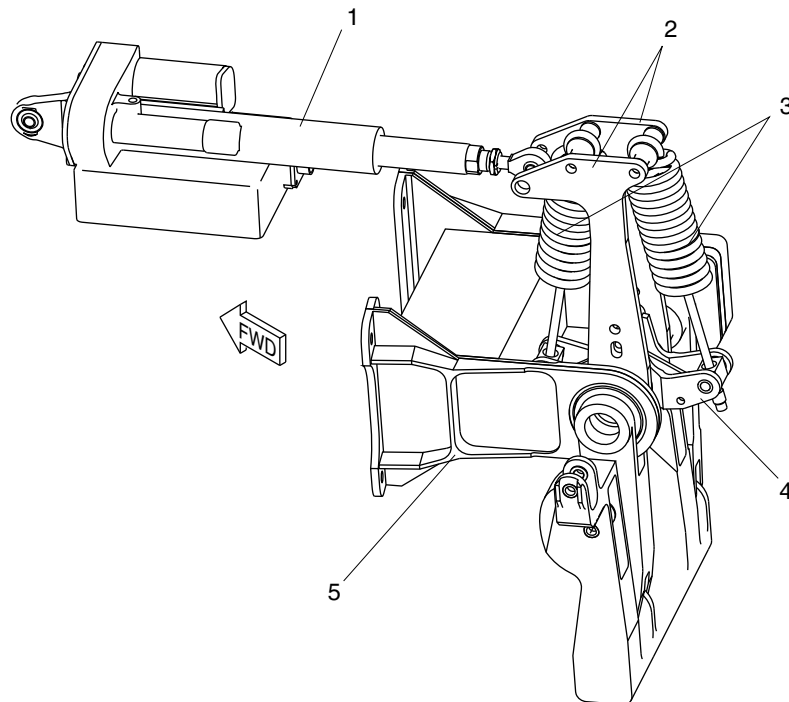
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Config 001
Page 7
Sep 05/2021



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OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Rudder Trim Actuator.
- 2. Trim Arms.
- 3. Feel Springs.
- 4. Summing Bar.
- 5. Mounting Frame.

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RUDDER FEEL AND TRIM UNIT PAGE 2
Figure 5

PSM 1-84-2A
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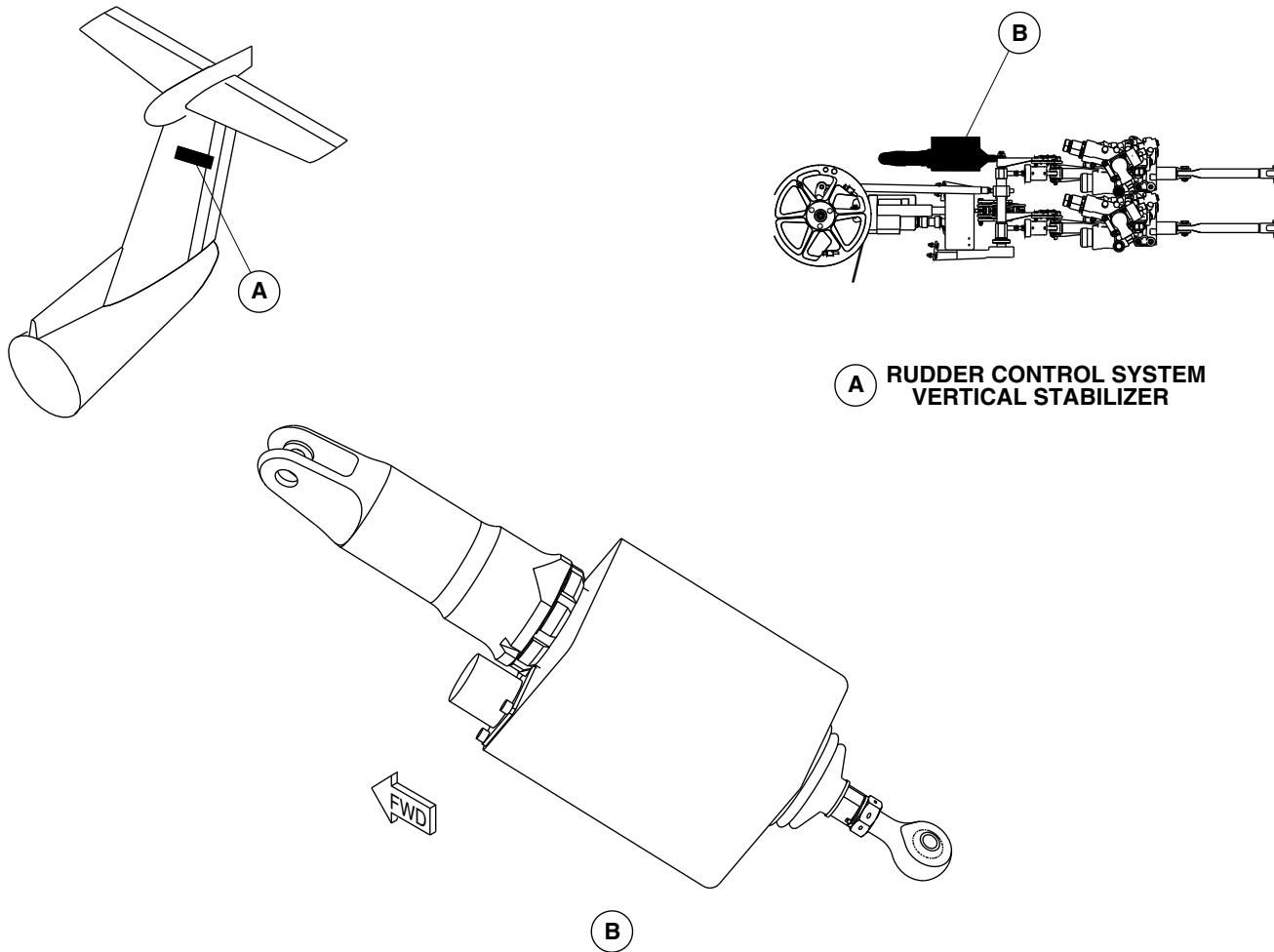
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Config 001
Page 8
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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YAW DAMPER
Figure 6

PSM 1-84-2A
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See first effectivity on page 2 of 27-22-00
Config 001

27-22-00

Config 001
Page 9
Sep 05/2021



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OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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PSM 1-84-2A
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Config 001

27-22-00

Config 001
Page 10
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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27-23-00-001

RUDDER SURFACES

Introduction

The rudder surfaces give directional control of the aircraft in flight, around the yaw, or normal axis.

General Description

The rudder is a primary flight control surface that is attached to the rear spar of the vertical stabilizer.

The rudder is divided into two sections:

- Rudder, Fore (27-23-06)
- Rudder, Trailing (27-23-01).

Detailed Description

[Refer to Figure 1.](#)

There are two rudder surfaces, the fore rudder and the aft or trailing rudder. The two surfaces extend almost the full length of the trailing edge of the vertical stabilizer. The rudders are attached to the aircraft through a series of hinges and safety hinges. Each rudder rotates on sealed bearings. The fore rudder is attached to the rear spar of the vertical stabilizer and is actuated by two Power Control Units (PCUs). The aft rudder is attached to the fore rudder by hinges and is actuated by pushrods so that the aft rudder deflects two times as far as the fore rudder.

Rudder, Fore

[Refer to Figure 2.](#)

The fore rudder is attached to the vertical stabilizer at four main hinge points. The upper and lower hinges have sealed, self-aligning ball bearings. The two center hinges have sealed, double-row ball bearings. Two safety hinges are supplied; one located above the upper main hinge point and one located below the lower main hinge point. The safety hinges carry no load when the main hinges are in good condition. As the main hinges wear, the safety hinges take an increasing share of the load.

There are extensions on the right side of the center hinge fittings, which act as attachment points for the rudder Power Control Units (PCUs). There is a fitting between the upper and center hinge fittings that provide the attachment point for the Linear Variable Digital Transducer (LVDT).

Bonding wires located near to the upper hinges of the fore rudder discharge static charge buildup from the rudder control surfaces to the vertical stabilizer.

Rudder, Trailing

[Refer to Figure 3.](#)

The trailing rudder is attached to the fore rudder rear spar with three main hinges and three safety hinges. The upper and lower main hinges have self-aligning ball bearings. The center main hinge has sealed double-row bearings. The three safety hinges have the same function as those of the safety hinges installed on the fore rudder. One safety hinge is located near the upper main hinge, one is located near the lower main hinge and other one is located below the center main hinge.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

The fore rudder upper hinge fitting attaches to a bracket on the vertical stabilizer rear spar. An extension on the right side of this bracket is the attachment point for the upper pushrod. The upper pushrod extends diagonally aft to a bracket on the upper hinge fitting on the left side of the trailing rudder. The fore rudder lower hinge fitting attaches to a bracket on the vertical stabilizer rear spar. An extension on the left side of this bracket is the attachment point for the lower pushrod. The lower pushrod extends diagonally aft to a bracket on the lower hinge fitting on the right side of the trailing rudder.

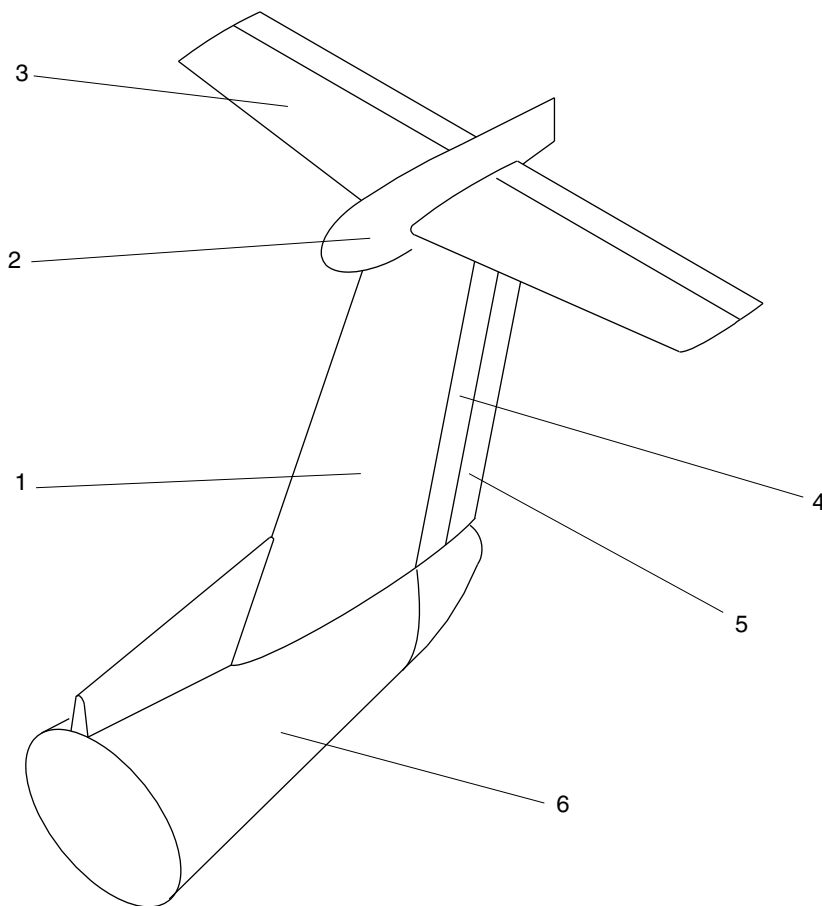
The geometrical arrangement of the pushrods makes sure that the trailing rudder deflects two times as much as the fore rudder. Therefore a relatively short PCU actuator movement results in large movement of the combined rudder surface.

Bonding wires that are located near the lower and upper hinges of the trailing rudder, discharge the static charge buildup from the rudder control surfaces to the vertical stabilizer.



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LEGEND

- 1. Vertical Stabilizer.
- 2. Stabilizer Fin Fairing.
- 3. Horizontal Stabilizer.
- 4. Fore Rudder.
- 5. Trailing Rudder.
- 6. Rear Fuselage Section.

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Rudder, Empennage
Figure 1

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Config 001

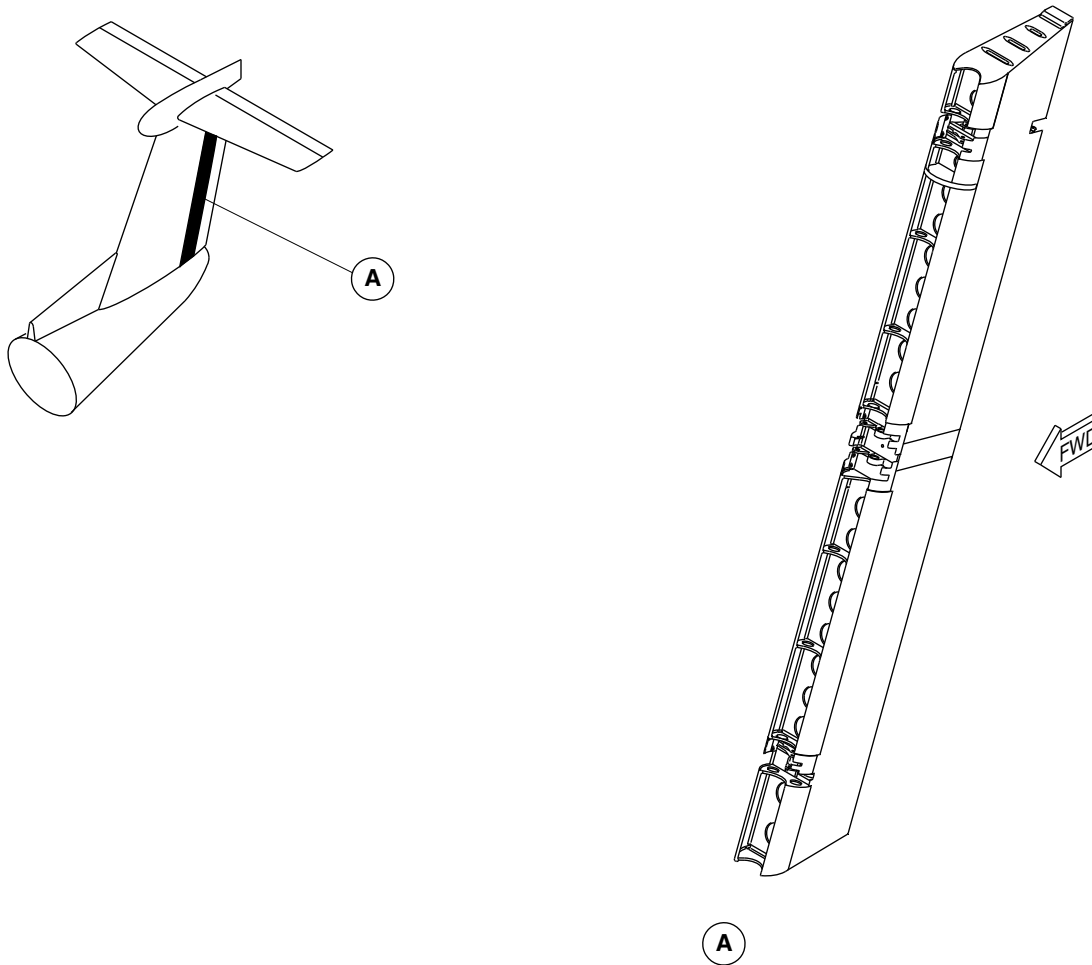
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Config 001
Page 4
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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Fore Rudder
Figure 2

PSM 1-84-2A
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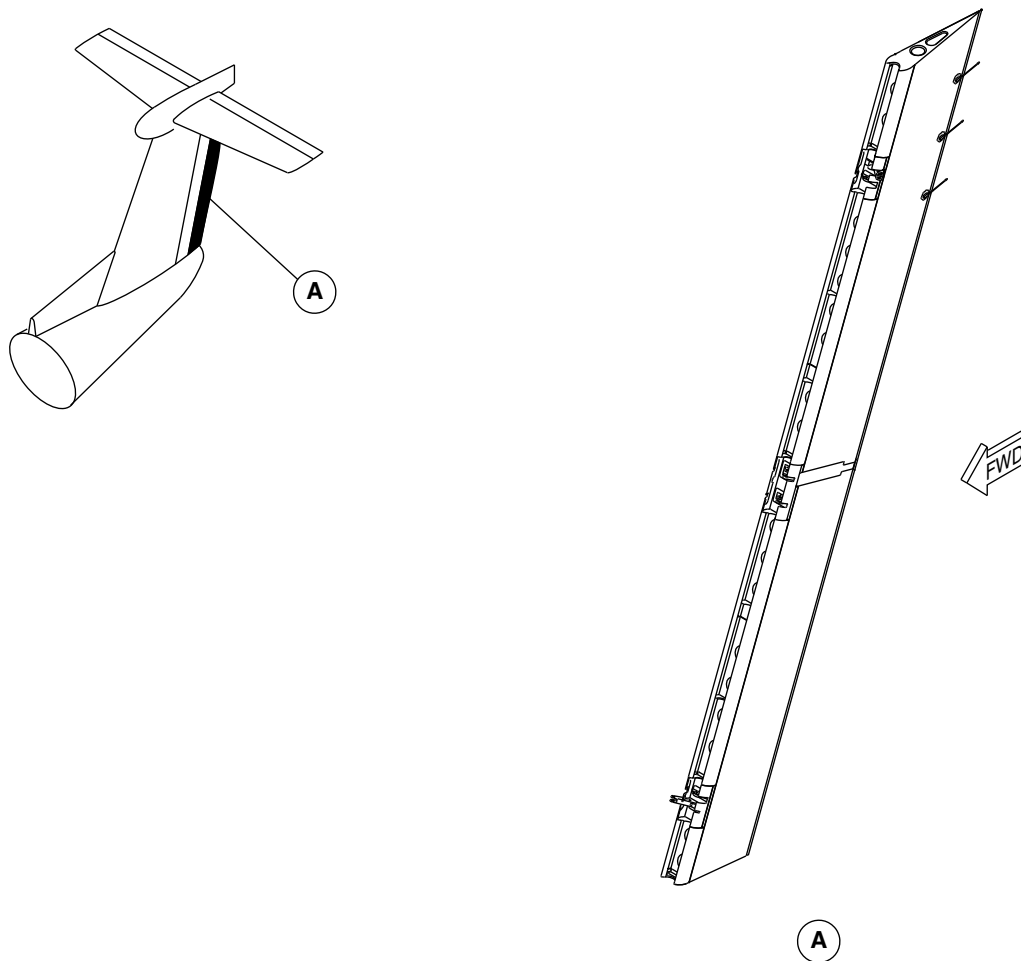
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Config 001
Page 5
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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Trailing Rudder
Figure 3

PSM 1-84-2A
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See first effectivity on page 2 of 27-23-00
Config 001

27-23-00

Config 001
Page 6
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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ELEVATOR CONTROL SYSTEM

Introduction

The elevator control system is a primary flight control system which supplies longitudinal (pitch) control of the aircraft during takeoff, flight and landing.

General Description

Longitudinal control is supplied by two independent, manually operated, hydraulically powered elevators. Either the pilot, the copilot or the autopilot can control the elevators.

The elevator control system is divided into the subsections that follow:

- Elevator System (27-31-00)
- Elevator Feel and Trim (27-32-00)
- Stall Warning/Stick Pusher (27-33-00)
- Elevator Surfaces (27-34-00)
- Pitch Disconnect System (27-38-00)

Detailed Description

[Refer to Figure 1.](#)

The elevator control system supplies longitudinal (pitch) control of the aircraft in flight. Fore and aft movement of either the pilot's or the copilot's control column supplies pitch commands to the hydraulically powered elevators. The autopilot also supplies pitch commands to the elevators (refer to SDS 27-31-00).

Two identical Pitch Feel and Trim Units (PFTUs), left and right, are located in the vertical stabilizer. They supply both pitch feel and pitch trim inputs to the elevator control system. The right PFTU is in parallel with the copilot's elevator controls and the left PFTU is in parallel with the pilot's elevator controls. Centering springs help to return the elevator control surfaces to a neutral position (refer to SDS 27-32-00).

The pilot's and copilot's control columns are mechanically connected to each other allowing the aircraft to be flown from either the pilot's or copilot's position. The two control columns can be disconnected from each other by the pitch disconnect mechanism in the event of a jam in either the pilot's or copilot's control circuit. The aircraft can then be flown using the remaining operational elevator control circuit. The pitch disconnect system is activated by pulling the pitch disconnect handle located on the pilot's side of the centre console in the flight compartment (refer to SDS 27-38-00).

Elevator System

[Refer to Figure 2.](#)

Pitch control of the aircraft is maintained by two mechanically controlled, hydraulically powered elevators located on the trailing edge of the left and right horizontal stabilizers.

PSM 1-84-2A
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See first effectivity on page 2 of 27-30-00
Config 001

27-30-00

Config 001
Page 2
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Fore and aft movement of the pilot's and copilot's control columns is transferred through two fully independent cable and pulley control circuits to the left and right elevator Power Control Units (PCUs). There are three identical hydraulic PCUs (outboard, centre and inboard) on each elevator. The outboard and centre PCUs on each elevator are active at all times while the inboard PCU is on standby. The No.1 hydraulic system supplies power to the left and right outboard PCUs. The No.2 hydraulic system supplies power to the left and right centre PCUs. The standby No.3 hydraulic system supplies power to the left and right inboard standby PCUs. A switch labelled STBY HYD PRESS in the flight compartment activates the inboard PCUs (refer to SDS 27-31-00).

Six PCU input bungees are installed in series with the six elevator PCUs. The input bungees prevent PCU runaway if the main control valve in the PCU jams. Two overload bungees are installed in the horizontal stabilizer; one in each elevator circuit. The overload bungee prevents any damage to the elevator control circuit in the event of a jam in one or more of the elevator PCUs.

[Refer to Figure 3.](#)

An ELEVATOR ASYMMETRY caution light comes on when there is a difference between the left and right elevator positions.

Position indication of the elevators is displayed on the Powered Flight Control System Indicator (PFCSI) shown on the Multi Function Display (MFD). Gust protection for the elevators is supplied by trapped hydraulic fluid within the PCUs when the system is depressurized.

Elevator Feel and Trim

[Refer to Figure 4.](#)

The input commands from the control columns are transferred to the elevator PCUs which, in turn, move the elevator surfaces. Therefore the pilot is unable to feel the aerodynamic forces acting on the elevator surfaces. The Pitch Feel and Trim Units (PFTUs) supply elevator control column forces which increase with column displacement as a function of airspeed and normal acceleration of the aircraft. Air Data Units (ADUs) send airspeed information to the FCECU. Artificial feel is produced by two pitch feel actuators attached to the PFTUs. As airspeed increases or decreases, the FCECU commands the pitch feel actuators to extend or retract respectively. This results in an increase or decrease in the amount of artificial feel acting on the control columns. If one pitch feel actuator fails, the elevator control system continues to operate using the operational actuator but with reduced artificial feel (refer to SDS 27-32-00).

[Refer to Figure 5.](#)

Two pitch trim actuators installed on top of the pitch feel and trim units supply elevator trim. The elevator trim actuator is controlled automatically by the autopilot or manually by the trim switches on the pilot's and copilot's control handwheels. Elevator trim position is shown in the flight compartment on the elevator trim indicator located on the left side of the centre console. The indicator displays elevator trim position as commanded by the trim switches on the pilot's and copilot's handwheels. The indicator is labelled "NU" for nose up, "ND" for nose down and "TO" for take-off. A white band next to the "TO" setting indicates the amount of elevator trim which is acceptable for take-off. An aural warning sounds if a take-off is attempted with the elevator trim position outside the take-off range.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Refer to Figure 3.

An ELEVATOR FEEL caution light comes on when there is a disagreement between the two elevator feel units.

A PITCH TRIM caution light comes on if there is a disagreement between the airspeed or trim inputs to the pitch trim actuator.

Stall Warning/Stick Pusher

Refer to Figures 6 and 7.

The left and right independent stall warning systems supply a warning of an impending stall by activating both stick shakers on the control columns. If either the left or right system senses an impending stall, both stick shakers activate. An electrically operated stick pusher system activated by the Stall Protection Module supplies forward pitch to prevent a deep stall. The electric actuator pushes both control columns forward if an uncorrected stall is sensed by both left and right systems.

The stall warning/stick pusher system supplies the flight crew with warnings of an impending stall through an audio warning and a mechanical shaking of the control columns. If the stall continues, the stick pusher applies a pitch force to push the control columns forward and decrease the angle of attack to prevent further penetration of the stall threshold (refer to SDS 27-33-00).

Elevator Surfaces

Refer to Figure 8.

There are two elevator surfaces, left and right, which are installed on the trailing edge of the left and right horizontal stabilizers, respectively. Three elevator PCU's are attached to each elevator

surface. An elevator LVDT is attached to the leading edge of each elevator surface (refer to SDS 27-34-00).

Pitch Disconnect System

Refer to Figure 9.

The pilot's and copilot's control columns are mechanically connected to each other through the pitch disconnect mechanism. If a jam occurs in either control circuit, the two control columns can be disconnected from each other by using the pitch disconnect handle located on the left side of the centre console. When the handle is in the engaged "IN" position, the pilot's and copilot's control columns are connected to each other by a clutch. When the handle is pulled out and rotated 90 degree, the clutch disengages and disconnects the two control columns. The pilot with the free control column retains pitch control. The aircraft can then be flown using the remaining operational elevator control circuit (refer to SDS 27-38-00).

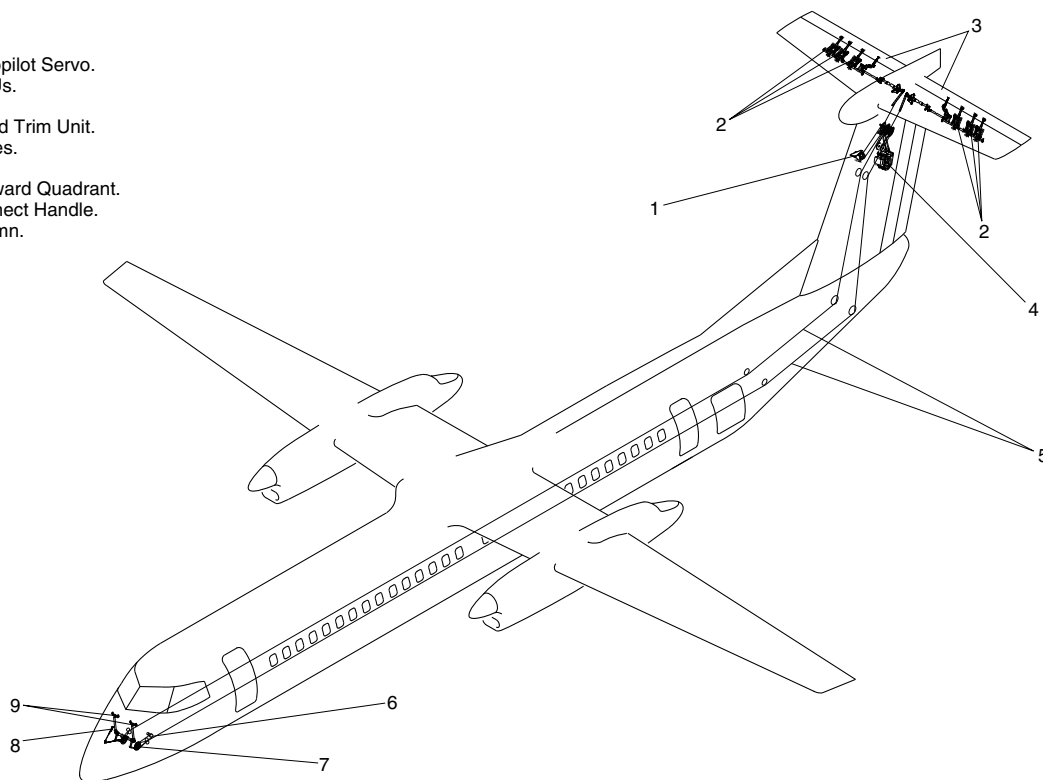


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

LEGEND

1. Elevator Autopilot Servo.
2. Elevator PCUs.
3. Elevators.
4. Pitch Feel and Trim Unit.
5. Control Cables.
6. Stick Pusher.
7. Elevator Forward Quadrant.
8. Pitch Disconnect Handle.
9. Control Column.



ELEVATOR CONTROL SYSTEM PAGE 1
Figure 1

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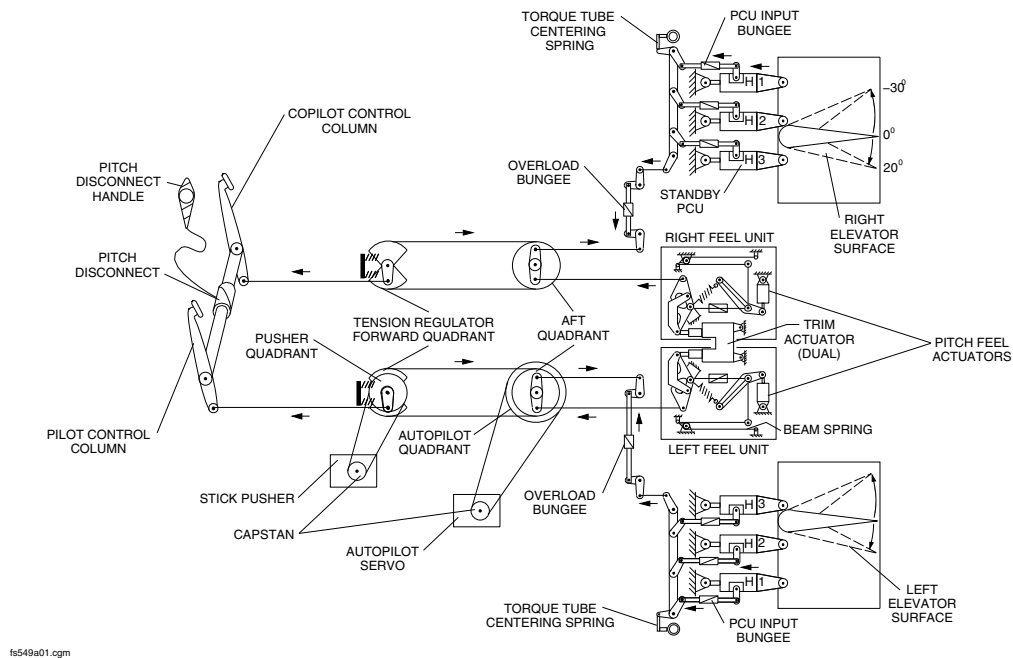
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Config 001
Page 5
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



ELEVATOR CONTROL SYSTEM PAGE 2
Figure 2

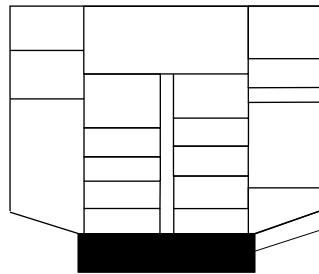
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Config 001
Page 6
Sep 05/2021



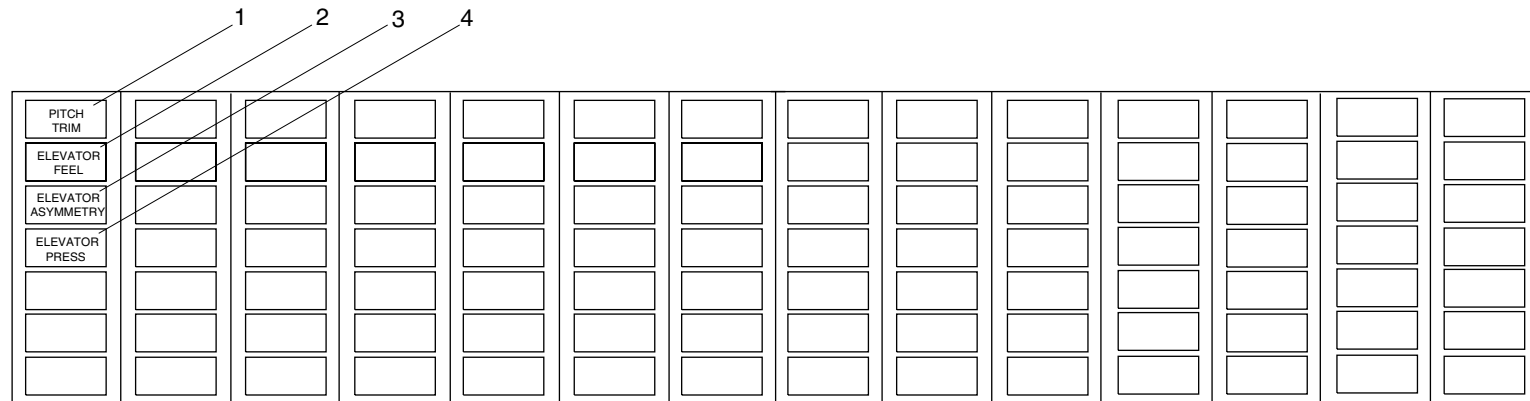
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



OVERHEAD CONSOLE

LEGEND

- 1 Pitch Trim (Amber).
- 2 Elevator Feel (Amber).
- 3 Elevator Asymmetry (Amber).
- 4 Elevator Pressure (Amber).



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ELEVATOR CAUTION LIGHTS
Figure 3

PSM 1-84-2A
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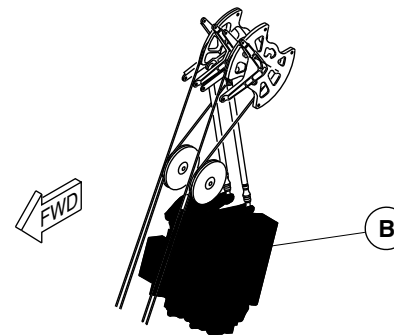
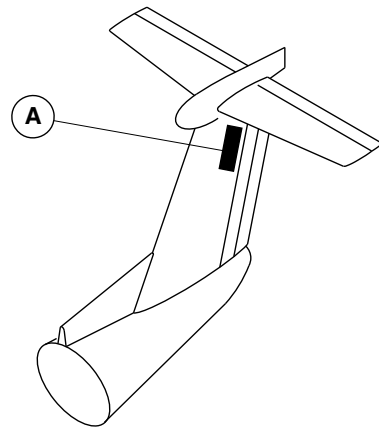
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Config 001
Page 7
Sep 05/2021

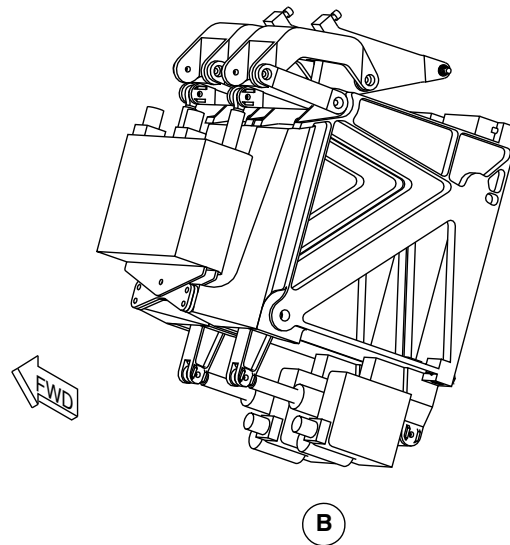


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A ELEVATOR CONTROL SYSTEM
– VERTICAL STABILIZER



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PITCH FEEL AND TRIM UNIT LOCATOR
Figure 4

PSM 1-84-2A
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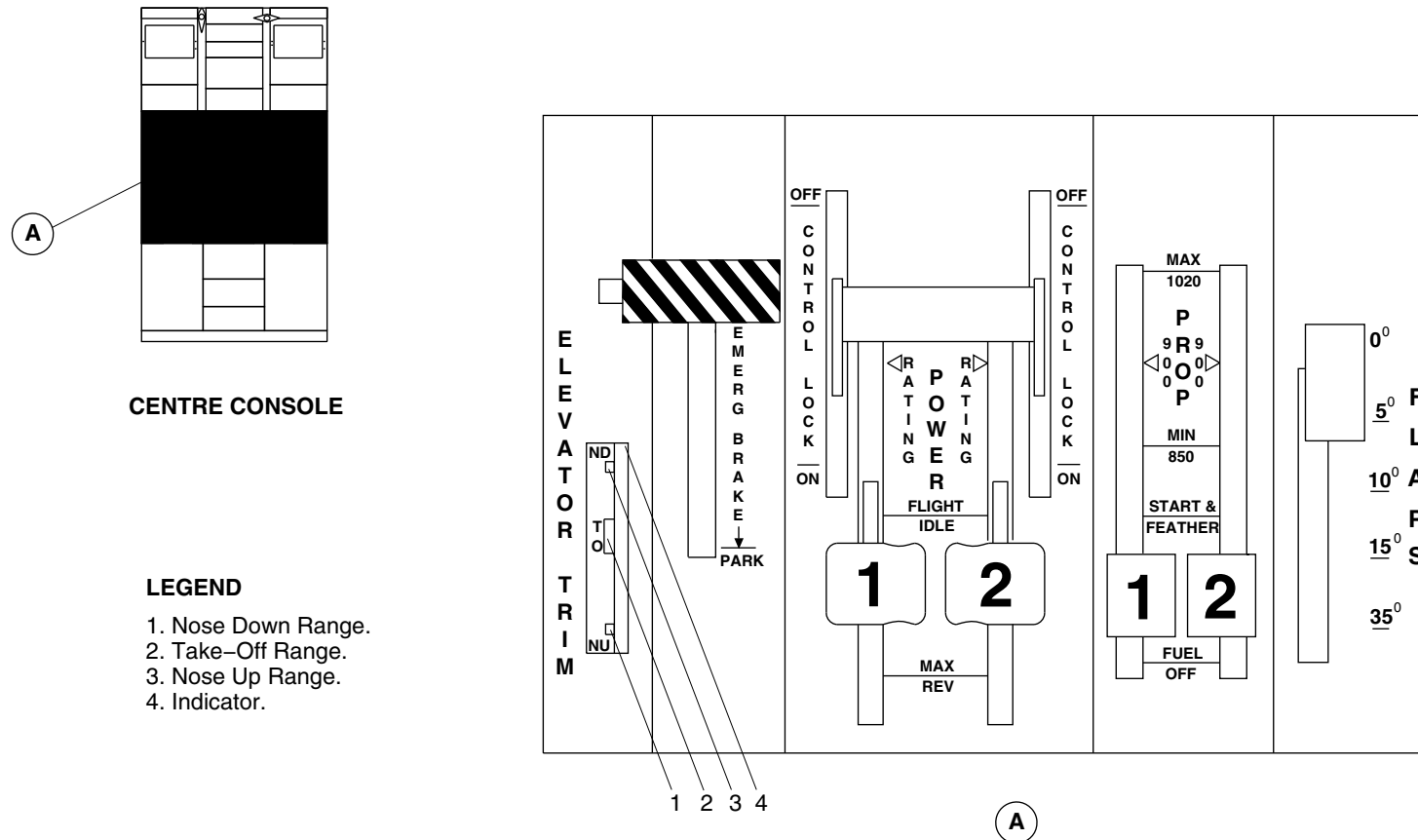
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Config 001
Page 8
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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ELEVATOR TRIM INDICATOR LOCATOR
Figure 5

PSM 1-84-2A
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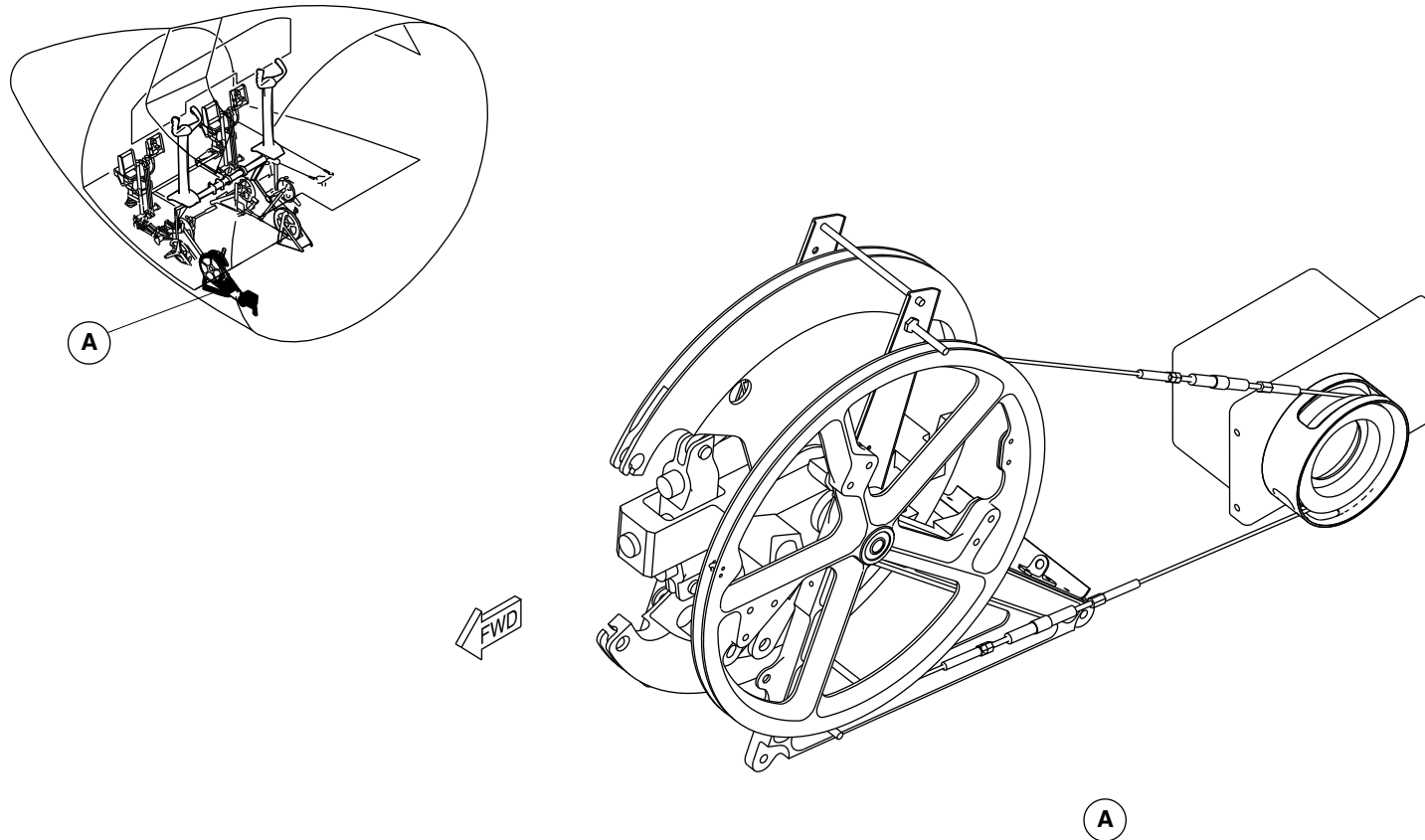
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Config 001
Page 9
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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STICK PUSHER LOCATOR
Figure 6

PSM 1-84-2A
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Config 001

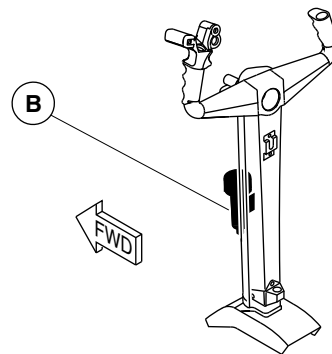
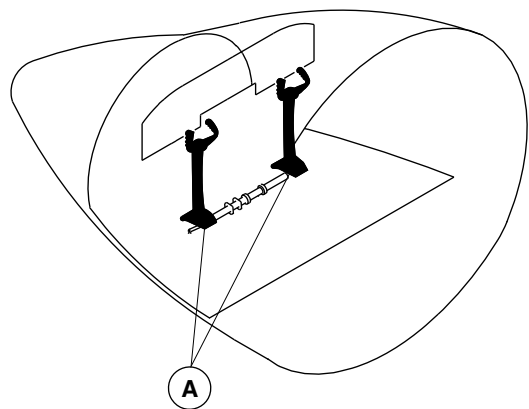
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Config 001
Page 10
Sep 05/2021



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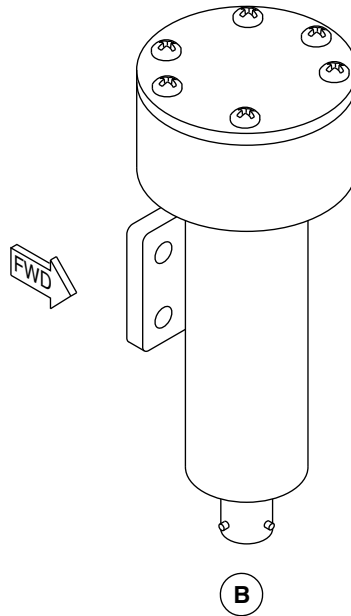
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NOTE

Pilot side shown.
Copilot side similar.

A CONTROL COLUMN ASSEMBLY



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STICK SHAKER LOCATOR
Figure 7

PSM 1-84-2A
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Config 001

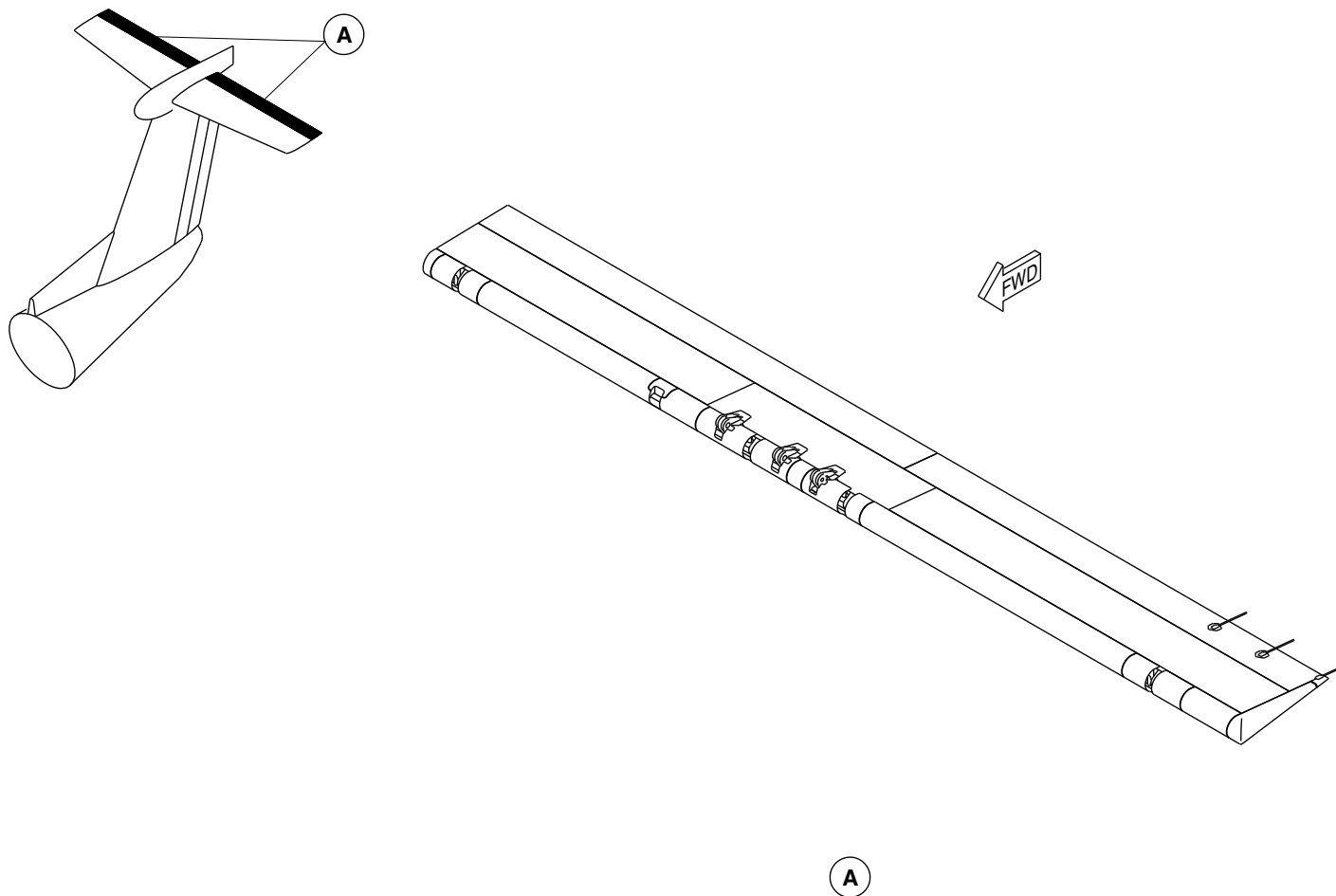
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Config 001
Page 11
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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ELEVATOR SURFACES LOCATOR
Figure 8

PSM 1-84-2A
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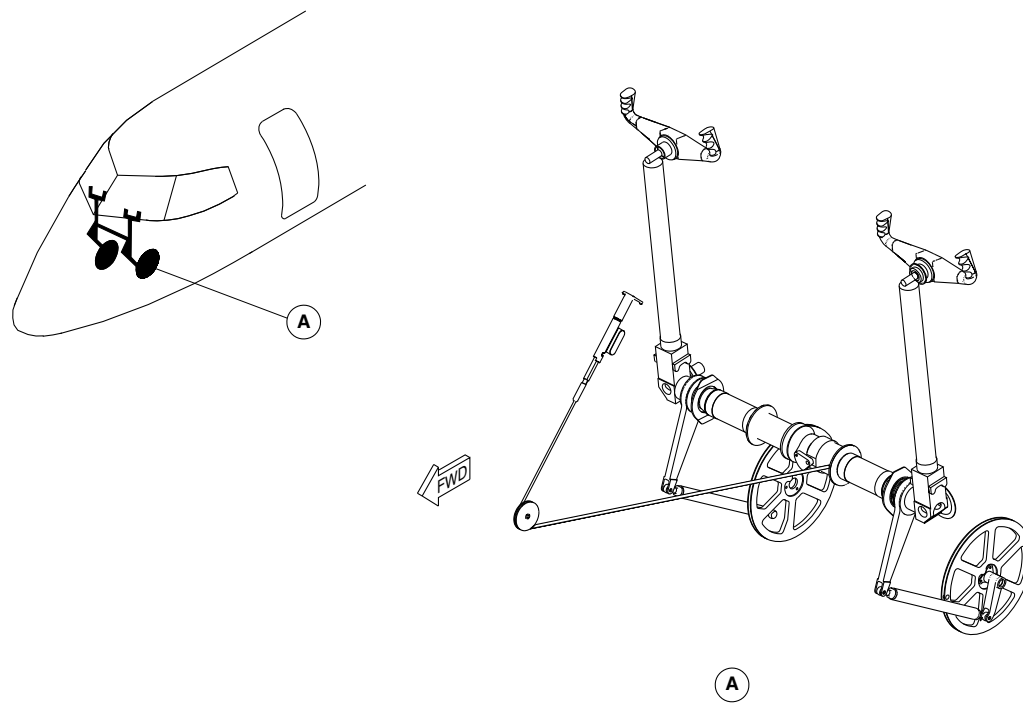
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Config 001
Page 12
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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PITCH DISCONNECT SYSTEM LOCATOR
Figure 9

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-30-00
Config 001

27-30-00

Config 001
Page 13
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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PSM 1-84-2A
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See first effectivity on page 2 of 27-30-00
Config 001

27-30-00

Config 001
Page 14
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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27-31-00-001

ELEVATOR SYSTEM

Introduction

The elevator system transmits the pitch commands from the pilot's and copilot's control columns to the elevator control surfaces. The elevator surfaces move in proportion to the amount of control column movement.

General Description

The left and right elevators are installed on the trailing edge of the left and right horizontal stabilizers, respectively. The elevators are moved down to pitch the aircraft nose down and moved up to pitch the aircraft nose up, around the lateral axis.

The elevators are positioned by commands from either the pilot control system or the Automatic Flight Control System (AFCS). Trim control and surface position indication is also supplied for each elevator.

Pitch commands to the elevators are supplied by the fore and aft movement of either the pilot's or copilot's control columns. The pitch commands are transmitted mechanically through:

- Control columns
- Cable tension regulators
- Cables

- Pushrods
- Pitch Feel and Trim Unit
- Power Control Units input torque tubes
- Elevator PCUs.

The PCUs then move the elevators using hydraulic system pressure.

The elevator system has the components that follow:

- Columns, Control (27-31-01)
- Handwheels (27-31-06)
- Quadrant, Left Forward (27-31-11)
- Quadrant, Right Forward (27-31-16)
- Regulators, Elevator Tension (27-31-21)
- Quadrant, Left Aft (27-31-26)
- Quadrant, Right Aft (27-31-31)
- Unit/Actuators, Elevator Power Control (27-31-36)
- Compensator (27-31-41)
- Tubes, Torque (27-31-46)
- Device, Centering (27-31-51)
- Bungees, Overload (27-31-56)
- LVDT, Elevator Position (27-31-61).

Detailed Description

Fore and aft movement of the pilot's and copilot's control columns, is transmitted simultaneously to the pilot's and copilot's elevator control

PSM 1-84-2A

EFFECTIVITY:

See first effectivity on page 2 of 27-31-00

Config 001

27-31-00

Config 001

Page 2

Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

circuits. The two elevator control circuits are routed independently of each other on the left and right side of the aircraft. This independent routing prevents loss of manual pitch control in the event of an engine rotor burst. The two elevator circuits are routed from the flight compartment, below the cabin floor and up the vertical stabilizer to the left and right elevator surfaces.

The pilot's and copilot's control columns are connected to each other by the pitch disconnect system, so that the aircraft can be flown from either the pilot's or copilot's position (refer to SDS 27–38–00). The two control columns can be disconnected from each other if a jam occurs in either the pilot's or copilot's control circuit. The aircraft is then flown using the unjammed elevator control circuit. The pitch disconnect system has a pitch disconnect handle on the pilot's side of the center console. When the PITCH DISC handle is pulled out to the limit and turned through 90 degrees clockwise or counterclockwise, the two control columns are disconnected from each other.

The pushrods supply the pitch control movement of the pilot's and the copilot's control columns, to the left/right forward elevator quadrants. The forward elevator quadrants are installed below the flight compartment floor.

On aircraft with ModSum 4–309210 OR SB84–31–33 OR SB84–31–39 OR SB84–31–40 incorporated [Refer to Figure 1.](#)): The pushrods are replaced with force sensors. There are two sensors, one for each control column. The sensors supply the pilot and the copilot inputs from the control columns to the forward elevator quadrants. They transmit a signal in proportion to the input force applied on control columns. The signal transmitted from sensors is recorded by the Flight Data Recorder.

The two forward elevator quadrants (left and right) also act as cable tension regulators. The left forward (pilot's) elevator quadrant has input from the stick pusher system (refer to SDS 27–33–00). From the forward elevator quadrants, the elevator cables are routed aft below the flight compartment and cabin floor. The cables then are routed through the rear fuselage and upwards to terminate at the two aft (left and right) elevator quadrants in the vertical stabilizer.

The aft quadrants transfer the pitch control movement through the overload bungees, pushrods and bellcranks to the PCU input torque tubes. The left aft (pilot's) elevator quadrant also has input from the Automatic Flight Control System (AFCS) through a cable, capstan and servo actuator. The torque tubes then transfer the pitch control movement through the PCU input bungees to the elevator PCUs which are attached to the elevator control surfaces. Extension or retraction of the PCUs moves the elevator surfaces according to the pitch commands from the pilot or the autopilot.

Each elevator control cable circuit passes through a cable seal in the rear pressure dome to prevent loss of cabin pressure.

There are three identical elevator Power Control Units (PCUs) connected to each elevator surface. The outboard PCUs are powered by the No. 1 hydraulic system, the center PCUs are powered by the No.2 hydraulic system. The inboard PCUs are powered by the No.3 standby hydraulic system. The outboard and center elevator PCUs are normally active. The inboard elevator PCUs are activated only when the No. 1 and/or No. 2 hydraulic systems fail.

An elevator PCU compensator installed in the return line of the No. 3 hydraulic system, keeps a positive pressure on the elevators when there is no system hydraulic pressure available. This retained pressure gives hydraulic gust lock protection from strong winds when the aircraft is parked or being taxied. The retained pressure also

PSM 1–84–2A

EFFECTIVITY:

See first effectivity on page 2 of 27–31–00

Config 001

27–31–00

Config 001

Page 3

Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

gives elevator flutter suppression in flight when there is no system hydraulic pressure.

Elevator position is shown on the pilot's Multi Function Display (MFD) by means of Linear Variable Differential Transformers (LVDTs) attached to each elevator surface. If a difference occurs between the left and right elevator surface position indications, the ELEVATOR ASYMMETRY caution light comes on in the flight compartment. For the caution light to come on, the difference in elevator position must be greater than:

- $6.0 \pm 1.0^\circ$ for airspeed less than 185 kts (on aircraft without MS 4-113392)
- $4.0 \pm 1.0^\circ$ for airspeed more than 185 kts (on aircraft without MS 4-113392)
- $6.0 \pm 1.0^\circ$ for airspeed less than 215 kts (on aircraft with MS 4-113392)
- $4.5 \pm 1.0^\circ$ for airspeed more than 215 kts (on aircraft with MS 4-113392).

Columns, Control

[Refer to Figure 2.](#)

The pilot's and copilot's control columns are of conventional design. The pilot's control column operates the left elevator through the left elevator circuit. The copilot's control column operates the right elevator through the right elevator circuit. Each column is attached to a lateral torque tube with bolts, which is installed in bearings below the flight compartment floor. The torque tube has a clutch, which can be disconnected in the event of a jam in either elevator control circuit to allow for individual control column operation of the elevators.

A stick shaker for the stall warning system is attached to the forward face of each control column. The stick pusher actuator is connected to a quadrant installed on left forward quadrant located below the flight compartment floor (refer to SDS 27-33-00).

Handwheels

[Refer to Figure 3.](#)

A hand operated control wheel is pivot-mounted to the top of each control column. A pitch trim split switch is located in the center of the outboard handgrip of each control handwheel. The pitch trim switches are split in two; both halves of the switches must be activated for the pitch trim command to proceed. The switches give manual control of the elevator pitch trim. They are thumb-operated switches, which are spring-loaded to the center-off position. When the switches are pushed forward, a nose-down trim is commanded and when the switches are pulled aft, a nose-up trim is commanded.

Quadrant, Left Forward

[Refer to Figures 4 and 5.](#)

The left forward quadrant is located below the flight compartment floor. The quadrant transmits fore and aft movement of the pilot's control column to the left aft quadrant. There are springs installed in the left forward quadrant to supply tension regulation of the cable system. The stick pusher quadrant is directly attached to the left forward quadrant with input from the stick pusher actuator. A rig pin point is located in the quadrant.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Quadrant, Right Forward

Refer to Figures 6 and 7.

The right forward quadrant is located below the flight compartment floor. The quadrant transmits fore and aft movement of the copilot's control column to the right aft quadrant. There are springs installed in the right forward quadrant to supply tension regulation of the cable system. A rig pin point is located in the quadrant.

Regulators, Elevator Tension

Refer to Figures 8 and 9.

Tension regulator springs installed in each of the forward quadrants give correct tension of the elevator control cables. The elevator tension regulators adjust the cable tension according to ambient temperature.

Quadrant, Left Aft

Refer to Figures 10 and 11.

The left aft quadrant is located in the vertical stabilizer. The quadrant is made of aluminum and rotates on a sealed bearing. It transmits the pilot's pitch control commands from the left forward quadrant through a series of pushrods, torque tubes and linkages to the left elevator PCUs and to the left Pitch Feel and Trim Unit. The elevator autopilot quadrant is directly attached to the left aft quadrant with input from the autopilot servo. A rig pin point is located in the quadrant.

Quadrant, Right Aft

Refer to Figures 12 and 13.

The right aft quadrant is located in the vertical stabilizer. The quadrant is made of aluminum and rotates on a sealed bearing. It transmits the copilot's pitch control commands from the right forward quadrant through a series of pushrods, torque tubes and linkages to the right elevator PCUs and to the right Pitch Feel and Trim Unit. A rig pin point is located in the quadrant.

Unit/Actuators, Elevator Power Control

Refer to Figure 14.

The elevator control system has six identical elevator PCUs; three attached to each elevator surface and the horizontal stabilizer rear spar. Each elevator PCU is a hydro-mechanical servo actuator, which receives pilot input commands from the PCU input bungees and the PCU input torque tubes. Pilot input commands are transmitted to the elevator PCU Main Control Valve (MCV). The MCV supplies metered aircraft hydraulic system pressure to the PCU actuator. The PCU actuator rod end is attached to the elevator control surface by a pin.

Compensator

Refer to Figure 15.

An elevator PCU compensator maintains fluid within both inboard PCU's, to give gust damping on the ground. It compensates for changes in fluid volume due to thermal expansion and contraction. The compensator is installed in the return line of the No.3 hydraulic system and locks hydraulic pressure in the No. 3 elevator PCU. This prevents structural damage to the elevator surfaces during windy



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

conditions. The compensator also maintains a positive pressure to supply flutter suppression of the elevator surfaces in the event the PCUs are depressurized in flight.

Tubes, Torque, Mass Balance

[Refer to Figures 16 and 17.](#)

There are two PCU input torque tubes. One is installed in the pilot elevator control circuit in the left horizontal stabilizer. The other is installed in the copilot elevator control circuit in the right horizontal stabilizer. The PCU input torque tubes transmit pilot commands to the elevator PCUs through the PCU input bungees. A rig pin point is located in the outboard bellcrank of each PCU input torque tube. A removable mass balance is attached to each (left hand and right hand) torque tube assembly.

Device, Centering

[Refer to Figure 18.](#)

A centering spring is installed in each PCU input torque tube, to assist the springs of the Pitch Feel and Trim Unit. The Pitch Feel and Trim Unit centers the two elevator control circuits after the pilot releases an input command. When either control column is moved fore or aft, the input command is transmitted through the elevator control circuit to the PCU input torque tube. The tube then rotates in either direction to transmit the input commands to the elevator PCUs. When the PCU input torque tube rotates, a load is placed on the centering spring causing the spring to increase torsion. After the pilot releases the input command, the load exerted by the centering spring causes the control column to return to the neutral position.

Bungees, Overload

[Refer to Figures 19 and 20.](#)

The elevator control system has eight overload bungees. Two bungees are located in the elevator control circuit; one between the right aft quadrant and its related torque tube and the other between the left aft quadrant and its related torque tube. The other six overload bungees are installed in series between the two PCU input torque tubes and the six elevator PCUs. The overload bungees contract or expand to compensate for structural deflection or thermal expansion or contraction. This prevents structural damage to the elevator surfaces or the surrounding structure. The overload bungees also prevent any damage to the elevator control circuit in the event of a jam in one or more of the elevator PCUs.

LVDT, Elevator Position

[Refer to Figures 21 , 22 , 23 and 24.](#)

The LVDT is a single channel, single sensor LVDT. It senses the elevator surface position and transmits the position data to the Flight Control Electronic Control Unit (FCECU) and to the Powered Flight Control Surface Indicator (PFCSI) on the pilots MFD. There are two LVDTs located in the horizontal stabilizer; one attached to each elevator surface.

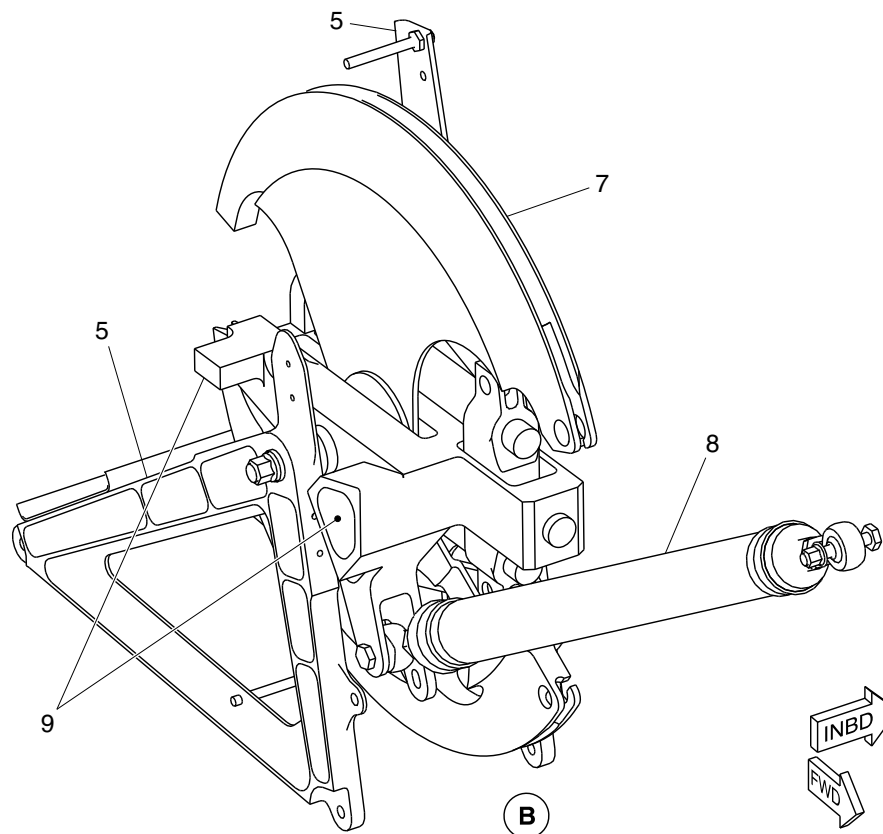


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

LEGEND

- 5. Mounting bracket.
- 7. RH cable tension regulator (quadrant).
- 8. RH push rod force sensor.
- 9. RH tension regulator secondary stop.



POST SB 84-31-33
POST SB 84-31-39
POST SB 84-31-40
POST MODSUM 4-309210

cg0008a02.dg, av/hp, apr24/2012

Elevator Left Forward Quadrant / Right Forward Quadrant
Figure 1 (Sheet 2 of 2)

PSM 1-84-2A
EFFECTIVITY:
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Config 001

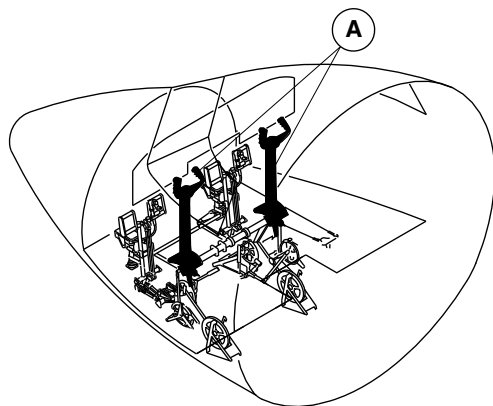
27-31-00

Config 001
Page 8
Sep 05/2021



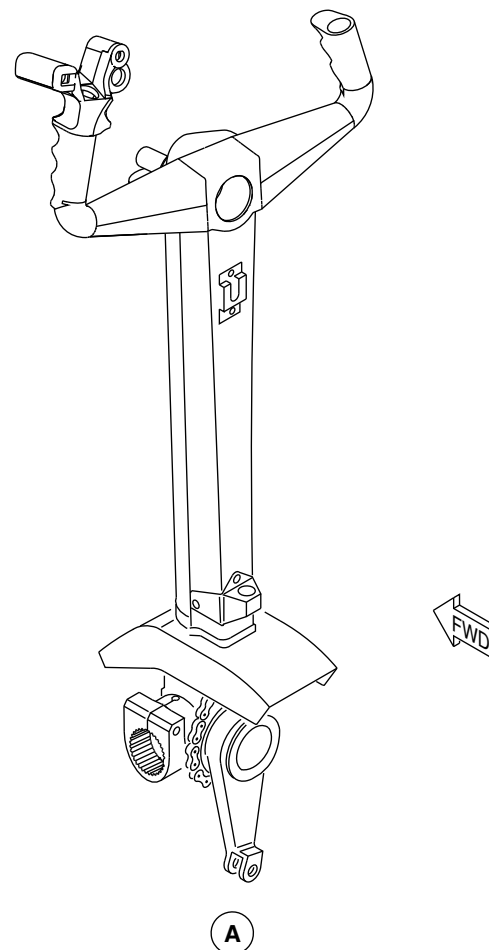
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NOTE

Pilot Control Column Shown.
Copilot Similar.



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ELEVATOR CONTROL COLUMN
Figure 2

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-31-00
Config 001

27-31-00

Config 001
Page 9
Sep 05/2021

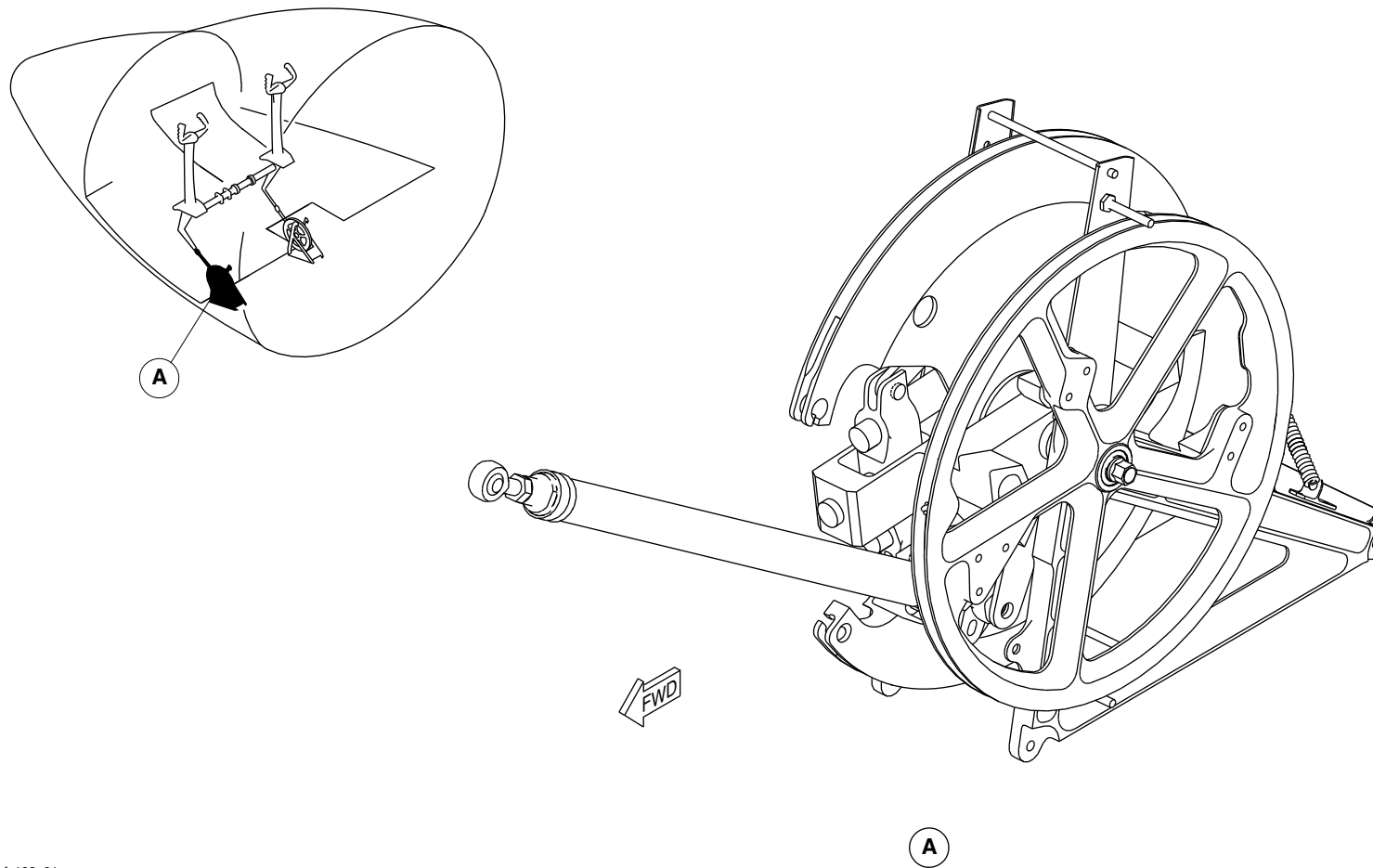


Print Date: 2025-04-22



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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ELEVATOR LEFT FORWARD QUADRANT PAGE 1
Figure 4

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-31-00
Config 001

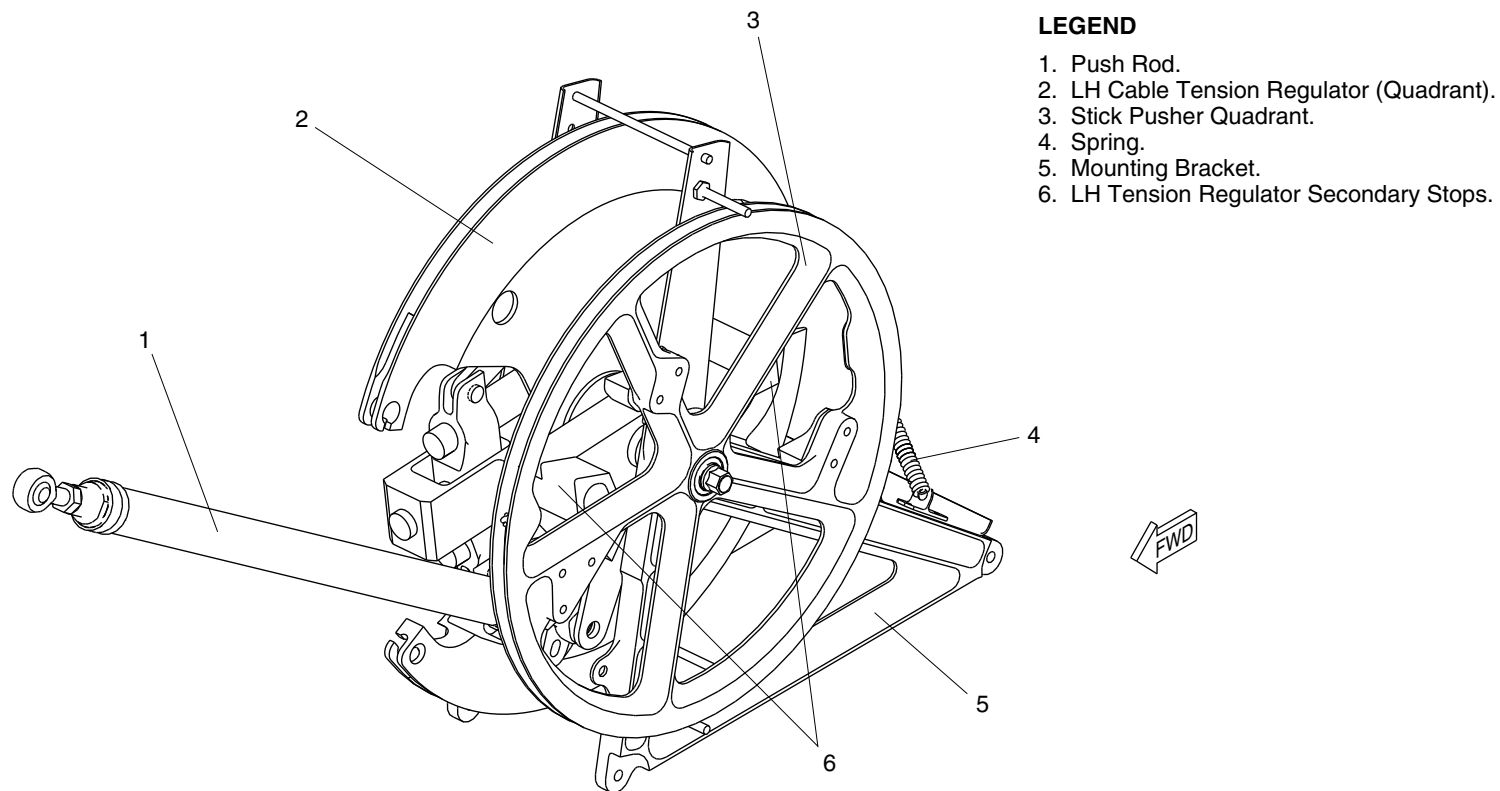
27-31-00

Config 001
Page 11
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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ELEVATOR LEFT FORWARD QUADRANT PAGE 2
Figure 5

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-31-00
Config 001

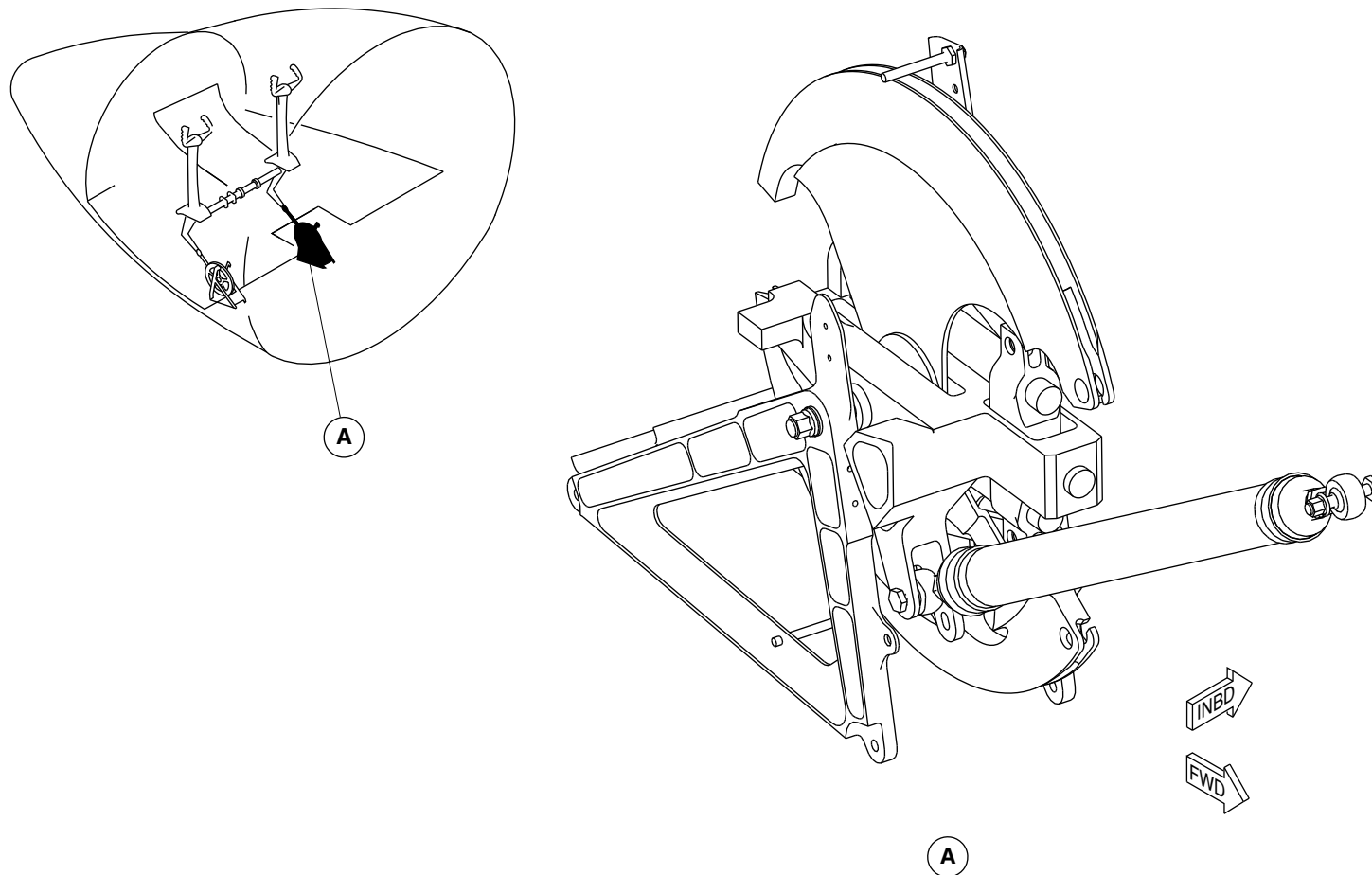
27-31-00

Config 001
Page 12
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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ELEVATOR RIGHT FORWARD QUADRANT PAGE 1
Figure 6

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-31-00
Config 001

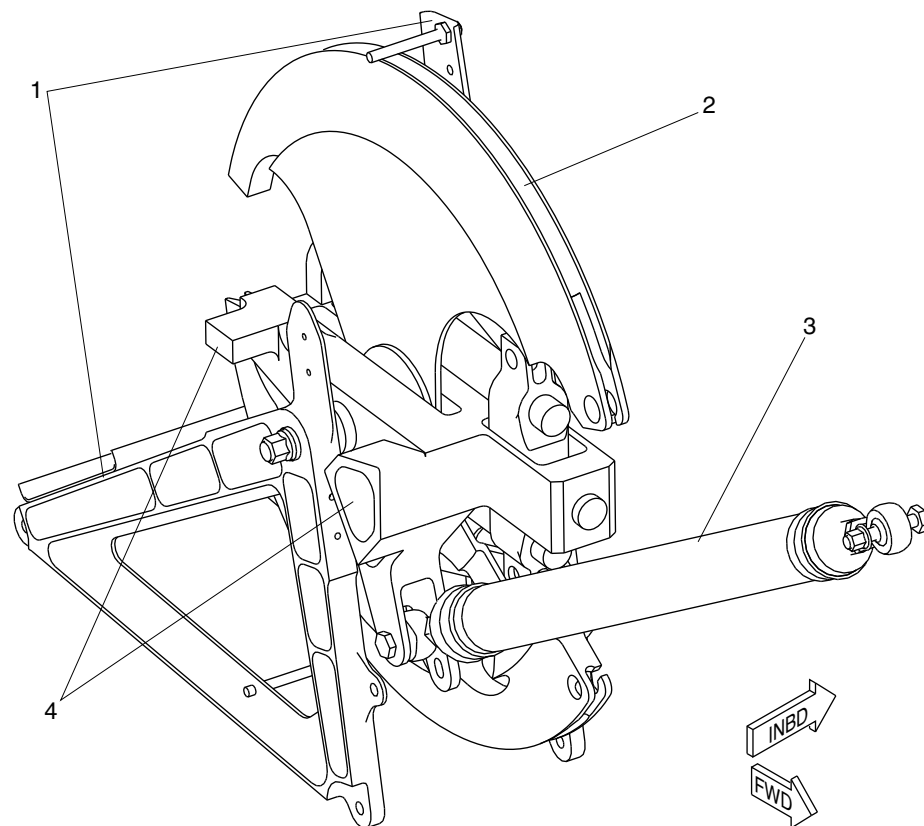
27-31-00

Config 001
Page 13
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Mounting Bracket.
- 2. Quadrant.
- 3. Push Rod.
- 4. Secondary Stops.

fs114a01.cgm

ELEVATOR RIGHT FORWARD QUADRANT PAGE 2
Figure 7

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-31-00
Config 001

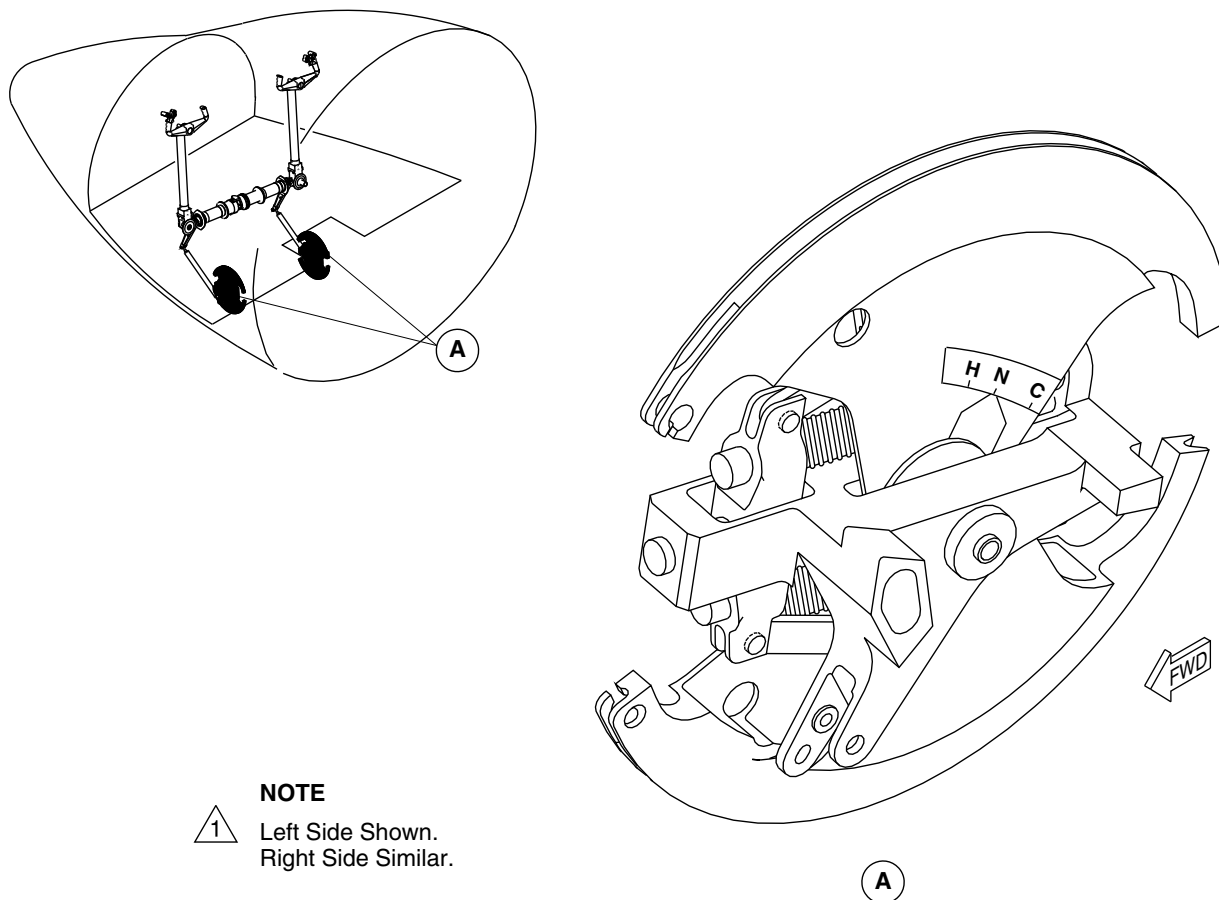
27-31-00

Config 001
Page 14
Sep 05/2021



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NOTE

1

Left Side Shown.
Right Side Similar.

fs235a01.cgm

ELEVATOR TENSION REGULATOR PAGE 1
Figure 8

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-31-00
Config 001

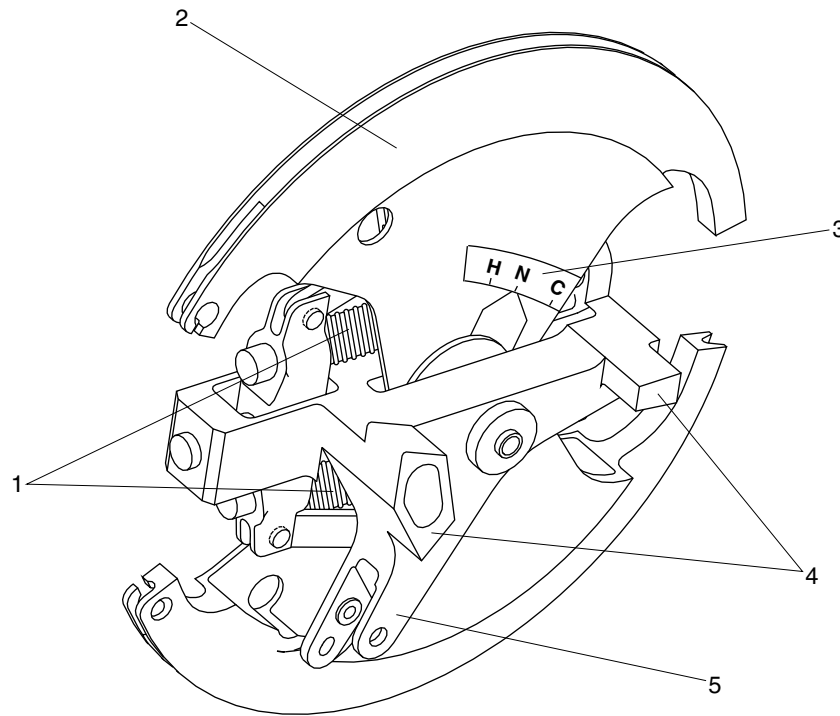
27-31-00

Config 001
Page 15
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Tension Spring.
- 2. LH Cable Tension Regulator (Quadrant).
- 3. Tension Indicator.
- 4. Quadrant Secondary Stop.
- 5. Push Rod Attachment.

fs236a01.cgm

ELEVATOR TENSION REGULATOR PAGE 2
Figure 9

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-31-00
Config 001

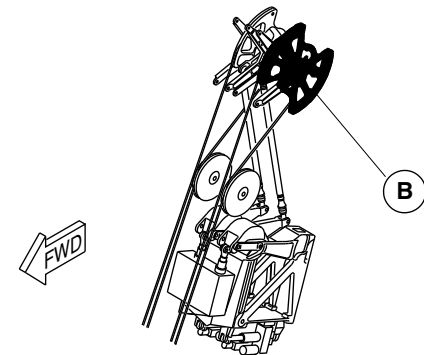
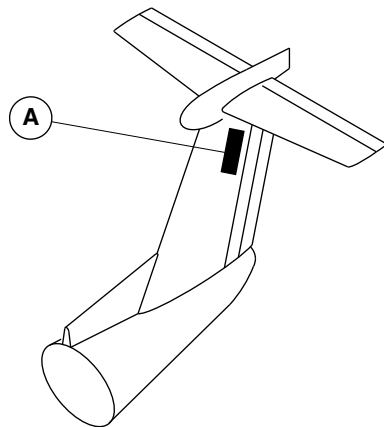
27-31-00

Config 001
Page 16
Sep 05/2021

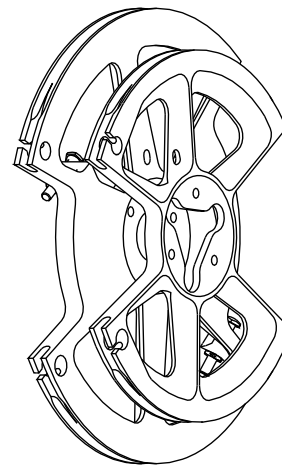


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A ELEVATOR CONTROL SYSTEM
– VERTICAL STABILIZER



B

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ELEVATOR LEFT AFT QUADRANT PAGE 1
Figure 10

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-31-00
Config 001

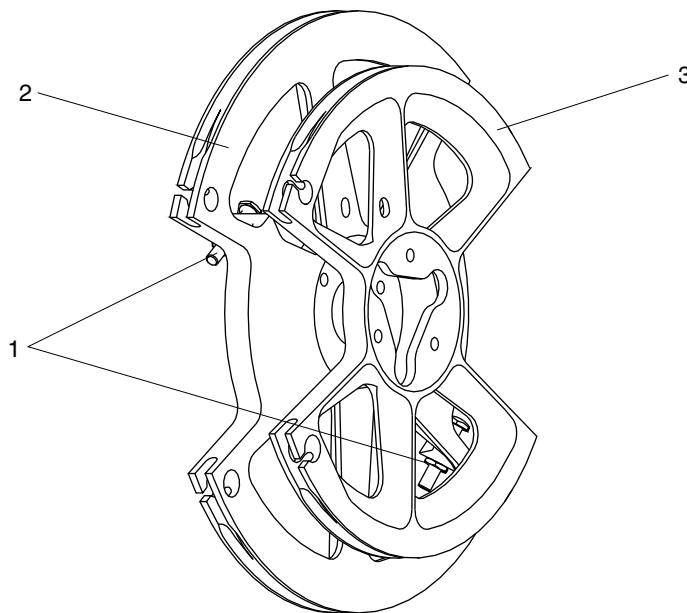
27-31-00

Config 001
Page 17
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Quadrant Primary Stop.
- 2. Elevator Autopilot Quadrant.
- 3. Left Aft Quadrant.

fs180a01.cgm

ELEVATOR LEFT AFT QUADRANT PAGE 2
Figure 11

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-31-00
Config 001

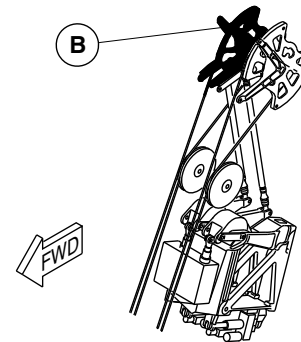
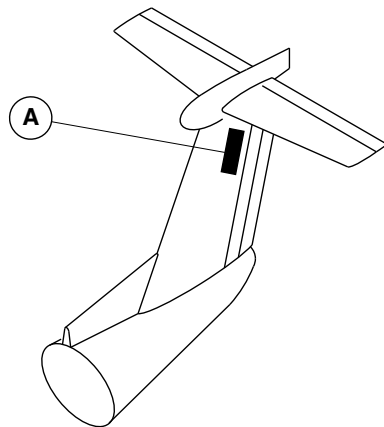
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Config 001
Page 18
Sep 05/2021

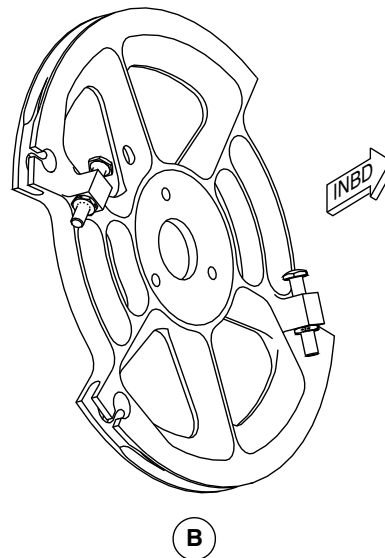


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



A ELEVATOR CONTROL SYSTEM
– VERTICAL STABILIZER



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ELEVATOR RIGHT AFT QUADRANT PAGE 1
Figure 12

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-31-00
Config 001

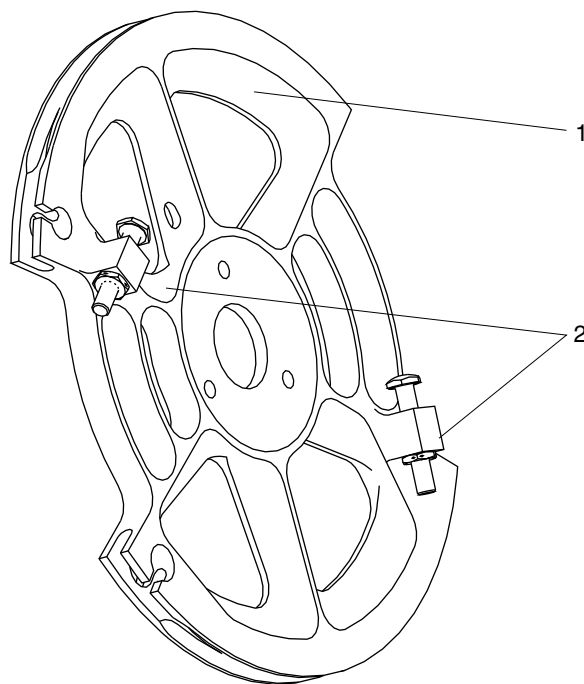
27-31-00

Config 001
Page 19
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Right Aft Quadrant.
- 2. Quadrant Primary Stop.

fs182a01.cgm

ELEVATOR RIGHT AFT QUADRANT PAGE 2
Figure 13

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-31-00
Config 001

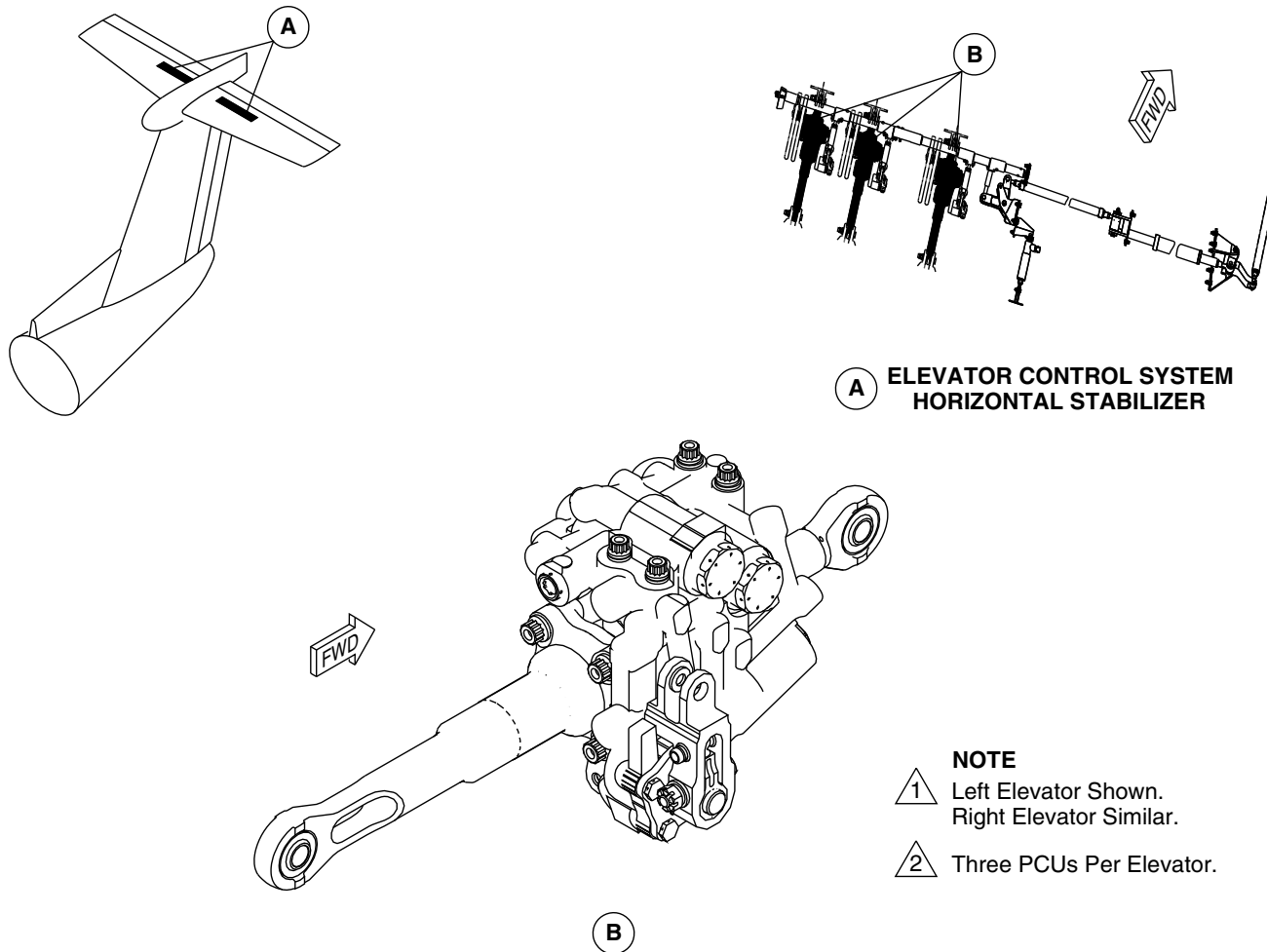
27-31-00

Config 001
Page 20
Sep 05/2021



DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



A ELEVATOR CONTROL SYSTEM
HORIZONTAL STABILIZER

- NOTE**
- 1 Left Elevator Shown.
Right Elevator Similar.
 - 2 Three PCUs Per Elevator.

ELEVATOR POWER CONTROL UNIT (PCU)
Figure 14

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-31-00
Config 001

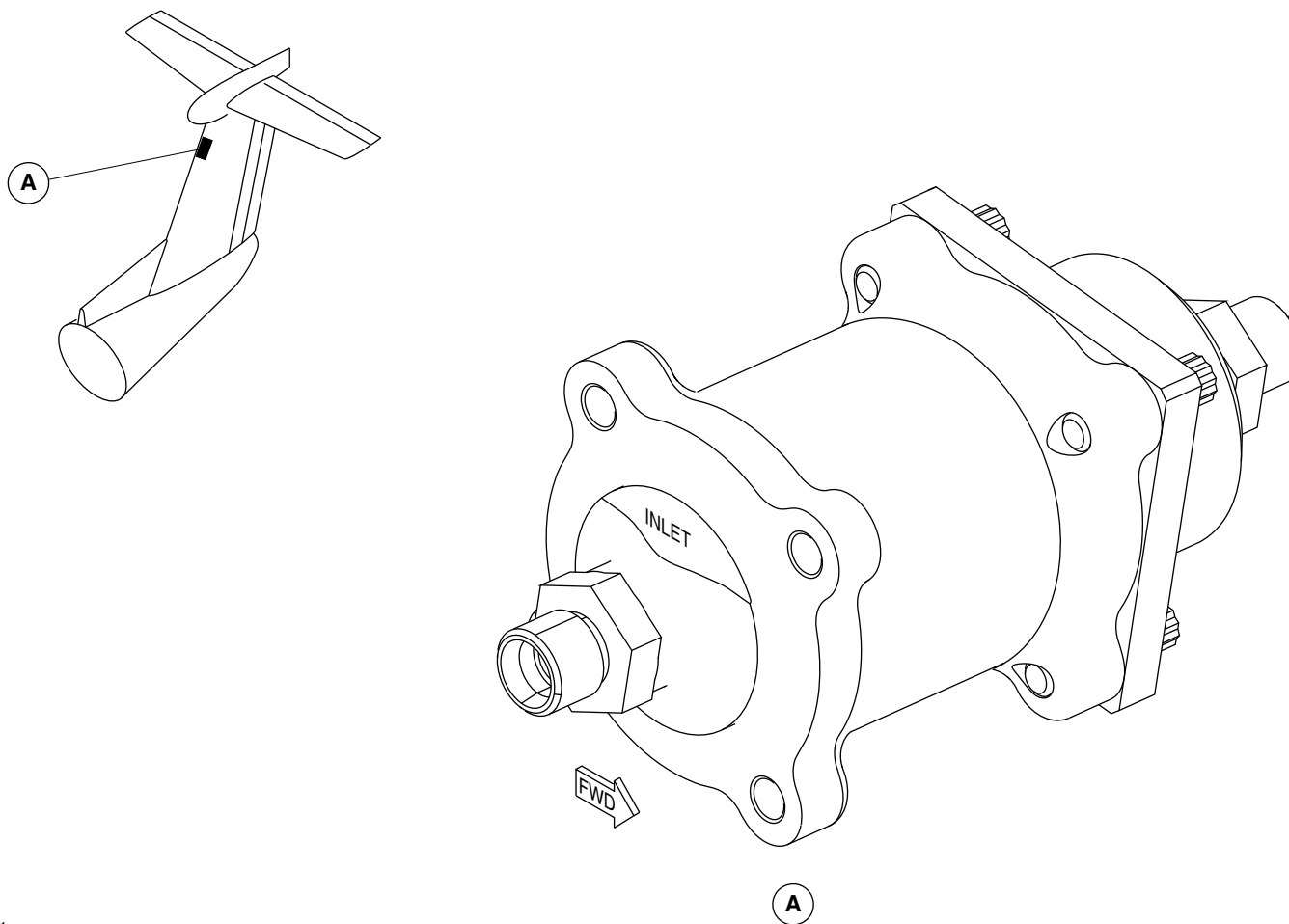
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Config 001
Page 21
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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COMPENSATOR
Figure 15

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-31-00
Config 001

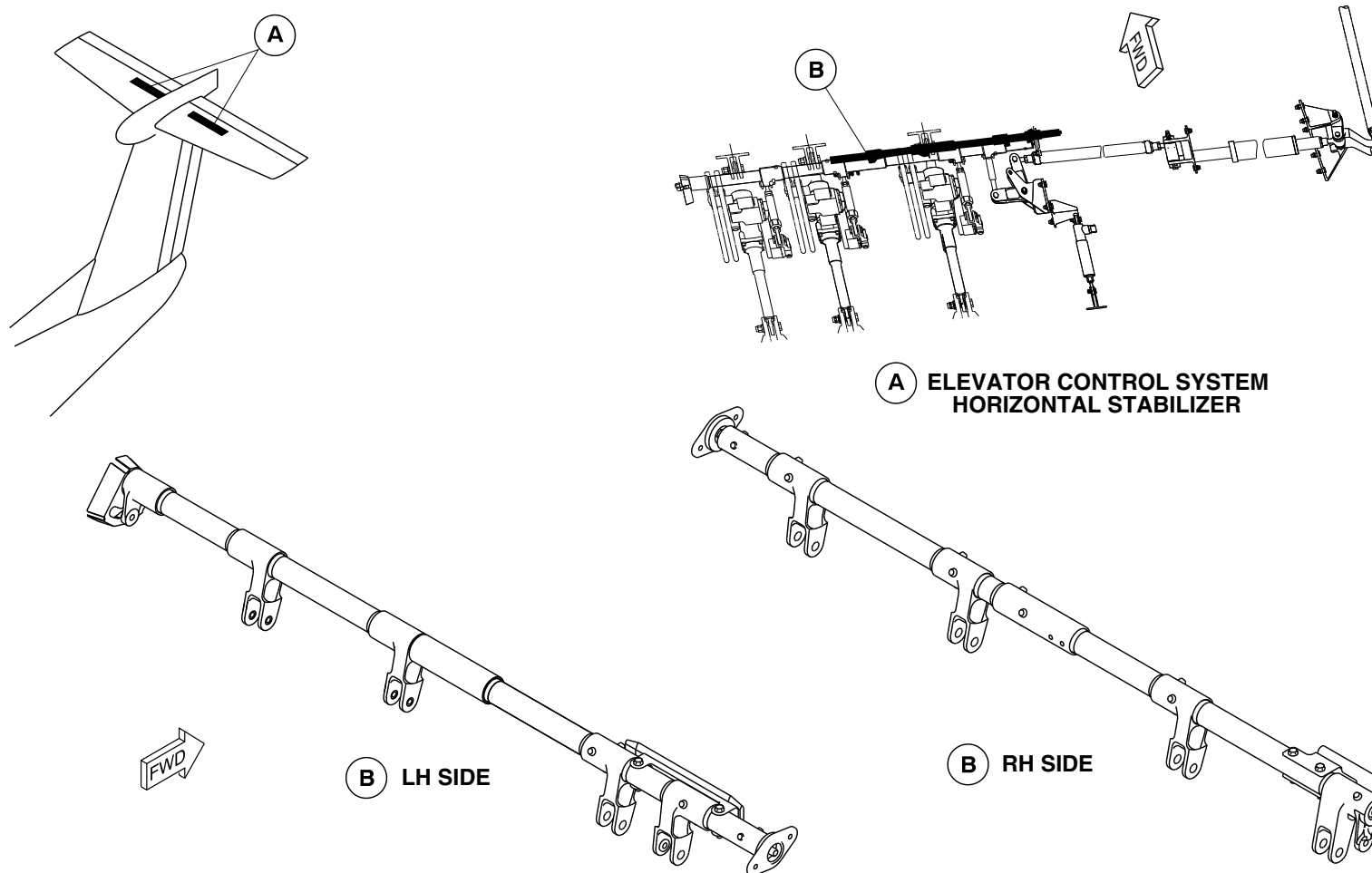
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Config 001
Page 22
Sep 05/2021



DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



fs411a01.dg, gv, 27/05/05

Torque Tubes
Figure 16

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-31-00
Config 001

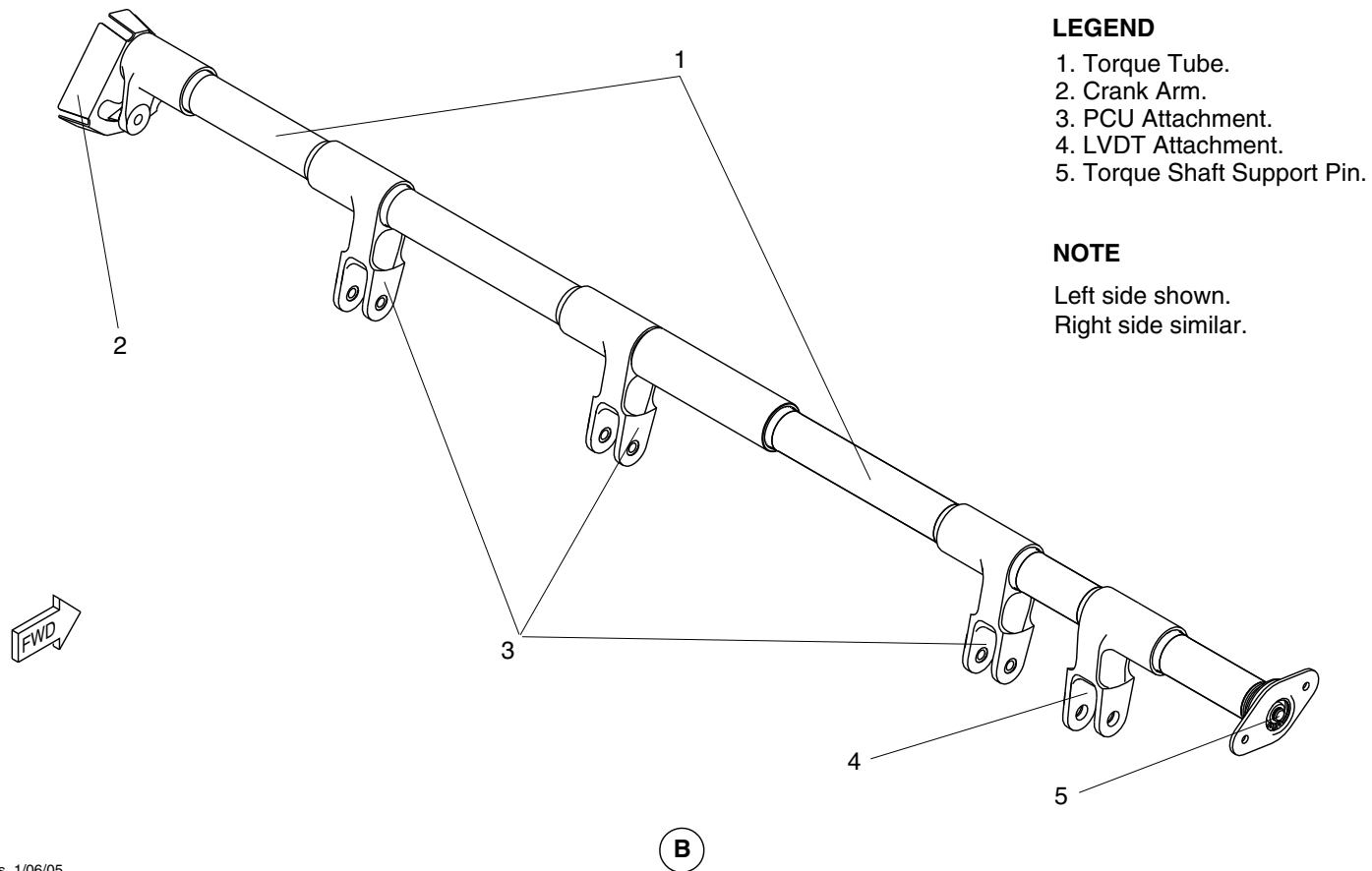
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Config 001
Page 23
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



fs412a01.dg, kms, 1/06/05

TORQUE TUBES
Figure 17

PSM 1-84-2A
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See first effectivity on page 2 of 27-31-00
Config 001

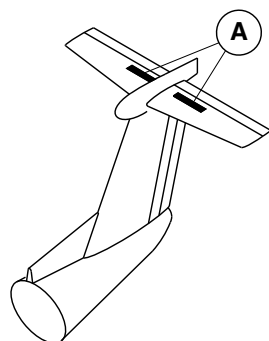
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Config 001
Page 24
Sep 05/2021

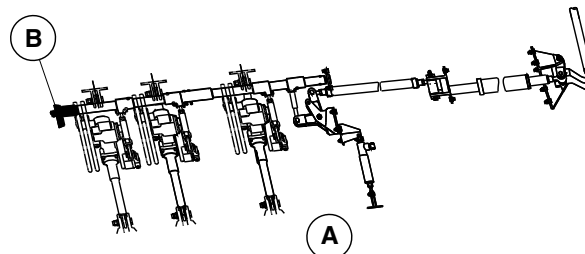


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



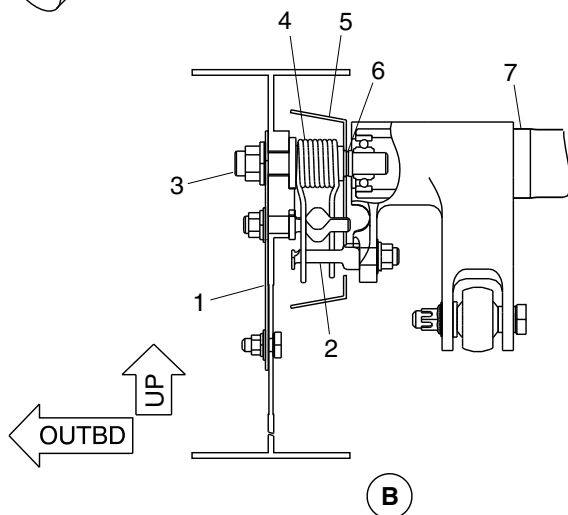
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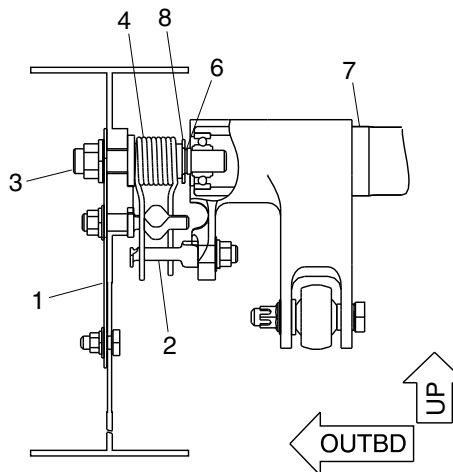
ELEVATOR CONTROL SYSTEM HORIZONTAL STABILIZER

LEGEND

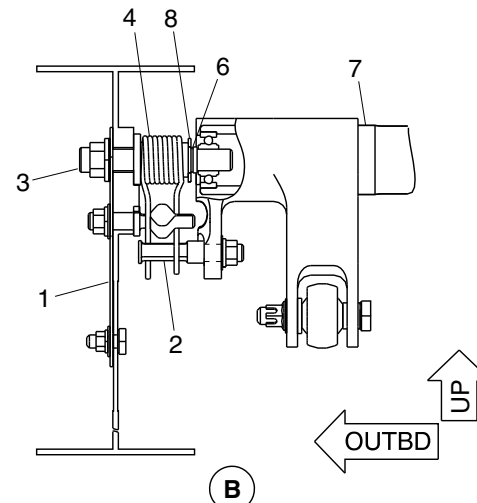
1. Adjusting plate.
2. Pin drive.
3. Torque shaft support pin.
4. Spring.
5. Cover.
6. Retaining ring.
7. Torque tube.
8. Washer.



**PRE MODSUM 4-121967
PRE MODSUM 4-113482
PRE SB84-27-31
PRE IS 4Q2750000**



**POST IS 4Q2750000
PRE SB84-27-31**



**POST MODSUM 4-121967
POST MODSUM 4-113482
POST SB84-27-31**

fs410a01.dg, rc, sep28/2013

Centering Device
Figure 18

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-31-00
Config 001

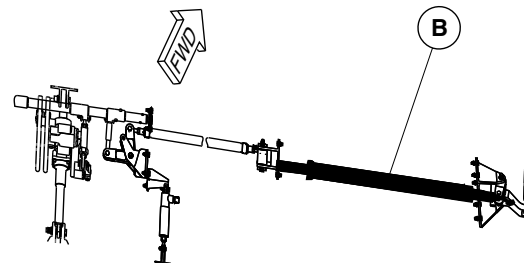
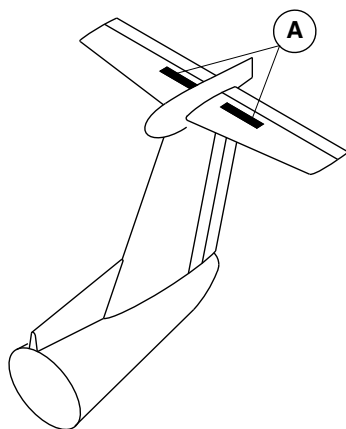
27-31-00

Config 001
Page 25
Sep 05/2021

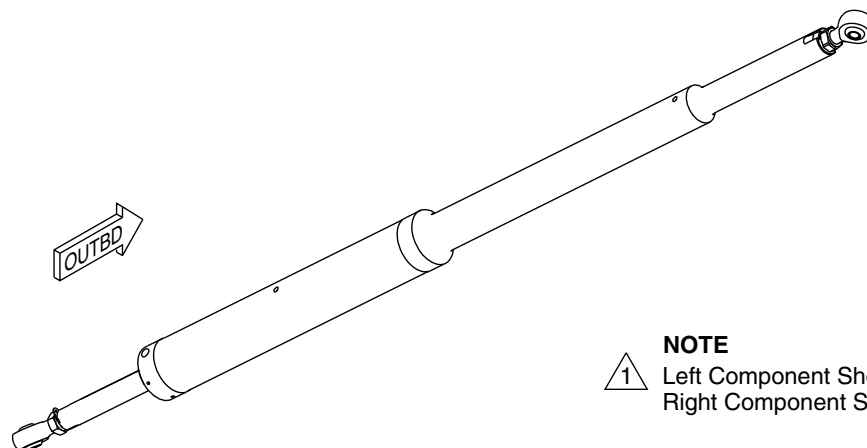


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



A ELEVATOR CONTROL SYSTEM
HORIZONTAL STABILIZER



NOTE
1 Left Component Shown.
Right Component Similar.

B

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OVERLOAD BUNGEE
Figure 19

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-31-00
Config 001

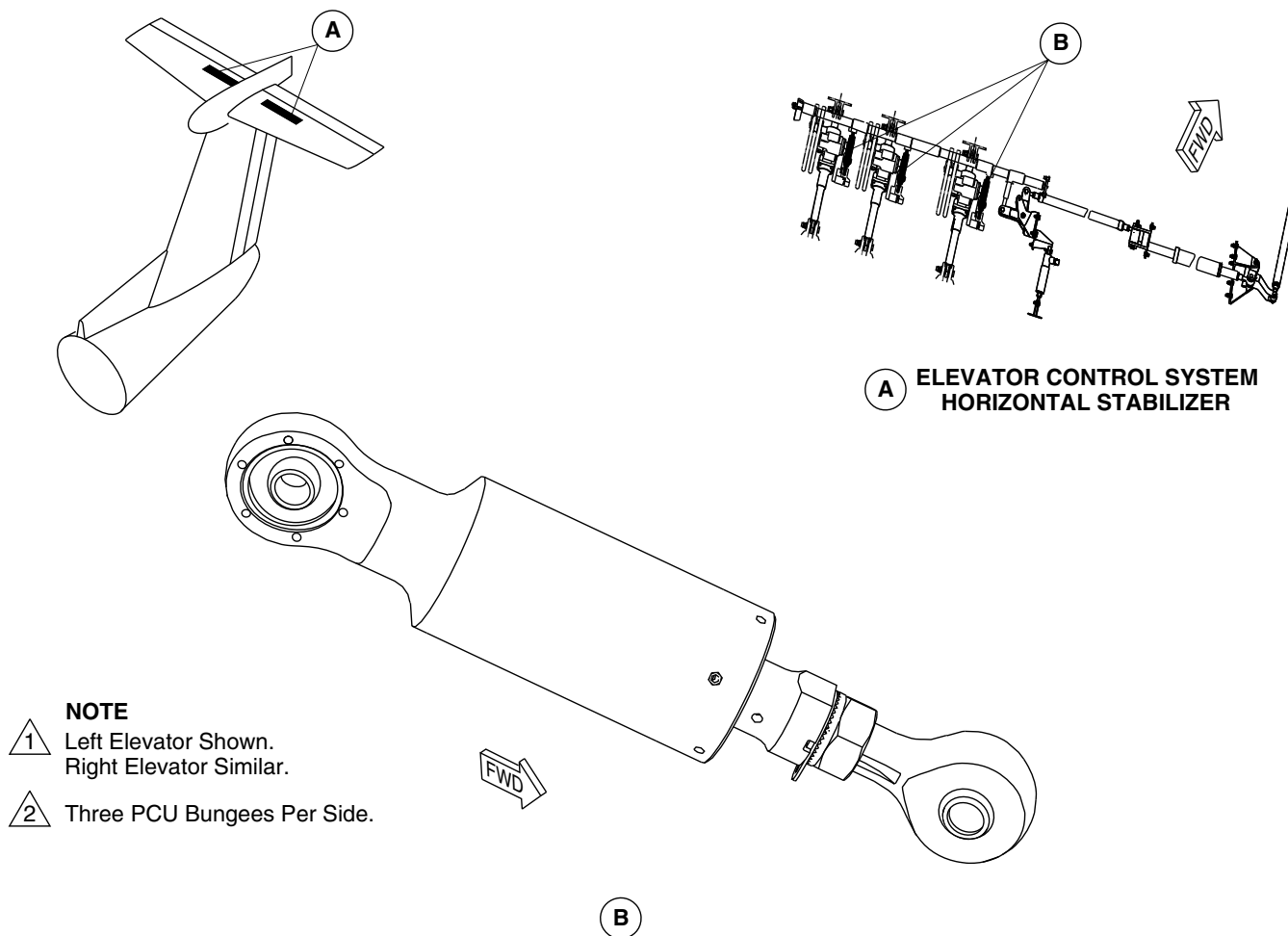
27-31-00

Config 001
Page 26
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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PSM 1-84-2A
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See first effectivity on page 2 of 27-31-00
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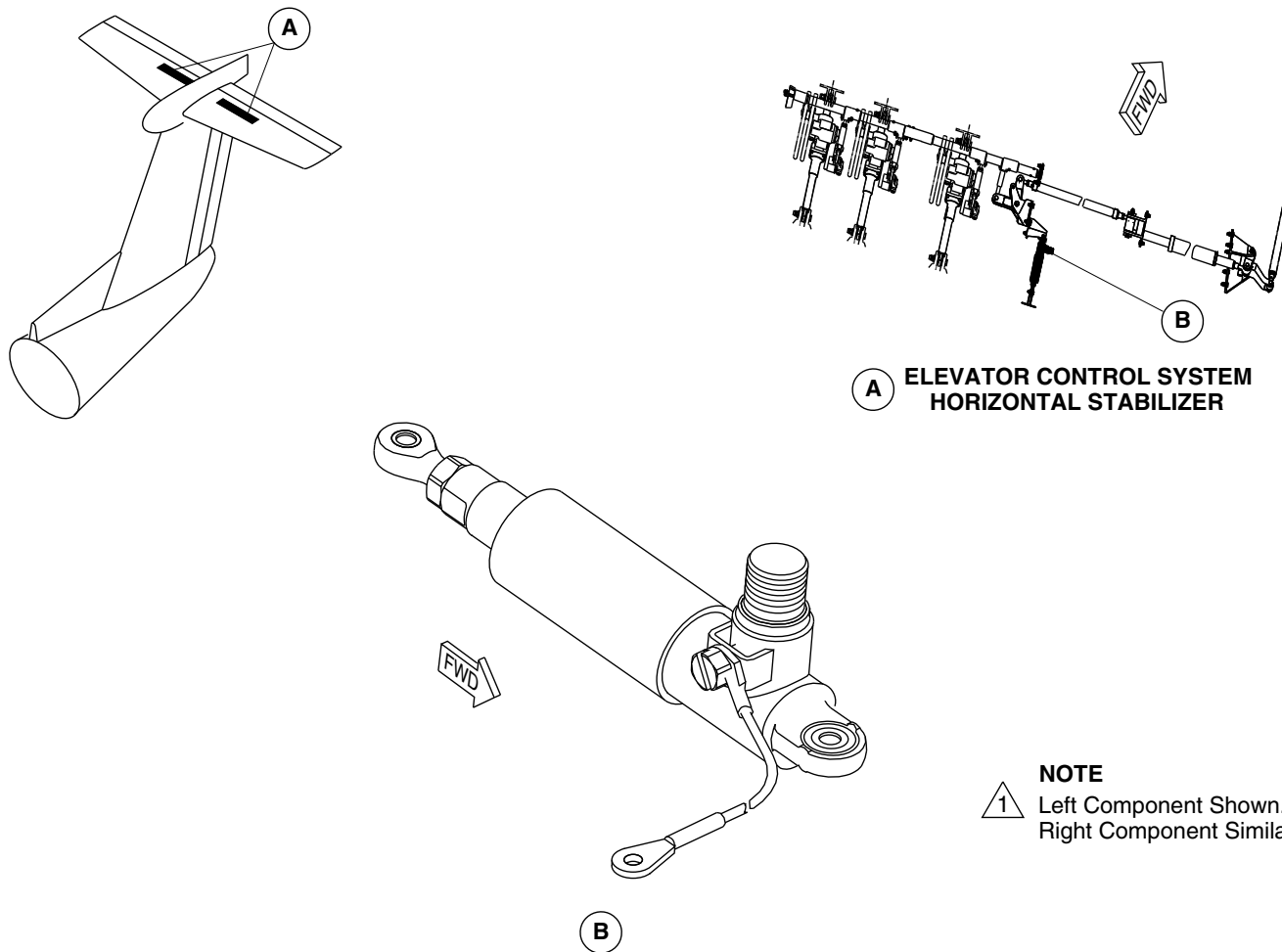
27-31-00

Config 001
Page 27
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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ELEVATOR POSITION LVDT
Figure 21

PSM 1-84-2A
EFFECTIVITY:
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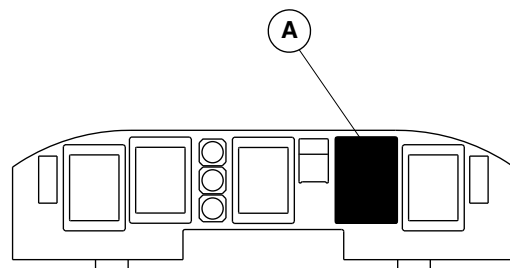
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Config 001
Page 28
Sep 05/2021



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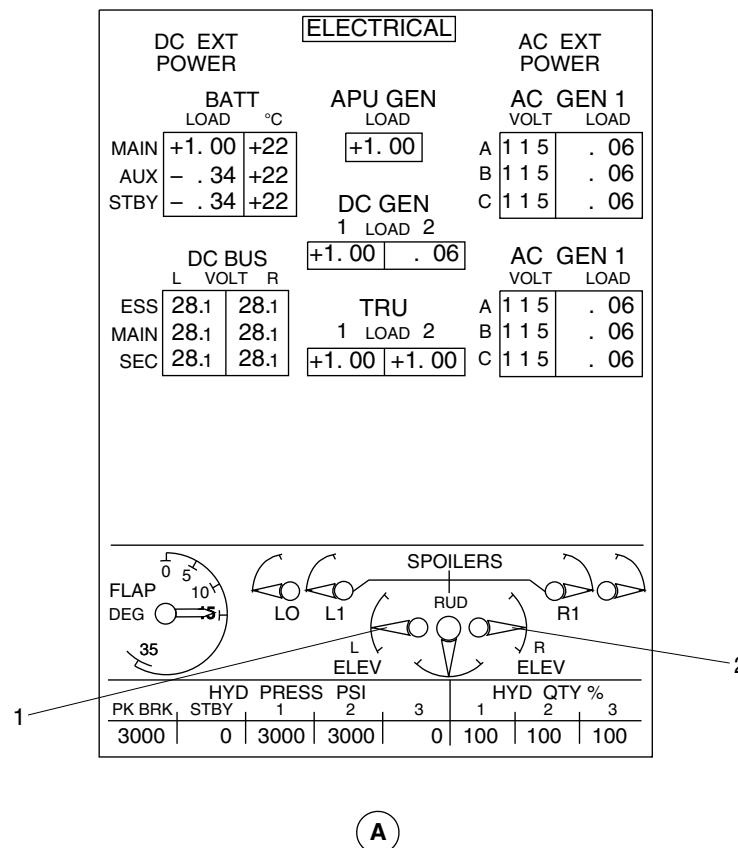
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Left Elevator Position Indication.
2. Right Elevator Position Indication.



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MULTIFUNCTION DISPLAY / ELEVATOR POSITION PAGE 1
Figure 22

PSM 1-84-2A
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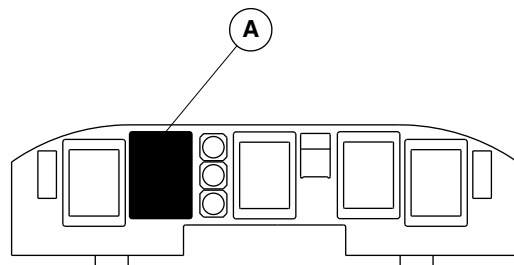
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Page 29
Sep 05/2021



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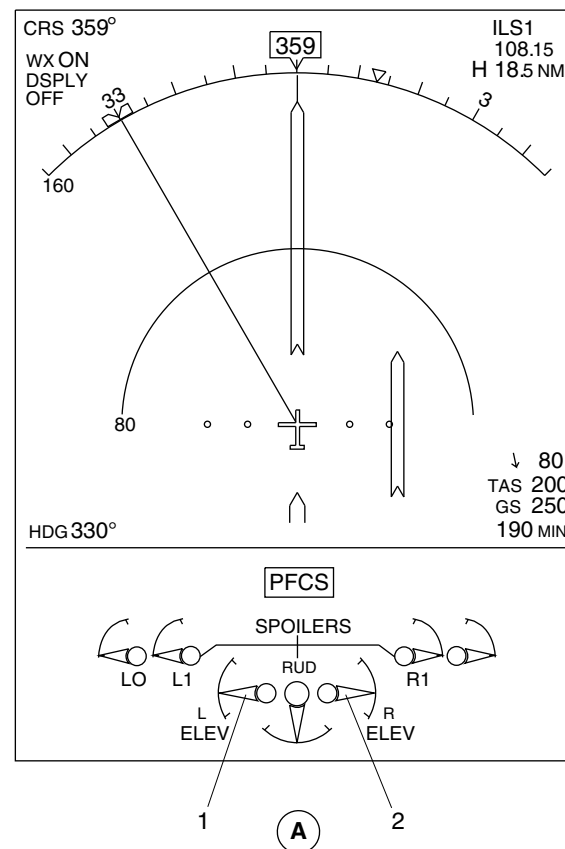
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Left Elevator Position Indication.
2. Right Elevator Position Indication.



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MULTIFUNCTION DISPLAY / ELEVATOR POSITION PAGE 2
Figure 23

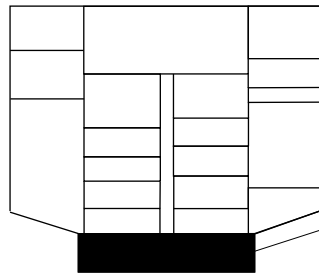
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27-31-00

Config 001
Page 30
Sep 05/2021



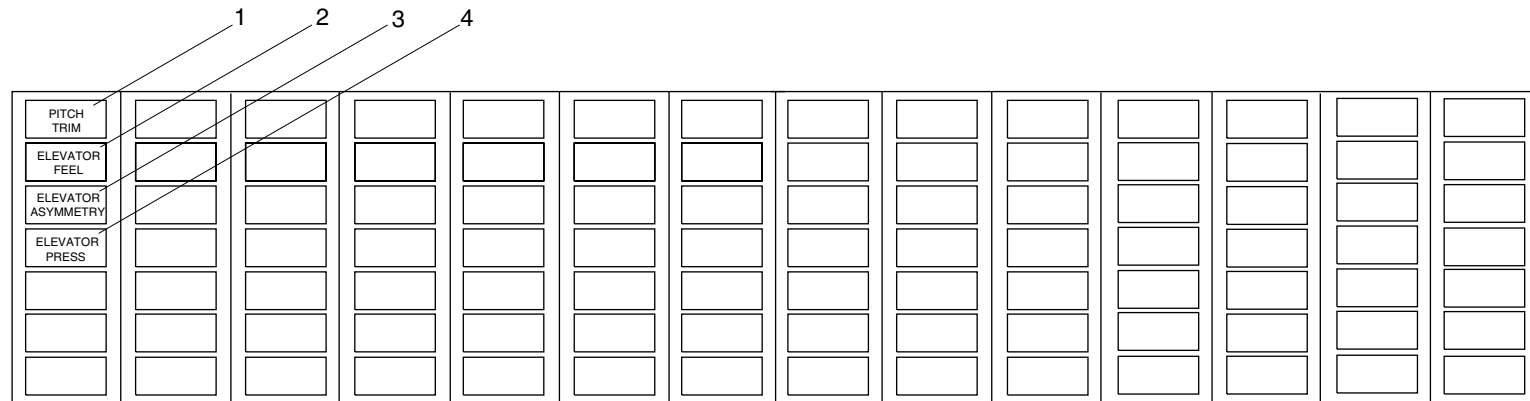
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



OVERHEAD CONSOLE

LEGEND

- 1 Pitch Trim (Amber).
- 2 Elevator Feel (Amber).
- 3 Elevator Asymmetry (Amber).
- 4 Elevator Pressure (Amber).



A

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ELEVATOR CAUTION LIGHTS
Figure 24

PSM 1-84-2A
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See first effectivity on page 2 of 27-31-00
Config 001

27-31-00

Config 001
Page 31
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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PSM 1-84-2A
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Config 001

27-31-00

Config 001
Page 32
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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27–32–00–001

ELEVATOR FEEL AND TRIM

Introduction

Artificial feel is used in the hydro–mechanical flight control system to give the pilot simulated feedback from the aerodynamic forces which are applied upon the flight surfaces. This artificial feel acting on the control column helps prevent the pilot from over controlling the aircraft. Trimming of the elevator control system gives the aircraft longitudinal stability during flight. Trimming counters the effects of changes to the aircraft center of gravity, airspeed and lift during flap extension or retraction.

General Description

Longitudinal or pitch trim is done by trimming both elevator surfaces.

The dual Pitch Feel and Trim Unit (PFTU), located near the aft elevator quadrants in the vertical stabilizer gives both pitch trim and pitch feel inputs into the elevator control system.

The elevator feel and trim system has the components that follow:

- Units, Pitch Feel and Trim (27–32–01)
- Actuators, Pitch Trim (27–32–06)
- Actuators, Pitch Feel (27–32–11)
- Indicator, Elevator Trim (27–32–16)
- Shutoff Switchlight, Elevator Trim (27–32–19).

Detailed Description

The elevator control system has a column centering device connected to a dual pitch trim actuator. This device, known as the dual pitch feel and trim unit, supplies an artificial feel to the control columns depending upon the airspeed and acceleration. The dual pitch feel and trim unit is in parallel with the pitch control linkages located near the elevator aft quadrants in the vertical stabilizer. Each pitch feel and trim unit has a feel actuator which positions a spring to produce the desired amount of feel at the control column.

A pitch trim actuator is installed on top of each pitch trim unit and gives elevator trim. The pitch trim actuators are controlled automatically by the autopilot, or manually by the NOSE DN/NOSE UP split trim switches on the pilot's and copilot's handwheels. Elevator trim position is shown in the flight compartment on the ELEVATOR TRIM indicator on the left side of the centre console.

A flap compensation pitch trim system gives an automatic pitch trim input to compensate for changes in lift during flap extension or retraction. This trim limits the longitudinal forces on the airframe.

Units, Pitch Feel and Trim

[Refer to Figures 1 and 2.](#)

There are two PFTUs incorporated into a single component that is installed in the vertical stabilizer. The right PFTU is in parallel with the copilot's elevator control circuit while the left PFTU is in parallel with the pilot's circuit. In the trim system, each PFTU has a pitch trim actuator, a trim pushrod, a trim stabilizing linkage, a trim bellcrank and a PFTU pushrod. In the pitch feel system, each PFTU has a pitch feel actuator, a feel bellcrank, a feel pushrod, a feel stabilizing linkage, a balance spring and a load limit bungee.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Actuators, Pitch Trim

Refer to Figures 3 and 4.

The pitch trim actuators are installed in an electro-mechanical unit that includes two DC motors, two motor control cards, one set of gears, one driveshaft, two actuators and two Linear Variable Differential Transformers (LVDTs). The two motors are installed in-line so that only one motor is needed at any time to operate the two actuators. Either motor rotates a single shaft which drives the gear set. The gears operate both the left and right pitch trim actuators at the same time.

The Autopilot Flight Control System (AFCS) supplies an automatic pitch trim signal to the elevator control system during flap extension or retraction. The flap auto-trim signal is processed through the Flight Control Electronic Control Unit (FCECU), the Flap Control Unit (FCU) and the Flight Guidance Module (FGM). The flap position transducers give flap position and speed feedback signals to the FCU which transmits the flap feedback signals to the FGM. The FGM analyzes the flap feedback signals and sends an autopilot pitch trim signal to the FCECU which gives command signals to the pitch trim actuators.

The pitch trim actuators then supply a trim input to the elevator control system through the two PFTUs and the two elevator aft quadrants. The elevators are automatically trimmed whenever the flaps are moved. The pitch trim signals sent by the FGM during flap extension and retraction occur in both manual and autopilot aircraft modes. This reduces the aerodynamic longitudinal forces acting on the airframe.

Pitch trim signals from the NOSE DN/NOSE UP split trim switches on the control handwheels, or from the autopilot, are prioritized by the

FCECU in this order: pilot, copilot and autopilot. The trim signal with the highest priority controls the pitch trim actuator.

The FCECU controls the elevator pitch trim rate according to the airspeed of the aircraft. At airspeeds below 150 KIAS (278 km/h), the trim actuators operate in high speed mode (0.75 degrees/sec). At airspeeds greater than 250 KIAS (463 km/h), the trim actuators operate in low speed mode (0.1 degrees/sec). The FCECU adjusts the trim rate between these two values for airspeeds between 150 and 250 KIAS (278 and 463 km/h). The FCECU receives airspeed inputs from the Air Data Units (ADUs).

The PITCH TRIM caution light, controlled by the FCECU, comes on if:

- One pitch trim motor, or its control card fails.
- Pitch trim “runaway” is sensed.
- Airspeed No.1 disagrees with Airspeed No.2.
- Pitch trim input commands disagree.

Actuators, Pitch Feel

Refer to Figures 5 and 6.

The elevator control system gives artificial feel to the two control columns using the left and right pitch feel actuators located in the PFTU. The pitch feel actuators extend or retract as commanded by the FCECU in relation to the airspeed of the aircraft. As the aircraft airspeed increases or decreases, the two pitch feel actuators are commanded to extend or retract respectively. This gives an increase or decrease in artificial feel.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Each pitch feel actuator is a single electro-mechanical unit which has one DC motor, one motor controller circuit card, one set of gears, one actuator and one LVDT.

Each motor controller operates its related motor based on control loop commands from the FCECU. The left motor controller receives its commands from the FCECU channel 1, and the right motor controller receives its commands from the FCECU channel 2.

Each FCECU channel supplies three signals to its related motor controller:

- Enable “ON or OFF”
- Motor direction
- Motor speed.

The two LVDTs located in the right and left pitch feel actuators supply feedback signals to the FCECU. The feedback signals let the FCECU monitor the position and speed of the pitch feel actuators, completing the control loop

The FCECU supplies extend or retract signals to the pitch feel actuators based on airspeed and acceleration information. The airspeed information is supplied by the ADUs and the acceleration information is supplied by the Attitude and Heading Reference System (AHRS). The data from the ADUs and AHRS is shared between channels 1 and 2 of the FCECU so that an identical signal is sent to each pitch feel actuator. This reduces mistrack or asymmetry between the pitch feel actuators.

Both pitch feel actuators are normally in the active mode with both enable signals from the FCECU in the “ON” position. Each motor operates its related actuator. The failure of a pitch feel actuator is sensed by the FCECU through the LVDT feedback signals. The

FCECU then switches the enable signal of the failed actuator to the “OFF” position. The FCECU continues to supply the enable “ON” signal to the other actuator. The aircraft operates with degraded pitch feel.

The ELEV FEEL caution light, controlled by the FCECU, comes on if:

- Left or right Pitch Feel Actuator fails
- Airspeed No.1 and Airspeed No.2 disagree
- Acceleration No.1 and Acceleration No.2 disagree.

Indicator, Elevator Trim

[Refer to Figures 7 and 8.](#)

The elevator trim indicator is an analog type gauge located in the flight compartment on the center console. The LVDTs located in the pitch trim actuators send their signals to the FCECU which then supplies a signal to the elevator trim indicator. The indicator displays elevator trim position as commanded by the trim switches on the pilot's and copilot's control handwheels. The indicator is labelled “NU” for nose up, “ND” for nose down and “TO” for take-off. A white band next to the “TO” setting shows the amount of elevator trim that is acceptable for take-off. An aural warning sounds if the power levers are advanced above Flight Idle plus 12 degrees, for take-off, with the elevator trim position outside the take-off range.

Shutoff Switchlight, Elevator Trim

[Refer to Figure 9.](#)

There are two elevator trim shutoff switchlights. One switchlight is installed on the left glareshield panel for the pilot and the other switchlight is on the right glareshield panel for the co-pilot. If the



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

FCECU detects that the manual pitch command continues for more than three seconds, the FCECU will send a signal to the Advisory Light Control Unit (ALCU). This causes the pilot's or the co-pilot's elevator trim shut off PUSH OFF switchlight to come on. When the pilot or co-pilot pushes the switchlight, the elevator trim is disabled.

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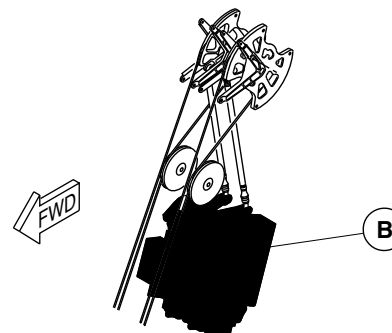
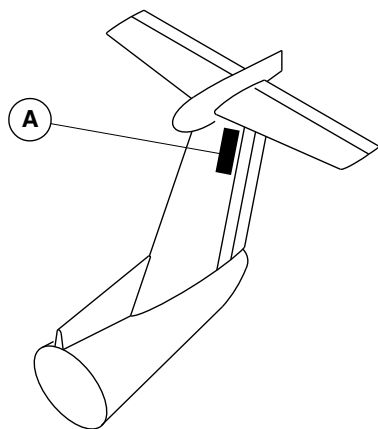
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Page 5
Sep 05/2021

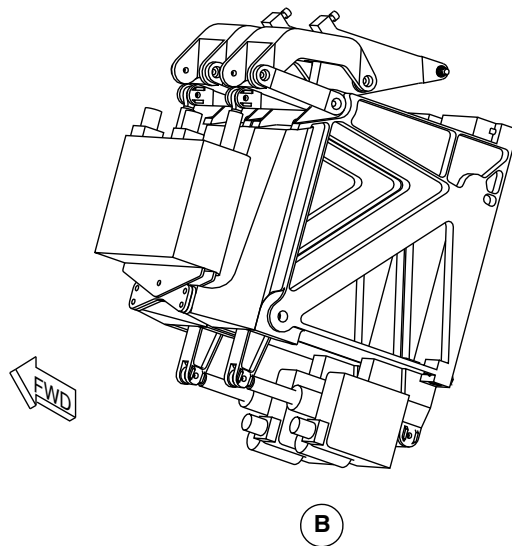


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



A ELEVATOR CONTROL SYSTEM
– VERTICAL STABILIZER



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PITCH FEEL AND TRIM UNITS PAGE 1
Figure 1

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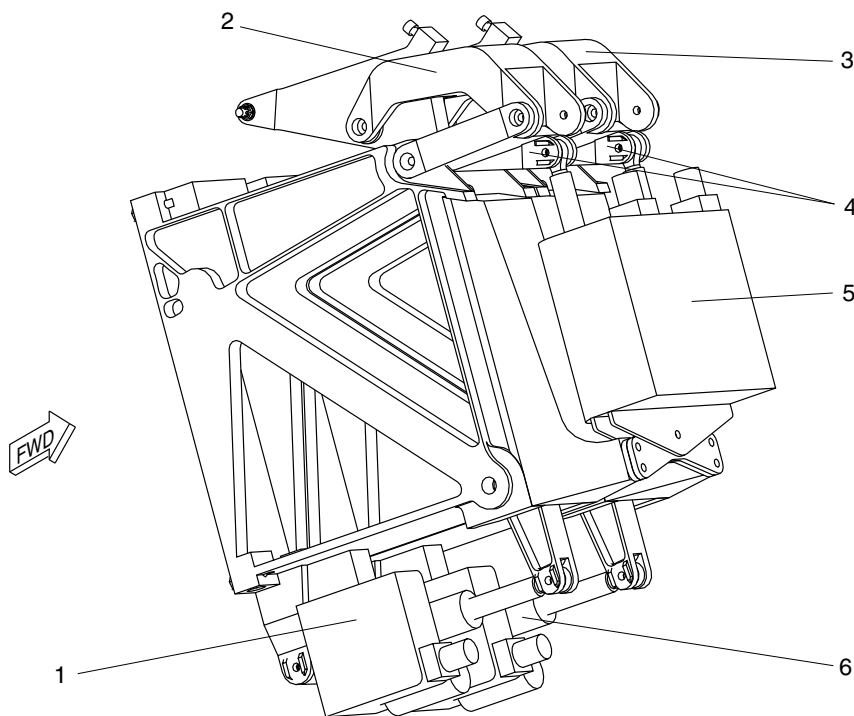
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Config 001
Page 6
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Pitch Feel Actuator RH.
- 2. Trim Bellcrank RH.
- 3. Trim Bellcrank LH.
- 4. Trim Push Rods.
- 5. Trim Actuators.
- 6. Pitch Feel Actuator LH.

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PITCH FEEL AND TRIM UNITS PAGE 2
Figure 2

PSM 1-84-2A
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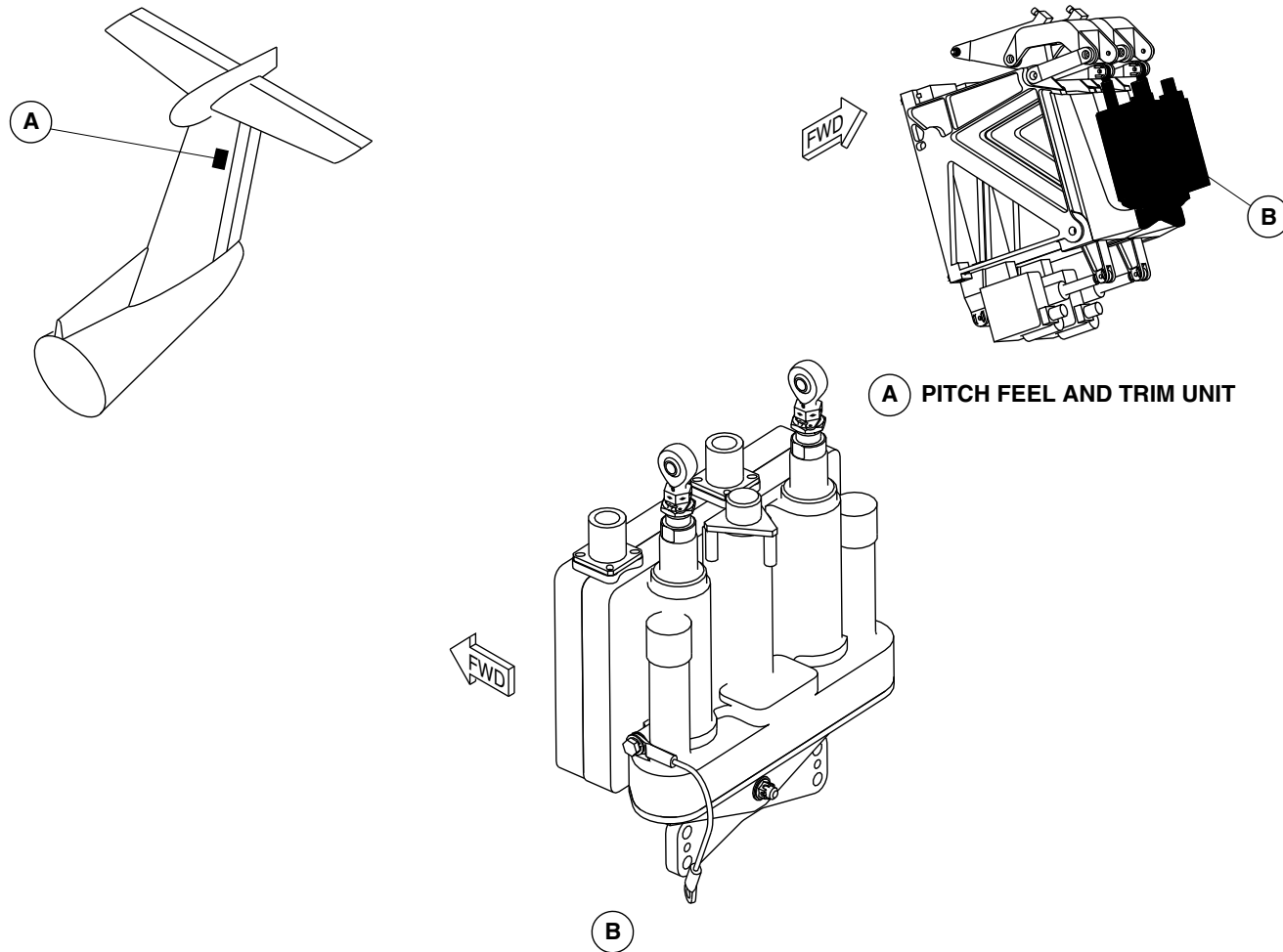
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Config 001
Page 7
Sep 05/2021



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PITCH TRIM ACTUATORS PAGE 1
Figure 3

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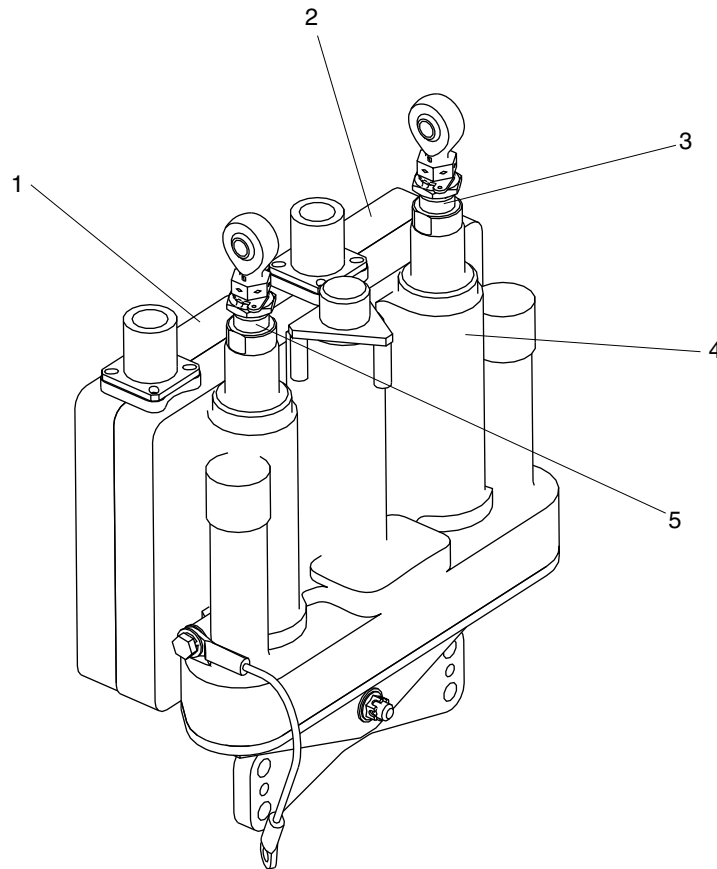
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Config 001
Page 8
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Motor Controller Circuit Card LH.
- 2. Motor Controller Circuit Card RH.
- 3. Pitch Trim Actuator RH.
- 4. Actuator Housing.
- 5. Pitch Trim Actuator LH.

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PITCH TRIM ACTUATORS PAGE 2
Figure 4

PSM 1-84-2A
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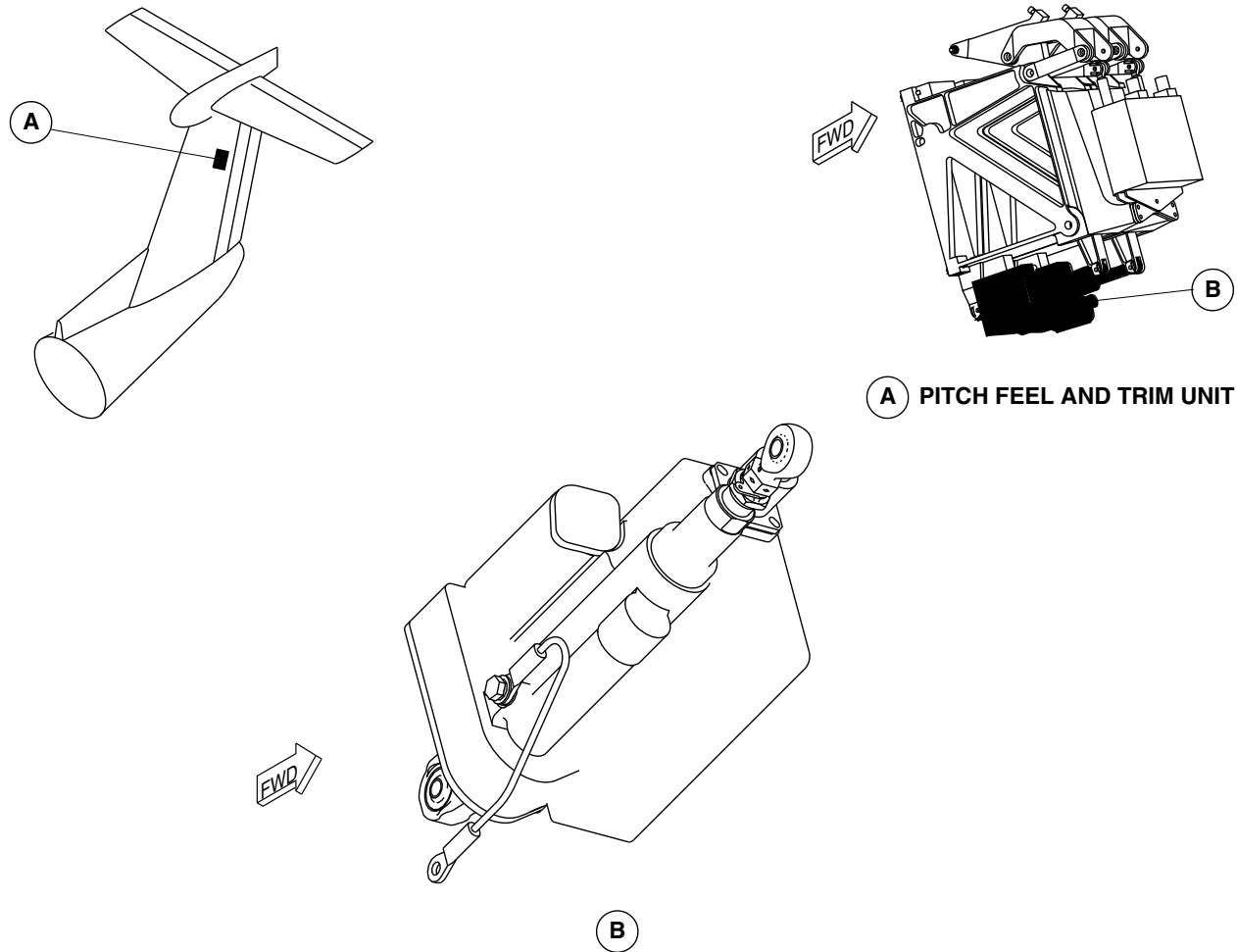
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Config 001
Page 9
Sep 05/2021



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PITCH FEEL ACTUATOR PAGE 1
Figure 5

PSM 1-84-2A
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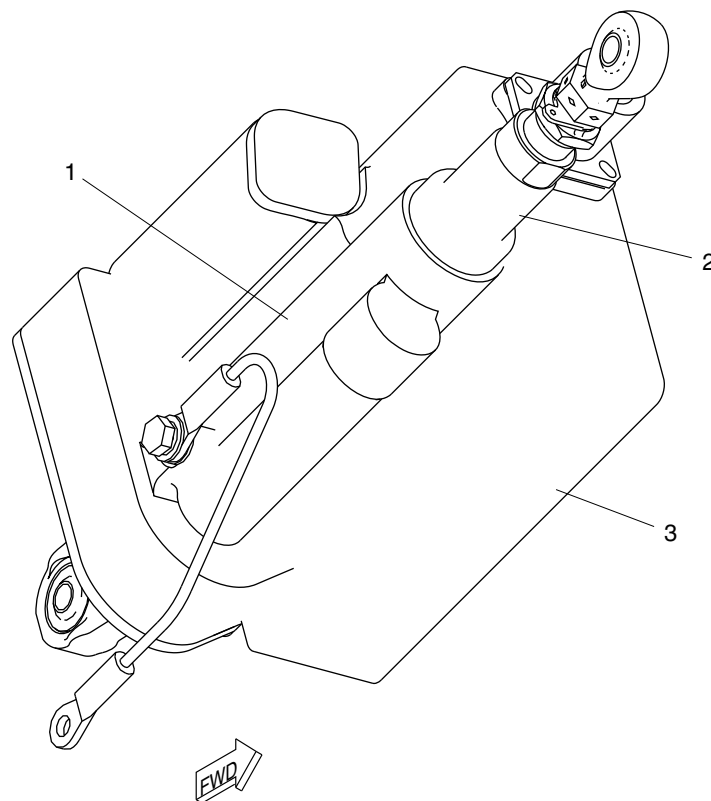
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Config 001
Page 10
Sep 05/2021



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LEGEND

- 1. Actuator Housing.
- 2. Pitch Feel Actuator.
- 3. Motor Controller Circuit Card.

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PITCH FEEL ACTUATOR PAGE 2
Figure 6

PSM 1-84-2A
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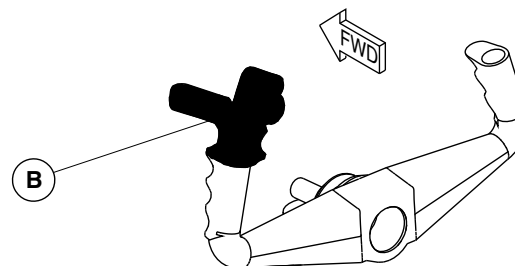
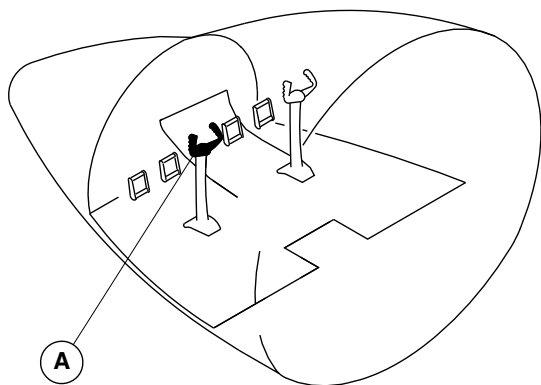
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Config 001
Page 11
Sep 05/2021



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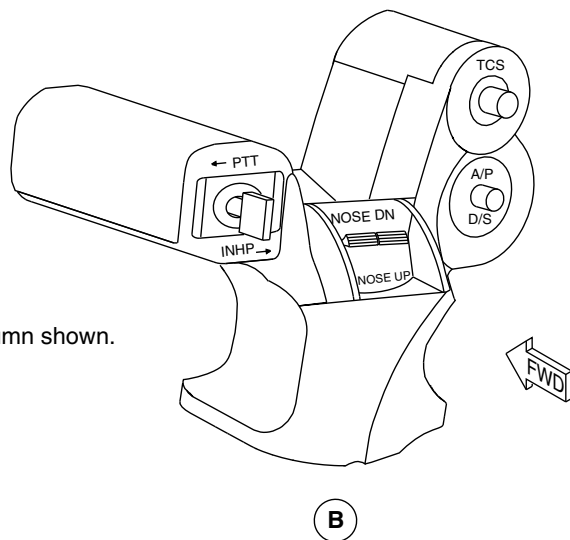
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



A PILOT CONTROL COLUMN

NOTE

Pilot Control Column shown.
Copilot similar.



B

PITCH TRIM SWITCH
Figure 7

PSM 1-84-2A
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Config 001

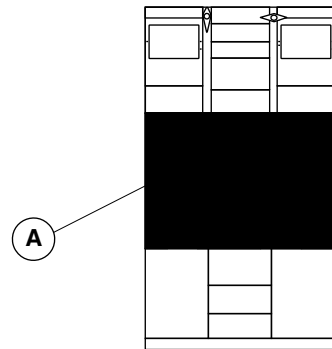
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Config 001
Page 12
Sep 05/2021



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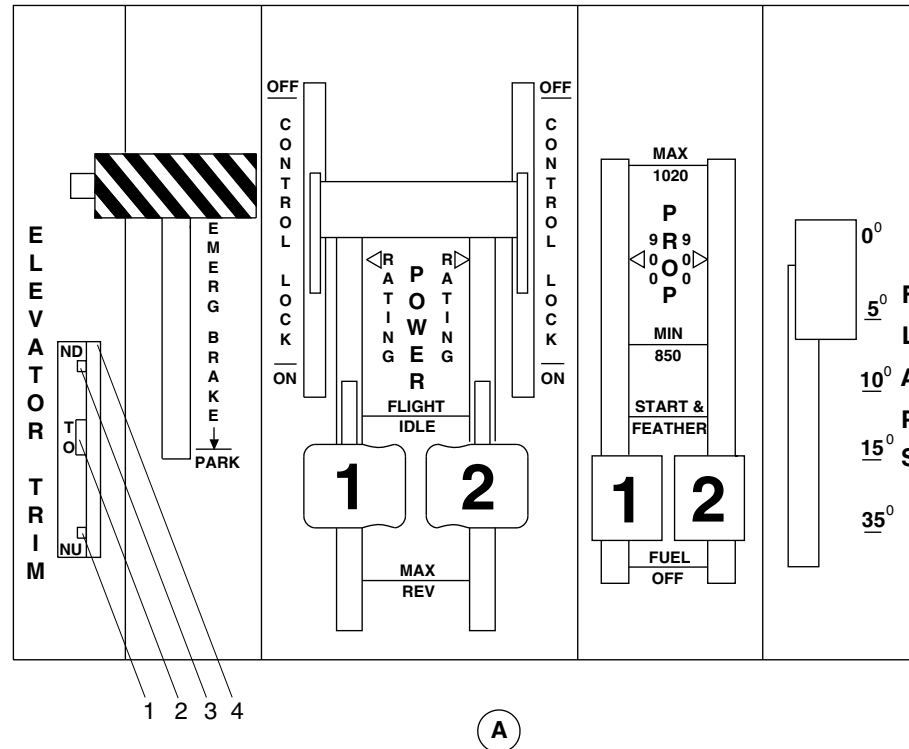
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CENTRE CONSOLE

LEGEND

1. Nose Down Range.
2. Take-Off Range.
3. Nose Up Range.
4. Indicator.



ELEVATOR TRIM INDICATOR
Figure 8

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Config 001

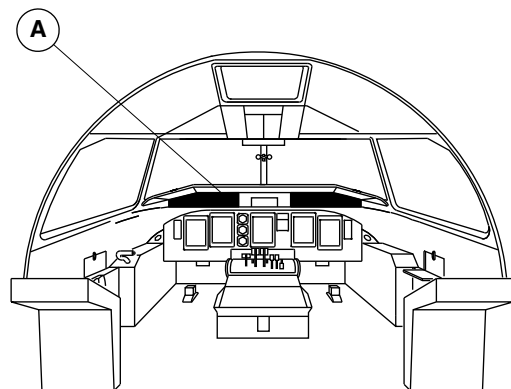
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Config 001
Page 13
Sep 05/2021



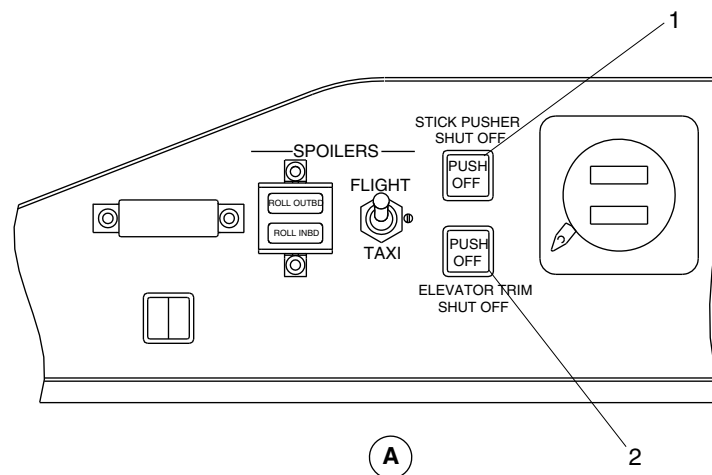
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LEGEND

1. Stick Pusher Shut Off Switchlight.
2. Elevator Trim Shut Off Switchlight.



NOTE

LH Glareshield shown.
RH Glareshield similar.

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ELEVATOR TRIM PUSHOFF SWITCHLIGHTS
Figure 9

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-32-00
Config 001

27-32-00

Config 001
Page 14
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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STALL WARNING/STICK PUSHER

Introduction

The Stall Protection System (SPS) gives an indication that the aircraft is near a stall condition and when the aircraft is in a stall condition, it gives a positive input to help the flight crew regain control.

General Description

[Refer to Figure 1.](#)

The SPS has the two primary functions that follow:

- Stick shakers
- Stick pusher.

Stick Shakers: The SPS uses stick shakers to give a tactile and aural warning of a near stall condition.

Stick Pusher: When the aircraft is in a stall condition, the SPS uses a stick pusher to move the elevators to reduce the wing angle of attack.

The SPS has the components that follow:

- Modules, Stall Protection (27–33–01)
- Actuator, Stick Pusher (27–33–06)
- Capstan, Stick Pusher (27–33–11)

- Quadrant, Stick Pusher (27–33–16)
- Shaker, Stick (27–33–21)
- Transducer, Angle of Attack (27–33–26)
- Switchlight, Stick Pusher Release (27–33–31)
- Switch, Pre–Flight Test.

The Stall Protection Modules (SPM1, SPM2) transmit data to the components that follow:

- Flight Guidance Module (FGM1, FGM2)
- Stick shakers
- Stick pusher
- Caution and Warning Annunciator Lights Panel
- Flight Data Concentrator (FDC)
- Ground Proximity Warning System (GPWS)
- Central Diagnostics System (CDS)

The SPM1, SPM2, FGM1, FGM2, and the FDC are located in its related Integrated Flight Cabinet (IFC1, IFC2).

Detailed Description

The SPS functions in the modes that follow:

- Initialization (INIT)
- Power–On Self Test (POST)
- Operational (OPER)
- Maintenance

PSM 1–84–2A
EFFECTIVITY:
See first effectivity on page 2 of 27–33–00
Config 001

27–33–00

Config 001
Page 2
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

- Teleloading.

Initialization (INIT) Mode: The SPS functions in the initialization mode after an electrical power interruption to initialize the hardware and software.

If the SPS does not receive a maintenance or teleloading request when the aircraft is on the ground, it will then function in the POST mode.

POST: The SPS does a POST after a time long-term power interruption while the aircraft is on the ground.

The POST senses SPS hardware malfunctions that are not continuously monitored in the operational mode or apparent to the flight crew.

The POST tests the parameters that follow:

- Program memory
- Data memory
- Continuous monitoring
- Power supplies HW monitoring
- Discrete inputs
- Discrete outputs
- Module validity hardware logic
- ARINC 429 inputs
- ARINC 429 outputs
- Memory partitioning hardware monitoring
- Time partitioning hardware monitoring

- I/O sequencer
- Analog reference voltages.
- Analog inputs.

The results of the 25 POST is stored in the Built In Test Equipment (BITE) and is sent to the CDS.

The SPM1 or SPM2 does a POST twice before declaring a malfunction. If the POST senses a malfunction, the SPM1 or SPM2 will prevent the related stick shaker and the stick pusher from operating.

If there is a long-term power interruption while the aircraft is in flight, the SPM will not do a POST. It will use the stored POST results to determine its status.

Operational Mode: The SPS uses the SPM1 or SPM2 to control their related Stick Shakers. The SPM1 and SPM2 function together to control the stick pusher.

In a stall condition, a STICK PUSHER SHUT OFF switchlight selection can be made to manually stop a stick push or the pilots can override the stick push using the control columns. The stick shakers will continue to operate.

The SPS operation mode is divided into the sub-modes that follow:

- Pre-flight
- In flight.

Pre-Flight Test: The pilot does a stall warning pre-flight test before each flight. The STALL WARN toggle switch located on the pilot side console is set to the TEST1 and then to the TEST2 position to test the SPM1 and SPM2. The switch must be held in the appropriate position for the entire test.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

In-Flight Operational Mode: The SPS operates when the calibrated airspeed is less than 215 kt (399 km/h) and the Above Ground Level (AGL) altitude is more than 200 ft (61 m).

The SPM1 and SPM2 use the parameters that follow to calculate when the aircraft is near a stall condition:

- Angle of Attack (AOA) data
- Flap position
- Mach number
- Engine torque
- Icing status.

The SPM1 and SPM2 are used to calculate the stick pusher operating angle. It uses the parameters that follow:

- AOA data
- Flap position
- Mach number
- Power lever angle
- Condition lever angle
- Icing status.

The SPM1 and SPM2 calculate when to start and cancel the stick shakers and stick pusher operation.

When the SPM1 and SPM2 operate their related stick shakers, a signal to the Automatic Flight Control System (AFCS) disengages the autopilot. In addition, a signal to the GWPS prevents the PULL UP audible alerts.

The SPM1 and SPM2 use their related AOA1 and AOA2 transducer angle input for stick shaker calculations. The SPS uses the average of AOA1 and AOA2 transducer angle inputs for the stick pusher calculations.

The SPM1 will use the same input as the SPM2 and the SPM2 will use the same input as SPM1 if any related data inputs that follow have malfunctioned:

- AOA data
- Flap position
- Mach number
- Engine torque
- Icing status
- Power lever angle
- Condition lever angle.

If no data is available, the SPM1, SPM2 will then use default data.

An AOA1 or AOA2 malfunction is the only signal that causes the stick pusher from operating.

For the stick pusher calculations, the SPM1 and SPM2 uses the average of the AOA1 and AOA2 inputs. The stick pusher will not operate if any of the conditions that follow is correct:

- The AOA1 or AOA2 transducer has malfunctioned
- There is a difference between the AOA1 and AOA2 inputs.

The SPMs will not use the input of an AOA transducer that is not heated.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

When the aircraft is in icing conditions, the lift supplied by the wings can be decreased enough to cause the aircraft to stall at a lower AOA. The stall operating angle of the stick shaker and stick pusher is set to a relatively lower AOA when in icing conditions. The SPS receives icing condition status data from the selection of the propeller anti-ice switch. On aircraft with ModSum 4-126089, the SPS receives icing condition status data from the selection of the INCR position of the REF SPEEDS switch on the ICE PROTECTION panel.

When the propeller anti-ice switch is turned ON, the SPS changes its calculations because of the reduced performance limits of the aircraft. On aircraft with ModSum 4-126089, when the REF SPEEDS switch on the ICE PROTECTION panel is set to INCR position, the SPS changes its calculations because of the reduced performance limits of the aircraft. The Electronic Instrument System (EIS) shows an INCR REF SPEEDS message.

Maintenance Mode: The BITE uses the CDS to give the condition of the component. It saves faults in a Non Volatile Memory (NVM) for reporting to line and shop maintenance.

The maintenance mode is selected from one or the other ARCDU during maintenance operations. The SPM does a test when it receives a maintenance discrete command through the IOM1 while the air speed is less than 50 kt (93 km/h) and the aircraft is on the ground.

The BITE allows aircraft maintenance personnel to:

- Do fault isolation and return to service testing after completing maintenance actions.
- Access failure reports from last or previous flight legs.
- Get the avionics status report.

- Get the part number of an LRU.

The BITE modes monitor the condition of the component as follows:

- POST
- Continuous Monitoring.

POST: The POST checks the condition of the component at power-up or after a long power interruption, more than 200 milliseconds.

Continuous Monitoring: The continuous monitoring checks the status of the component in flight. It records faults in a NVM for later troubleshooting using the CDS.

When energized, the SPS does the different types of monitoring that follow:

- Internal monitoring
- External input monitoring
- Stick shaker monitoring
- Stick pusher monitoring.

Internal Monitoring: The SPM1 and SPM2 continuously do the self monitoring tests to make sure that it can do its functions. It tests the parameters that follow:

- Secondary power supplies
- Continuous monitoring
- Memory partitioning
- Stack overflow
- Job overflow

PSM 1-84-2A

EFFECTIVITY:

See first effectivity on page 2 of 27-33-00

Config 001

27-33-00

Config 001
Page 5
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

- Bus error
- I/O sequencer validity
- CPU Board/EXT board ARINC 429 dialogue
- SPM cross monitoring.

An SPM cross monitoring malfunction prevents stick pusher operation. All other continuous self monitoring test malfunctions prevent stick shaker and stick pusher operations.

External Input Monitoring: The SPM1 and SPM2 monitor the external inputs that follow:

- AOA transducers
- AOA transducer corrections
- Flap Position Indicator Unit (FPIU) flap position data
- Propeller de-icing selected data
- Proximity Sensor Electronics Unit (PSEU) Weight On Wheels (WOW) data
- Air Data Unit (ADU) data
- Attitude and Heading Reference System (AHRS) data
- Radio Altimeter (RA) data
- Full Authority Digital Engine Control (FADEC) Engine data.

AOA Transducer Monitoring: The SPM1 and SPM2 monitor the transducer parameters that follow:

- Excitation
- Range

- Consistency between the sine and cosine values
- Heater malfunction.

The SPM1 and SPM2 check the difference between the AOA1 and AOA2 transducers when the calibrated airspeed is greater than 60 kt (111 km/h) and the aircraft is not in a stall condition.

The AOA transducer angles are critical parameters for the SPS functions. If an AOA transducer malfunctions, each SPM will use the remaining transducer to make its stick shaker calculation.

If the two AOA transducers malfunction, the SPM will prevent stick shaker operation. The stick shakers will continue to function when the SPMs sense a difference between the two AOA transducer values.

The SPMs will prevent stick pusher operation if any of the conditions that follow are true:

- The AOA transducer monitor stops sensing that an AOA transducer has malfunctioned
- The comparison monitor senses a difference between the two AOA transducers.

AOA Transducer Correction Monitoring: The SPM1 and SPM2 compare the AOA transducer correction values from the Aircraft Configuration Module (ACM). The AOA transducer corrections are non-critical parameters for the SPS because they have a minor effect on the stick shaker and stick pusher operation. When the AOA transducer correction values do not agree, the stick shakers and stick pusher continue to operate in flight.

Malfunctions of non-critical parameters are not shown in flight. The #1 STALL SYST FAIL or #2 STALL SYST FAIL caution light will come on 30 seconds after the aircraft has landed.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

FPIU Flap Position Data Monitoring: The FPIU data is monitored by each SPM. The flap position data is non-critical for the SPS. If the related flap position data to a SPM is not correct, it will use data from the other flap position input.

If the two flap position inputs malfunction, the stick shakers and stick pusher continue to operate in flight.

Propeller De-Icing Selected Data Monitoring: The SPMs compare the PROPELLER DE-ICING SELECTED discrete signals. The propeller anti-icing selected data is considered non-critical for the SPS. If the two discrete signals disagree, the stall SPMs will operate as if a de-icing selection was made.

REF SPEEDS SELECTED Data Monitoring: On aircraft with ModSum 4-126089, the SPMs compare the REF SPEEDS SELECTED discrete signals. The REF SPEEDS SELECTED data is considered non-critical for the SPS. If the two discrete signals disagree, the stall SPMs will operate as if a de-icing selection was made.

The stick shakers and stick pusher continue to operate in flight.

Malfunctions of non-critical parameters are not shown in flight. The #1 STALL SYST FAIL or #2 STALL SYST FAIL caution light will come on 30 seconds after the aircraft has landed.

PSEU WOW Data Monitoring: Each SPM compares the WOW1 and WOW2 discrete signals from the PSEU. If the WOW1 and WOW2 discrete signals disagree, the SPM will use true airspeed and radio altitude to sense the in flight or on ground status of the aircraft.

ADU Data Monitoring: The ADU1 and ADU2 are monitored by each SPM. The ADU data are non-critical for the SPS. If the ADU air data to its related SPM is not correct, it will use data from the other ADU.

For single ADU data failure, the SPM do not inhibit the stick shakers or stick pusher, and do not indicate a failure until 30 seconds after the aircraft has landed.

For dual ADU failure, the SPM cannot perform the AOA estimation monitoring, so the stick pusher is inhibited and the PUSHER SYST FAIL light is illuminated. The stick shaker will remain active in-flight for a dual ADU failure.

Malfunctions of non-critical parameters are not shown in flight. The #1 STALL SYST FAIL or #2 STALL SYST FAIL caution light will come on 30 seconds after the aircraft has landed.

AHRS Data Monitoring: The AHRS1 and AHRS2 data are monitored by each SPM. The AHRS data is non-critical for the SPS. If the AHRS data to the related SPM is not correct, it will use data from the other AHRS.

For single AHRS data failure, the SPM do not inhibit the stick shakers or stick pusher, and do not indicate a failure until 30 seconds after the aircraft has landed.

For dual AHRS failure, in-flight, the PUSHER SYST FAIL light will illuminate and the pusher function is inhibited. The stick shaker will remain active in-flight for a dual AHRS failure.

Malfunctions of non-critical parameters are not shown in flight. The #1 STALL SYST FAIL or #2 STALL SYST FAIL caution light will come on 30 seconds after the aircraft has landed.

RA Data Monitoring: Each SPM compares the AGL altitude and the RA TRIP input discrete signal. If the AGL altitude and the RA TRIP discrete signal disagree, the SPS makes the radio altitude invalid.

FADEC Engine Data Monitoring: The FADEC engine data is monitored by each SPM. The SPMs check the engine torque parameters.

PSM 1-84-2A

EFFECTIVITY:

See first effectivity on page 2 of 27-33-00

Config 001

27-33-00

Config 001

Page 7

Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

The FADEC engine data is non-critical for the SPS. If the related engine data to a SPM has malfunctioned, it will use the other engine data. If the data from two engines has malfunctioned, the SPM will use default values.

The stick shakers and stick pusher continue to operate in flight.

Malfunctions of non-critical parameters are not shown in flight. The #1 STALL SYST FAIL or #2 STALL SYST FAIL caution light will come on 30 seconds after the aircraft has landed.

Stick Shaker Monitoring: The SPMs monitor the stick shakers. Each stall SPM monitors its related stick shaker for the malfunctions that follow:

- Current consumption less than 80 mA and more than 400 mA
- Open circuit
- Short circuit to ground.

Stick Pusher Monitoring: The monitoring of the stick pusher is done by the SPMs and the stick pusher actuator. Each SPM monitors its related stick pusher for the malfunctions that follow:

- Un-commanded stick push
- No stick push when commanded
- SPM stick pusher driver failure.

Un-commanded Stick Push: Each SPM checks that the PUSH signal from the stick pusher actuator is consistent with its command.

No Stick Push When Commanded: Each SPM checks that the PUSH signal from the stick pusher actuator is consistent with its command.

In addition, each SPM checks the FAIL signal from the stick pusher actuator.

SPM Stick Pusher Driver Failure: Each SPM stick pusher driver failure compares its stick pusher hardware command with its stick pusher software command.

The stick pusher actuator monitors the parameters that follow:

- Loss of the 28VDC Main Power input to the stick pusher actuator
- Difference between the PUSH1 and PUSH2 commands from the two SPS channels
- Selection of a stick pusher shut off switch
- Internal failures of the stick pusher actuator.

When a malfunction is sensed, the stick pusher actuator sets its FAIL signal output discrete signal.

Teleloading Mode: The electronic instrument system uses the teleloading mode to connect the operational software serial data to the SPS through the Portable Maintenance Access Terminal (PMAT).

These conditions must be met to access the teleloading mode:

- Aircraft WOW
- A "data load request" discrete.

The SPS does a software configuration check before teleloading begins. This ensures that the software version to be teleloaded is consistent with the equipment part number.

The SPS restarts after a complete and successful teleload. It initiates a complete POST within 10 seconds.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Note: Electrical power interruptions to the SPMs have different results depending on the duration.

A short term power interruption that continues for more than 75 milliseconds and less than 200 milliseconds will cause the SPM to reset. It will not start a POST.

A long term power interruption will cause the SPM to reset. If the aircraft is on the ground, it will start a POST.

Refer to Figure 2.

The STALL WARN switch is set to the TEST1 position to test the functions that follow:

- The RA stick pusher inhibit is functioning
- #1 STALL SYST FAIL caution light comes on
- PUSHER SYST FAIL caution light comes on
- The related stick shaker operates.

NOTE

A stick push does not occur.

The STALL WARN switch is set to the TEST2 position to test the functions that follow:

- The RA stick pusher inhibit is functioning
- #2 STALL SYST FAIL caution light comes on
- PUSHER SYST FAIL caution light comes on
- The related stick shaker operates.

NOTE

A stick push does not occur.

Refer to Figure 3.

The SPS operates when the conditions that follow are correct:

- Calibrated airspeed is less than 215 kt (399 km/h)
- AGL altitude is more than 200 ft (61 m)
- Stick pusher PUSH OFF annunciator switch is not pushed to the OFF position.

When the aircraft is near a stall condition, an electrical motor inside the stick shaker turns an unbalanced mass at approximately 1600 rpm. This causes the control columns to vibrate. In addition to this tactile warning, the stick shaker motor, audible stick shaker striker assembly and the rattling of the chain mechanism inside the control column creates a loud noise.

When one stick shaker is operating, its vibration is transmitted through the torque tubes and the push/pull rod to the other control column.

The stick pusher will move the control columns to decrease the angle of attack of the wing when the aircraft is in a stall condition. The pilots can override the stick pusher clutch by applying an opposite breakout 36 kilogram-force (kgf) to the control column. After breakout, a 30 kilograms-force (kgf) is needed to continue to override the clutch.

The stick pusher PUSH OFF switchlight can also be pushed to turn off the stick pusher.

NOTE

Note: A stick pusher malfunction can cause the SHUT OFF switchlight to turn on.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

If one stick pusher SHUT OFF switchlight or the other is pushed and latched, the PUSH OFF caption will go out on both annunciator switchlights. The stick pusher actuator will be shut off.

When the same switchlight is pushed again, it releases and the PUSH OFF captions in the two switchlights turn on.

Malfunctions of the SPS are shown with caution lights. Malfunctions sensed by the stick pusher actuator are shown on the stick pusher shut off annunciator switches located on the glareshield panel.

[Refer to Figure 4.](#)

The SPS malfunctions are shown with the caution lights that follow:

- #1 STALL SYST FAIL
- #2 STALL SYST FAIL
- PUSHER SYST FAIL.

The #1 STALL SYST FAIL and #2 STALL SYST FAIL caution lights come on immediately in flight when malfunctions that prevent stick shaker and stick pusher operation are sensed.

Malfunctions of non-critical parameters are not shown in flight. The #1 STALL SYST FAIL or #2 STALL SYST FAIL caution light will come on 30 seconds after the aircraft has landed.

A stick shaker motor malfunction is not shown in flight. The #1 STALL SYST FAIL or #2 STALL SYST FAIL caution light will come on 30 seconds after the aircraft has landed.

Any of the conditions that follow will cause the PUSHER SYST FAIL caution light to come on:

- Any failure that causes the #1 STALL SYST FAIL or #2 STALL SYST FAIL caution lights to turn on

- A failure of the stick pusher actuator.

The PUSHER SYST FAIL caution light comes on when a stick pusher SHUT OFF switchlight selection is made. The PUSHER SYST FAIL light will go out when the two stick pusher SHUT OFF switchlights are not set and no malfunctions exist.

The Left Main and Right Main busses supply electrical power through 5 Ampere circuit breakers and the Prime Power Supply Modules (PPSM 1, PPSM 2) to the related SPM and stick shaker motors.

The left main and right main busses supply electrical power through 7 ½ ampere circuit breakers to the stick pusher and the related stick pusher SHUT OFF switchlight.

The SPM interface with the components that follow:

- AOA1, AOA2 Transducers
- Other SPMs
- ACM
- FPIU
- Propeller De-Icing
- Input/Output Processors (IOP1, IOP2)
- PSEU
- ADU1, ADU2
- Attitude and Heading Reference Units (AHRU1, AHRU2).

AOA1, AOA2 Transducers: Each AOA transducer has 2 synchro-resolvers. One supplies AOA transducer angle data to its related SPM. The other synchro-resolver supplies AOA transducer data to the other SPM.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

The SPMs supply a 7.5 Vac, 2500 Hz excitation voltage for the synchro-resolvers.

The outputs of the AOA Transducers are AC voltages that relate to the sine and cosine of the transducer angles. The SPMs use the voltages to calculate the AOA.

Each SPM receives the heating status of the AOA1 and AOA2 transducers.

Other SPMs: Each SPM transmits its validity status data on an ARINC 429 data bus to the other SPM.

ACM1, ACM2: Each SPM receives installation correction values for each AOA Transducer from the ACM1 and ACM2. The installation correction values adjust for errors in the mounting of the AOA Transducers.

To calculate the body AOA, SPM1 and SPM2 correct the angles that follow:

- The 10 degree AOA transducer installation offset
- Any specific installation error represented by the AOA transducer corrections stored in the ACM1 and ACM2.

FPIU: Each SPM receives two analog DC signals from the left and right channels of the FPIU. The signals relate to the left and right flap positions.

In addition, each SPM receives a FLAP HEALTHY discrete signal from the left and right channels of the FPIU when the system is serviceable.

Propeller De-Icing: The Propeller De-Icing Switch supplies a discrete signal to the two SPMs. The SPS will use one de-icing selected status discrete or the other to calculate the operating angle.

REF SPEEDS: On aircraft with ModSum 4–126089, the selection of the INCR position of the REF SPEEDS switch on the ICE PROTECTION panel supplies a REF SPEEDS SELECTED discrete signal to the two SPMs. The SPS will use one REF SPEEDS SELECTED status discrete or the other to calculate the operating angle.

IOP1, IOP2: Each SPM receives digital data from the RA and FADEC through IOP1 and IOP2.

Each SPM sends data to the IOP1 and IOP2 to cause the Electronic Instruments System (EIS) to show an INCR REF SPEEDS message when a propeller de-icing selection is made.

Each SPM receives engine data through the IOP1 and IOP2 from the four FADEC channels. The SPMs use engine torque in the stick shaker operating angle calculations. It uses power lever angle and condition lever angle in the stick pusher operating angle calculations.

PSEU: Each SPM receives WOW1 and WOW 2 discrete signals from the PSEU. The two nose gear WOW proximity switches are used to set in-flight or on-ground status.

ADU1, ADU2: Each SPM receives data from the ADU1 and ADU2. The SPMs use the Mach number and the true airspeed to calculate the operating angle. The SPS uses the Calibrated Airspeed (CAS) to prevent stick pusher operation at airspeeds greater than 215 kt (399 km/h). It prevents AOA transducer comparison monitoring at airspeeds below 60 kt (111 km/h).

AHRS1, AHRS2: Each Stall SPM receives data from the two AHRS. It uses the parameters that follow:

- Roll angle
- Pitch angle

PSM 1–84–2A

EFFECTIVITY:

See first effectivity on page 2 of 27–33–00

Config 001

27–33–00

Config 001

Page 11

Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

- Pitch rate
- Normal acceleration.

RA1, RA2: The SPM1 receives an RA TRIP discrete signal from RA1 when the AGL altitude is less than 200 ft (61 m). The SPM2 receives an RA TRIP discrete signal from RA1 in a single RA aircraft configuration when the altitude is less than 200 ft (61 m) AGL. It receives an RA TRIP discrete signal from RA2 in a dual RA aircraft configuration when the altitude is less than 200 ft (61 m) AGL. The SPS uses these RA TRIP discrete signals to prevent stick pusher operation below 200 ft (61 m) AGL.

For the pre-flight test, each SPS gives an RA TEST ENABLE discrete signal and an analog RA TEST SIGNAL voltage to test the RA inhibit function. A separate test enable discrete signal is supplied from the pre-flight test switch. The RA must receive all the test input to do the test.

Each SPM receives AGL radio altitude through the IOP1 and IOP2. It uses the AGL radio altitude and true airspeed to calculate in flight status of the aircraft when the WOW inputs from the PSEU disagree.

AFCS: Each SPM sends a discrete signal to the AFCS to disengage the autopilot when the aircraft angle of attack has reached the stick shake condition.

Stick Shakers: Each SPM uses two wires to connect it to its related stick shaker. One wire is used to supply 28 Vdc and the other is used for return. This configuration allows the SPMs to command and monitor the stick shaker motors.

Stick Pusher: Each SPM supplies a PUSH command to the stick pusher actuator. SPM1 gives the PUSH1 command to the stick pusher actuator motor and SPM2 gives the PUSH2 command to the stick pusher actuator clutch.

The stick pusher actuator sends a PUSH discrete signal to each SPM to give indication of a stick push.

The stick pusher actuator supplies a FAIL signal to the SPMs to give indication of the conditions that follow:

- Stick pusher actuator internal monitoring has malfunctioned
- Stick pusher SHUT OFF annunciator switch is set.

Caution and Warning Panel: The caution and warning annunciator lights panel will turn on the #1 STALL SYST FAIL light when the SPM1 malfunctions. The #2 STALL SYST FAIL light turns on when the SPM2 malfunctions. The PUSHER SYST FAIL light comes when one SPM or the other malfunctions.

Flight Data Recorder (FDR): The SPS sends data through IOP1 to the FDR. Each SPM sends the data that follows:

- Corrected AOA1 and AOA2
- Stick shaker healthy status
- Stick pusher healthy status.

GPWS: Each SPM sends a discrete signal to the GPWS to prevent the PULL UP aural warning when the aircraft is in a stall condition.

CDS: Each SPM sends malfunction data through the IOP1 to the CDS. When the aircraft is on the ground, the CDS is used to reset or test the SPS.

Pusher SHUT OFF annunciator switchlights: The pilot stick pusher SHUT OFF switchlights supply a PILOT RELEASE signal to the stick pusher actuator to prevent stick pusher operation. The copilot stick pusher SHUT OFF annunciator switch supplies a CO-PILOT



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

RELEASE signal to the stick pusher actuator to prevent stick pusher operation.

The stick pusher is connected to the elevator control system with a mechanical cable. When commanded, the stick pusher turns until it touches a mechanical stop, corresponding to a fixed column position.

Note: The stick shaker, stick pusher actuator and capstan are passively cooled. The stick pusher actuator can operate against a stalling load for only a short time before overheating. Accordingly, the SPS contains software protection to limit the stick pusher operation to a maximum of 15 seconds, with a subsequent time delay of 1 minute before the stick pusher can be commanded again.

Modules, Stall Protection

Refer to Figure 5.

The SPM1, SPM2 has the two electronic circuit boards that follow:

- Extension board
- Central Processing Unit (CPU) board.

One part of the extension board is for the SPS and the other part is used by GPWS for input/output processing.

The SPM uses discrete signals and ARINC 429 interfaces between the Central Processing Unit (CPU) and external boards.

The SPMs make a reference voltage for AOA transducer resolvers that is 7 Vac, 2500 Hz.

Each SPM is guided into the IFC 1, IFC 2 using a guide rail and guide pins. The top and bottom guide pins at the IFC backpanel make sure that the module and the backpanel come together precisely in the correct location.

The SPMs are held in place using front locking levers. This gives easy installation and removal of the Line Replaceable Unit (LRU) without tools.

Actuator, Stick Pusher

Refer to Figure 6.

The stick pusher actuator turns an output pinion to move a stick pusher capstan that moves the control columns forward when the aircraft is in a stall condition.

The stick pusher actuator uses two 28 Vdc electrical power sources. One power source supplies motor current and the other supplies electrical power to the monitor circuits.

The stick pusher actuator has the parts that follow:

- Mounting plate
- Electromechanical disconnect clutch
- Permanent magnet DC drive motor
- Gear train
- Power electronics assembly
- Printed circuit board assembly.

The stick pusher actuator needs two independent PUSH1 and PUSH2 input signals from the SPM1, SPM2 to operate. The PUSH1 signal operates the motor and the PUSH2 signal engages its electromechanical clutch.

A pilot STICK PUSHER SHUT OFF switchlight selection prevents stick pusher motor operation and disengages its clutch. A copilot



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

STICK PUSHER SHUT OFF switchlight selection prevents motor operation.

A pilot or copilot STICK PUSHER SHUT OFF switch selection prevents stick pusher operation regardless of the mode of the PUSH1 and PUSH2 input signals.

The motor current is limited to control the stick pusher actuator output torque. The output torque is nominally set to eight Newton metres (Nm), which is equivalent to approximately 32 kilograms (kg) of column force.

The stick pusher actuator continuously monitors its electromechanical and electronic components when electrical power is supplied. The stick pusher actuator outputs a PUSH signal when the actuator is supplying a stick push. The stick pusher actuator outputs a FAIL signal when the internal monitoring circuit senses a malfunction of its electromechanical or electronic components.

The stick pusher actuator weighs 2.8 lb (1.27 kg) and is 5.25 in. (134 mm) high, 3.78 in. (96 mm) wide and 3.60 in. (91 mm) long. It is attached to the stick pusher capstan with four mounting bolts.

Capstan, Stick Pusher

[Refer to Figure 7.](#)

The stick pusher assembly has a stick pusher actuator connected to a stick pusher capstan. The stick pusher actuator gives the actual stick push when commanded by the SPM1 and SPM2. The stick pusher capstan gives the mechanical interface with the aircraft elevator control system.

The stick pusher capstan is a mechanical assembly that takes an input from a gear pinion and gives an output to the elevator control

system cables. The stick pusher capstan has an override slip clutch that lets the cable pulley turn if the stick pusher actuator cannot be moved because of a mechanical malfunction.

The stick pusher capstan consists of the following:

- Housing
- Shaft
- Input gear
- Override clutch
- Pulley
- End cap.

The override clutch is a dual friction surface slip clutch that is adjustable with a nut.

The stick pusher capstan is attached to the aircraft and connected to the elevator control system with a cable. The stick pusher actuator is attached to the stick pusher capstan. The stick pusher actuator may be removed from the aircraft without removing the stick pusher capstan.

The stick pusher capstan weighs 1.8 lb (0.82 kg). It is attached to the aircraft with four bolts that screw into floating nut–plates on the stick pusher capstan. Two aircraft cables are attached to the capstan pulley using swaged ball fittings that anchor to pockets in the pulley.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Quadrant, Stick Pusher

Refer to Figure 8.

The stick pusher quadrant receives input from the stick pusher capstan to operate the elevator control cables so that the control column is pushed forward when a stall condition is sensed.

It has springs that supply tension regulation of the control cable system.

The stick pusher quadrant is located below the left flight compartment floor.

Shaker, Stick

Refer to Figure 9.

The stick shaker has a housing that contains an electrical motor that turns an unbalanced weight.

The stick shaker is approximately 5.12 in. (132 mm) long and 2.44 in. (62 mm) in diameter. It is attached to the control column using a bracket.

Transducer, Angle of Attack (AOA)

Refer to Figure 10.

The AOA1, AOA2 Transducers are used to measure the local angle of attack.

Each AOA transducer has a fixed body that is attached to the aircraft and a transducer that is moved by the relative airflow.

Each AOA Transducer has the components that follow:

- Transducer
- Shaft, bearings, and gears
- Counterweight
- Damper
- Two resolvers.

The transducer is connected to a shaft that turns freely.

A counterweight and a damper are attached to the shaft to balance the transducer.

The damper is used to reduce interference and wear on the movable parts. The damper is electromagnetic device that has a rotor with magnets and a stator. The damper has mobile stops that make contact with fixed stops in the body of the AOA transducer. It has a turning limit of 43 degrees in a clockwise or counterclockwise direction.

The AOA transducer has two resolvers to send angular data to the SPMs. Each resolver uses 7 Vac, 2500 Hz excitation power.

The AOA transducer has a self-regulated heater to prevent icing. The body is not heated. If the AOA transducer is not being heated, the SPMs will not use its input.

Each AOA transducer weighs 2.0 lb (0.91 kg). It is attached to the aircraft with five mounting screws and two locating pins. When installed, the vane protrudes 3.78 in. (96 mm) from the aircraft.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Switchlight, Stick Pusher Release

Refer to Figure 11.

The stick pusher shut off PUSH OFF switchlights come on when the stick pusher actuator malfunctions.

If one stick pusher shut off PUSH OFF switchlight or the other is pushed, the stick pusher actuator will not operate and the PUSH OFF caption will go off.

The two stick pusher shut off PUSH OFF switchlights are attached to the glareshield panel assembly. Each switchlight is located in the field of view of each pilot.

Switch, Pre-Flight Test

The STALL WARN pre-flight test switch is used to test the SPS.

The pre-flight test switch is labelled as STALL WARN. It is a positive break, double pole, toggle switch with a lever that springs back to the center OFF position. The toggle lever is connected to a mechanism located in the body of the switch. The internal mechanism moves in the opposite direction as the toggle lever.

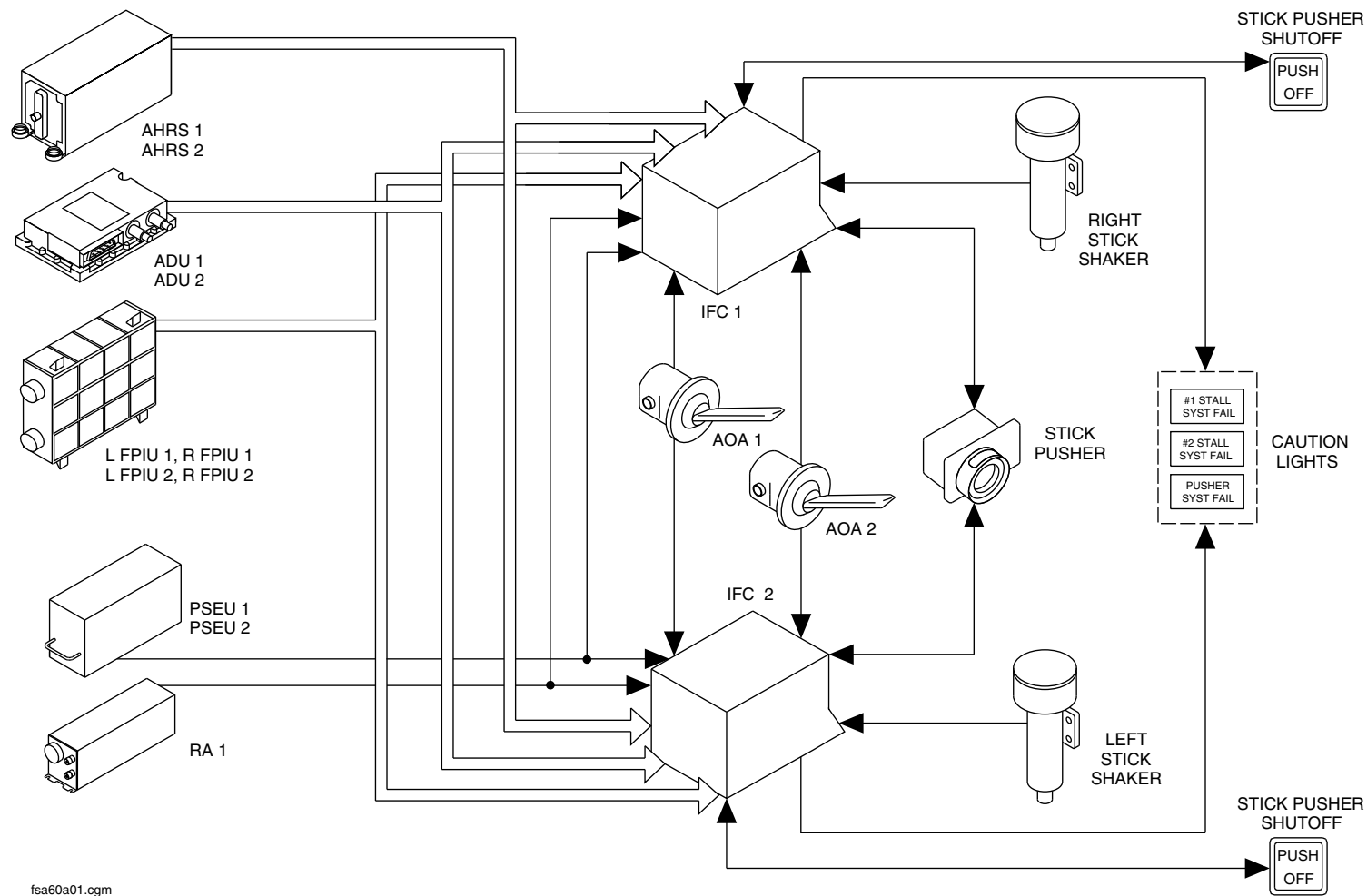
The pre-flight test switch has solder lug terminals for wiring connections.

The pre-flight test switch is attached to the pilot's side console panel with a mounting nut and lock washer.



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



SPS BLOCK DIAGRAM
Figure 1

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-33-00
Config 001

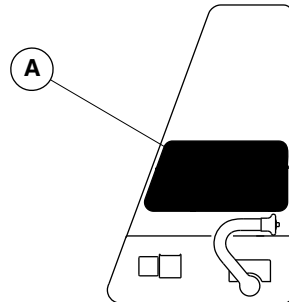
27-33-00

Config 001
Page 17
Sep 05/2021



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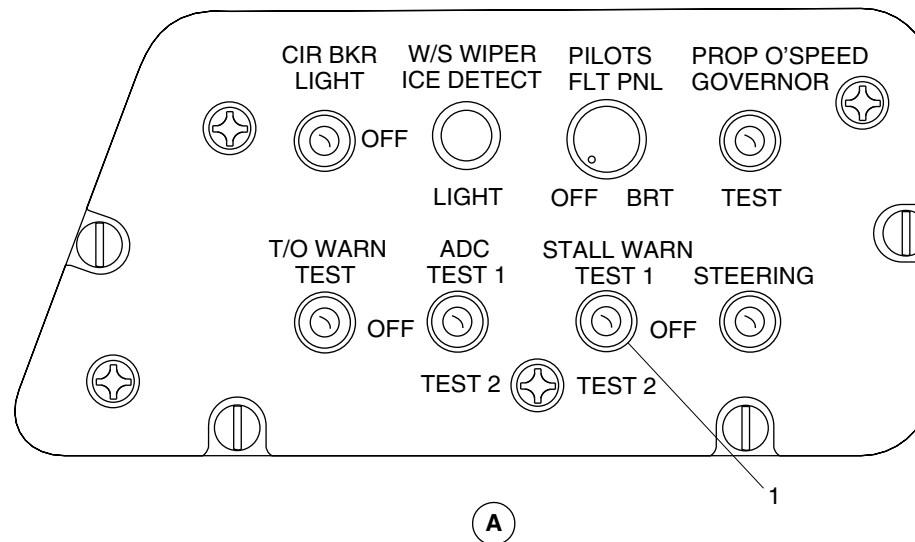
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEFT SIDE CONSOLE

LEGEND

1. Stall Warn Test Toggle Switch.



fs985a01.cgm

SPS PRE-FLIGHT TEST SWITCH
Figure 2

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-33-00
Config 001

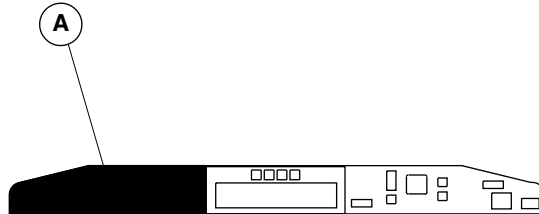
27-33-00

Config 001
Page 18
Sep 05/2021



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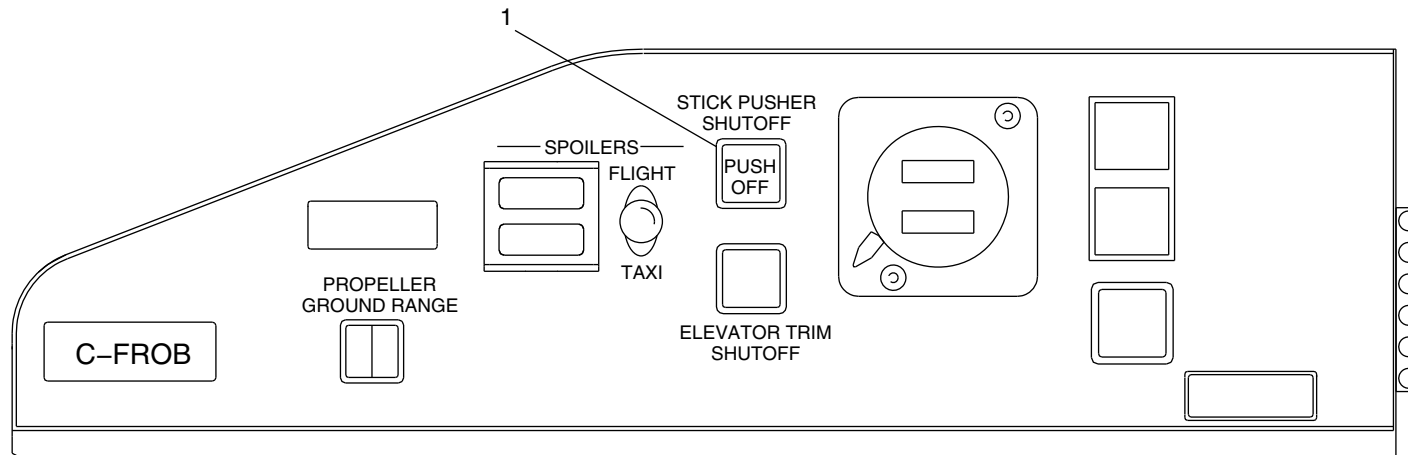
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



GLARESHIELD PANEL

LEGEND

1. Stick Pusher Shutoff Annunciator Switch.



A

NOTE

Pilot side shown.
Copilot side similar.

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SPS STICK PUSHER RELEASE SWITCHLIGHT
Figure 3

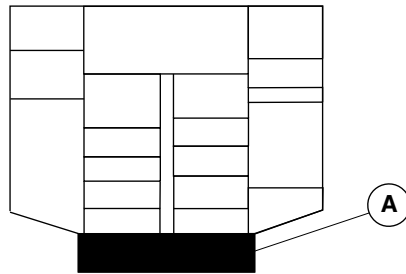
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Config 001

27-33-00

Config 001
Page 19
Sep 05/2021



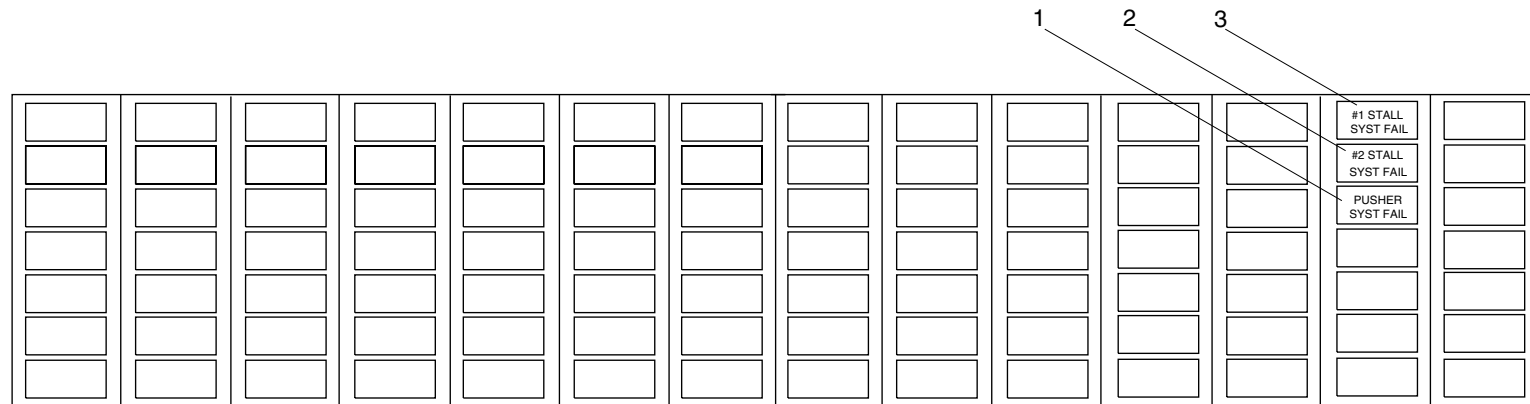
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OVERHEAD CONSOLE

LEGEND

1. Pusher System Fail (Amber).
2. # 2 Stall System Fail (Amber).
3. # 1 Stall System Fail (Amber).



A

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SPM CAUTION LIGHT INDICATIONS
Figure 4

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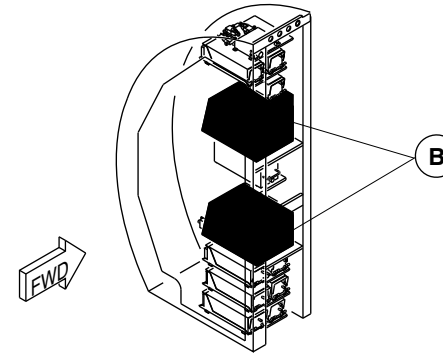
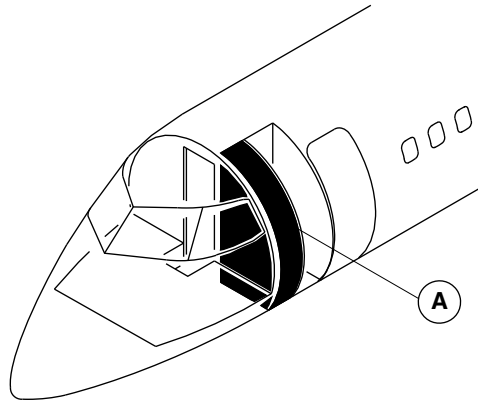
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Config 001
Page 20
Sep 05/2021

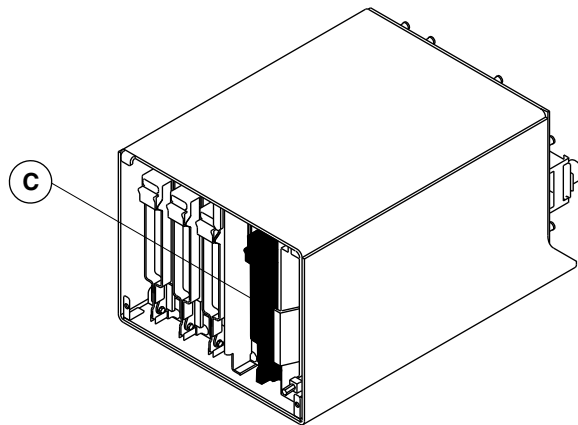


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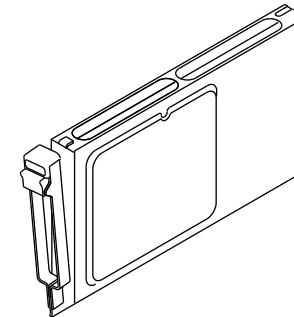


A AVIONICS RACK ELECTRICAL
INSTALLATION



B INTEGRATED FLIGHT CABINET

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C

STALL PROTECTION MODULES LOCATOR Figure 5

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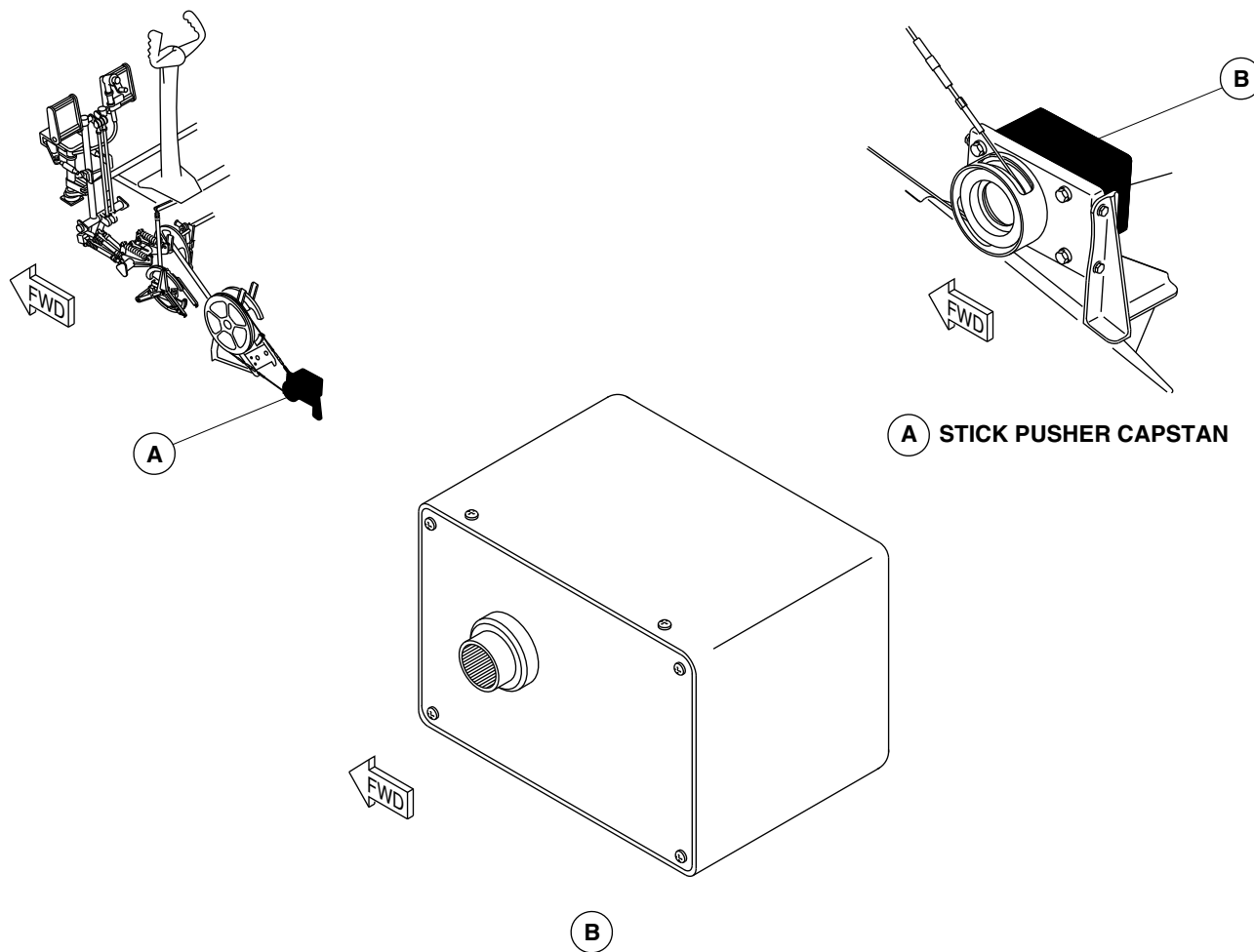
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Config 001
Page 21
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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STICK PUSHER ACTUATOR LOCATOR
Figure 6

PSM 1-84-2A
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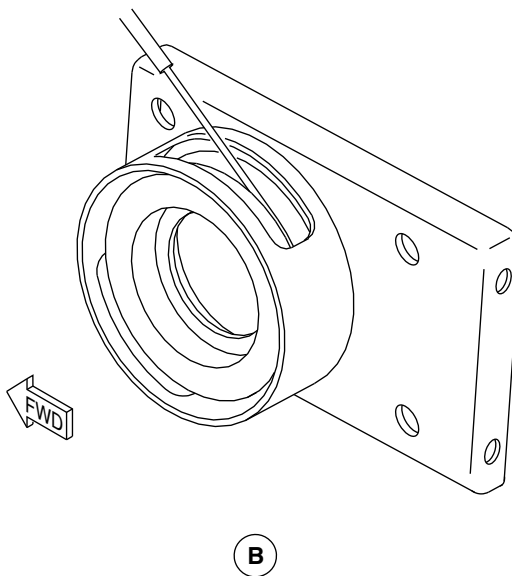
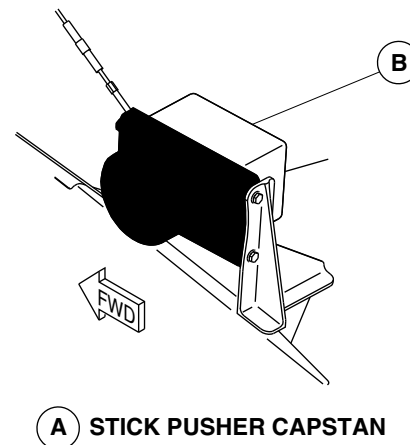
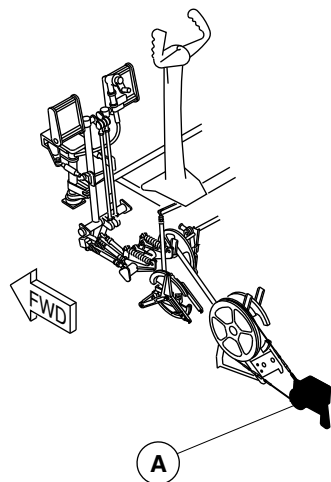
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Config 001
Page 22
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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STICK PUSHER CAPSTAN LOCATOR
Figure 7

PSM 1-84-2A
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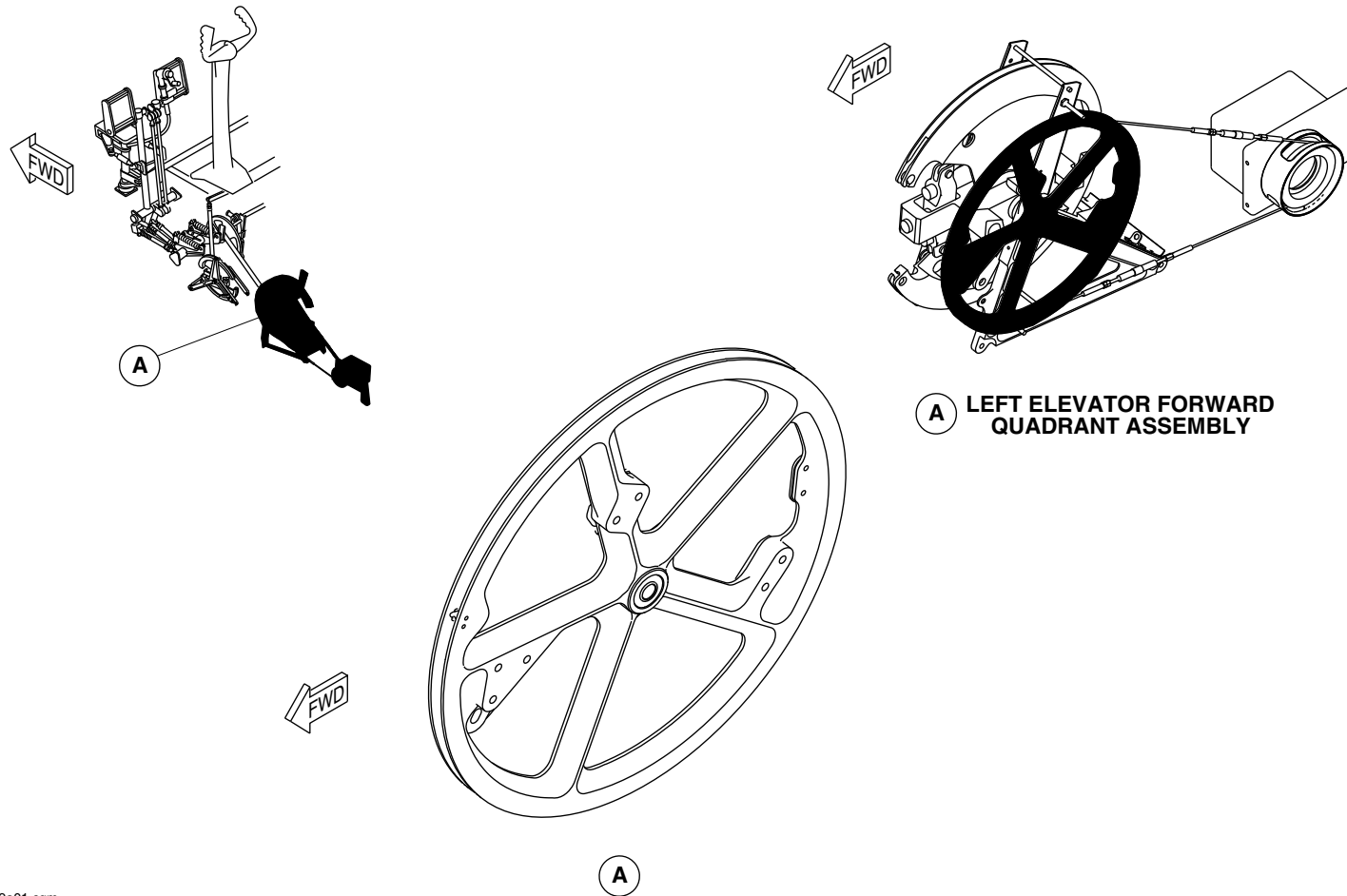
27-33-00

Config 001
Page 23
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



STICK PUSHER QUADRANT LOCATOR
Figure 8

PSM 1-84-2A
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Config 001

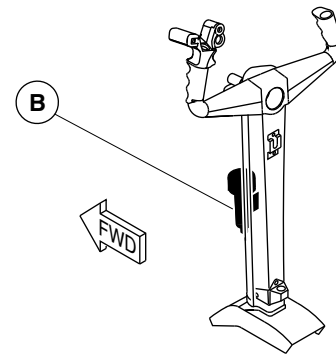
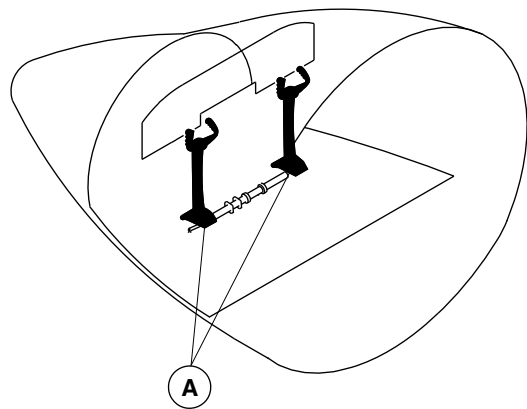
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Config 001
Page 24
Sep 05/2021



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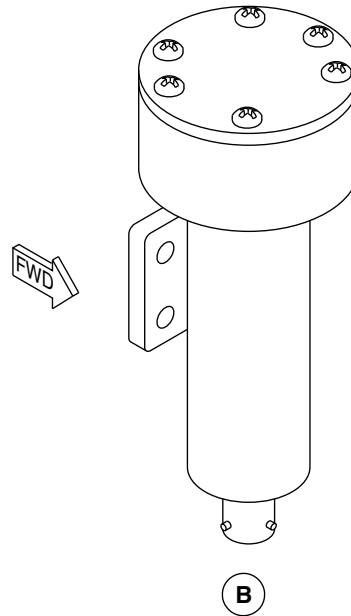
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NOTE

Pilot side shown.
Copilot side similar.

A CONTROL COLUMN ASSEMBLY



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STICK SHAKER LOCATOR
Figure 9

PSM 1-84-2A
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Config 001

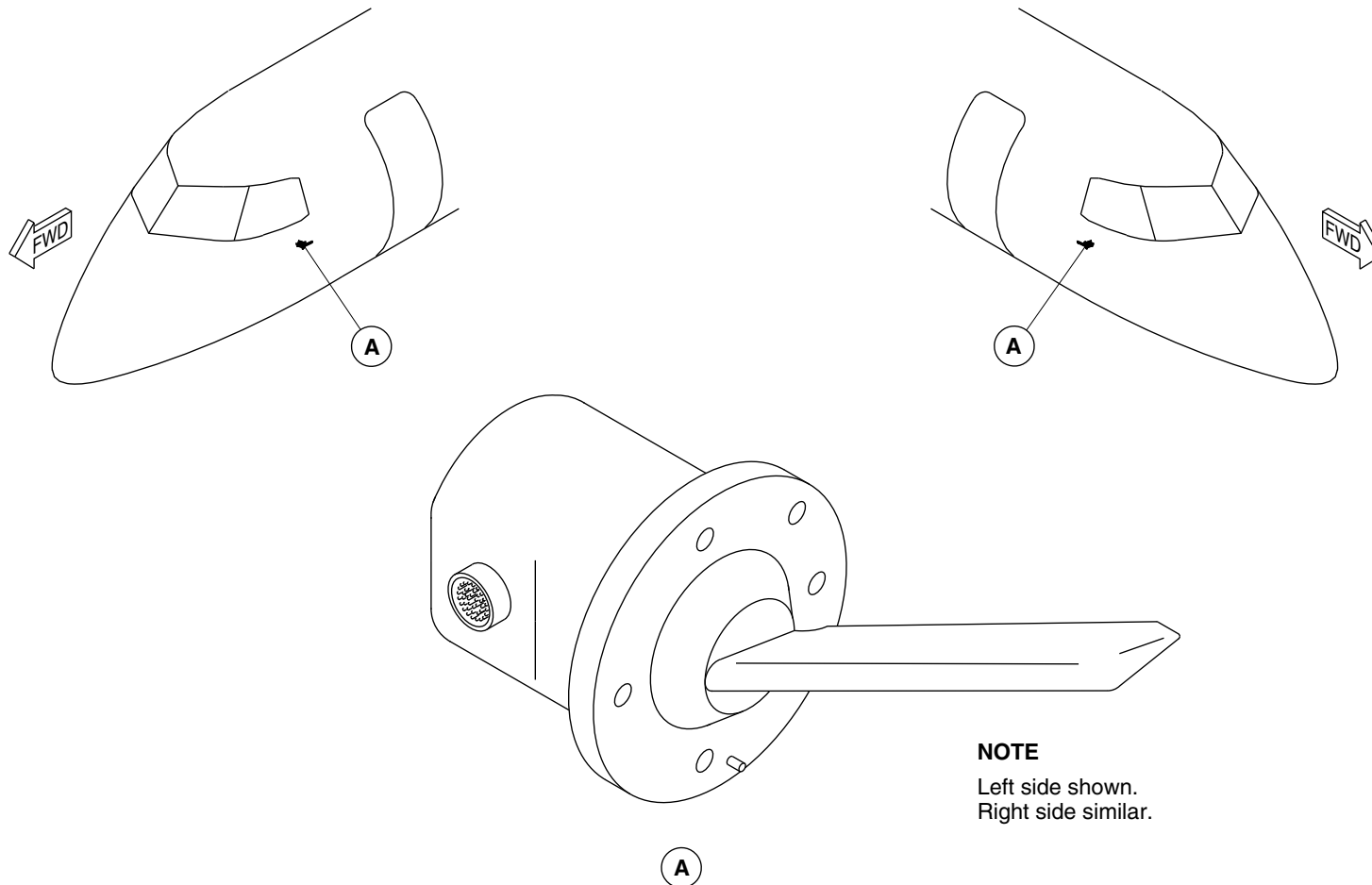
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Config 001
Page 25
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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ANGLE OF ATTACK TRANSDUCER LOCATOR
Figure 10

PSM 1-84-2A
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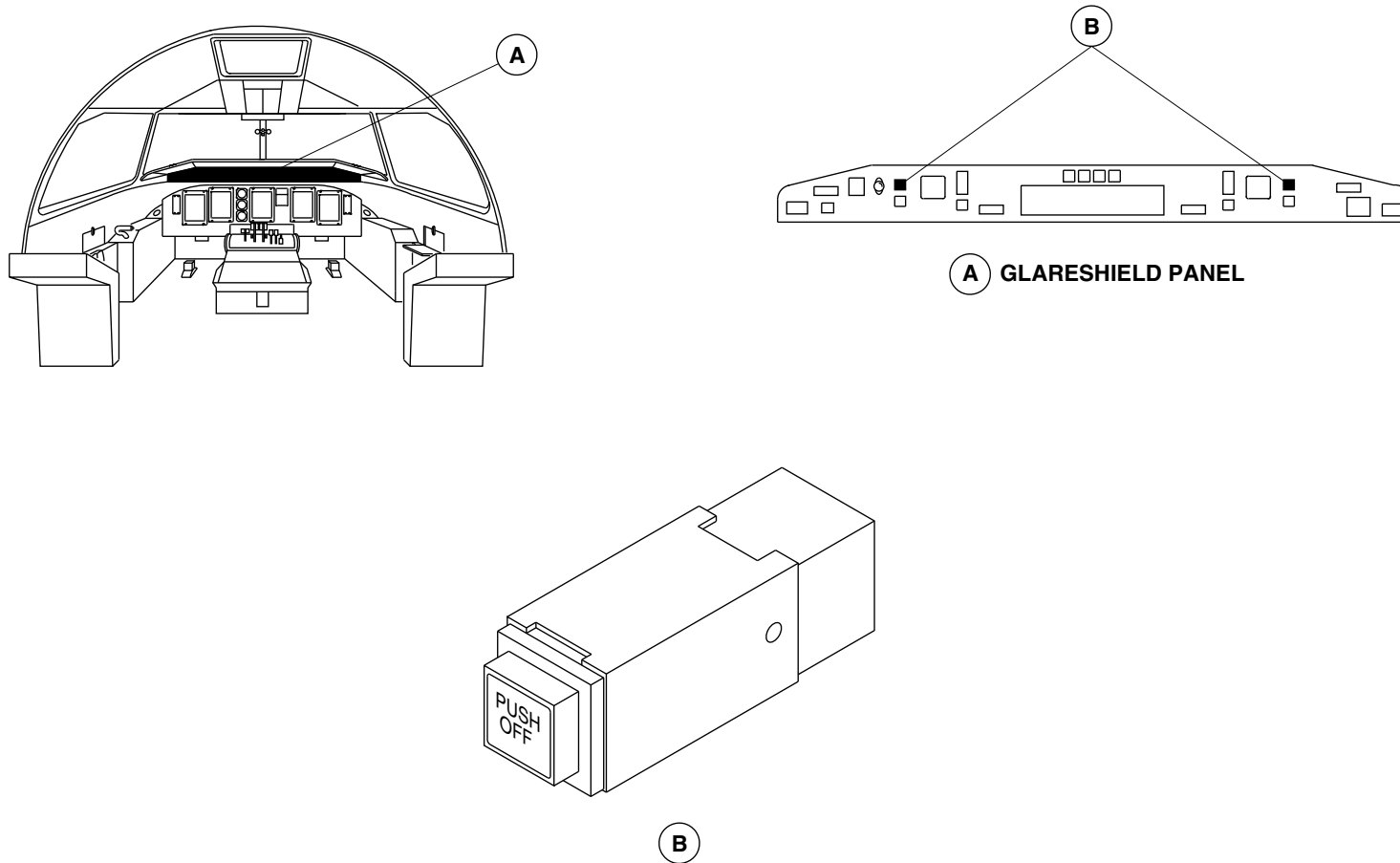
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Config 001
Page 26
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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SPS STICK PUSHER RELEASE SWITCHLIGHT LOCATOR
Figure 11

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-33-00
Config 001

27-33-00

Config 001
Page 27
Sep 05/2021



DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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PSM 1-84-2A
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See first effectivity on page 2 of 27-33-00
Config 001

27-33-00

Config 001
Page 28
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

27–34–00–001

ELEVATOR SURFACES

Introduction

The elevators are primary flight control surfaces which are attached to the horizontal stabilizer. The elevator surfaces give longitudinal control of the aircraft in flight, around the lateral axis.

General Description

Refer to Figures 1 and 2.

There are two elevator surfaces, left and right, which are installed on the trailing edge of the left and right side of the horizontal stabilizer. Three elevator Power Control Units (PCUs) are attached to each elevator surface. A Linear Variable Differential Transformer (LVDT) is attached to the front spar of each elevator surface to supply position information to the Powered Flight Controls Surface Indicator (PFCSI).

Detailed Description

Refer to Figures 3 , 4 and 5.

The left and right independent elevator surfaces are attached to the horizontal stabilizer, by four hinges using low torque plain roller bearings. There is an inboard and an outboard hinge fitting and two more hinge fittings near the centre of the control surface front spar. The three PCU attachment fittings are located near the centre hinge

points. The LVDT attachment fitting is located near the inboard PCU attachment point (refer to SDS 55–20–00).

Each elevator has a span of 174.0 in. (4.42 m) with a constant chord of 20.5 in. (520 mm). The elevator surface travel is 30 degrees up and 20 degrees down. Flutter suppression is given by a retained stiffness in the PCUs.

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Config 001

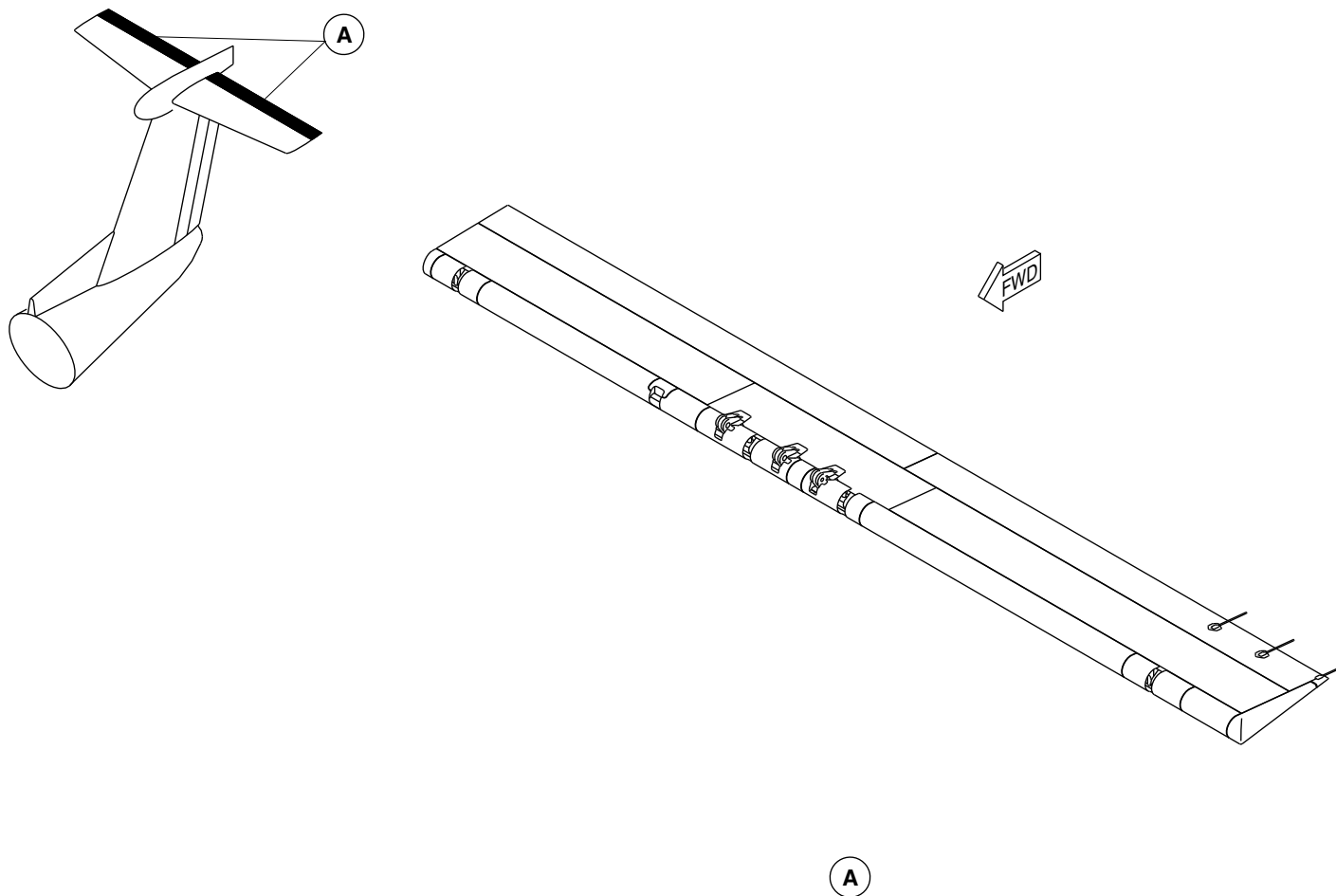
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Config 001
Page 2
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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ELEVATOR SURFACE PAGE 1
Figure 1

PSM 1-84-2A
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See first effectivity on page 2 of 27-34-00
Config 001

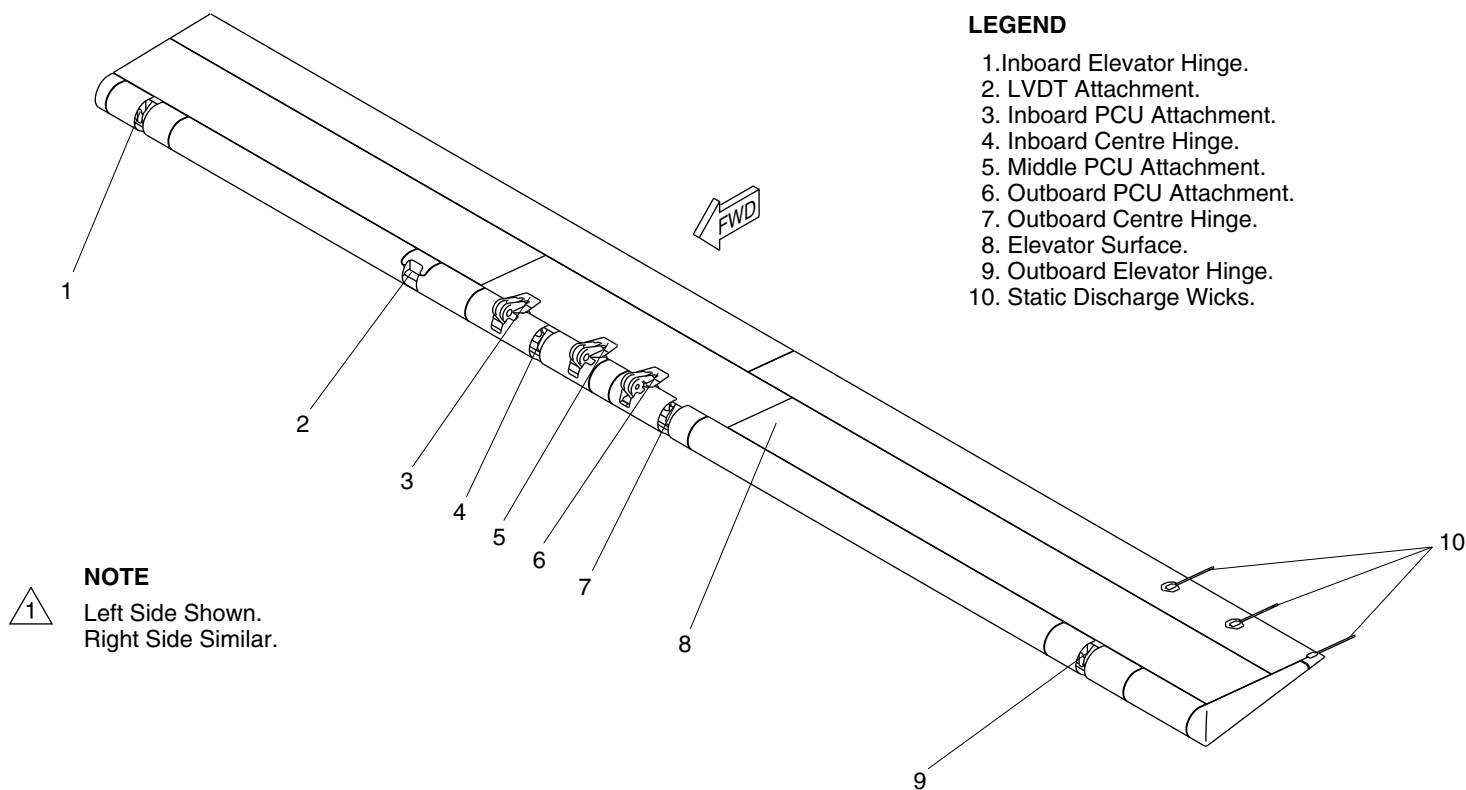
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Config 001
Page 3
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Inboard Elevator Hinge.
2. LVDT Attachment.
3. Inboard PCU Attachment.
4. Inboard Centre Hinge.
5. Middle PCU Attachment.
6. Outboard PCU Attachment.
7. Outboard Centre Hinge.
8. Elevator Surface.
9. Outboard Elevator Hinge.
10. Static Discharge Wicks.

NOTE
1 Left Side Shown.
Right Side Similar.

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ELEVATOR SURFACE PAGE 2
Figure 2

PSM 1-84-2A
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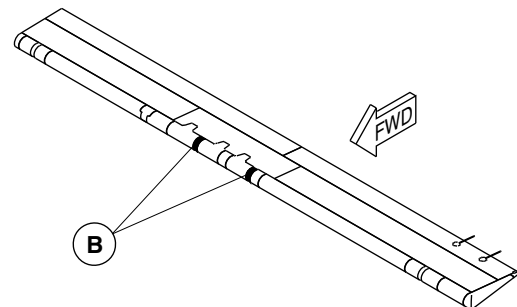
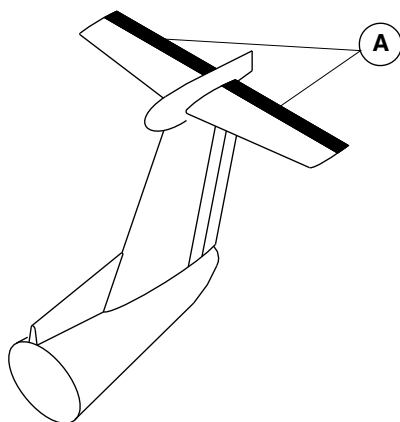
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Config 001
Page 4
Sep 05/2021



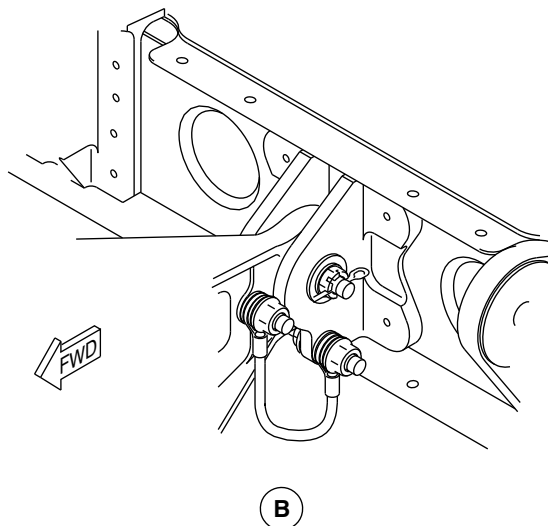
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A ELEVATOR SURFACE

- NOTE**
- 1 Inboard Component Shown.
Outboard Component Similar.
 - 2 Left Elevator Shown.
Right Elevator Similar.



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ELEVATOR SURFACE CENTRE HINGES
Figure 3

PSM 1-84-2A
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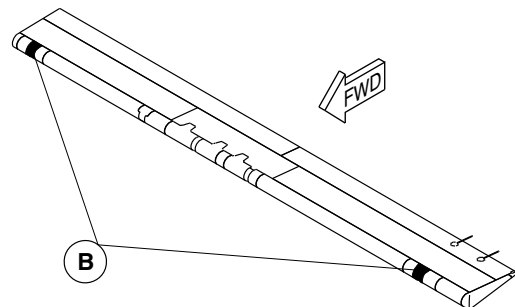
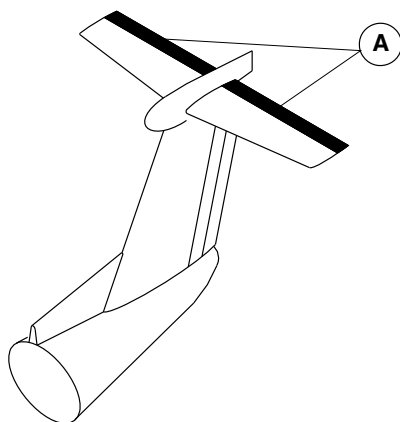
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Config 001
Page 5
Sep 05/2021

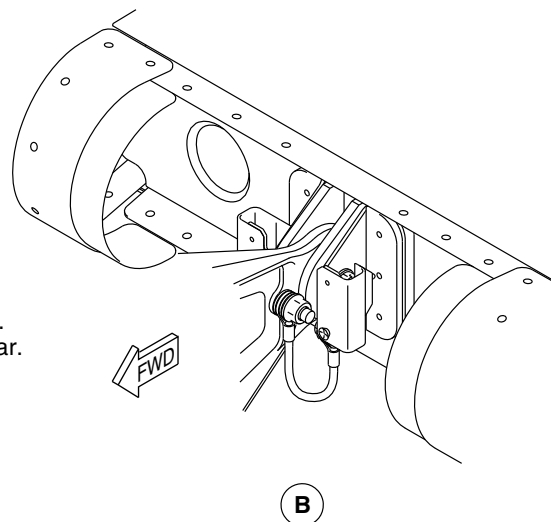


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A ELEVATOR SURFACE



NOTE

- 1 Inboard Component Shown.
Outboard Component Similar.
- 2 Left Elevator Shown.
Right Elevator Similar.

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ELEVATOR SURFACE INBOARD AND OUTBOARD HINGES
Figure 4

PSM 1-84-2A
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See first effectivity on page 2 of 27-34-00
Config 001

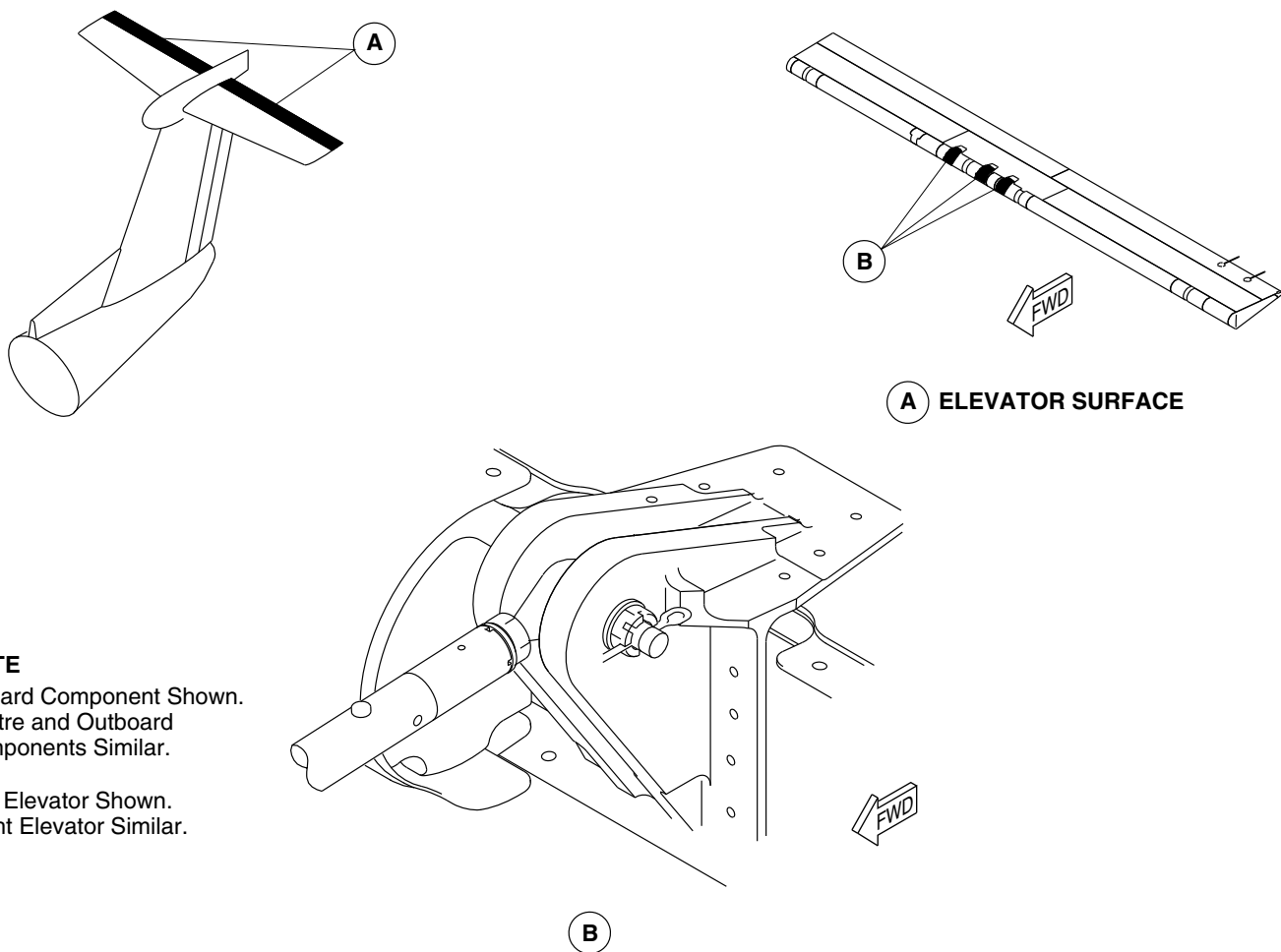
27-34-00

Config 001
Page 6
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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ELEVATOR SURFACE PCU ATTACHMENT
Figure 5

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-34-00
Config 001

27-34-00

Config 001
Page 7
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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PSM 1-84-2A
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See first effectivity on page 2 of 27-34-00
Config 001

27-34-00

Config 001
Page 8
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

27-38-00-001

PITCH DISCONNECT SYSTEM

Introduction

The pitch disconnect system lets the left and right elevator control circuits be disconnected from each other, in the event a jam occurs in either control circuit. The remaining operational circuit controls the pitch of the aircraft around the lateral axis.

General Description

The pilot's control column operates the left elevator circuit and the copilot's control column operates the right elevator circuit. The two control columns are connected to each other so that either column can operate both elevator circuits at the same time. This lets the aircraft be flown from either the pilot's or copilot's position. If a jam occurs in either control circuit, the PITCH DISC handle in the flight compartment is pulled to disconnect the left elevator circuit from the right elevator circuit. The pilot with the free control column will have pitch control.

The pitch disconnect system has the components that follow:

- Handle, Pitch Disconnect (27-38-01)
- Mechanism, Clutch-Pitch Disconnect (27-38-06).

Detailed Description

Refer to Figures 1 and 2.

The left and right elevator control circuits are connected to each other by a torque tube installed below the flight compartment floor. This lets the elevators be operated from either the pilot's or copilot's control columns. The torque tube connects the two control columns through a clutch mechanism. The pitch disconnect system has a vertically mounted PITCH DISC handle located on the left side of the center console. The handle is connected to the clutch mechanism by a cable routed below the flight compartment floor. When the PITCH DISC handle is pulled and turned 90 degrees clockwise or counterclockwise, the clutch mechanism is disengaged and the two control columns can operate independently of each other.

Handle, Pitch Disconnect

Refer to Figure 3.

The PITCH DISC handle is located in the flight compartment, on the left side of the center console. The handle is used to separate the left and right elevator controls in case of a jammed system. When the handle is in the engaged "IN" position, the clutch and clutch plate are kept in positive contact with each other by the clutch spring. When the handle is in this position, the pilot's and copilot's control columns are connected to each other. The PITCH DISC handle has a cable routed below the flight compartment floor and is connected to the clutch. When the handle is pulled out to the limit and turned 90 degrees clockwise or counterclockwise, the clutch is disengaged which disconnects the two control columns from each other. The pilot with the free control column will have pitch control.

PSM 1-84-2A

EFFECTIVITY:

See first effectivity on page 2 of 27-38-00

Config 001

27-38-00

Config 001

Page 2

Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Adjustment of the pitch disconnect system is by a turnbuckle installed in the cable.

Mechanism, Clutch–Pitch Disconnect

Refer to Figures 4 , 5 , 6 and 7.

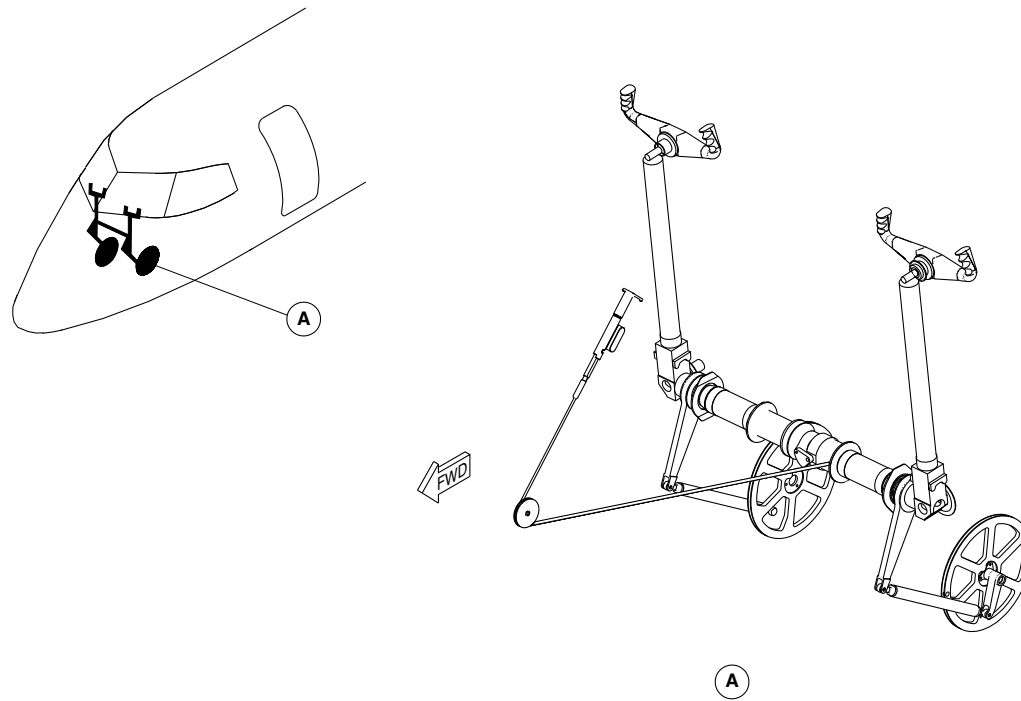
The clutch mechanism is located in the torque tube connecting the two control columns below the flight compartment floor. It includes a clutch lever, cams, a cam follower, clutch springs, clutch and clutch plate assembly. When the PITCH DISC handle is pulled to the limit and turned 90 degrees clockwise or counterclockwise, the cable operates the clutch lever and cam assembly. The cam followers withdraw the spring loaded clutch from the clutch plate. This disconnects the two columns allowing the pilot with the free column to have pitch control. A pin in the shaft of the pitch disconnect handle rides in a slot in the shaft housing. When the handle is pulled out and turned 90 degrees, the pin moves into the slot to keep the handle locked in the disengaged position.

To re-engage the pitch disconnect mechanism move the pilot's and copilot's control columns to the neutral position. Turn the PITCH DISC handle 90 degrees and lower to the engaged position. When the control columns are properly aligned, the handle and the clutch are returned to the engaged position by the clutch springs.



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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PITCH DISCONNECT CLUTCH SYSTEM PAGE 1
Figure 1

PSM 1-84-2A
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See first effectivity on page 2 of 27-38-00
Config 001

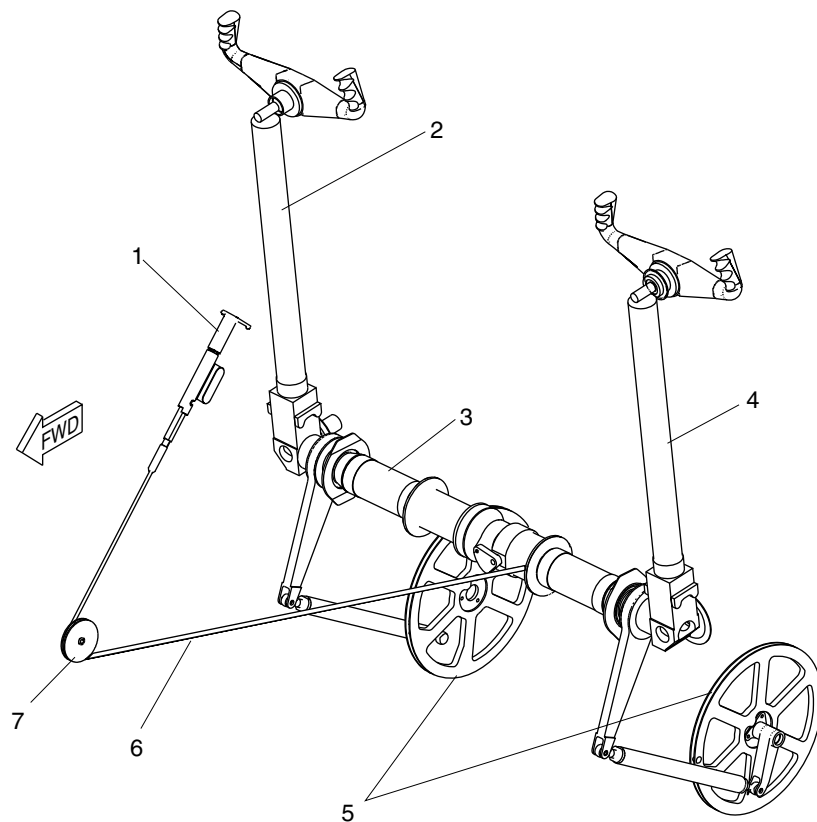
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Config 001
Page 4
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Pitch Disconnect Handle.
2. Copilot Control Column.
3. Torque Tube.
4. Pilot Control Column.
5. RH and LH Forward Elevator Quadrants.
6. Pitch Disconnect Cable.
7. Pulley.

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PITCH DISCONNECT CLUTCH SYSTEM PAGE 2
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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Config 001

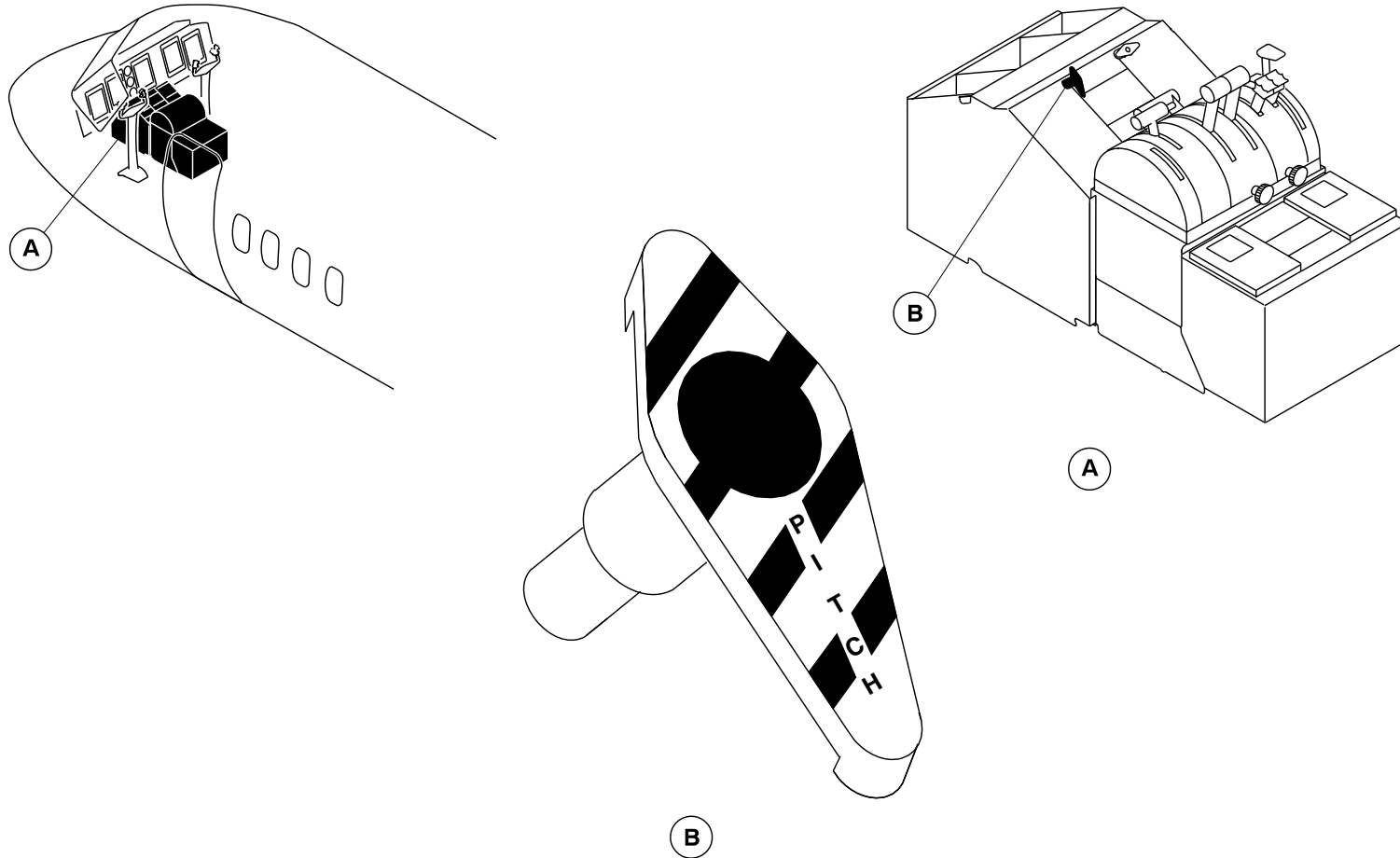
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Config 001
Page 5
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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Pitch Disconnect Handle
Figure 3

PSM 1-84-2A
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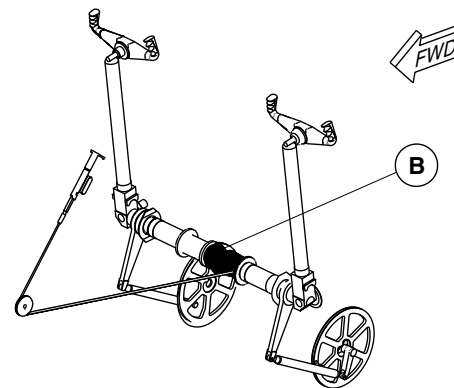
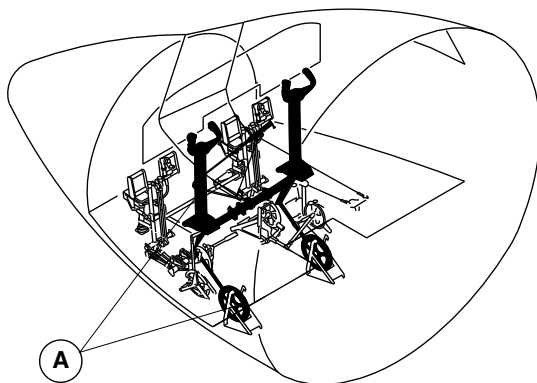
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Config 001
Page 6
Sep 05/2021

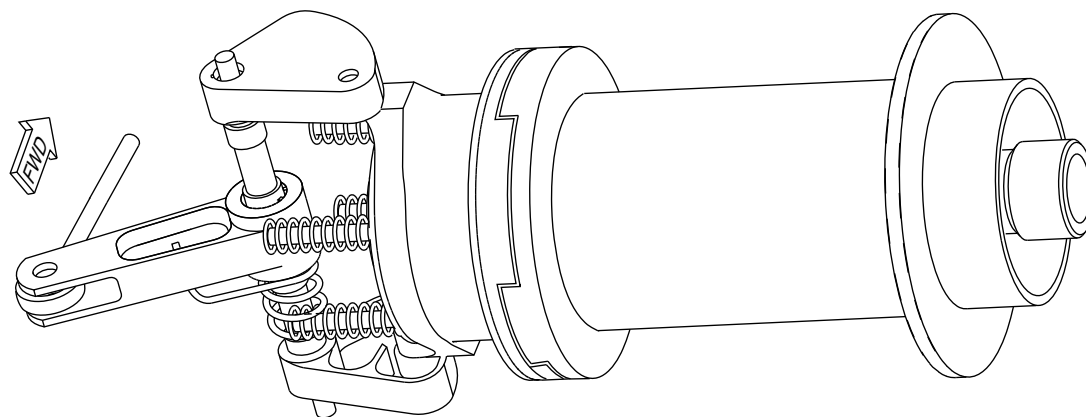


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



A PITCH DISCONNECT SYSTEM



B

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PITCH DISCONNECT CLUTCH MECHANISM PAGE 1
Figure 4

PSM 1-84-2A
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See first effectivity on page 2 of 27-38-00
Config 001

27-38-00

Config 001
Page 7
Sep 05/2021

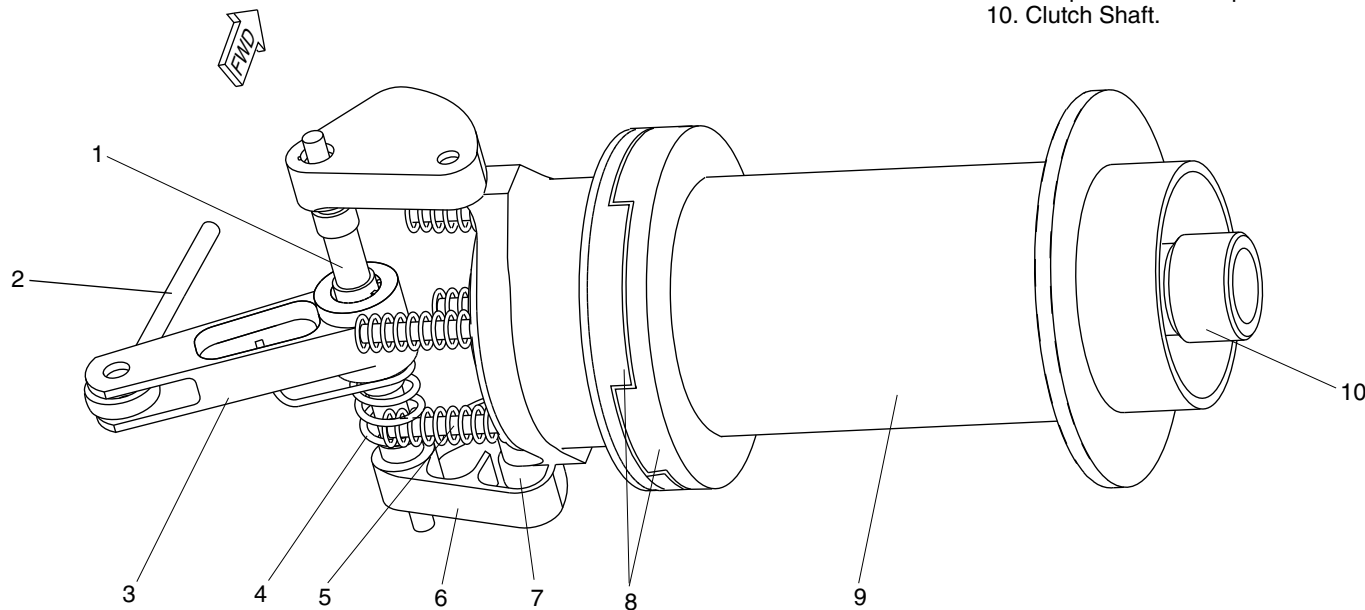


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

LEGEND

1. Cam Shaft.
2. Cable Attachment.
3. Cable Attachment Lever.
4. Lever Return Spring.
5. Clutch Return Spring (4 pcs.).
6. Actuator Cam.
7. Cam Roller.
8. Clutch Plates.
9. Torque Tube To Copilot's Column.
10. Clutch Shaft.



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PITCH DISCONNECT CLUTCH MECHANISM PAGE 2
Figure 5

PSM 1-84-2A
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See first effectivity on page 2 of 27-38-00
Config 001

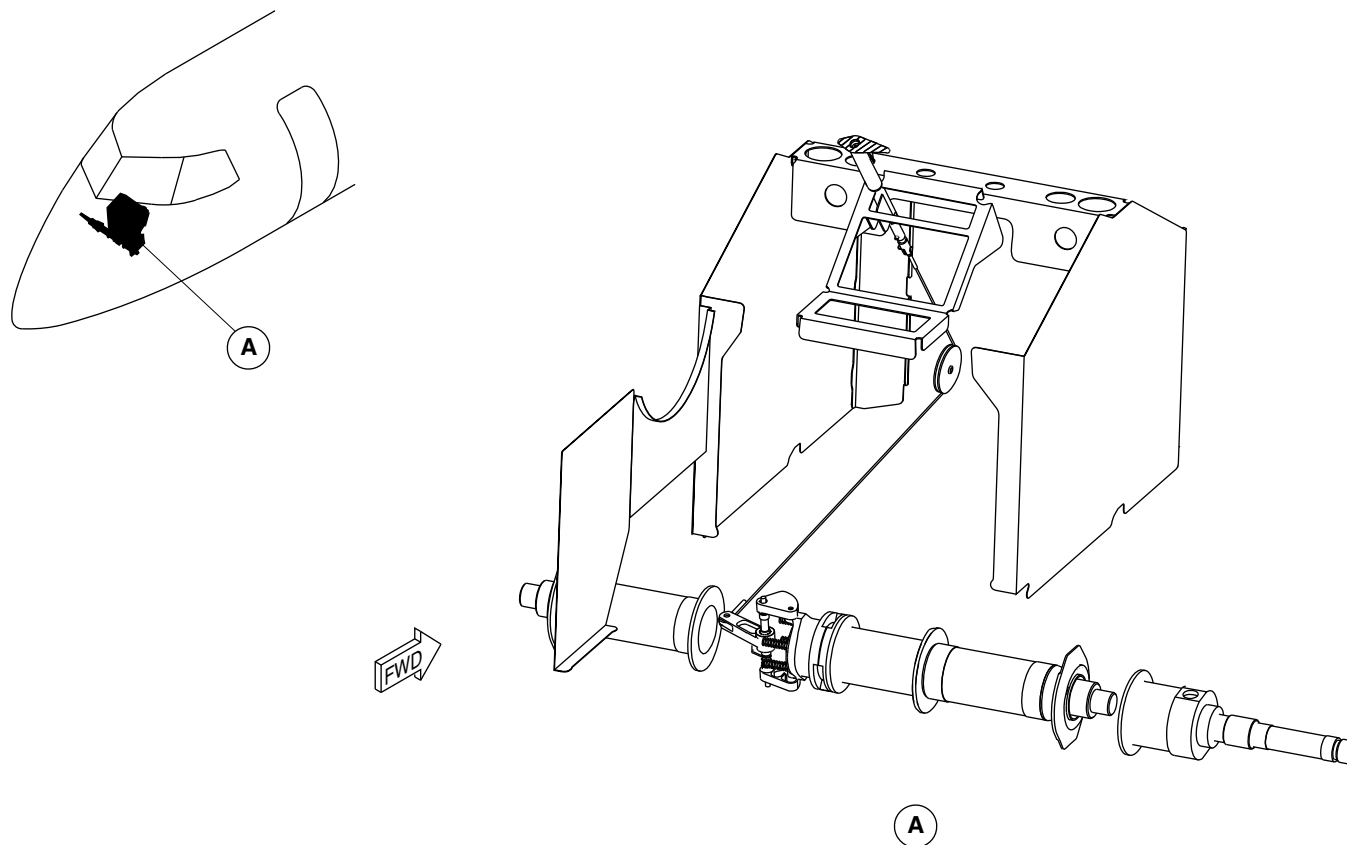
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Config 001
Page 8
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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PITCH DISCONNECT CLUTCH SYSTEM LOCATOR
Figure 6

PSM 1-84-2A
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See first effectivity on page 2 of 27-38-00
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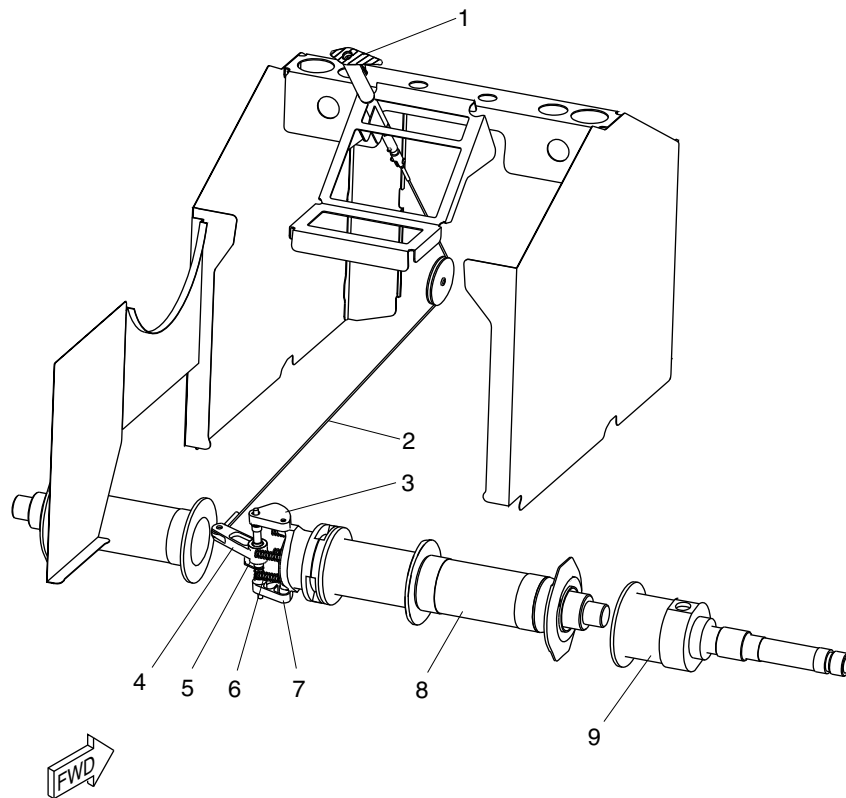
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Config 001
Page 9
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Pitch Disconnect Handle – Disengaged.
2. Cable.
3. Cam.
4. Clutch Lever – Disengaged.
5. Clutch Lever Spring.
6. Clutch Springs.
7. Cam Follower.
8. Torque Tube.
9. Clutch Housing.

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PITCH DISCONNECT CLUTCH SYSTEM DETAIL
Figure 7

PSM 1-84-2A
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See first effectivity on page 2 of 27-38-00
Config 001

27-38-00

Config 001
Page 10
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

27-50-00-001

WING FLAPS

Introduction

The wing flaps supply increased lift during take-off and landing.

General Description

Each wing has an inboard and an outboard flap surface. The inboard flap on each wing is installed between the fuselage structure and the engine nacelle inboard structure. The outboard flap on each wing is installed between the engine nacelle outboard structure and the aileron (refer to SDS 27-54-00). Indication of flap position is given by an indicator on the copilot's Multi Function Display (refer to SDS 27-53-00).

The wing flap system can be divided into the sections that follow:

- Flap Control System (27-51-00)
- Flap Drive System (27-52-00)
- Flap Position Indication (27-53-00)
- Flap Surfaces (27-54-00)

Detailed Description

[Refer to Figure 1.](#)

The flap surfaces are electronically controlled and hydraulically operated. The flaps are extended when the flap lever is moved in its

quadrant. The flap quadrant is located in the power quadrant on the right side of the center console. The flap quadrant has five positions corresponding to the five possible flap positions: 0, 5, 10, 15 and 35 degrees.

[Refer to Figure 2.](#)

The flap control system is interfaced with the Caution and Warning system through two caution lights: FLAP POWER and FLAP DRIVE (refer to SDS 27-52-00).

The flap selector lever is mechanically connected to the rudder restrictor on the right rudder forward quadrant.

On aircraft with SB84-27-20 or ModSum 4-901129 incorporated, a spring is attached between the flap selector lever and the center console. The spring helps to hold the weight of the rudder restrictor mechanism to make the movement of the selector lever easier.

To select a flap position, the flap lever is moved to the desired gate. A trigger located below the flap lever knob must be pulled up before the lever can be moved and released before the new position is reached. The flap lever is mechanically forced into each gate, so it is not possible to pass a flap position without first stopping at that gate.

[Refer to Figure 3.](#)

The Flap Control Unit (FCU) starts the actuation when the flap lever reaches the first gate. When the flap lever moves, it mechanically operates the flap lever Rotary Variable Differential Transducer (RVDT) which signals the FCU that the flap lever is in a new position. An Arming Switch is located on the flap lever. Movement of the flap selector lever causes the arming switch to send an enabling signal to the FCU. When the FCU receives the enabling signal, it commands the Flap Power Unit (FPU) to start moving the flaps to the newly selected position (refer to SDS 27-51-00).

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-50-00
Config 001

27-50-00

Config 001
Page 2
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

When the selected flap position has been reached, the FCU stops any further flap movement. The flap drive system keeps the flaps in the selected position (refer to SDS 27-52-00).

PSM 1-84-2A
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See first effectivity on page 2 of 27-50-00
Config 001

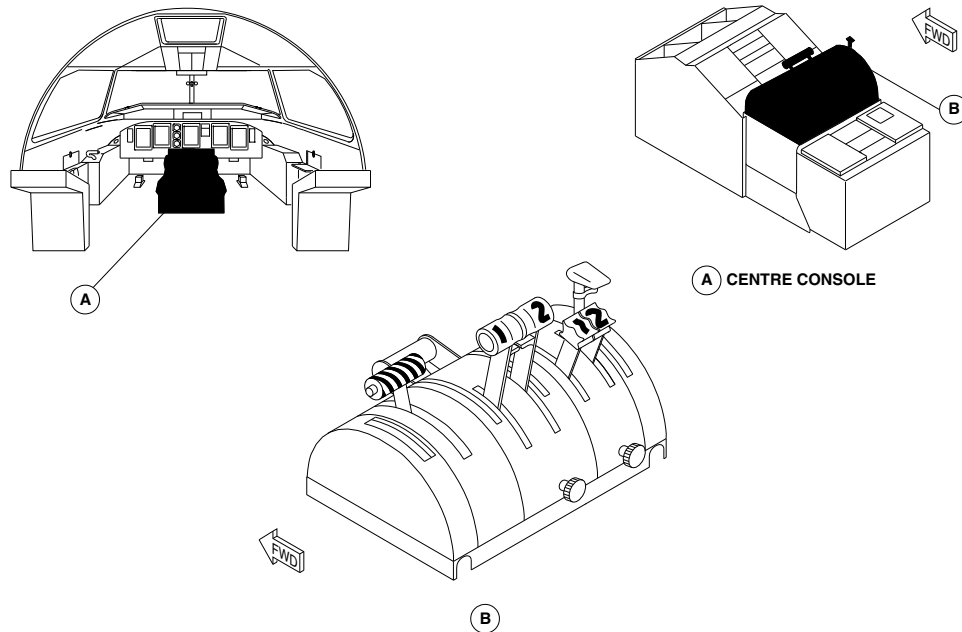
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Config 001
Page 3
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



4761a001 Power Quadrant Locator.iso

POWER QUADRANT
Figure 1

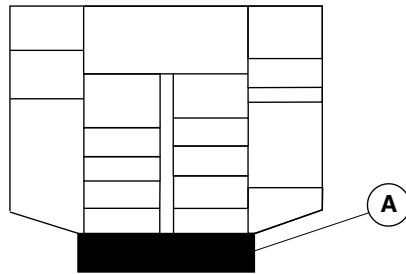
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See first effectivity on page 2 of 27-50-00
Config 001

27-50-00

Config 001
Page 4
Sep 05/2021



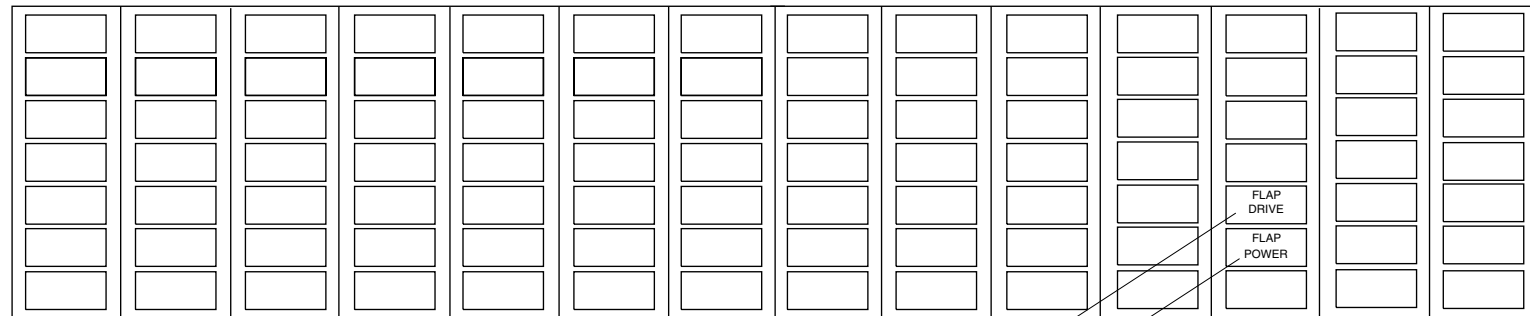
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



OVERHEAD CONSOLE

LEGEND

1. Flap Drive (Amber).
2. Flap Power (Amber).



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FLAP SYSTEM CAUTION LIGHTS
Figure 2

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-50-00
Config 001

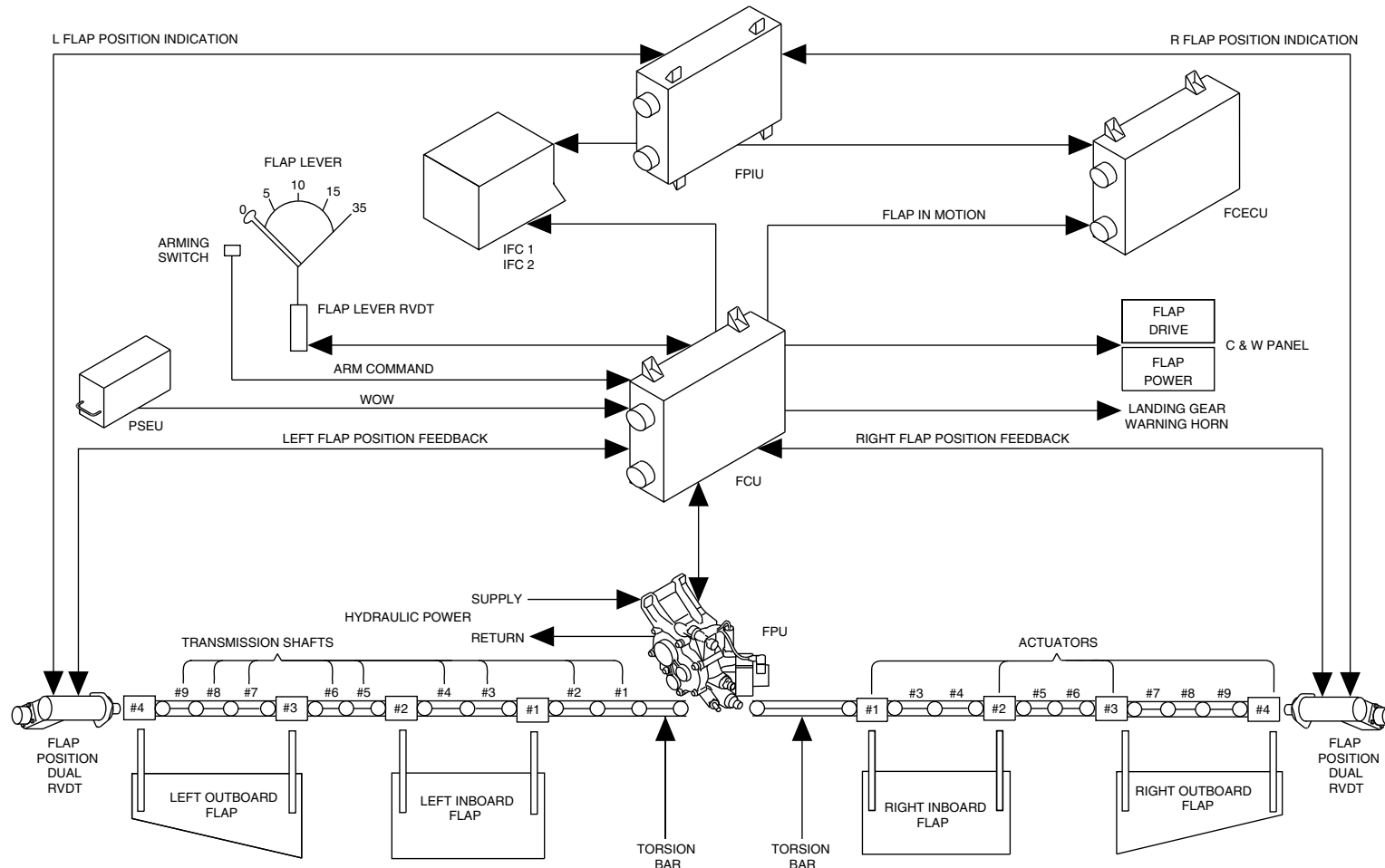
27-50-00

Config 001
Page 5
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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WING FLAP SYSTEM
Figure 3

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-50-00
Config 001

27-50-00

Config 001
Page 6
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

27-51-00-001

FLAP CONTROL SYSTEM

Introduction

The flap control system uses a digital position control loop to operate the flap surfaces by modulating the flow to the hydraulic motor of the flap power unit.

General Description

The flap control system is an electronically controlled, hydro-mechanical system. It monitors flap position and selection signals and produces the output signals to the Flap Power Unit (FPU) to control flap movement.

The flap control system is divided into the components that follow:

- Unit, Flap Control (FCU) (27-51-01)
- Lever/Quadrant, Flap Selector (27-51-06)
- Transducer, Flap Lever (27-51-11).

Detailed Description

[Refer to Figures 1 and 2.](#)

The flap selector lever moves in a quadrant with five detent gates at: 0, 5, 10, 15 and 35 degrees. When the flap position has been selected, the flap lever transducer, installed on the flap selector lever, senses the selected flap position and sends a signal to the

Flap Control Unit (FCU). An arming switch located on the flap lever transducer reads the new position of the flap selector lever. If the selector lever is in one of the five detent gates, the arming switch sends an enabling signal to the FCU to let flap deployment proceed.

The FCU receives a pressure signal from the Flap Power Unit (FPU) and flap position signals from the two wing tip flap position transducers. When the FCU receives these signals, it sends a signal to the FPU to operate the flap actuators and move the flaps. The FCU, the Flap Position Indication Unit (FPIU) and the flap position transducers monitor the movement of the flaps (refer to SDS 27-53-00). When the flaps reach the selected flap position, the FCU sends a signal to stop any further flap movement.

The flaps are held in the selected position by the irreversible ballscrew actuators. The electronic monitoring of the flaps makes sure that all the flaps move together to prevent flap asymmetry and non-selected flap movement.

Unit, Flap Control (FCU)

[Refer to Figures 3 and 4.](#)

The FCU is a microprocessor-based controller.

The FCU performs the functions that follow:

- Processes the signals from the flap position and flap lever transducers to monitor the actual flap position and controls the flap movement to the new position
- Receives pressure ON/OFF signal from the pressure switch in the flap power unit
- Receives the enabling signal from the arming switch on the flap lever transducer



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

- Performs internal checks to monitor the FCU for faults (Built In Test)
- Receives the Weight On Wheels signal to inhibit the power to the Built In Test (BIT) mode during flight
- Performs internal checks to monitor the FCU for faults
- Performs asymmetry and non–selected flap movement monitoring
- Supplies the ON/OFF power signal to the flap power unit solenoid valve
- Supplies controlled current to the FPU electro–hydraulic solenoid valve to actuate the system
- Supplies the FLAP POWER and FLAP DRIVE signals to the aircraft indicating and monitoring system
- Supplies the flap selected position to the indication and monitoring system
- Supplies the system status information through the ARINC 429 link and the RS422 serial port.

The flap control unit is located below the cabin floor in the HIRF enclosure. The HIRF box is not installed on aircraft with ModSum 4–153780 or 4–126136 incorporated.

Lever/Quadrant, Flap Selector

The flap selector lever and quadrant are located in the flight compartment on the right side of the center console. The flap selector lever moves in a detented cam from 0 degrees (full forward) through detents at 5, 10, 15 to a 35 degree setting (full aft). The lever has a spring–loaded cam follower, which engages the cam detents

at each selected position. The cam follower is released from the detents by a trigger on the lever.

The flap selector lever mechanically operates the flap lever transducer, which sends the flap position demand signal to the FCU. The arming switch is located on the flap selector lever. The arming switch circuit senses the position of the flap selector lever. If the lever is in one of the five detented positions, the arming switch sends an enabling signal to the FCU, which lets the flaps move to the selected position.

The flap selector lever is mechanically connected to the rudder restrictor mechanism, which restricts the rudder to half travel whenever the flaps have been selected to the 0 degree position.

On the aircraft with the ModSum 4–309202 installed, a modified flap lever quadrant is used. This flap lever quadrant is installed with a microswitch on the flap 35 degree detent. When the flap lever is selected to the landing flap position, it activates Enhance Ground Proximity Warning System (EGPWS) Mode 1 inhibit. Further, if the flap lever is moved from the flap 35 degree position, the operation of the microswitch causes the EGPWS to return to the standard mode automatically.

Transducer, Flap Lever

The flap lever transducer is a single section, five–wire type Rotary Variable Differential Transformer (RVDT). The RVDT is mechanically operated by the flap selector lever and is electrically supplied by the FCU.

The RVDT is designed to sense the angular position of the flap selector lever. When a new flap position has been selected, the RVDT senses the position of the flap selector lever and sends the



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

position data to the FCU to start movement of the flaps to the selected position.

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-51-00
Config 001

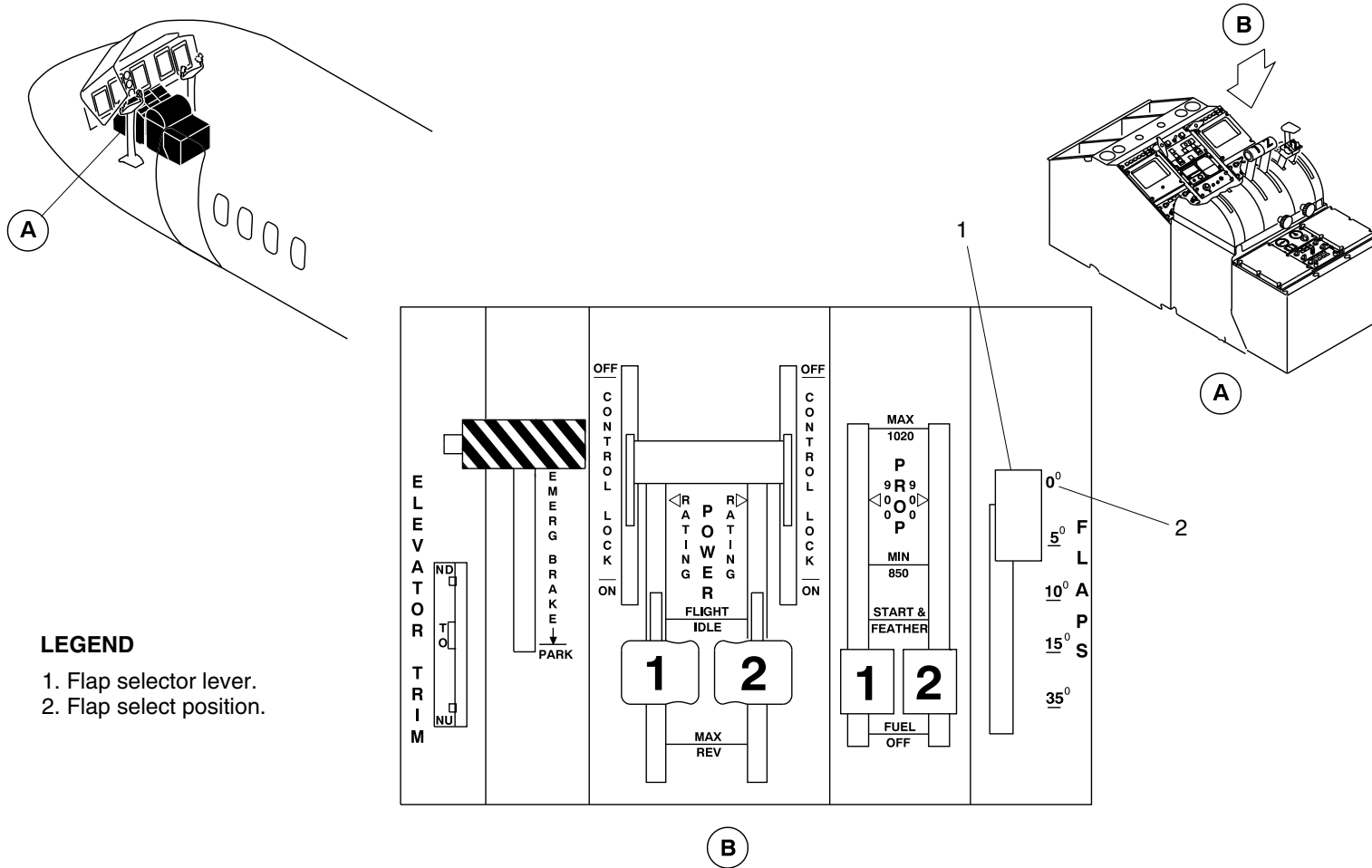
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Config 001
Page 4
Sep 05/2021



DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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Flap Selector Lever Detail
Figure 1

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-51-00
Config 001

27-51-00

Config 001
Page 5
Sep 05/2021



The diagram illustrates the flap control system for a Boeing 737-400, showing the flow of hydraulic power and electrical signals between various components.

Key Components and Connections:

- Flap Control Unit (FCU):** The central hub for flap control, receiving inputs from the Flap Lever RVDT, WOW, and PSEU. It sends outputs to the FPIU, FCEU, and the Flap Positioning Units (FPU).
- Flap Lever RVDT:** Provides feedback to the FCU based on the Flap Lever position (0, 5, 10, 15, 35 degrees).
- WOW (Wing Overload Warning) Switch:** Provides a signal to the FCU.
- PSEU (Proximity Switch Electronic Unit):** Provides a signal to the FCU.
- FPIU (Flap Position Indication Unit):** Receives signals from the FCU and provides L Flap Position Indication.
- FCEU (Flap Control Electronic Unit):** Receives signals from the FCU and provides R Flap Position Indication.
- Flap Positioning Unit (FPU):** Receives hydraulic power from the FCU and controls the actuators for the flaps.
- Actuators:** Control the movement of the flaps, including the Left Outboard Flap, Left Inboard Flap, Right Inboard Flap, and Right Outboard Flap.
- Transmission Shafts:** Connect the actuators to the flaps.
- Flap Position Dual RVDT:** Provides feedback to the FCU for the outboard flaps.
- Flap Drive and Flap Power:** Provide power to the flaps.
- Landing Gear Warning Horn:** Activated by the FCU when the flaps are in motion.

Hydraulic Power Flow:

- Supply hydraulic power to the FPU.
- Return hydraulic power from the FPU.
- Hydraulic power is distributed to the actuators for the flaps.

Electrical Signal Flow:

- Flap Lever RVDT to FCU.
- WOW to FCU.
- PSEU to FCU.
- FCU to FPIU and FCEU.
- FPIU to L Flap Position Indication.
- FCEU to R Flap Position Indication.
- FCU to FPU.
- FPU to actuators.
- Actuators to transmission shafts.
- Transmission shafts to flaps.
- Flaps to Flap Position Dual RVDT.
- Flap Position Dual RVDT to FCU.
- FCU to Flap Drive and Flap Power.
- FCU to Landing Gear Warning Horn.

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-51-00
Config 001

Config 001
Page 6
Sep 05/2021



The diagram illustrates the installation of the HIRF box in the cabin floor assembly. It includes a top view of the cabin floor layout, a side view of the floor assembly, and two detailed views of the HIRF box (B) and the cabin floor assembly (C).

Top View: A plan view of the cabin floor showing the layout of seats and the location of the HIRF box (A) in the center aisle.

Side View: A side view of the cabin floor assembly showing the HIRF box (B) installed in the floor.

HIRF Box (B): A detailed view of the HIRF box, showing its internal components and the location of the HIRF box (C) in the floor assembly.

Cabin Floor Assembly (C): A detailed view of the cabin floor assembly, showing the HIRF box (B) installed in the floor.

Labels:

- A**: HIRF BOX
- B**: HIRF BOX
- C**: CABIN FLOOR ASSEMBLY

Modifications:

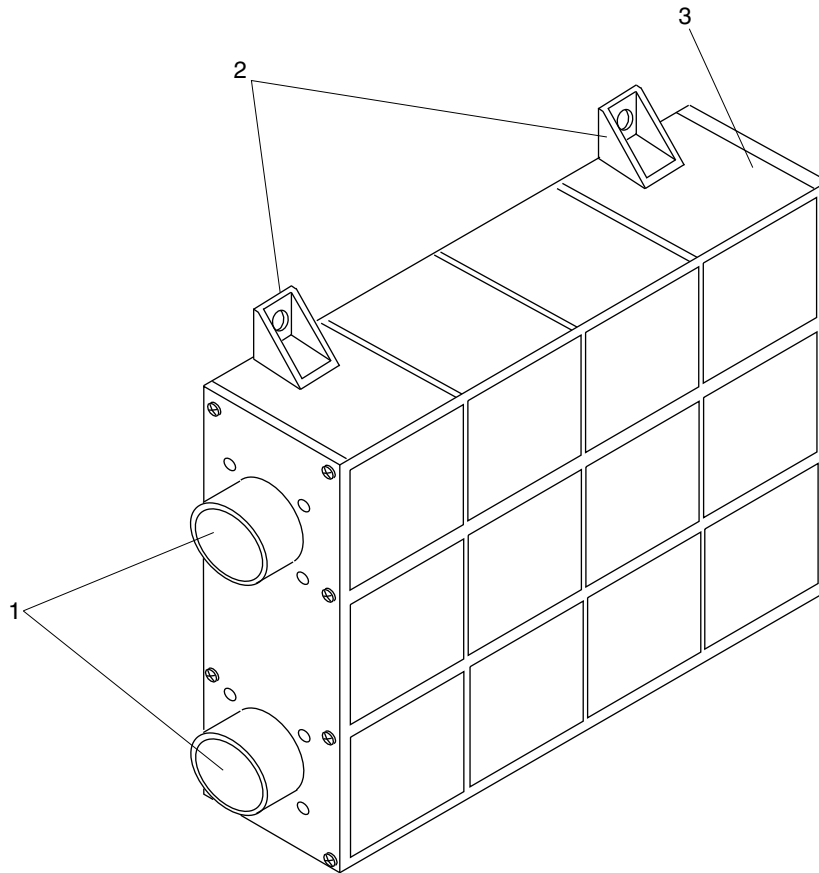
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- PRE MODSUM 4-126136**
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- POST MODSUM 4-126136**

Print Date: 2025-04-22



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Electrical Connectors.
- 2. Mounting Bracket.
- 3. Flap Control Unit Box.

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FLAP CONTROL UNIT (FCU) PAGE 2
Figure 4

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-51-00
Config 001

27-51-00

Config 001
Page 8
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

27-52-00-001

FLAP DRIVE SYSTEM

Introduction

The flap drive system moves the flaps from the 0 degree position (full up) to the 35 degree position (full down) or to the intermediate 5, 10 or 15 degree positions.

General Description

Refer to Figure 1.

The flap drive system is electronically controlled and hydro-mechanically operated. It is designed to move the four flap sections together to the position set by the flap selector lever. When the flaps have moved into the selected position, the flap drive system makes sure that the flaps stay in that position.

The flap drive system has the components that follow:

- Unit, Flap Power (FPU) (27-52-01)
- Shafts, Transmission (27-52-06)
- Bars, Torsion (27-52-11)
- Actuators, Flap (27-52-16)
- Tracks, Flap (27-52-21).

Detailed Description

The hydraulically powered Flap Power Unit (FPU) turns a series of transmission shafts installed on the wing aft spars. The driveshafts operate eight irreversible ballscrew flap actuators, which move the flaps along tracks located under the wings and on the sides of the fuselage and the engine nacelles.

The FPU receives flap actuation signals from the Flap Control Unit (FCU). It also sends feedback signals to let the FCU monitor flap movement.

Unit, Flap Power (FPU)

Refer to Figures 2 and 3.

The Flap Power Unit (FPU) is located on the wing aft spar, to the right of the aircraft centreline. The FPU converts aircraft hydraulic power to rotary mechanical power to operate the flap actuators, raising or lowering the flaps, as selected.

The FPU has a hydraulic module and a gearbox.

The hydraulic module includes the components that follow:

- Inlet filter
- Solenoid and shut-off valve
- Two-stage Electro-Hydraulic Servo Valve (EHSV)
- Hydraulic motor
- Pressure switch.

The gearbox has a two-stage spur reduction gear train housed in an aluminum casing.

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-52-00
Config 001

27-52-00

Config 001
Page 2
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Hydraulic pressure is supplied from the No. 1 hydraulic system to the FPU through a filter and a restrictor at the inlet port. A return line carries the hydraulic fluid back to the system reservoir. A plastic drain line carries any internal fluid leakage from the FPU to an overboard drain.

The solenoid valve receives an electrical on/off signal from the FCU, which hydraulically arms the circuit. The pressure switch located downstream of the shut-off valve monitors the pressure. The FCU commands the Electro-Hydraulic Servo Valve (EHSV) to actuate the flap drive system. The EHSV controls the flow of pressurized hydraulic fluid to the flap drive system so that the flaps are operated at a controlled speed without the need of a flow regulator.

Shafts, Transmission

Refer to Figures 4 , 5 , 6 , 7 and 8.

The transmission shaft system transmits the mechanical power from the Flap Power Unit (FPU) to the eight flap actuators. There are 16 transmission shafts located along the aft spars of the wings. Nine shafts extend outward from the FPU along the left wing aft spar to engage with each of the four flap actuators on the left wing. Seven shafts extend outward from the FPU along the right wing aft spar to engage with each of the four flap actuators on the right wing. All 16 transmission shafts are aluminum tube assemblies of different lengths.

The transmission shafts have different configurations. Some shafts have a universal joint at one end and a self-aligning bearing support at the other end, while some shafts have a universal joint at both ends. The ends of each shaft are externally or internally splined as is necessary for connection to the adjacent shaft.

Bars, Torsion

Refer to Figures 9 and 10.

There are two torsion bars installed on the aircraft. Each torsion bar is an aluminum alloy shaft with external splined connections. The two shafts transfer the rotary movement of the FPU to the transmission shafts and actuators on the wings.

The two torsion bars transmit a high speed, low torque rotating force from the FPU to the transmission shafts and flap actuators. The right torsion bar has a universal joint at both ends of the shaft to allow for wing deflections and connects the FPU to the No.1 flap actuator. The left torsion bar has a support and bearing assembly at one end of the shaft and a universal joint at the other end and connects the FPU to the #1 transmission shaft.

Actuators, Flap

Refer to Figures 11 , 12 , 13 and 14.

There are eight flap actuators installed on the aircraft, four on each wing. The flap actuators are mechanical units designed to convert high speed, low torque rotary drive to a low speed, high torque linear drive. The flap actuators move the flap surfaces to a selected position and keep the selected position against the aerodynamic forces on the flap surfaces. The flap actuators receive the mechanical input from the transmission shafts or the torsion bars through a spline connection and transfer this force to the next actuator by a through shaft. The No.1 and No.2 flap actuators on each wing are identical and operate the inboard flap section. The No.3 and No.4 flap actuators on each wing are constructed slightly different from each other and operate the outboard flap section.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Each flap actuator includes the components that follow:

- No-back/torque limiter device
- Worm gear
- Recirculating ballscrew.

The no-back/torque limiter device limits the load transferred from the transmission shaft to the flap actuator and supplies irreversibility in both directions of rotation. This keeps the flaps in the selected position regardless of the aerodynamic loads. The rotary motion of the worm gear disengages the no-back device and the mechanical drive is transmitted directly to the ballscrew. The ballscrew changes the rotary motion into linear motion and moves the attached flap surfaces in the flap tracks. Once the movement of the flap surface has stopped the no-back device reengages to supply irreversibility. Maximum flap travel is limited by the use of non-jamming end stops.

[Refer to Figure 15.](#)

If a minor flap failure occurs, the FCU sends a FLAP DRIVE caution light to the caution panel but the flap system may still operate. The caution light will go out if the failure goes off. The FLAP DRIVE caution light will come on for the faults that follow:

- Hydraulic pressure without a selection command (the FPU shutoff valve stays open after the end of flap movement).
- Flap stall. The flap panels stop before they get to the commanded position in case of too much load is applied to the actuators or seizure of the FPU or actuators. The condition is detectable by comparing the indication of the flap selected position and the flap position indicator.

- Internal EEPROM failure – “Flap position selected” monitor output from the FCU is not valid. The condition is detectable by comparing the flap selector lever position and the flap position indicator.

If a major flap failure occurs, the FCU opens the severing switch and commands the system to stop in a fail safe condition. The FLAP POWER caution light comes on in the C&W panel. This failure can only be reset with the FCU power on.

If a critical flap failure (assymetry or uncommanded movement) occurs, the FCU opens the severing switch and the fault latch relay. The system will be commanded to stop in the fail safe condition. The FLAP POWER caution light comes on. This failure can only be reset with the aircraft on the ground and through the applicable NVM reset button on the maintenance panel.

The FLAP POWER caution light will come on for the faults that follow:

- Flap assymetry
- Flap uncommanded movement
- Flap position transducer failure
- Flap lever transducer failure
- No hydraulic pressure after selection of command
- Select command and no arm signal
- Electrical power loss
- Servo valve drive failure
- Watch dog intervention
- Latch relay open status.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Tracks, Flap

The flap tracks control the incidence angle of the flaps during flap deployment.

[Refer to Figures 16 and 17.](#)

There are five flap tracks on each wing. Flap tracks #1 and #2 are cam type tracks, which support the inboard flap section. Flap track #1 is located on the side of the fuselage while flap track #2 is located on the inboard side of the nacelle. Flap tracks #3, #4 and #5 support the outboard flap section. Flap track #3 is a cam type track and is located on the outboard side of the nacelle. Flap tracks #4 and #5 are cantilever type tracks attached to the wing rear spar extending aft from the bottom of the wing surface.

The inboard flaps are installed with rollers at each end, which engage in tracks #1 and #2. The rollers give support for the flap and allow for smooth extension and retraction of the flap.

The outboard flaps are installed on the wing by tracks #3, #4 and #5. The outboard flaps have rollers, which engage in track #3 and roller carriage assemblies, which engage in tracks #4 and #5. The roller carriage assemblies use four different types of rollers to support the flaps. Deadweight rollers are used for support when the aircraft is on the ground. The three remaining rollers are used during flight. Sideload rollers prevent lateral movement of the flaps and “0 to 16 degree” rollers support the flaps to the 15 degree detent position. “16 to 35 degree” rollers support the flaps from the 15 degree detent position to full extension of 35 degrees.

In the event of roller failure, the outboard flap is retained to tracks #3 and #5 by failsafe cables and the inboard flap is retained to tracks #1 and #2 by failsafe fittings. Rubbing pads are supplied to prevent excessive side movement of the flaps in the tracks. In the fully

retracted position, the leading edge of the flap aligns with the upper flap shroud on the wing to give an aerodynamically smooth wing surface.

Training Information Points

The worm gear is lubricated for life with special grease: KLUBER ISOFLEX L3–600. The Ballscrew is lubricated with grease conforming to MIL–G–23827 through two accessible grease nipples.

PSM 1–84–2A

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See first effectivity on page 2 of 27–52–00

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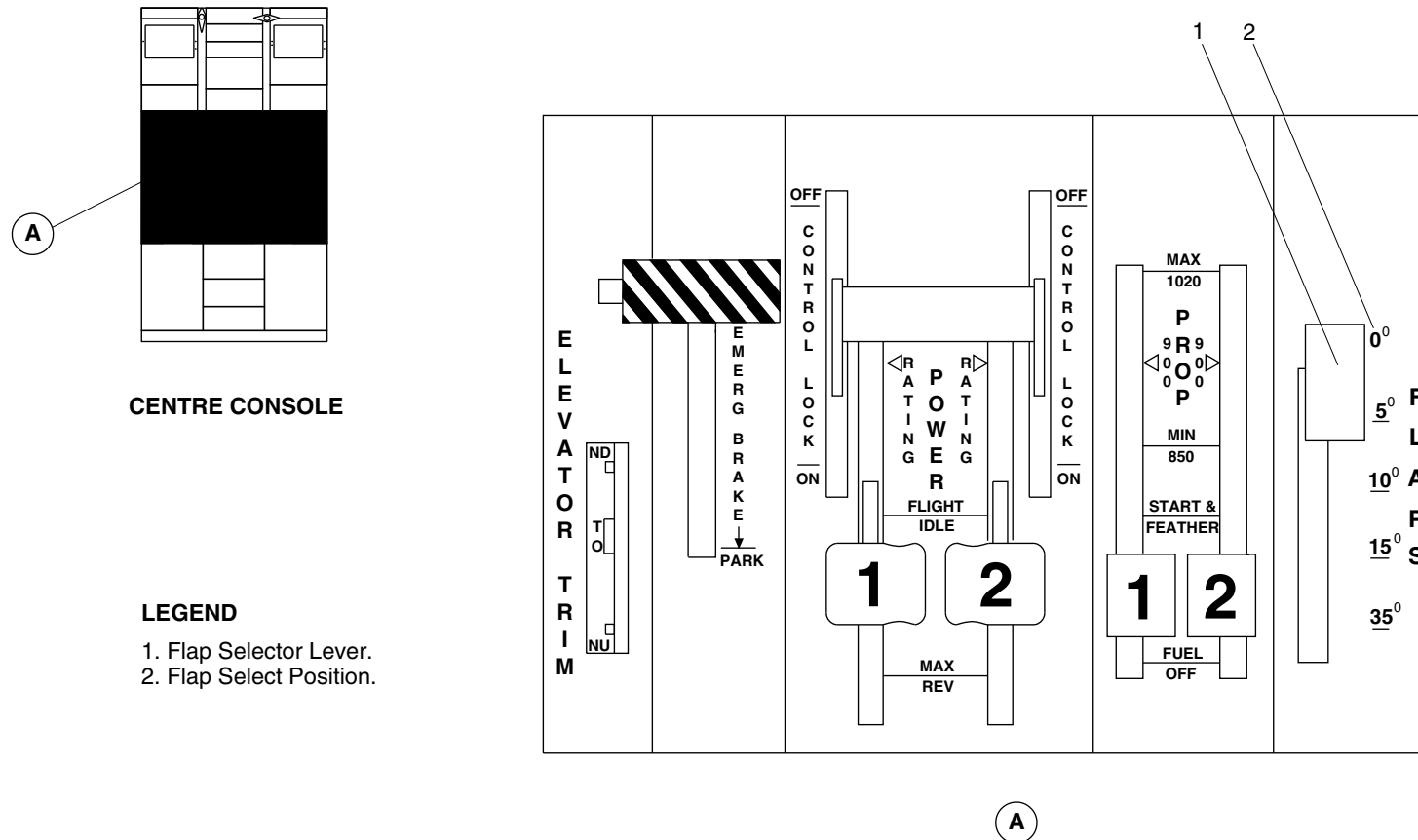
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Config 001
Page 5
Sep 05/2021



DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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FLAP SELECT LEVER
Figure 1

PSM 1-84-2A
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See first effectivity on page 2 of 27-52-00
Config 001

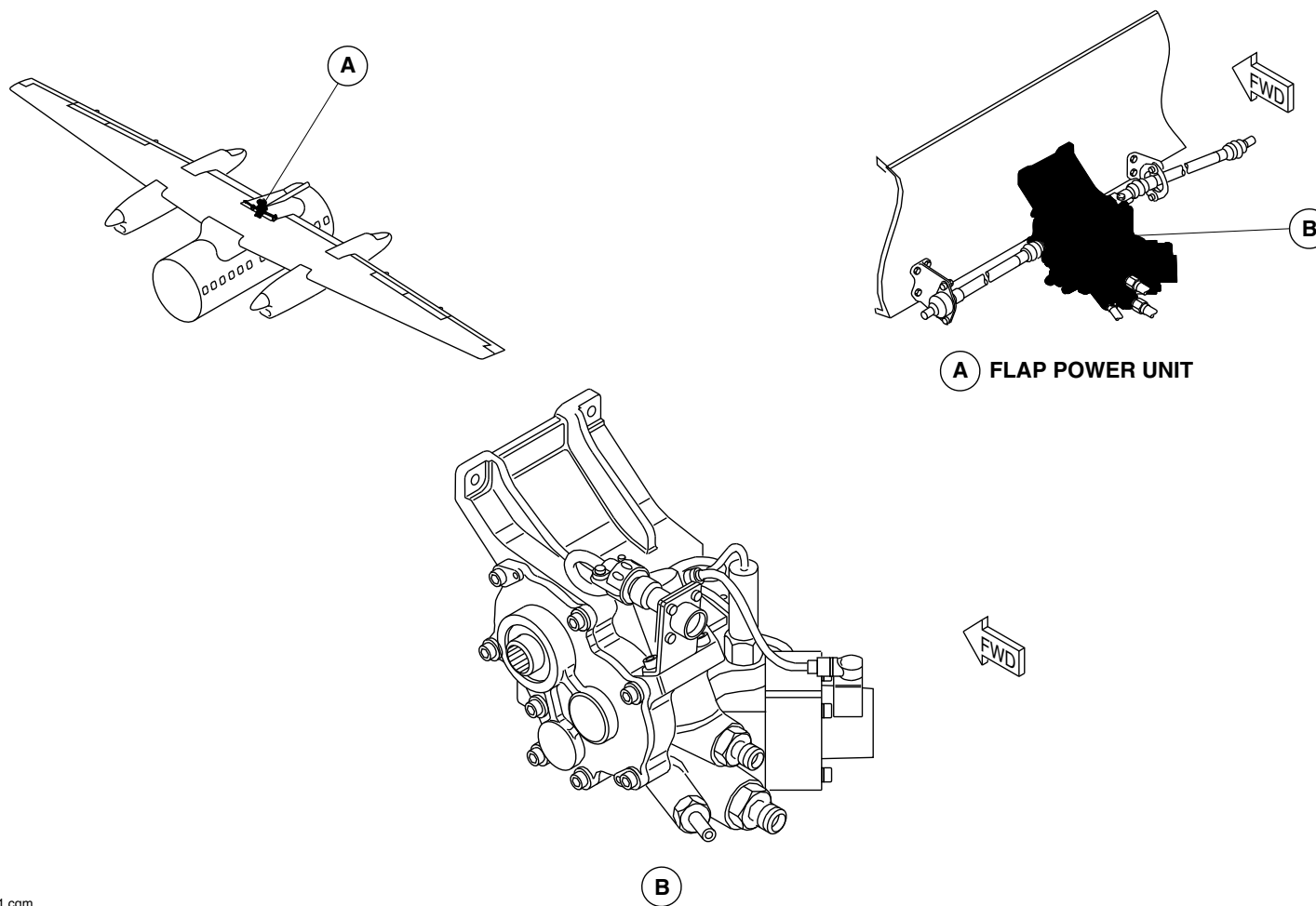
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Config 001
Page 6
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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FLAP POWER UNIT PAGE 1
Figure 2

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-52-00
Config 001

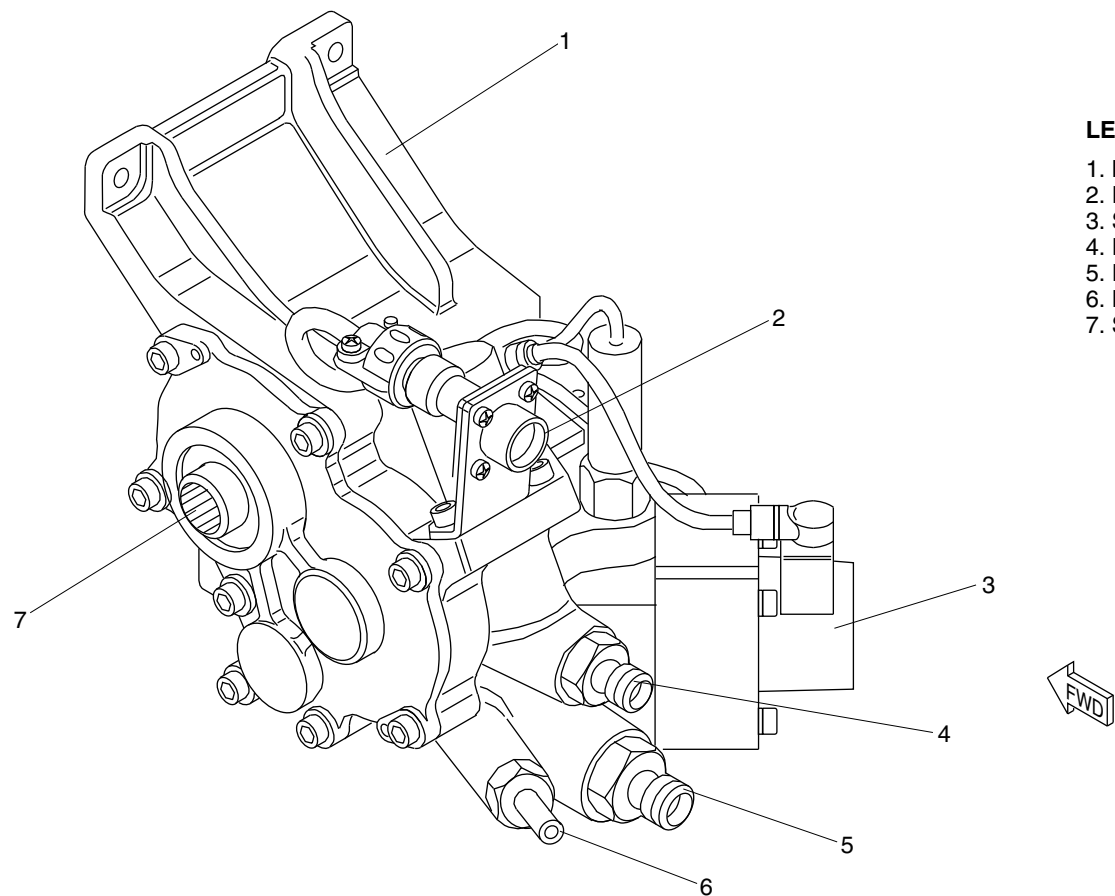
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Config 001
Page 7
Sep 05/2021



DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Mounting Bracket.
- 2. Electrical Connector.
- 3. Solenoid.
- 4. Pressure Port.
- 5. Return Port.
- 6. Drain Port.
- 7. Splined Coupling.

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FLAP POWER UNIT PAGE 2
Figure 3

PSM 1-84-2A
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See first effectivity on page 2 of 27-52-00
Config 001

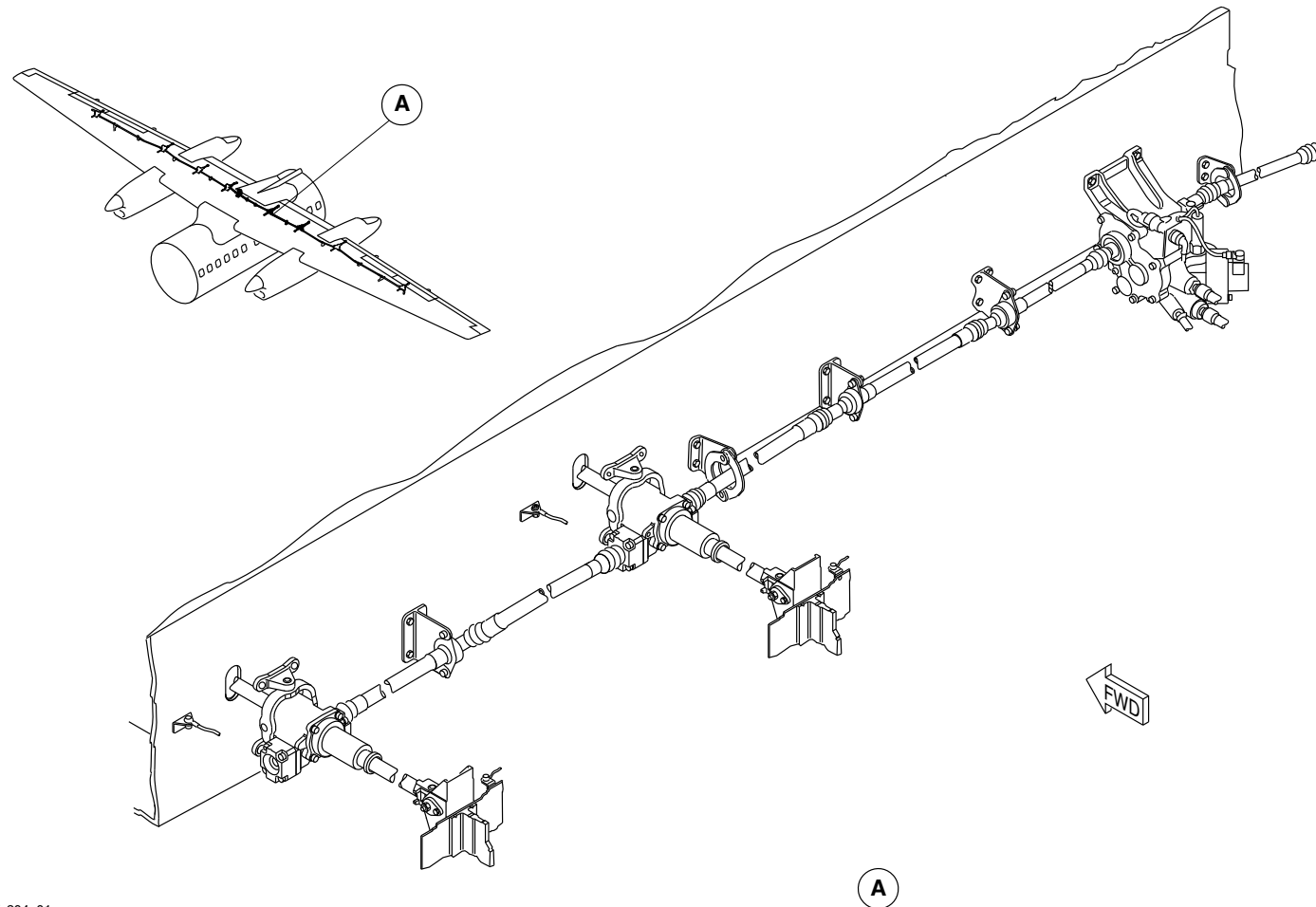
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Config 001
Page 8
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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FLAP TRANSMISSION SHAFTS PAGE 1
Figure 4

PSM 1-84-2A
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See first effectivity on page 2 of 27-52-00
Config 001

27-52-00

Config 001
Page 9
Sep 05/2021

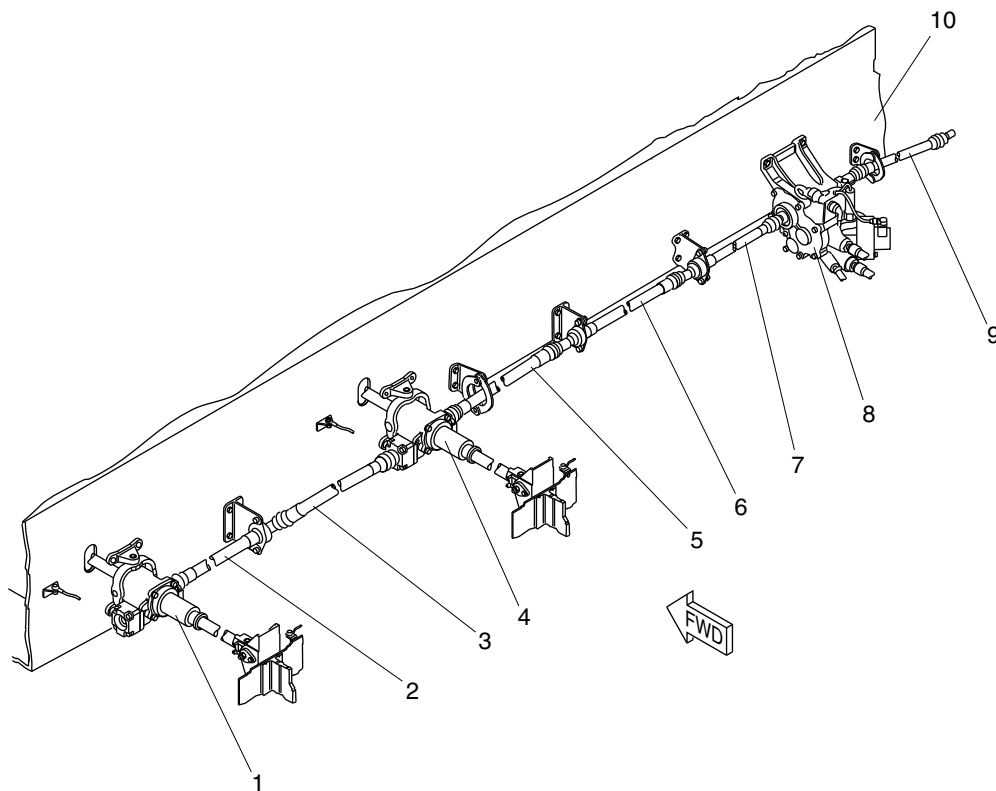


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LEGEND

1. Flap Actuator No. 2.
2. Flap Transmission Shaft No. 4.
3. Flap Transmission Shaft No. 3.
4. Flap Actuator No. 1.
5. Flap Transmission Shaft No. 2.
6. Flap Transmission Shaft No. 1.
7. Left Torsion Bar.
8. Flap Power Unit.
9. Right Torsion Bar.
10. Left Wing Rear Spar.



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FLAP TRANSMISSION SHAFTS PAGE 2
Figure 5

PSM 1-84-2A
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Config 001

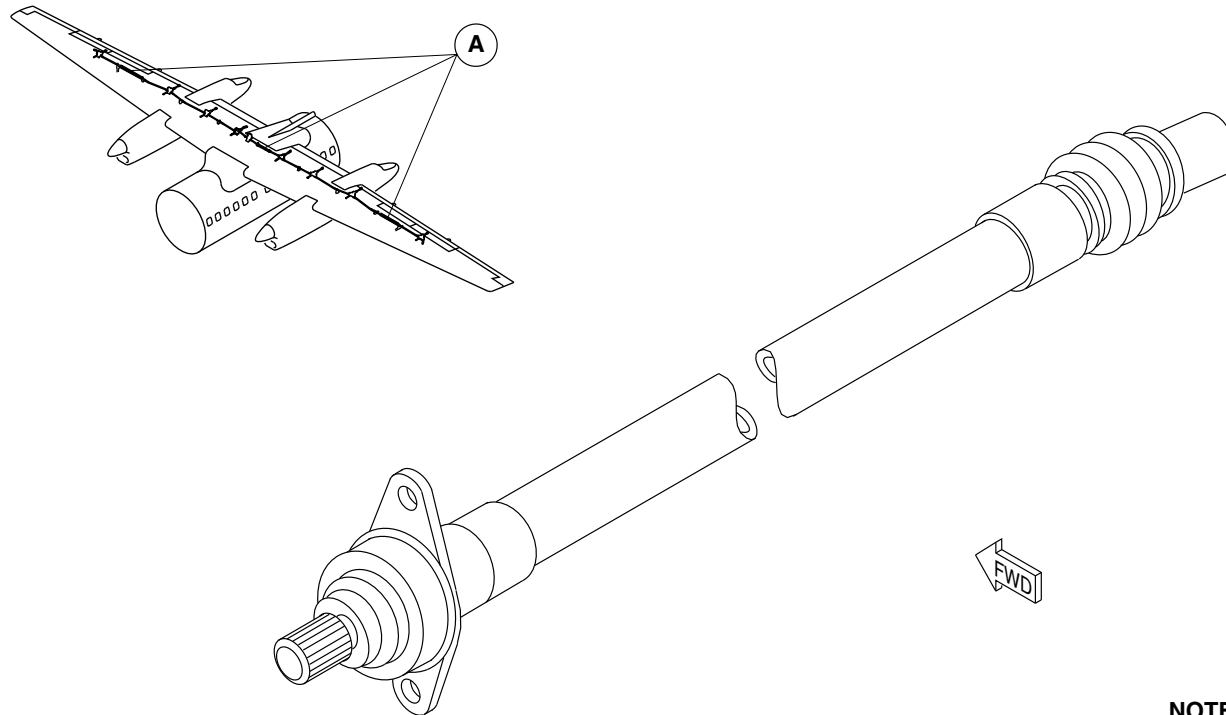
27-52-00

Config 001
Page 10
Sep 05/2021



DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



NOTE

No. 1 Shaft Shown.
No. 8 Shaft (LH and RH) Similar.

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FLAP TRANSMISSION SHAFTS PAGE 3
Figure 6

PSM 1-84-2A
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See first effectivity on page 2 of 27-52-00
Config 001

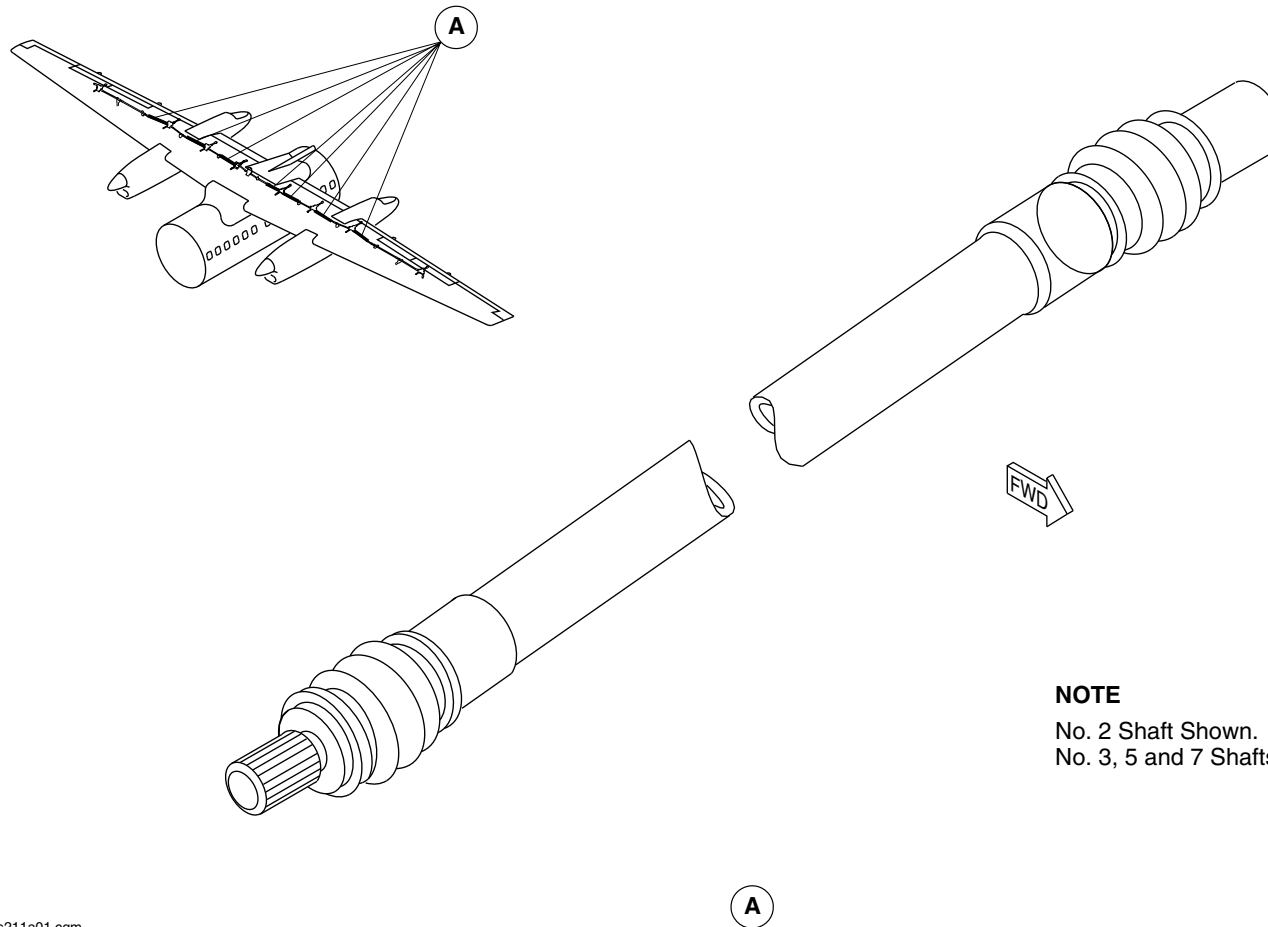
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Config 001
Page 11
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



NOTE

No. 2 Shaft Shown.
No. 3, 5 and 7 Shafts (LH and RH) Similar.

fs211a01.cgm

FLAP TRANSMISSION SHAFTS PAGE 4
Figure 7

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-52-00
Config 001

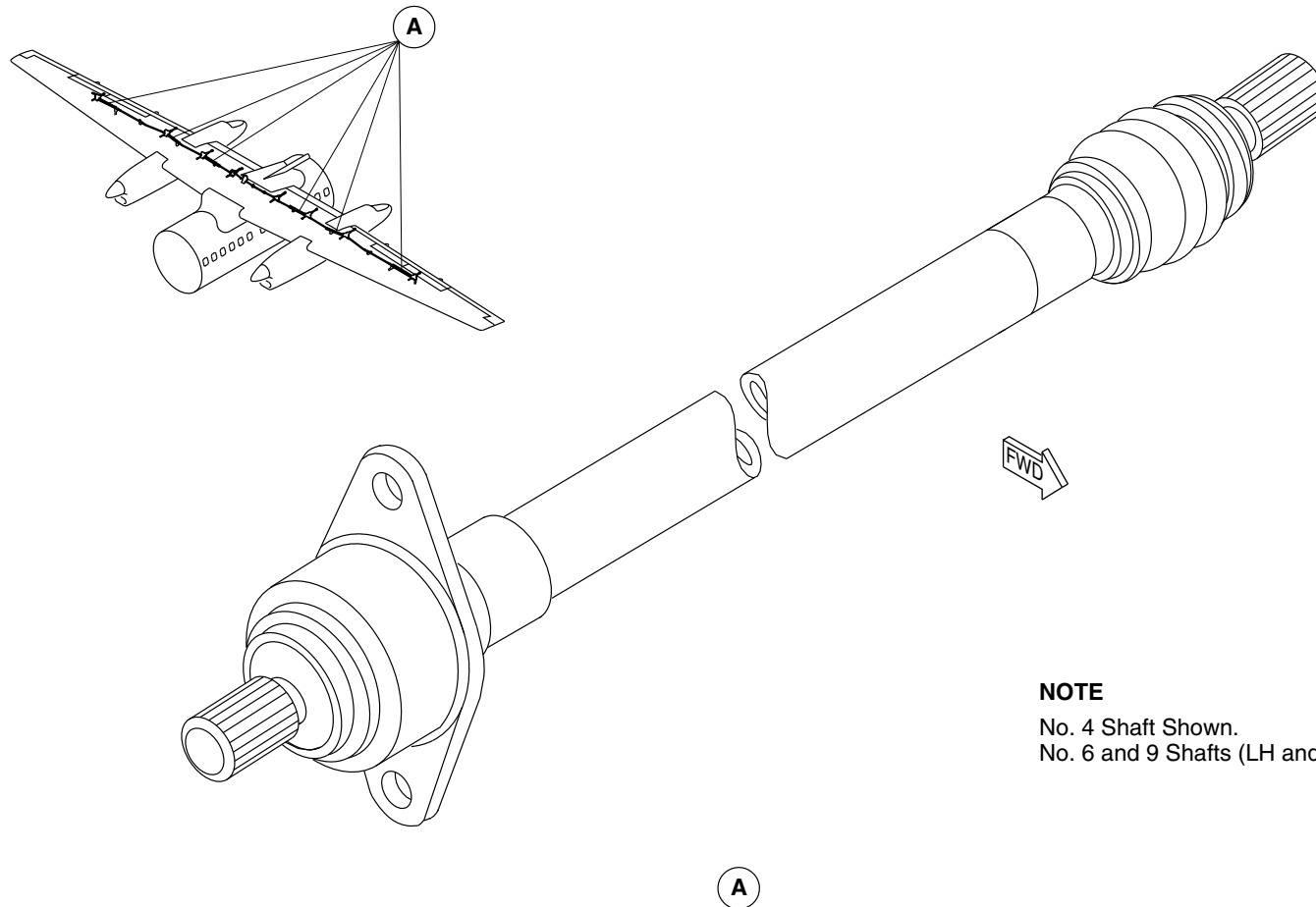
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Config 001
Page 12
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



fs210a01.cgm

NOTE

No. 4 Shaft Shown.

No. 6 and 9 Shafts (LH and RH) Similar.

FLAP TRANSMISSION SHAFTS PAGE 5
Figure 8

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-52-00
Config 001

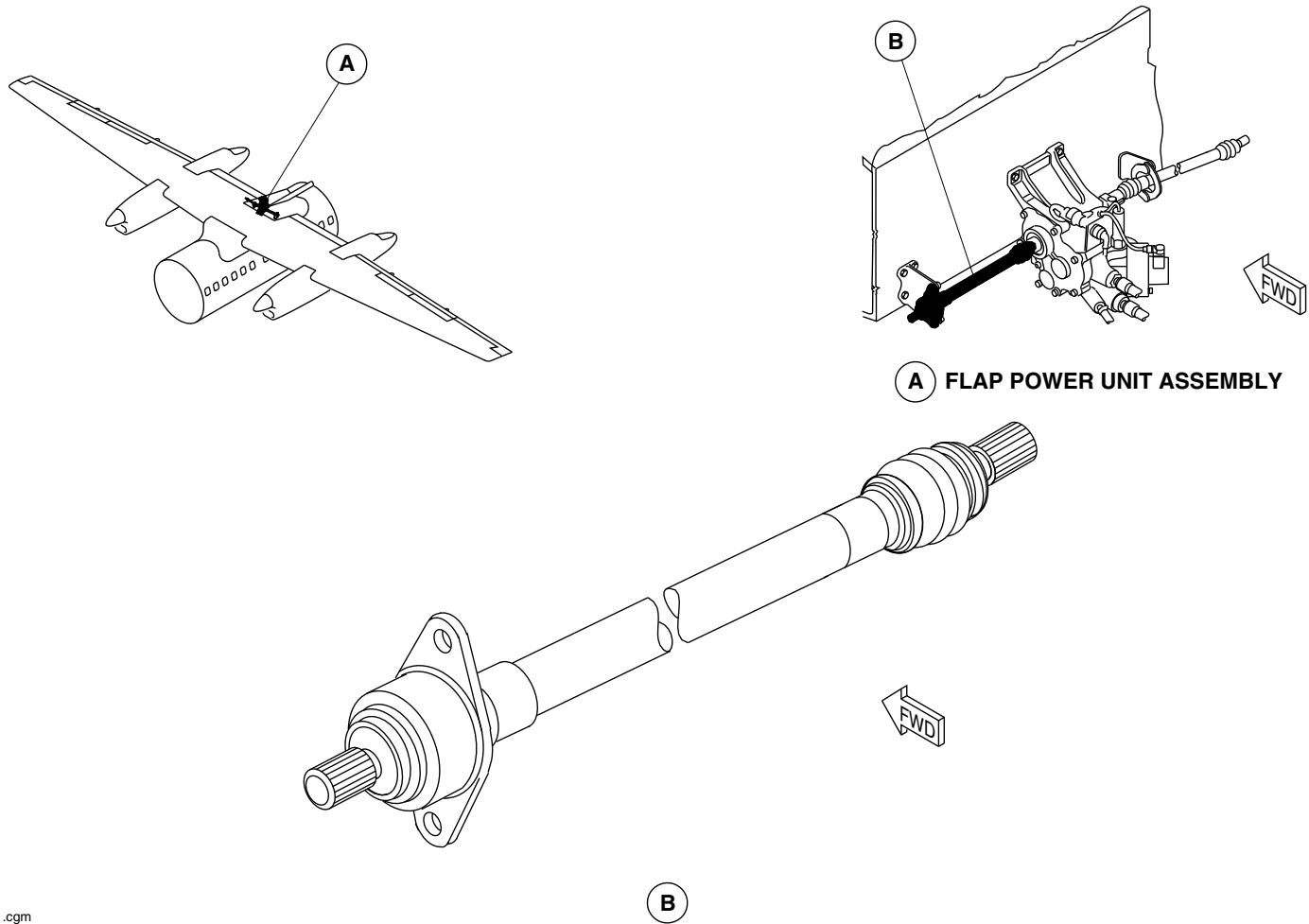
27-52-00

Config 001
Page 13
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



fs085a01.cgm

FLAP TORSION BAR PAGE 1
Figure 9

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-52-00
Config 001

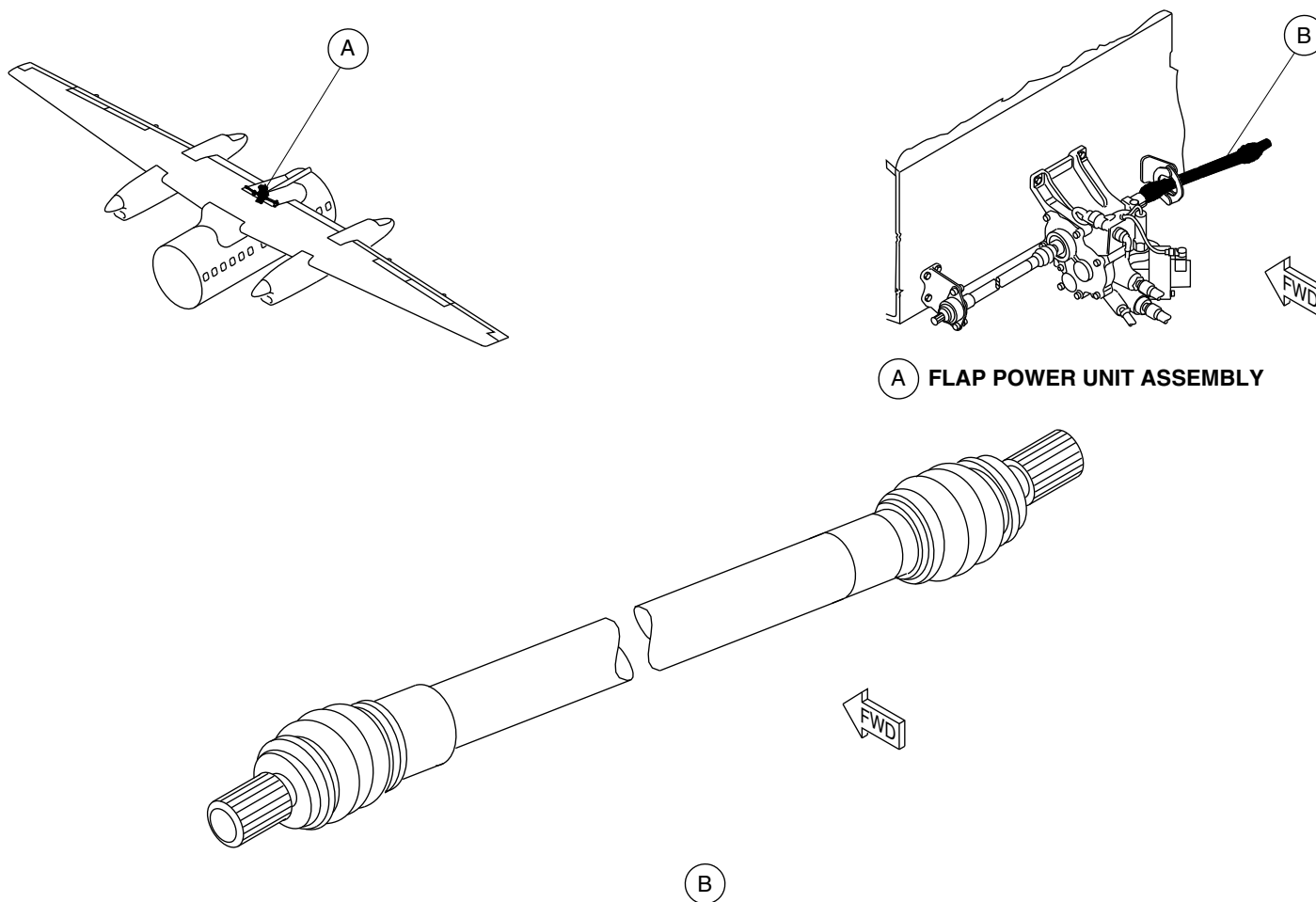
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Config 001
Page 14
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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FLAP TORSION BAR PAGE 2
Figure 10

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-52-00
Config 001

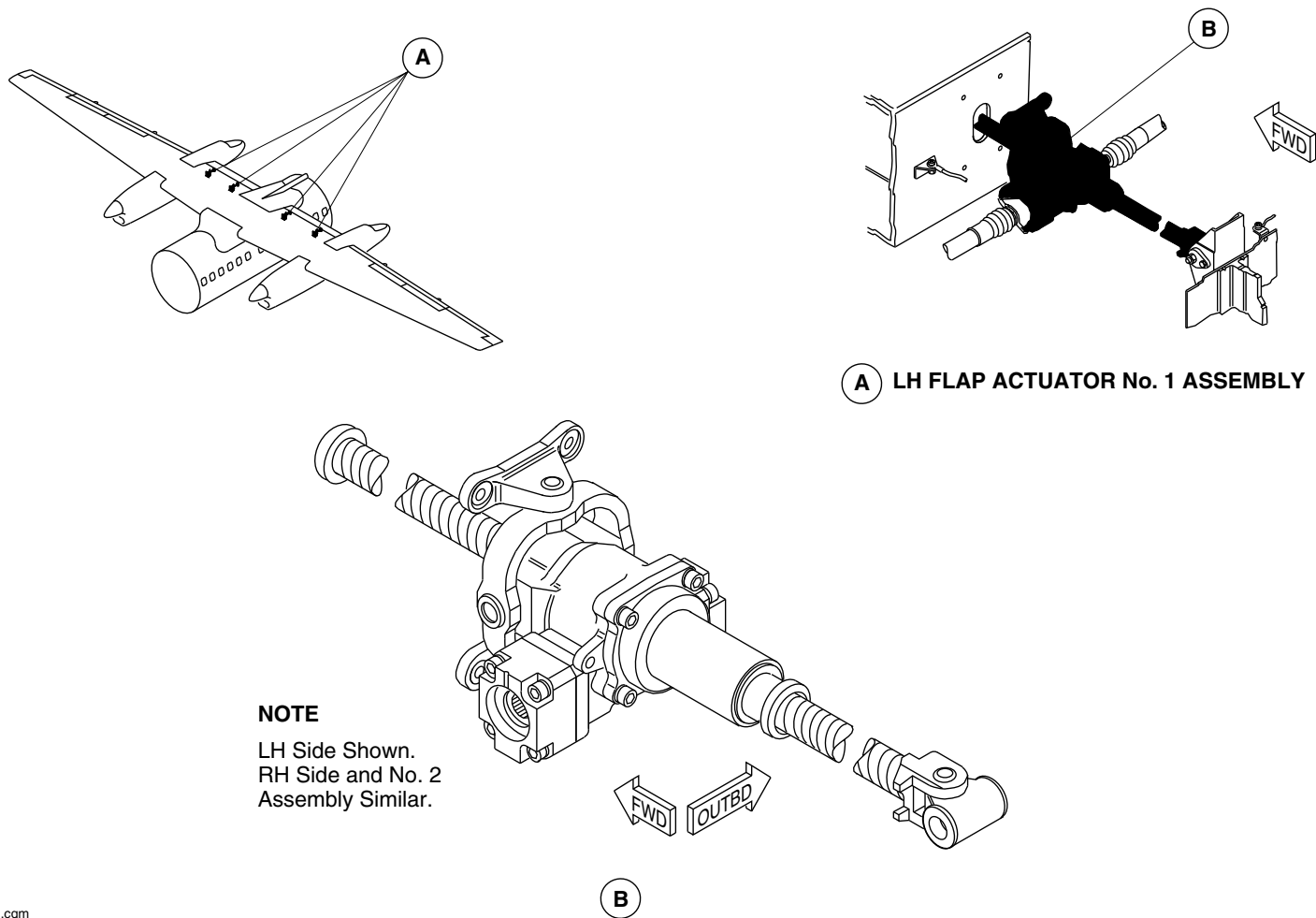
27-52-00

Config 001
Page 15
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



fs125a01.cgm

FLAP ACTUATORS PAGE 1
Figure 11

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-52-00
Config 001

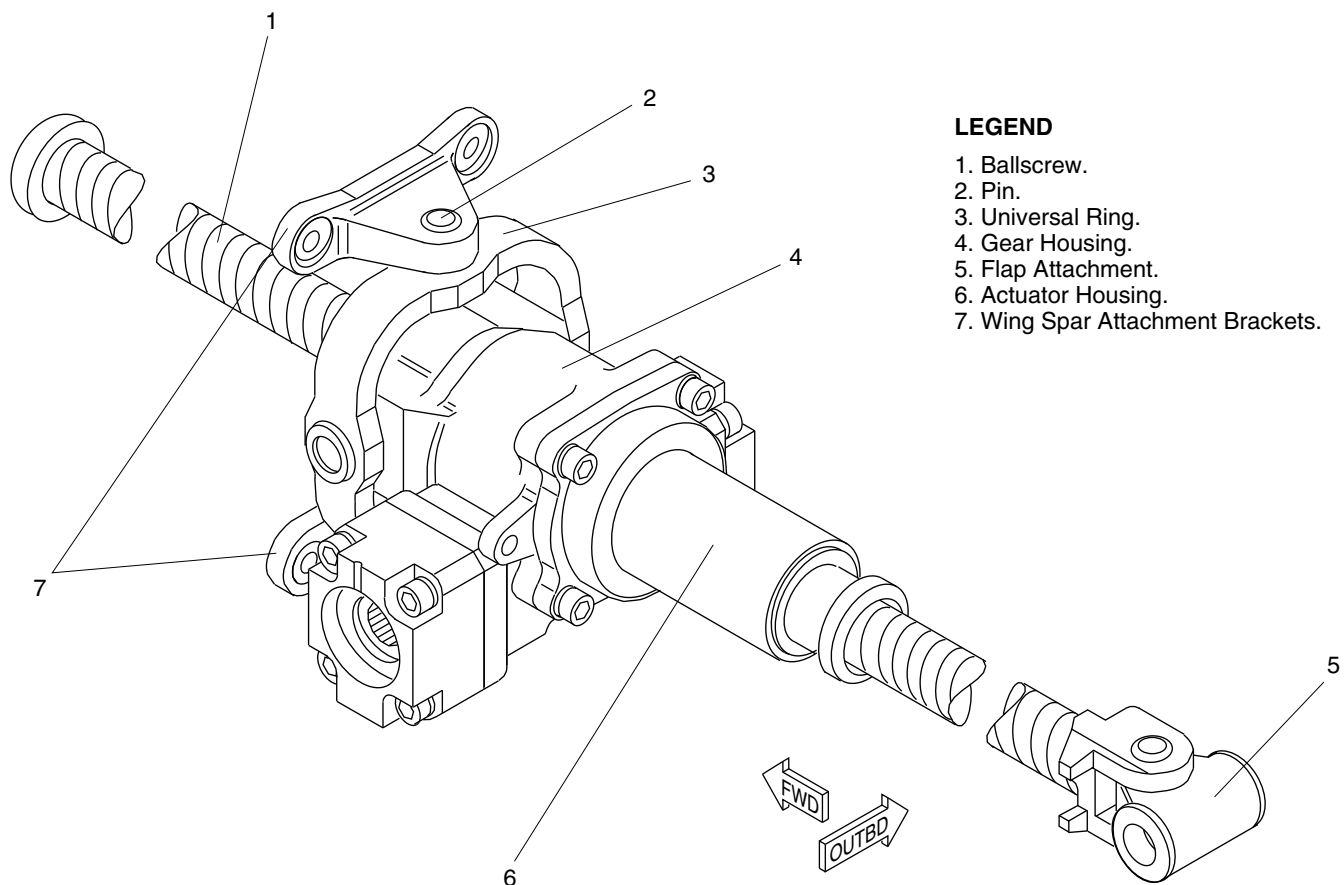
27-52-00

Config 001
Page 16
Sep 05/2021



DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Ballscrew.
- 2. Pin.
- 3. Universal Ring.
- 4. Gear Housing.
- 5. Flap Attachment.
- 6. Actuator Housing.
- 7. Wing Spar Attachment Brackets.

fs122a01.cgm

FLAP ACTUATOR PAGE 1
Figure 12

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-52-00
Config 001

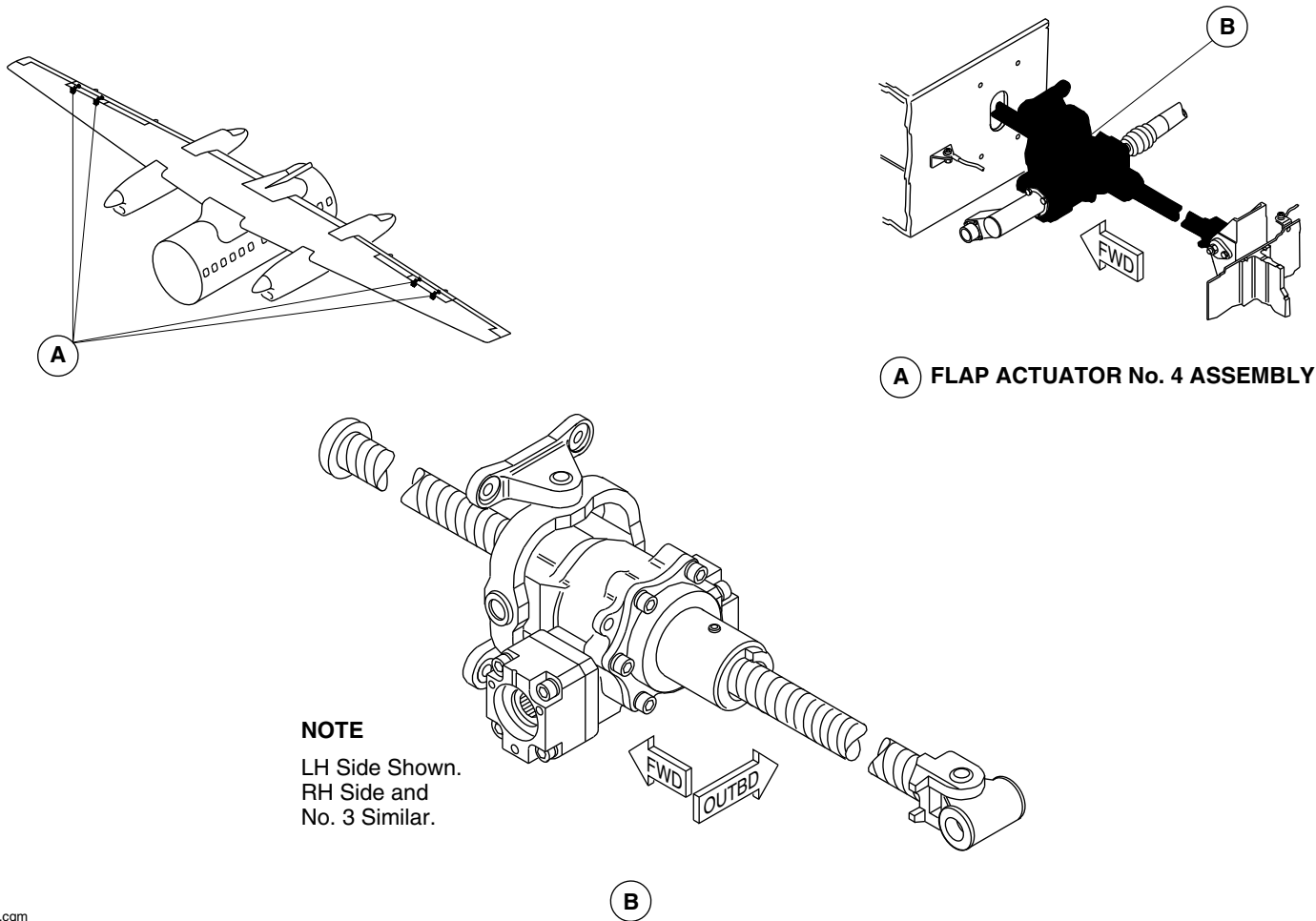
27-52-00

Config 001
Page 17
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



fs121a01.cgm

FLAP ACTUATORS PAGE 2
Figure 13

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-52-00
Config 001

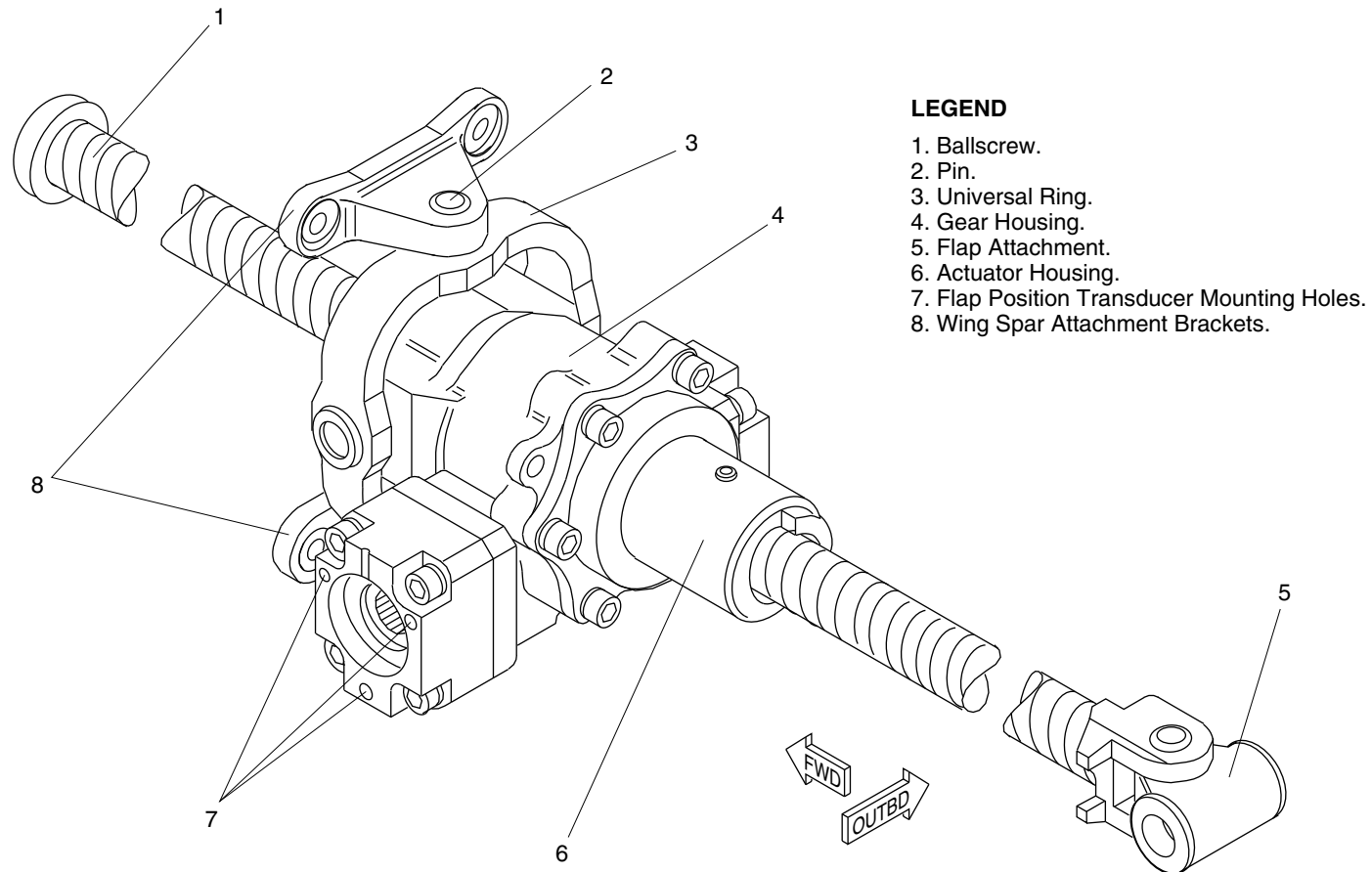
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Config 001
Page 18
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



fs126a01.cgm

FLAP ACTUATOR PAGE 2
Figure 14

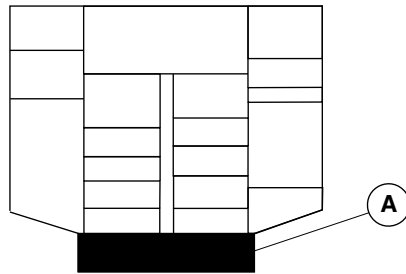
PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-52-00
Config 001

27-52-00

Config 001
Page 19
Sep 05/2021



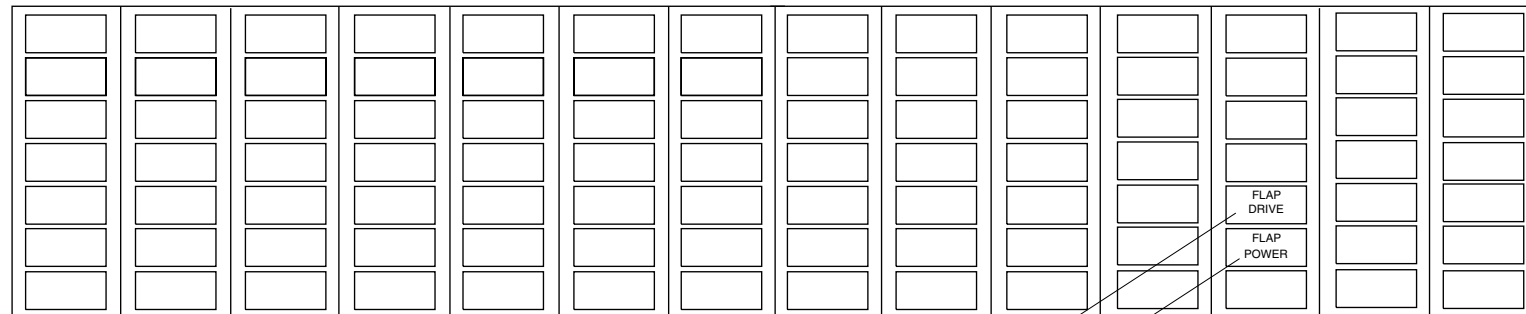
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



OVERHEAD CONSOLE

LEGEND

1. Flap Drive (Amber).
2. Flap Power (Amber).



fs531a01.cgm

FLAP SYSTEM CAUTION LIGHTS
Figure 15

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-52-00
Config 001

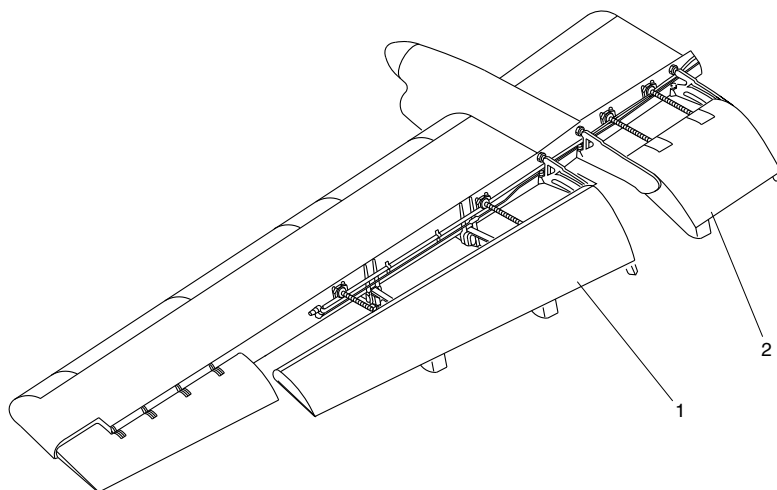
27-52-00

Config 001
Page 20
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Outboard Flap.
- 2. Inboard Flap.

NOTE

Left side shown.
Right side similar.

fs525a01.cgm

FLAP DRIVE / WING PAGE 1
Figure 16

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-52-00
Config 001

27-52-00

Config 001
Page 21
Sep 05/2021

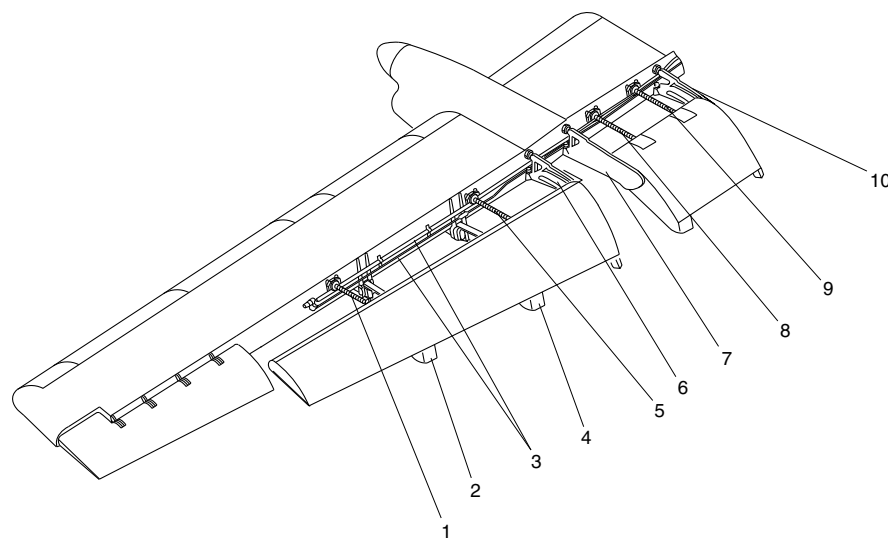


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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

LEGEND

1. No. 4 Flap Actuator.
2. No. 5 Flap Track.
3. Transmission Shafts.
4. No. 4 Flap Track.
5. No. 3 Flap Actuator.
6. No. 3 Flap Track.
7. No. 2 Flap Track.
8. No. 2 Flap Actuator.
9. No. 1 Flap Actuator.
10. No. 1 Flap Track.



fs542a01.cgm

FLAP DRIVE / WING PAGE 2
Figure 17

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-52-00
Config 001

27-52-00

Config 001
Page 22
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

27-53-00-001

FLAP POSITION INDICATOR

Introduction

The flap position indication system supplies flap position data to the flight compartment and to other aircraft systems for indication and monitoring purposes.

General Description

The flap position indication system supplies data to the Flap Control Unit (FCU), the Flight Control Electronic Control Unit (FCECU) and the Integrated Flight Cabinets (IFCs). The data is used for flap position monitoring and recording.

The flap position indication system has the components that follow:

- Unit, Flap Position Indication (FPIU) (27-53-01)
- Transducer, Flap Position (27-53-06)

Detailed Description

The flap position indication system has a Flap Position Indication Unit (FPIU) located below the cabin floor aft of the High Intensity Radio Frequency (HIRF) enclosure. The HIRF box is not installed on aircraft with ModSum 4-153780 or 4-126136 incorporated. The flap position indication system also has two flap position transducers (RVDTs) located on the outboard flaps.

The flap position indication system has two functions: [Refer to Figures 1 and 2.](#)

- Supplies indication of the current flap position to the copilot's Multi-Function Display indicator
- Monitors the position of the flaps during flap deployment and sends the position data to the FCU.

Unit, Flap Position Indication (FPIU)

[Refer to Figures 3 and 4.](#)

The FPIU receives inputs from the outer section (channel B) of the left and right wing tip flap position transducers (RVDTs). The FPIU then sends the output to the Indication and Monitoring System and to the avionics computers to perform the critical flight functions of flap position indication.

The FPIU is divided into two separate and independent analog channels on a single printed circuit board. The printed circuit board is contained in a case. The case is installed below the cabin floor and is attached to the aft side of the HIRF enclosure.

Each channel in the FPIU performs the functions that follow:

- Gives flap position indication to the IFCs and the FCECU
- Interfaces with, and supplies electrical excitation to the wing tip RVDTs
- Demodulates the signal from the wing tip RVDTs
- Performs an analog monitor of its applicable RVDT
- Supplies a signal to the Indication and Monitoring System.



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Transducer, Flap Position

[Refer to Figure 5.](#)

There are two flap position transducers. They are installed on the outboard flap actuator of each wing. Each flap position transducer is a two section Rotary Variable Differential Transformer (RVDT) consisting of channel A and channel B. Channel A is electrically supplied by the Flap Control Unit (FCU) and supplies flap position feedback to the flap control loop. Channel B is electrically supplied by the FPIU and supplies flap position data for indication purposes.

PSM 1-84-2A
EFFECTIVITY:
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Config 001

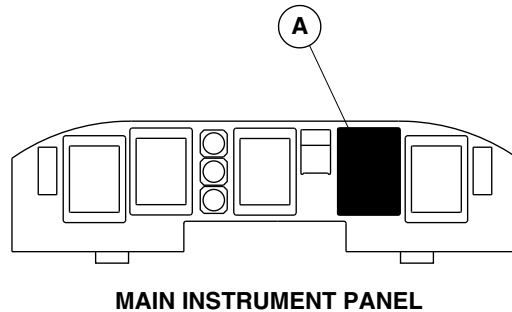
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Config 001
Page 3
Sep 05/2021



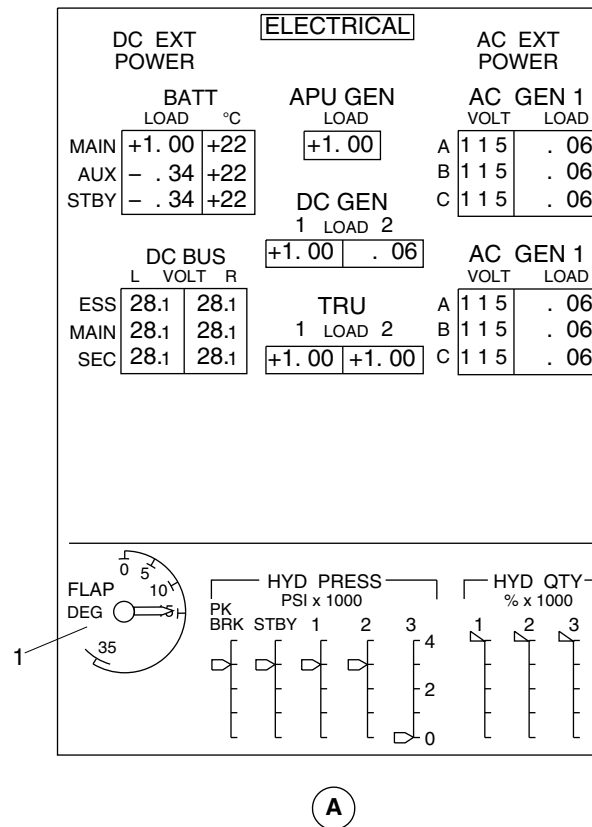
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OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Flaps Position Indication.



fs901a01.cgm

FLAP POSITION INDICATION / MULTIFUNCTION DISPLAY PAGE 1
Figure 1

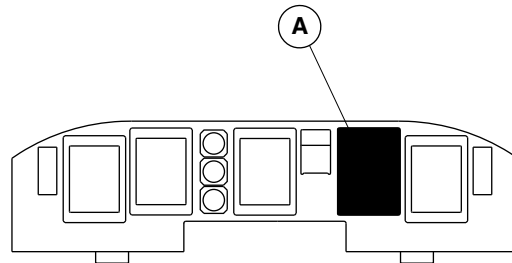
PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-53-00
Config 001

27-53-00

Config 001
Page 4
Sep 05/2021



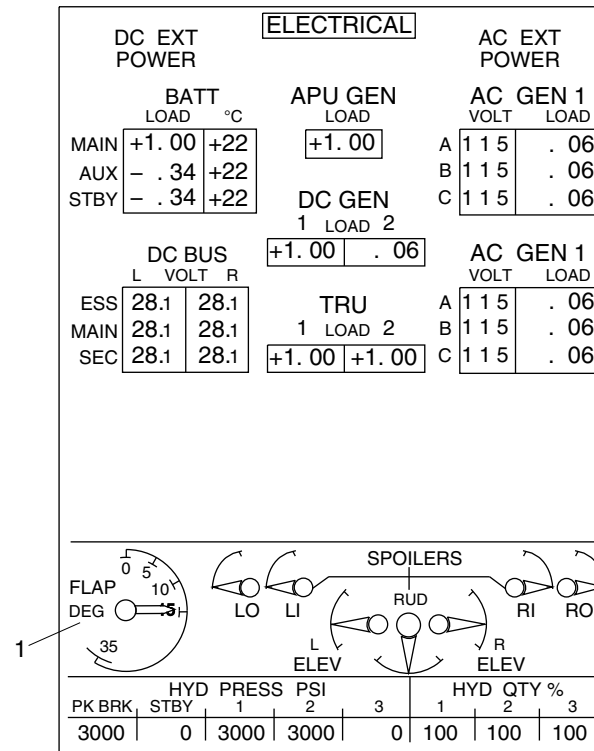
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



MAIN INSTRUMENT PANEL

LEGEND

1. Flaps Position Indication.



A

fs902a01.cgm

FLAP POSITION INDICATION / MULTIFUNCTION DISPLAY PAGE 2
Figure 2

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-53-00
Config 001

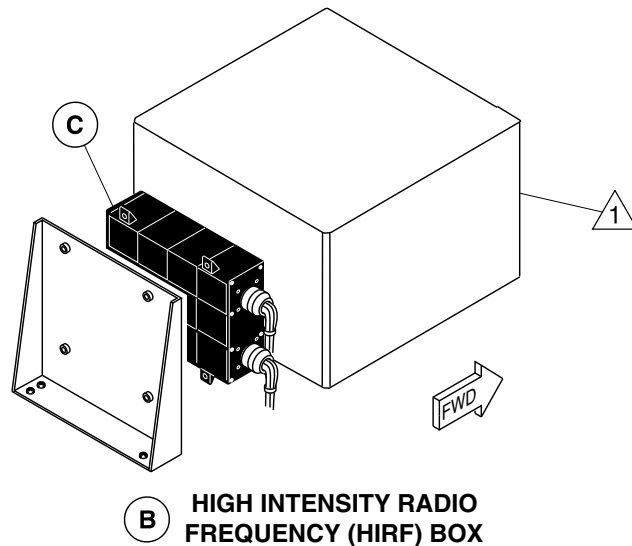
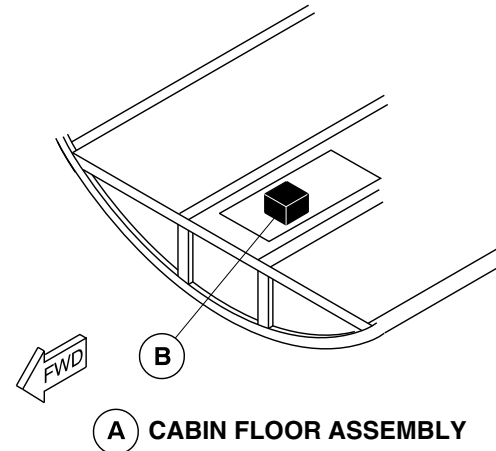
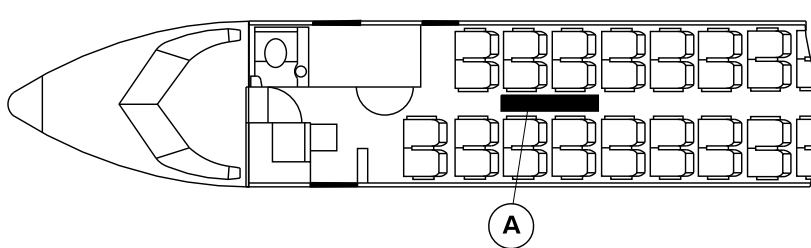
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Config 001
Page 5
Sep 05/2021



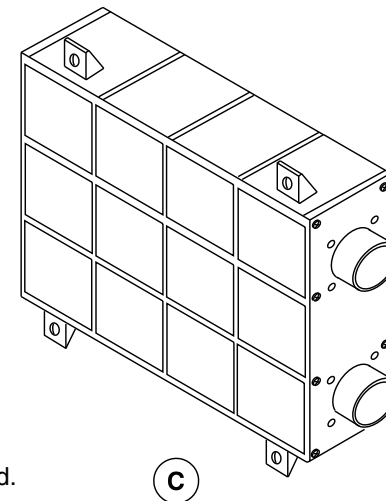
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



NOTE

1 The HIRF box is not installed on aircraft with
ModSum 4-153780 or 4-126136 incorporated.



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Flap Position Indication Unit (FPIU) Page 1
Figure 3

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-53-00
Config 001

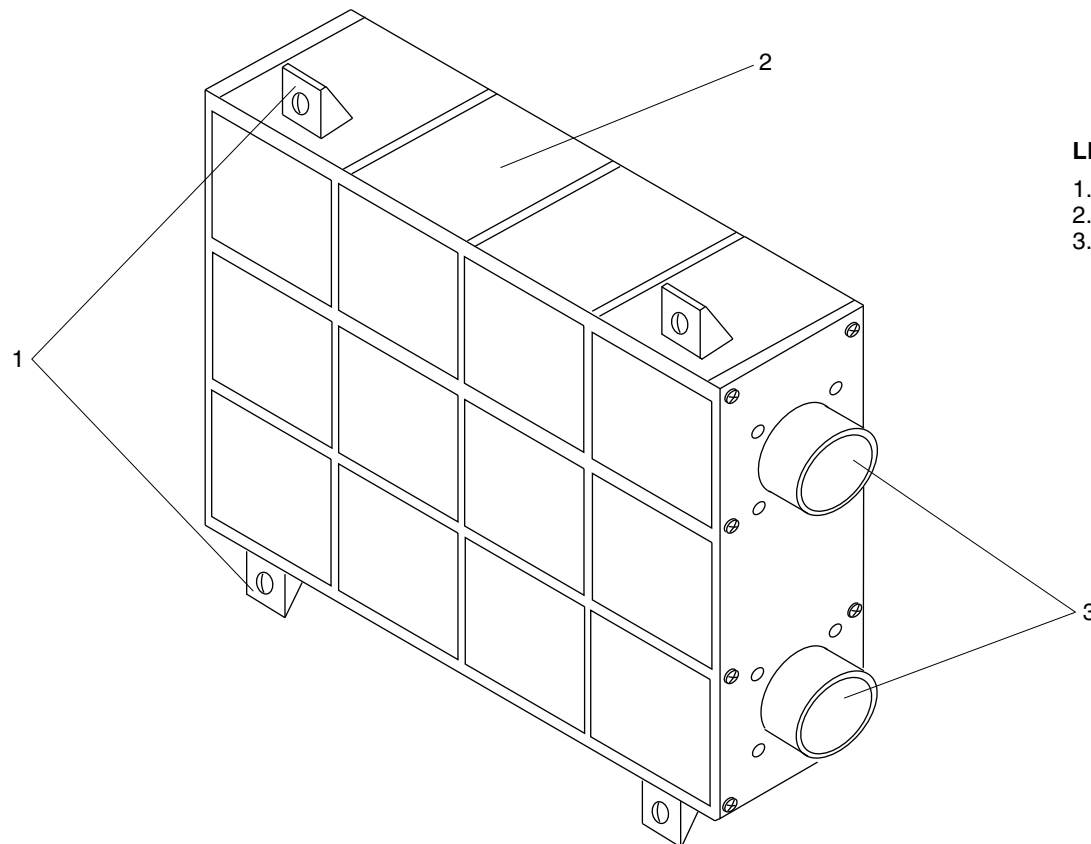
27-53-00

Config 001
Page 6
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Mounting Bracket.
- 2. Flap Position Indication Unit.
- 3. Electrical Connectors.

fs240a01.cgm

FLAP POSITION INDICATION UNIT (FPIU) PAGE 2
Figure 4

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-53-00
Config 001

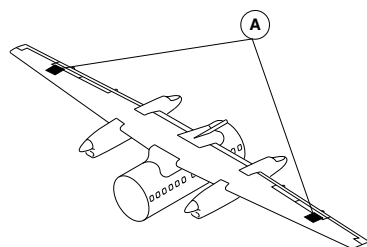
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Config 001
Page 7
Sep 05/2021



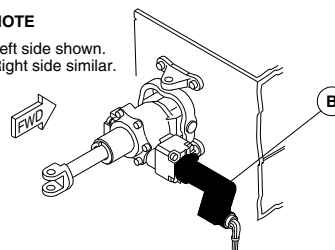
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OF CANADA LIMITED

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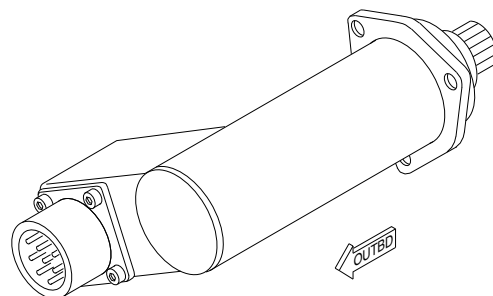


NOTE

Left side shown.
Right side similar.



A OUTBOARD FLAP ACTUATOR ASSEMBLY



B

fs080a01.cgm

FLAP POSITION TRANSDUCER
Figure 5

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-53-00
Config 001

27-53-00

Config 001
Page 8
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

27-54-00-001

FLAP SURFACES

Introduction

The flap surfaces extend inboard of the ailerons and aft from the trailing edges of both wings. The flaps give increased lift, when extended, during take off and landing

General Description

The flap surfaces include the components that follow:

- Flaps, Inboard (27-54-01)
- Flaps, Outboard (24-54-06).

Detailed Description

[Refer to Figure 1.](#)

The flaps are selected by the pilot during take-off and landing. When the flaps are retracted, they fit into the flap shroud along the trailing edge of the wing to form an aerodynamically clean wing surface.

There are two large chord flap sections on each wing. The inboard section has a constant chord of approximately 39 % of the overall wing chord. It is supported at each end by two hidden tracks; one on the fuselage and the other on the inboard side of the engine nacelle. The outboard section has a tapered chord which is also approximately 39% of the overall wing chord. It is supported by three

tracks; one on the outboard side of the engine nacelle and two at intermediate stations below the wing.

Flaps, Inboard

[Refer to Figure 2.](#)

The inboard flap is a single slotted, high lift Fowler action type flap. The inboard flap has a constant chord cross-section of approximately 39% of the total wing chord.

Flaps, Outboard

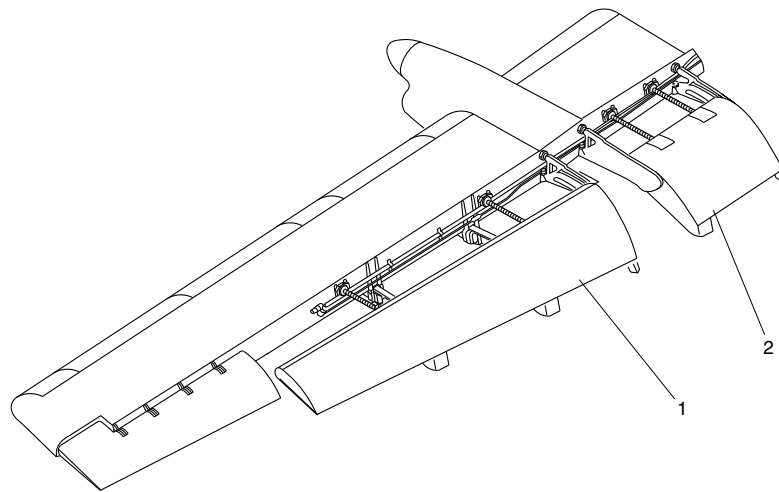
[Refer to Figure 3.](#)

The outboard flap is a single slotted, high lift Fowler action type flap. The outboard flap has a tapered chord cross-section of approximately 39% of the total wing chord.



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Outboard Flap.
- 2. Inboard Flap.

NOTE

Left side shown.
Right side similar.

fs525a01.cgm

FLAPS
Figure 1

PSM 1-84-2A
EFFECTIVITY:
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Config 001

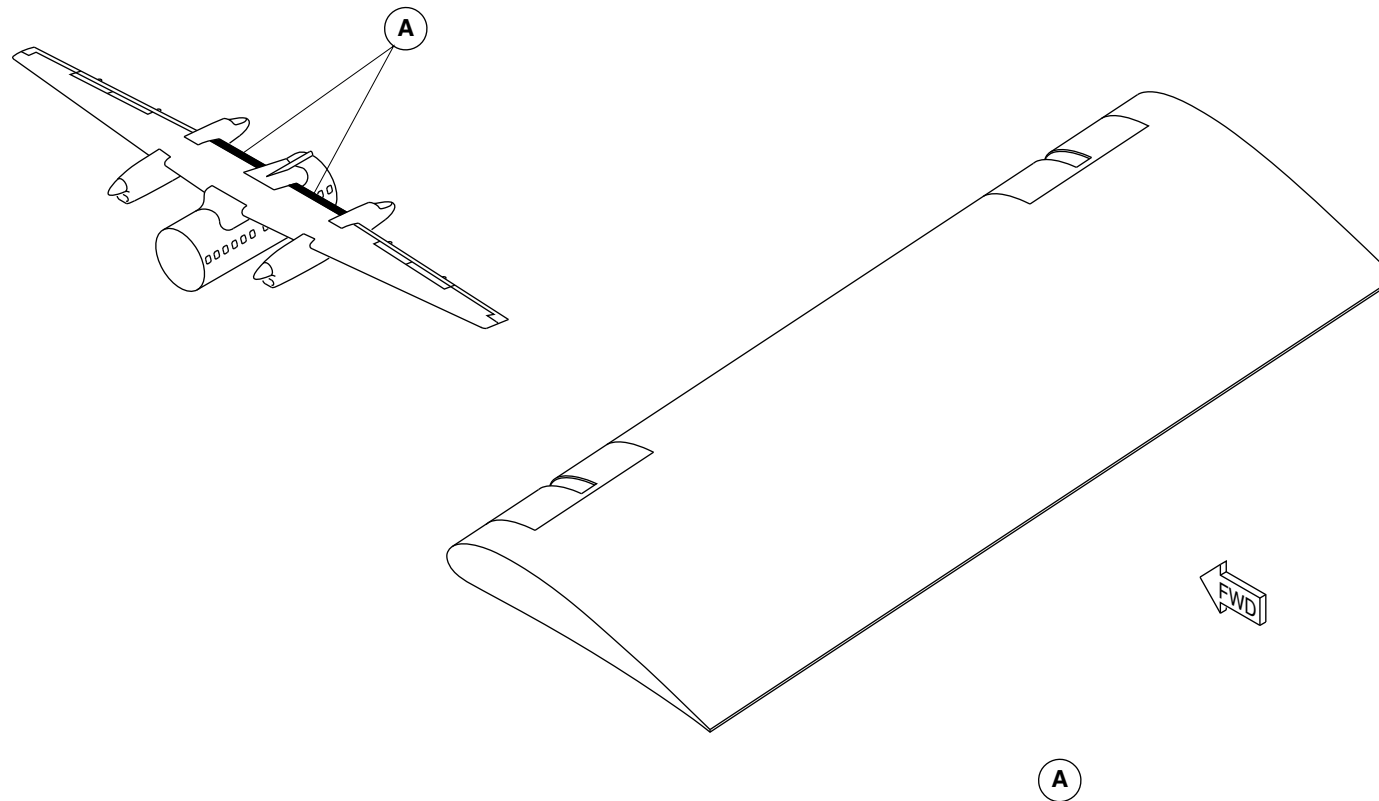
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Config 001
Page 3
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



fs188a01.cgm

NOTE
1 Left Side Shown.
Right Side Similar.

INBOARD FLAP
Figure 2

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-54-00
Config 001

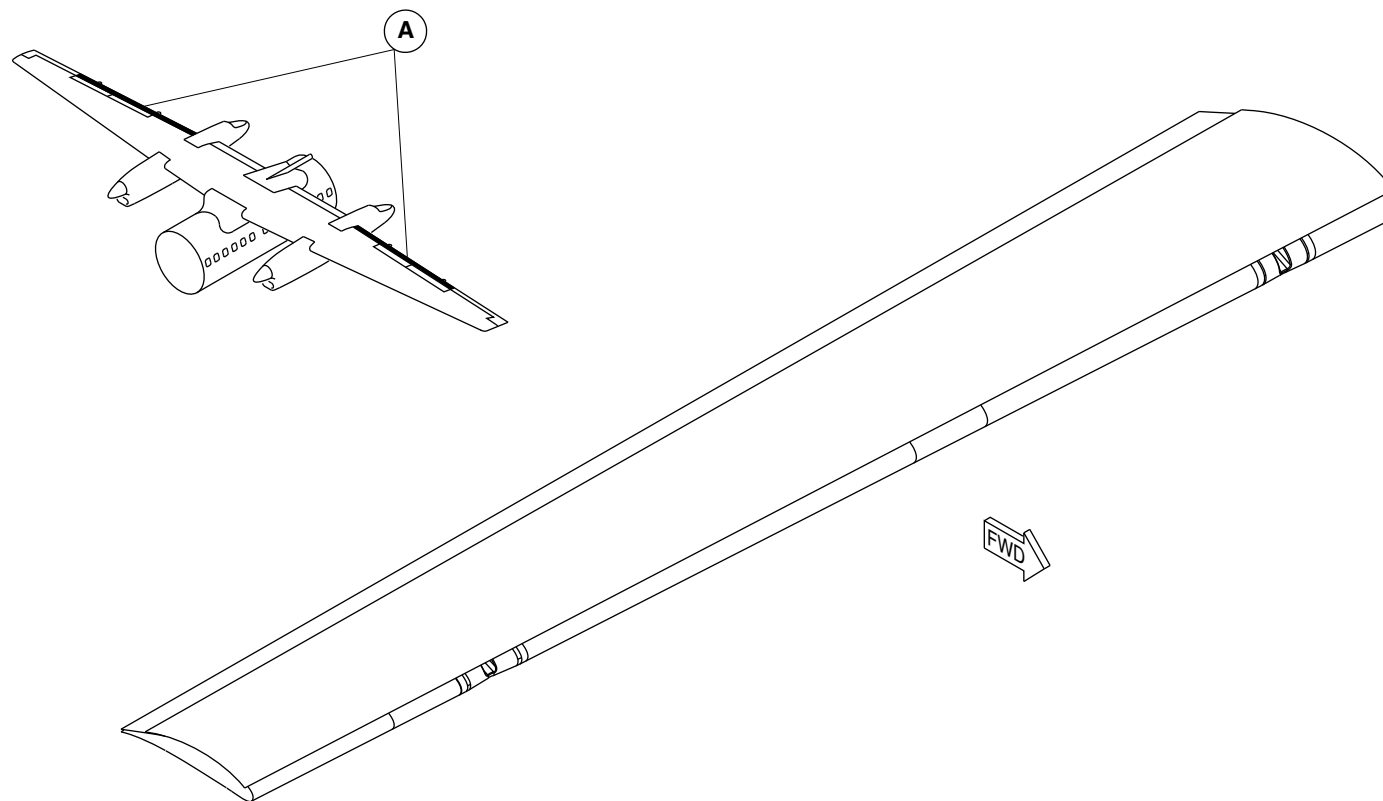
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Config 001
Page 4
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



fs241a01.cgm

A

1

NOTE

Right Side Shown.
Left Side Similar.

OUTBOARD FLAP
Figure 3

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-54-00
Config 001

27-54-00

Config 001
Page 5
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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See first effectivity on page 2 of 27-54-00
Config 001

27-54-00

Config 001
Page 6
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

27-60-00-001

SPOILER CONTROL SYSTEM

Introduction

The roll spoilers are a secondary flight control system. They are hydraulically powered and operate with the aileron control system to give lateral control of the aircraft in flight. The roll spoilers also decrease the lift made by the wings after the aircraft has landed.

General Description

There are two hydraulically powered roll spoilers on each wing. The roll spoilers operate with the ailerons to give lateral control of the aircraft in flight. Rotary movement of either the pilot's or the copilot's handwheels operates the ailerons and the spoilers together through an interconnect rod joining the two control columns. Roll spoilers only move in the up direction and in proportion to the movement of the upgoing aileron.

The Automatic Flight Control System (AFCS) can also operate the spoiler control system through the autopilot servo. The servo is connected to the forward aileron quadrant under the flight compartment floor (refer to SDS 27-10-00).

A roll disconnect system is supplied in the event a cable or linkage jam occurs (refer to SDS 27-16-00).

The No.1 hydraulic system supplies power to operate the inboard spoilers and the No.2 hydraulic system powers the outboard spoilers.

There are three modes of spoiler operation: Flight Mode, Ground Mode, and Taxi Mode.

- Flight mode: The spoiler panels respond to the inputs from the control handwheels or from the autopilot. At airspeeds less than 170 kt (315 km/h), both the inboard and outboard spoilers are operational. At airspeeds greater than 170kt (315 km/h), the Flight Control Electronic Control Unit (FCECU) locks out the outboard spoilers. The inboard spoilers remain operational at all airspeeds.
- Ground Mode: With the engine power levers at or below flight idle plus 12 degrees and the two Weight-On-Wheels signals (WOW1 and WOW2) supplied from the Proximity Sensor Electronic Unit (PSEU), the FCECU energizes all the lift dump valves. All four spoiler panels will fully extend and the lift made by the wings will decrease. The FLIGHT/TAXI switch must be in the FLIGHT position.
- Taxi Mode: With all the conditions identical to the ground mode and the FLIGHT/TAXI switch set to TAXI; the four spoiler panels will fully retract.

Detailed Description

The spoilers control system includes the sub-systems that follow:

- Roll Spoilers System (27-61-00)
- Ground Spoiler System (27-62-00)
- Spoiler Panels (27-63-00).

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-60-00
Config 001

27-60-00

Config 001
Page 2
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Roll Spoiler System

Refer to Figures 1 and 2.

Rotary motion of the pilot's handwheel operates a cable system connected to pushrods, levers, clutches, quadrants and Power Control Units (PCUs). The PCUs extend or retract the spoiler panels using hydraulic system pressure. The No.1 hydraulic system supplies power to operate the inboard spoilers and the No.2 hydraulic system supplies power to the outboard spoilers. Roll spoilers move only in the up direction and in proportion to the movement of the upgoing aileron.

The pilot's handwheel operates the spoilers through the spoiler control circuit. The copilot's handwheel operates the ailerons through the aileron control circuit. The two control circuits are joined through a control rod connected at the base of each control column. This allows the aircraft to be operated from either the pilot's or copilot's position. The ailerons and the roll spoilers can be disconnected from each other if a jam occurs in either system (refer to SDS 27-16-00). The aircraft can then be operated with limited roll control by using the remaining system.

Refer to Figures 3 and 4.

The FCECU controls the outboard spoiler lockout function. If the airspeed is increasing and exceeds 170 kt (315 km/h), the FCECU depressurizes the outboard spoiler PCUs. The FCECU also inhibits the ROLL SPLR OUTBD HYD caution light. If the airspeed is decreasing and falls below 165 kt (306 km/h), the FCECU pressurizes the outboard spoiler PCUs. The ROLL SPLR OUTBD HYD caution light will turn on if the No. 2 system hydraulic pressure falls below 900 psi. (6205 kPa).

If the FCECU fails to lock out the outboard spoilers above 185 kt (343 km/h), it will turn on the SPLR OUTBD caution light. If the FCECU inadvertently locks out the outboard spoilers below 150 kt (278 km/h), it will turn on both the SPLR OUTBD and the ROLL SPLR OUTBD HYD caution lights.

The FCECU receives its airspeed information from the Air Data Units (ADUs).

A spoiler cable sensor is installed on the fuselage spoiler cables. If the tension of the fuselage spoiler cables decreases to zero, the cable sensor will cause the spoilers to retract. Also, the ROLL SPLR INBD HYD and ROLL SPLR OUTBD HYD caution lights will come on.

Ground Spoiler System

Refer to Figure 5.

The roll spoiler system automatically changes from the flight mode to the ground mode after the aircraft has landed. The roll spoilers operate symmetrically on landing (overriding flight mode operation). This action decreases or "dumps" the lift from the wings and increases braking efficiency.

Ground mode operation occurs when:

- The Proximity Sensor Electronic Unit (PSEU) gives the Weight-On-Wheels (WOW1 and WOW2 respectively) signal to the Flight Control Electronic Control Unit (FCECU)
- Power levers 1 and 2 are at or below flight idle plus 12 degrees
- The spoiler FLIGHT/TAXI switch is in the FLIGHT position.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

The FCECU and the PSEU control the lift dump function. The FCECU left channel controls the two inboard lift dump valves and the right channel controls the two outboard lift dump valves. Each pair of lift dump valves are solenoid operated and are connected hydraulically in series. When both valves are open, hydraulic pressure is supplied to the extend side of the related spoiler PCU.

The FCECU sends a high side signal (+28 V) to the lift dump valve solenoids if the both power levers are at or below flight idle plus 12 degrees. The FLIGHT/TAXI switch must also be in the FLIGHT position. The PSEU sends a low side signal (ground) to the lift dump valve solenoids if the aircraft has landed and both Weight-On-Wheels (WOW1 and WOW2) are present. Therefore, if all the above conditions are met, all four spoilers will extend.

The FLIGHT/TAXI switch interfaces with the right channel of the FCECU. The signal is then shared with the left channel for redundancy. After the aircraft lands and the spoilers extend, the FLIGHT/TAXI switch is selected to the TAXI position. This action de-energizes the spoiler lift dump valves and the spoilers retract. After retraction, the FCECU performs a Built-in-Test (BIT). It energizes each lift dump valve solenoid sequentially to check if any valve is stuck in the open position.

When all four spoiler panels extend (greater than 65 degrees) in the ground mode, the FCECU sends a signal to turn on the spoiler advisory lights (ROLL OUTBD and ROLL INBD). The FCECU turns on the ground caution lights (ROLL SPLR OUTBD GND and ROLL SPLR INBD GND) if the related spoiler panels do not extend within one second after touchdown. The FCECU will also turn on the ground caution lights if the spoilers do not retract in the TAXI mode or if BIT finds a failure in the lift dump valves. For the ground caution lights to come on, all the appropriate conditions for ground spoiler deployment must be satisfied.

Spoiler Panels

[Refer to Figures 6 and 7.](#)

Four spoiler panels are installed on the upper surface of the wings forward of the outboard flaps. There are two inboard panels and two outboard panels. Each panel is joined to the upper wing structure through three hinge fittings. The hinge fittings are extended to create an airgap when the panels are deployed. The PCUs are attached to the centre hinge through a non-adjustable short link with self-aligning bearings at either end. Full spoiler deflection is 75 degrees.

PSM 1-84-2A

EFFECTIVITY:

See first effectivity on page 2 of 27-60-00

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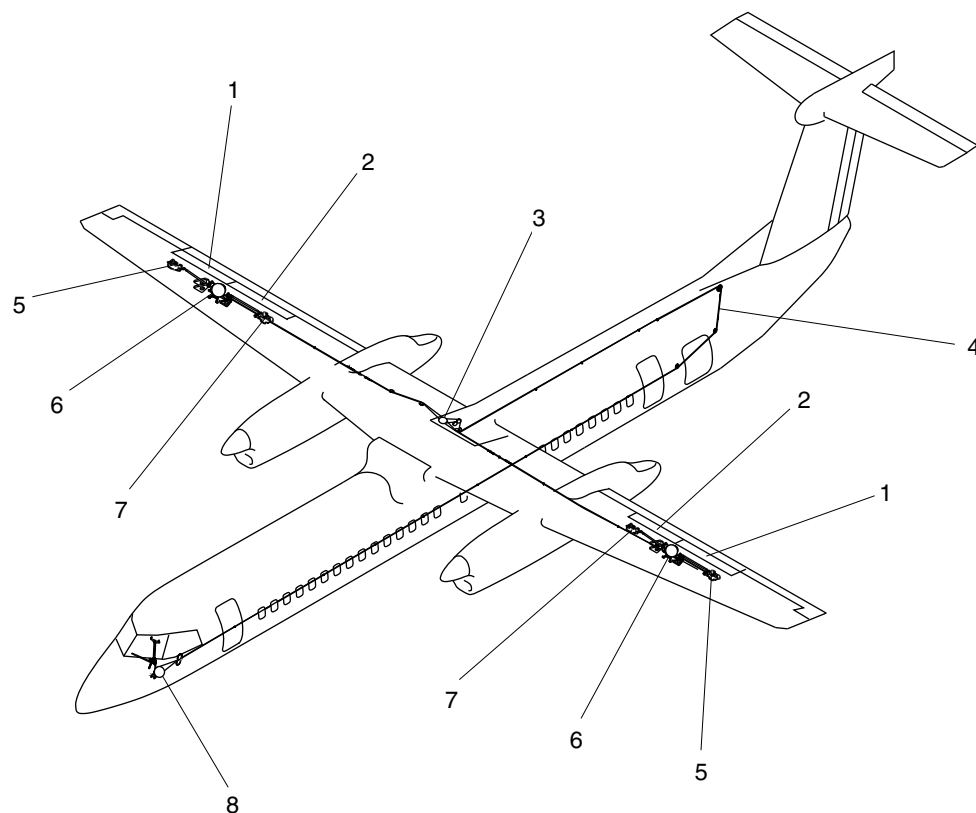
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Config 001
Page 4
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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LEGEND

1. Outboard Spoiler.
2. Inboard Spoiler.
3. Spoiler Splitter Quadrant.
4. Spoiler Control Cables.
5. Outboard Spoiler PCU.
6. Spoiler Clutch.
7. Inboard Spoiler PCU.
8. Spoiler Forward Quadrant.

SPOILER CONTROL SYSTEM PAGE 1
Figure 1

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-60-00
Config 001

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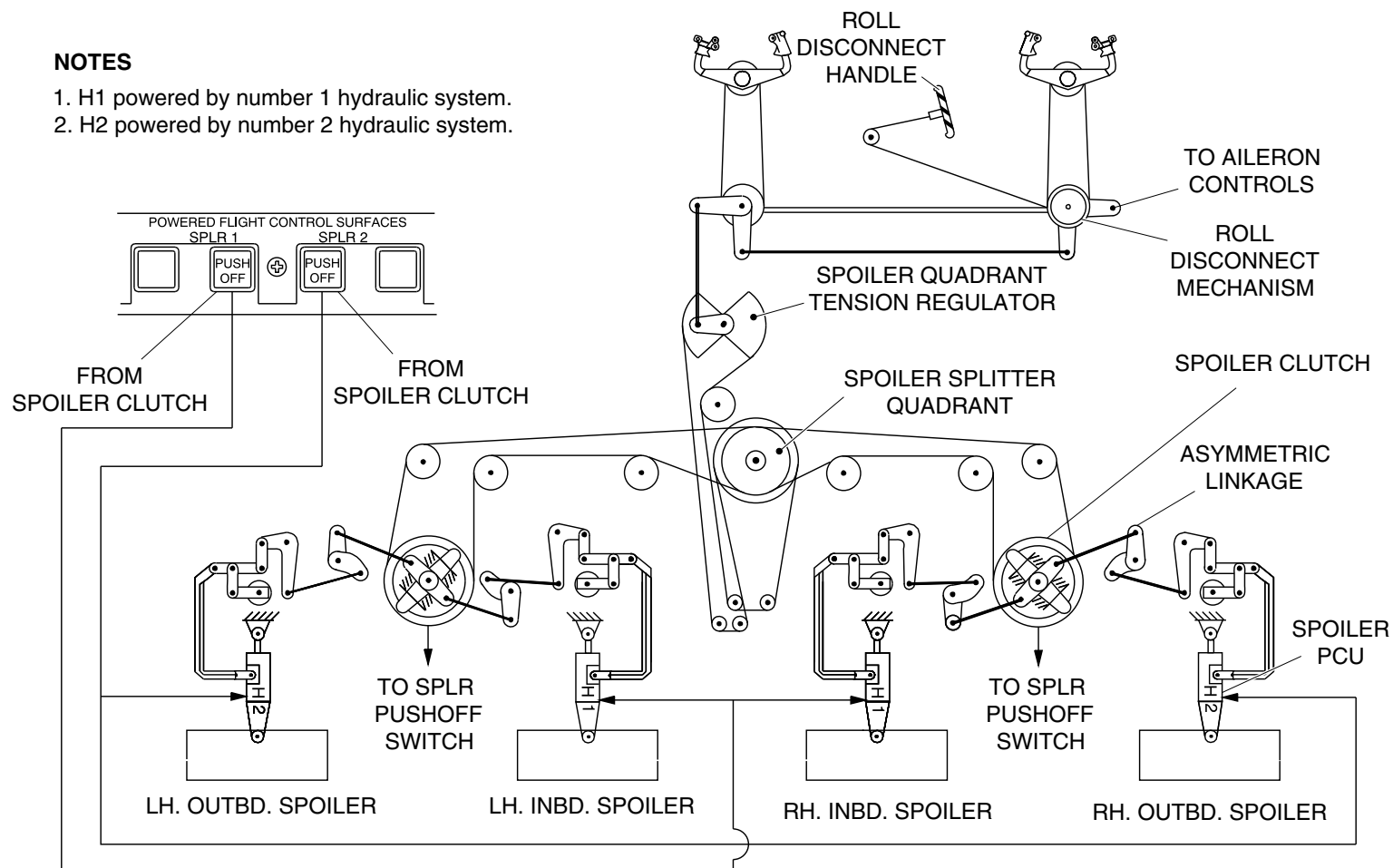
Config 001
Page 5
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

NOTES

1. H1 powered by number 1 hydraulic system.
2. H2 powered by number 2 hydraulic system.



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Spoiler Control System Page 2
Figure 2

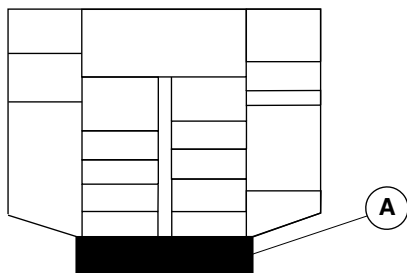
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27-60-00

Config 001
Page 6
Sep 05/2021



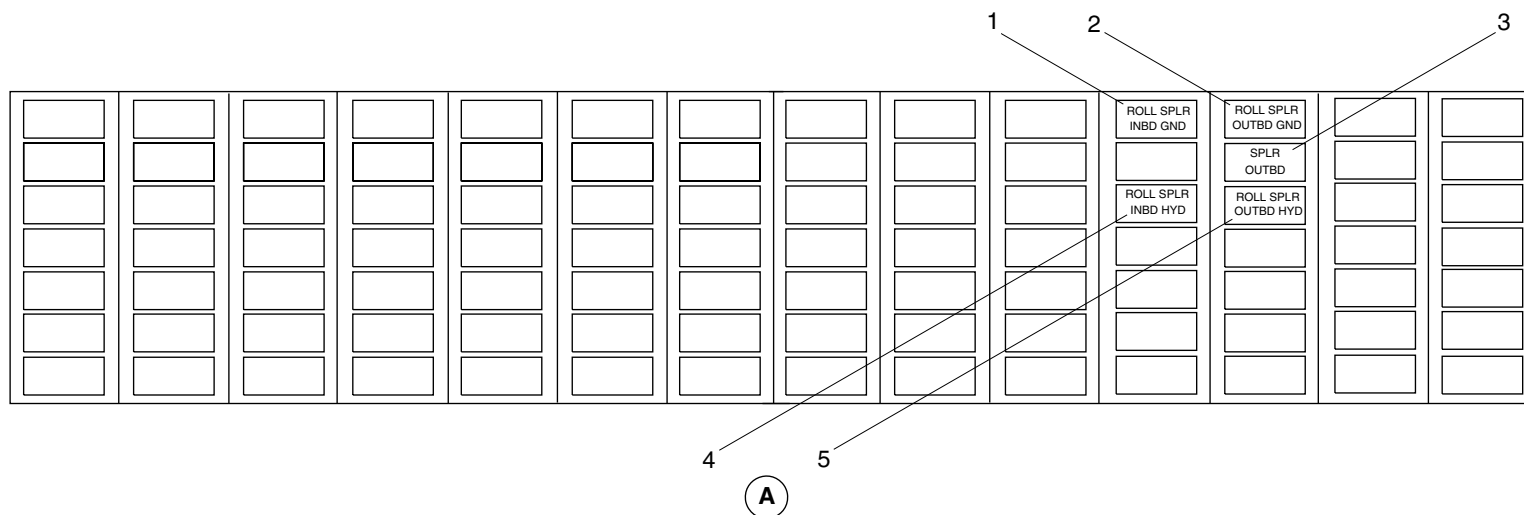
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



OVERHEAD CONSOLE

LEGEND

1. Roll Spoiler Inboard Ground (Amber).
2. Roll Spoiler Outboard Ground (Amber).
3. Spoiler Outboard (Amber).
4. Roll Spoiler Inboard Hydraulic (Amber).
5. Roll Spoiler Outboard Hydraulic (Amber).



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SPOILER CAUTION LIGHTS
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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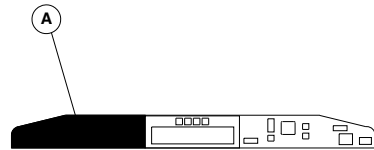
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Config 001
Page 7
Sep 05/2021



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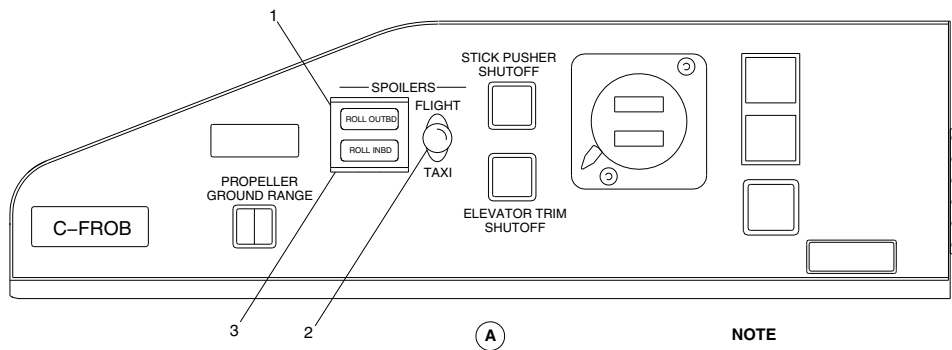
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



GLARESHIELD PANEL

LEGEND

1. Roll Outboard Annunciator Light.
2. Flight/Taxi Switch.
3. Roll Inboard Annunciator Light.



NOTE

Pilot side shown.
Copilot side similar.

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SPOILER ANNUNCIATOR LIGHTS
Figure 4

PSM 1-84-2A
EFFECTIVITY:
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Config 001

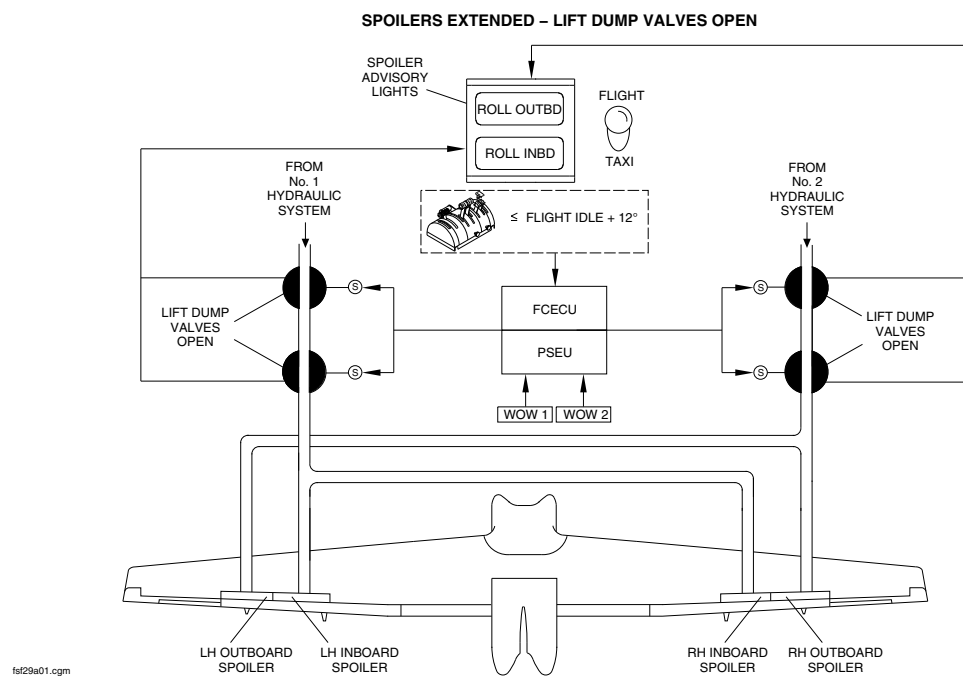
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Config 001
Page 8
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



GROUND SPOILER MODE
Figure 5

PSM 1-84-2A
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Config 001

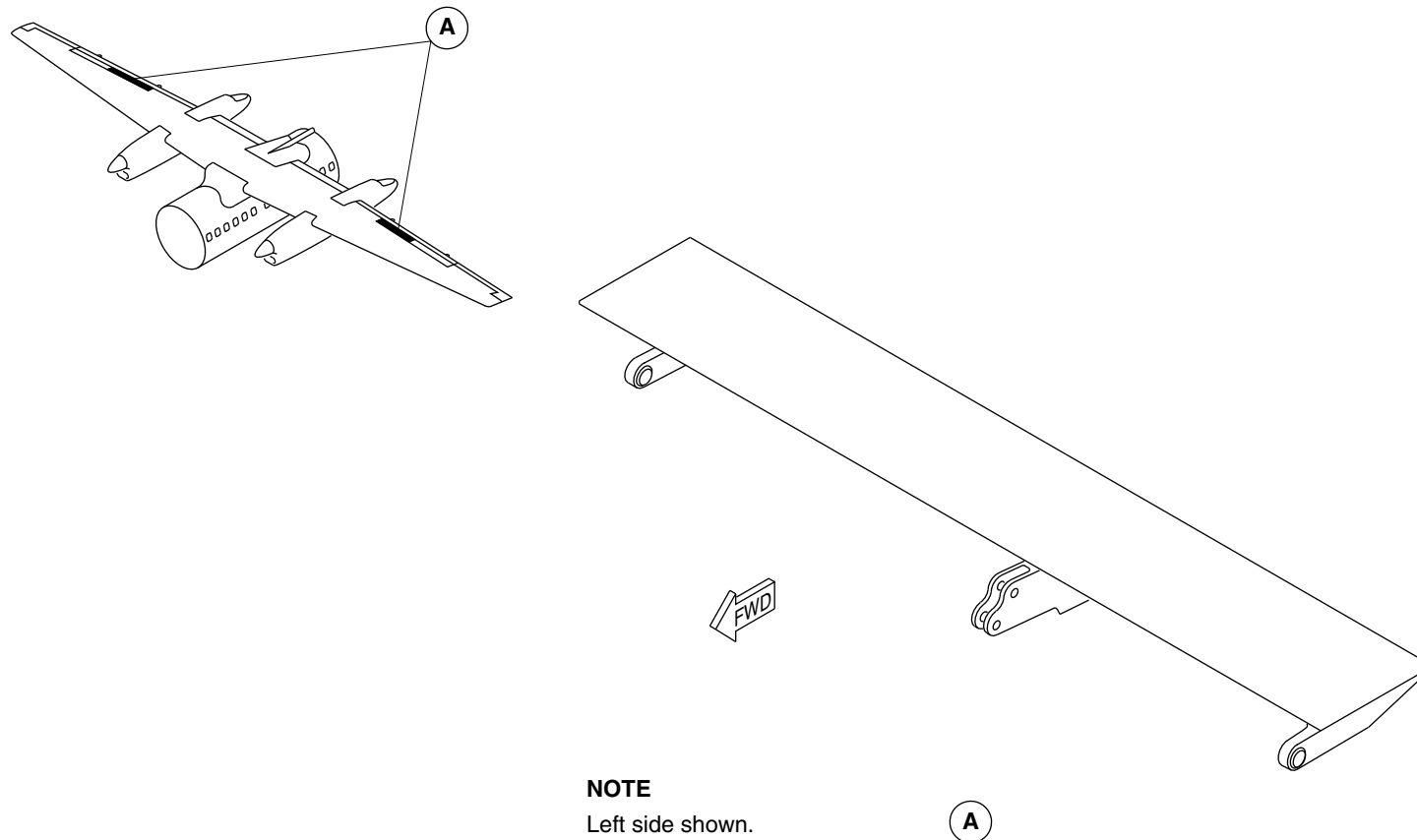
27-60-00

Config 001
Page 9
Sep 05/2021



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NOTE

Left side shown.
Right side similar.

fs091a01.cgm

INBOARD SPOILER PANELS
Figure 6

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-60-00
Config 001

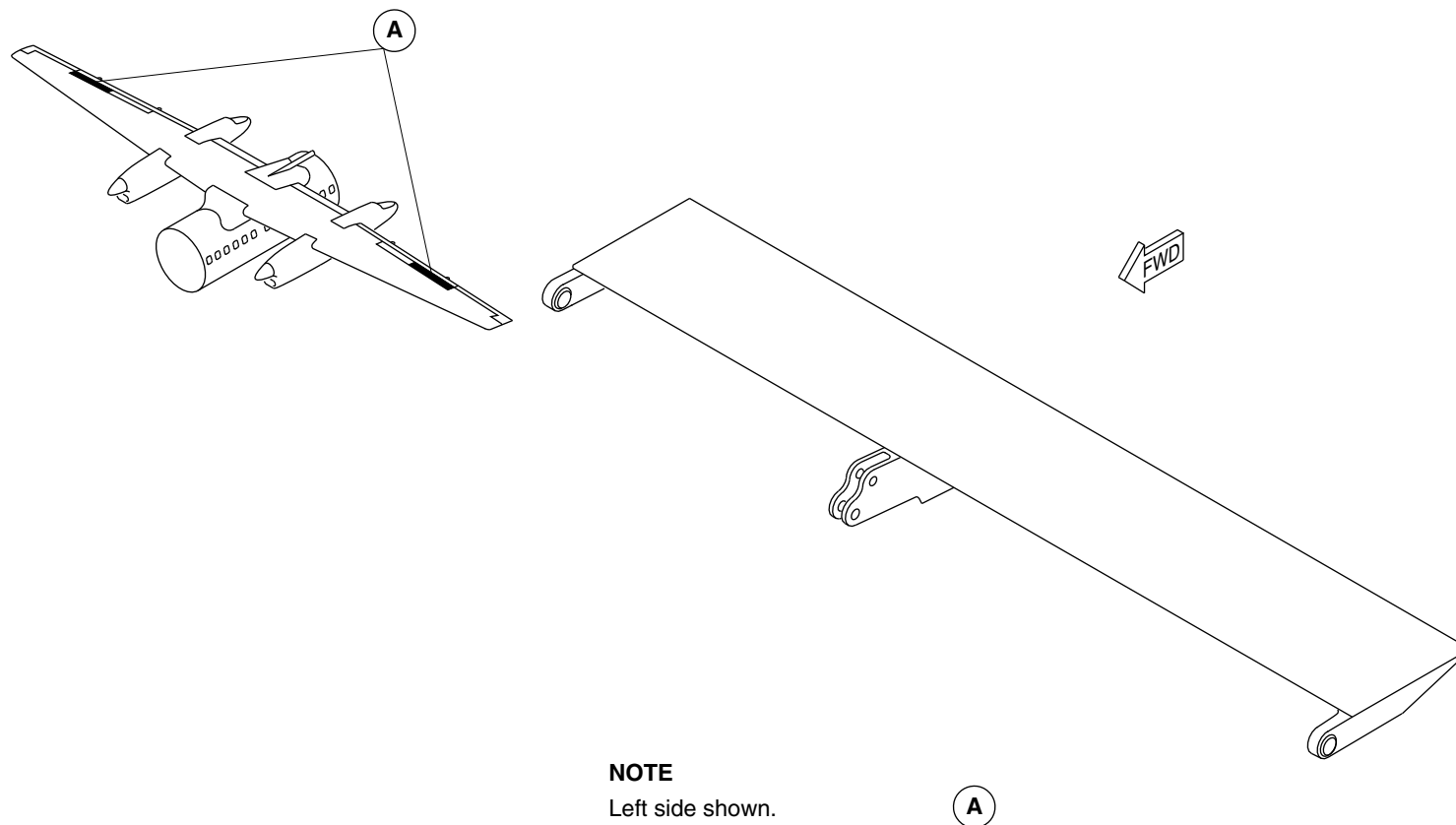
27-60-00

Config 001
Page 10
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



NOTE

Left side shown.
Right side similar.

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OUTBOARD SPOILER PANELS
Figure 7

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-60-00
Config 001

27-60-00

Config 001
Page 11
Sep 05/2021



DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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PSM 1-84-2A
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See first effectivity on page 2 of 27-60-00
Config 001

27-60-00

Config 001
Page 12
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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27-61-00-001

ROLL SPOILERS SYSTEM

Introduction

The roll spoilers are a secondary flight control system. They are hydraulically powered and operate with, and in proportion to, the aileron control system to give lateral control of the aircraft in flight. The roll spoilers also decrease the lift made by the wings on landing.

General Description

There are two hydraulically powered roll spoilers on each wing. The roll spoilers operate with the ailerons to give lateral control of the aircraft in flight. Rotary movement of either the pilot's or the copilot's control handwheels operates the ailerons and the spoilers at the same time through an interconnect rod joining the two control columns. The ailerons and the roll spoilers can be disconnected from each other if a jam occurs in either system. The aircraft can then be operated with limited roll control by using the unjammed system.

Movement of the handwheels operates a cable system connected to pushrods, levers, clutches, quadrants and Power Control Units (PCUs). The PCUs extend or retract the spoiler panels using hydraulic system pressure. The No.1 hydraulic system supplies power to operate the inboard spoilers and the No.2 hydraulic system supplies power to the outboard spoilers. Roll spoilers move up in proportion to the movement of the up going aileron.

Caution lights indicate failures and advisory lights indicate operational modes.

There are three modes of spoiler operation: Flight Mode, Ground Mode (refer to SDS 27-62-00), and Taxi Mode.

The spoilers control system includes the sub-systems that follow:

- Roll Spoilers System (27-61-00)
- Ground Spoilers System (27-62-00)
- Spoiler Panels (27-63-00).

Detailed Description

[Refer to Figures 1 and 2.](#)

The roll spoiler system, during flight, helps the ailerons to keep lateral control and, on the ground, supplies lift dump. Rotary movement of the control wheel is transmitted by a chain and push/pull rod, to the spoiler quadrant and tension regulator at the base of the pilot's control column.

On aircraft with ModSum 4-309210 OR SB84-31-33 OR SB84-31-39 OR SB84-31-40 incorporated ([Refer to Figure 3.](#): The push/pull rod in the roll spoiler system is replaced with a force sensor. The force sensor is installed between the left control column and the spoiler quadrant. The spoiler quadrant is installed below the floor, on the left side of the forward fuselage. The force sensor supplies a signal in proportion to the force applied to the handwheel. The signal transmitted by the force sensor is recorded by the Flight Data Recorder.

Control cables are routed from the quadrant to a splitter quadrant installed on the rear spar. Left and right cables, routed along the aft face of the rear spar on each wing, connect to the spoiler clutches.

PSM 1-84-2A

EFFECTIVITY:

See first effectivity on page 2 of 27-61-00

Config 001

27-61-00

Config 001

Page 2

Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

The clutches are connected to a series of pushrods, levers, and quadrants. They give mechanical inputs to both inboard and outboard spoiler actuators and operate as control quadrants in normal operation. The actuators extend or retract to deploy the spoiler panels. A clutch in the pulley of each mechanism can disengage and isolate either inboard or outboard spoilers if PCU linkage jamming occurs. The input to the applicable Power Control Unit (PCU) is cancelled by slippage in the clutch. When the clutch slips, the applicable roll spoiler advisory switchlight (SPLR1 or SPLR 2 PUSH OFF) comes on. Pushing the SPLR1 PUSH OFF switchlight depressurizes the inboard spoiler PCUs. Pushing the SPLR2 PUSH OFF switchlight depressurizes the outboard spoiler PCUs. With the PCUs depressurized, the spoilers retract to 0 degree position (flush with the wing surface).

[Refer to Figure 4.](#)

The No.1 hydraulic system supplies power to operate the inboard spoilers and the No.2 hydraulic system powers the outboard spoilers.

FLIGHT MODE

At speeds above 170 kt (315 km/h) only the inboard spoilers operate, the outboard spoilers are disabled by the right side channel of the Flight Control Electronic Control Unit (FCECU) energizing the No.2 actuator unload valve. At decreasing airspeeds below 165 kt (306 km/h), both inboard and outboard spoilers operate.

TAXI MODE

[Refer to Figure 5.](#)

At the end of the aircraft landing roll, the pilot can select the FLIGHT/TAXI switch to TAXI. This will energize the No.1 and No.2 actuator unload valves and all the spoilers will retract. The switch will

automatically move to FLIGHT when power levers are advanced to FLIGHT IDLE +12 degrees.

[Refer to Figure 6.](#)

If the No.1 hydraulic system fails, the ROLL SPLR INBD HYD caution light comes on. If the No.2 hydraulic system fails, the ROLL SPLR OUTBD HYD caution light comes on. The pilot will push the appropriate glareshield switchlight (SPLR1 or SPLR 2 PUSH OFF) to deactivate the failed spoiler system.

Spoiler Quadrant

[Refer to Figures 7 and 8.](#)

The spoiler quadrant is installed at the base of the pilots control column and connected to the interconnect rod. It includes a tension regulator, which maintains constant cable tension under all temperature conditions. The secondary control stops, and a rig pin hole to lock the quadrant in neutral are included in the quadrant.

Spoiler Splitter Quadrant

[Refer to Figures 9 and 10.](#)

The quadrant is attached to the aft face of the wing rear spar. It has two brackets and an assembly incorporating three pulleys. The larger lower pulley is the attachment point for the fuselage control cable. Roll inputs from the fuselage cable rotates all three pulleys, the middle and top pulley transfer this input to the wing control cables.



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Clutch Units

Refer to Figures 11 and 12.

The clutch units, one in each wing are installed centrally below the two spoiler panels. The clutch units operate the spoiler actuator input levers through two pushrods and one bellcrank. The units contain clutch mechanisms to allow for a jam in the controls for either spoiler panel, while keeping control over the remaining spoiler system. If the linkage jams, a force of approximately 27 lb (12.25 kg) will cause disengagement of the clutch and the clutch will slip. When the clutch slips it operates a microswitch, causing the inboard or outboard spoiler switchlight (SPLR1 or SPLR 2 PUSH OFF) to come on. Pressing the switchlight will depressurize the applicable spoiler system.

Inboard / Outboard Power Control Units

Refer to Figures 13 and 14.

The Power Control Units (PCUs) transfer mechanical input commands into hydraulic inputs. The input lever operates a servo valve which directs hydraulic pressure to the actuator piston. A command input moves the actuator rod with an attached feed back linkage. Actuator travel stops when the actuator rod position corresponds to the input lever position.

The PCU has four ports:

- Extend Port
- Retraction or Hold Down Port (Flight Spoiler Mode)
- Return Port
- Lift Dump Port (Ground Spoiler Mode).

The outboard spoiler PCU is powered by the No. 2 hydraulic system, No. 1 system powers the inboard spoiler PCU. Internal piston end stops limit maximum actuator travel during extension and retraction. The spoiler panels can extend to a maximum of 75 degrees. The PCU's are mounted on brackets attached to the wing rear spar.

Unload Valves

Refer to Figures 15 and 16.

The No.1 unload valve is located on the left side of the fuselage in the wing root area. The No.2 unload valve is located on the right side of the fuselage also in the wing root area. The No.1 unload valve controls both inboard spoiler PCU's, and the NO.2 unload valve controls both outboard spoiler PCU's.

In Flight Mode, the No.2 unload valve is operated electrically through the Flight Control Electronic Control Unit (FCECU), to prevent extension of the outboard spoilers when aircraft airspeed is above 170 kt. The No.2 unload valve can also be operated by depressing the roll spoiler switchlight SPLR 2 PUSH OFF. If the inboard spoiler PCU linkage jams, the No. 1 unload valve will prevent extension of the inboard spoilers when the SPLR 1 PUSHOFF switchlight is pushed. The unload valves can be energized by selecting the FLIGHT/TAXI switch to TAXI.

Linear Variable Differential Transformer (LVDT), Spoiler Position

Refer to Figures 17 and 18.

The LVDT gives position indication through the FCECU to the Powered Flight Controls Surface Indicator (PFCSI), in the flight compartment and to the Flight Data Recorder (FDR). The LVDT is



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

attached to a bracket on the aft face of the wing rear spar and a bracket on the spoiler panel.

Spoiler Cable Sensor

[Refer to Figure 19.](#)

A spoiler cable sensor will automatically retract the spoilers if there is a failure of fuselage spoiler cable.

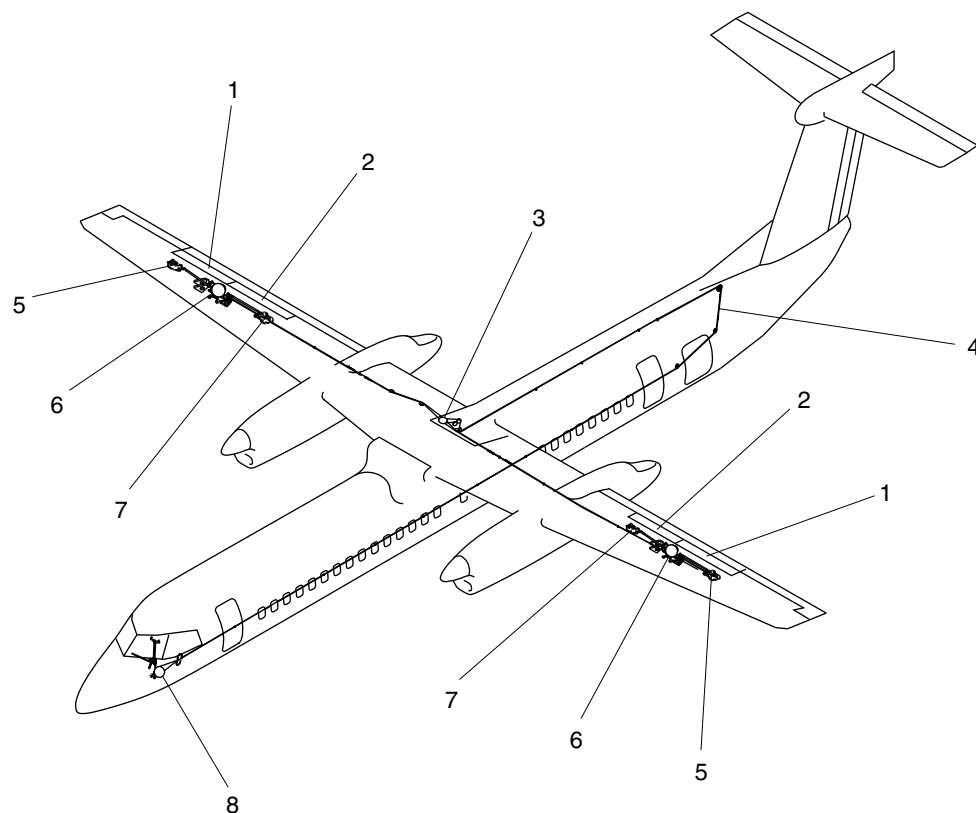
A spoiler cable sensor is installed on the fuselage spoiler cables aft of the spoiler quadrant below the flight compartment floor. The spoiler cable sensor–assembly has a roller arm, a tension spring, a latch spring and switches.

If the tension of the fuselage spoiler cables decreases to zero, the tension spring will pull the roller arm against a stop. In this position, the end of the roller arm will operate the switches. The normally–open switches are closed. This gives a signal to the inboard and outboard unload valves to close. The spoiler PCU extend pressure is removed and the hold–down pressure moves all the spoilers to the retracted position. Also, the ROLL SPLR INBD HYD and ROLL SPLR OUTBD HYD caution lights will come on. The latch spring will latch the roller arm and keep the spoiler sensor switches in the closed position until it is released by maintenance personnel on the ground.



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LEGEND

1. Outboard Spoiler.
2. Inboard Spoiler.
3. Spoiler Splitter Quadrant.
4. Spoiler Control Cables.
5. Outboard Spoiler PCU.
6. Spoiler Clutch.
7. Inboard Spoiler PCU.
8. Spoiler Forward Quadrant.

SPOILER CONTROL SYSTEM PAGE 1
Figure 1

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-61-00
Config 001

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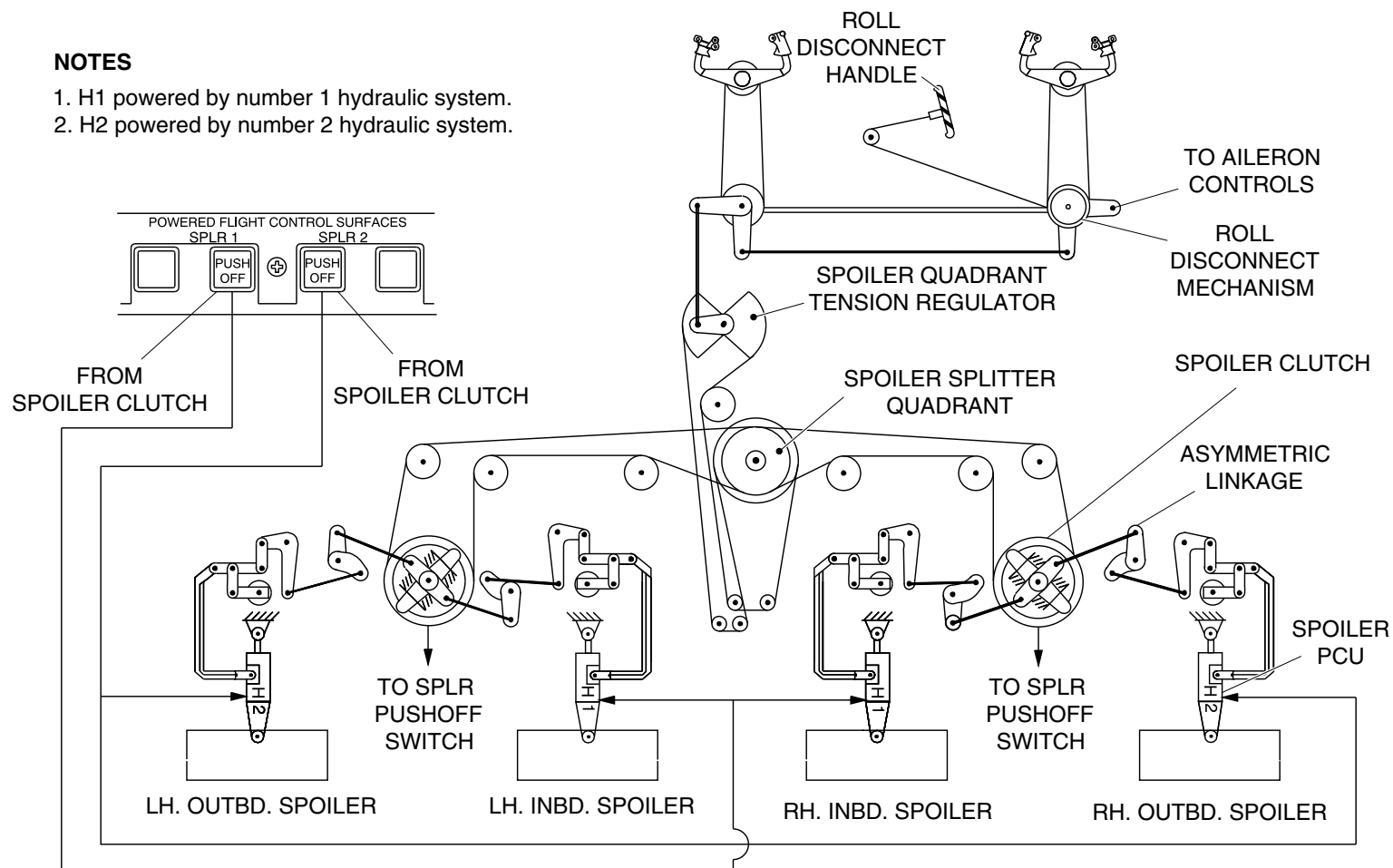
Config 001
Page 6
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

NOTES

1. H1 powered by number 1 hydraulic system.
2. H2 powered by number 2 hydraulic system.



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Spoiler Control System Page 2
Figure 2

PSM 1-84-2A
EFFECTIVITY:
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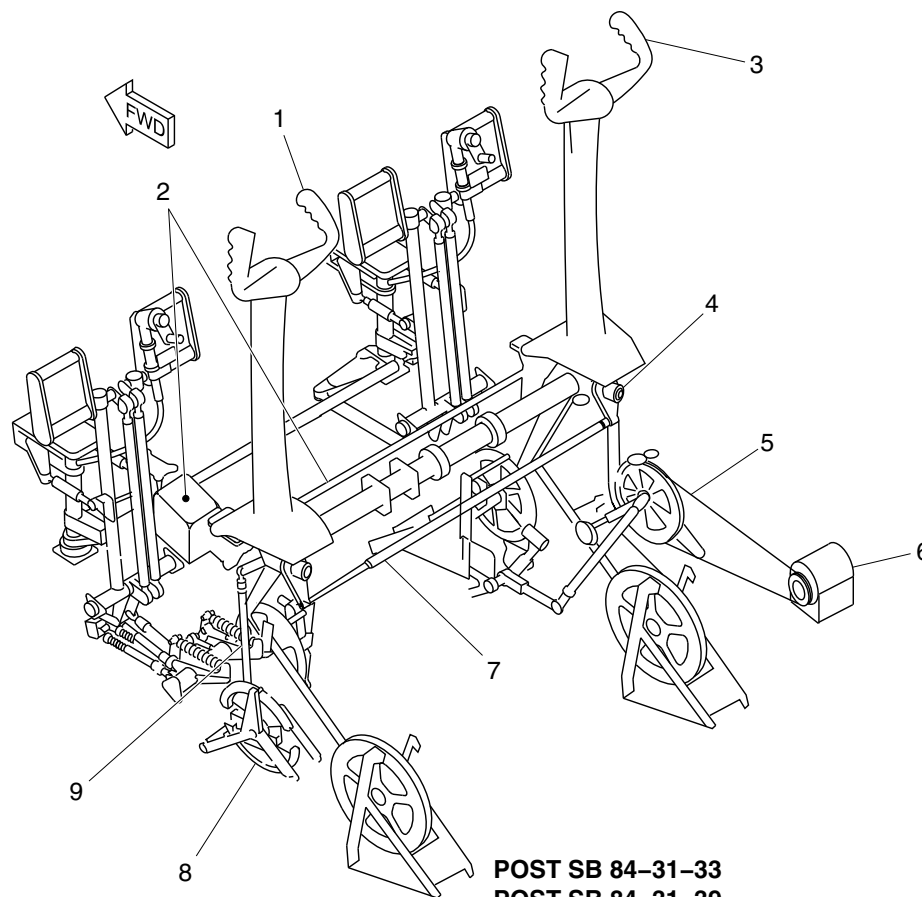
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Config 001
Page 7
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Pilot handwheel.
2. Aileron trim and centering unit.
3. Copilot handwheel.
4. Roll disconnect clutch.
5. Aileron forward quadrant.
6. Aileron autopilot servo.
7. Interconnect rod.
8. Spoiler forward quadrant.
9. Push/Pull rod – force sensor.

POST SB 84-31-33
POST SB 84-31-39
POST SB 84-31-40
POST MODSUM 4-309210

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Flight Controls / Spoiler System
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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Config 001

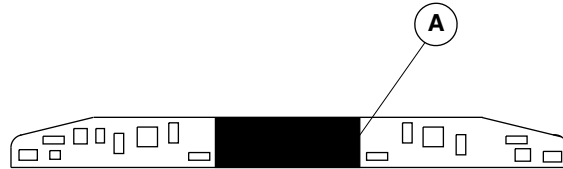
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Config 001
Page 8
Sep 05/2021



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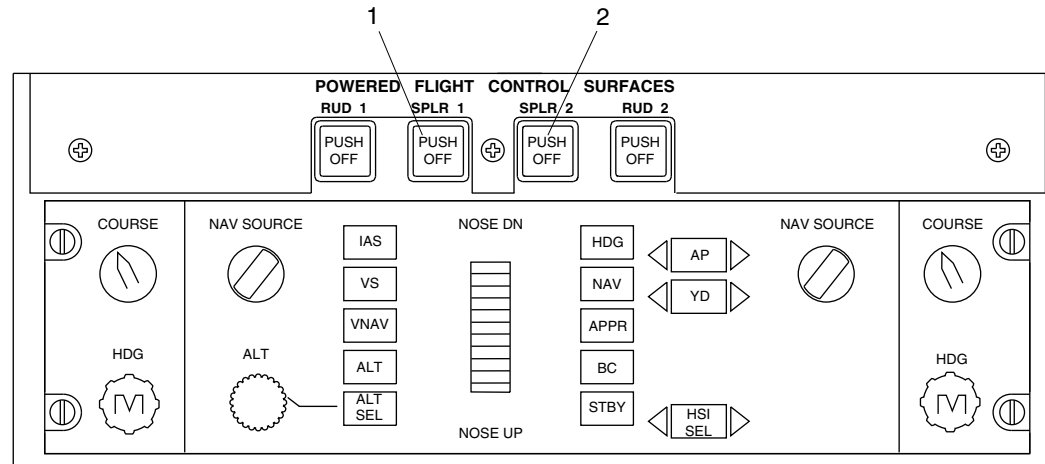
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



GLARESHIELD PANEL

LEGEND

1. Spoiler 1 Push Off Switchlight.
2. Spoiler 2 Push Off Switchlight.



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SPOILER SWITCHLIGHTS
Figure 4

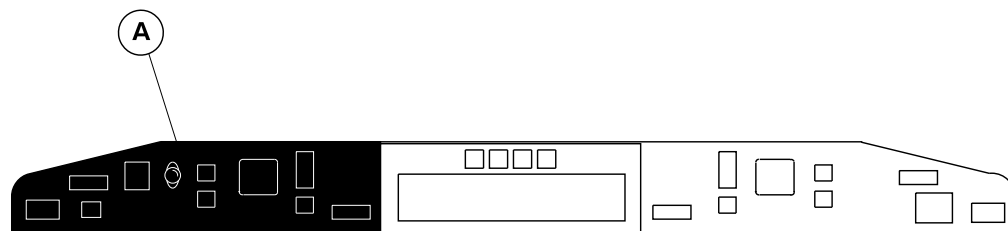
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Config 001

27-61-00

Config 001
Page 9
Sep 05/2021



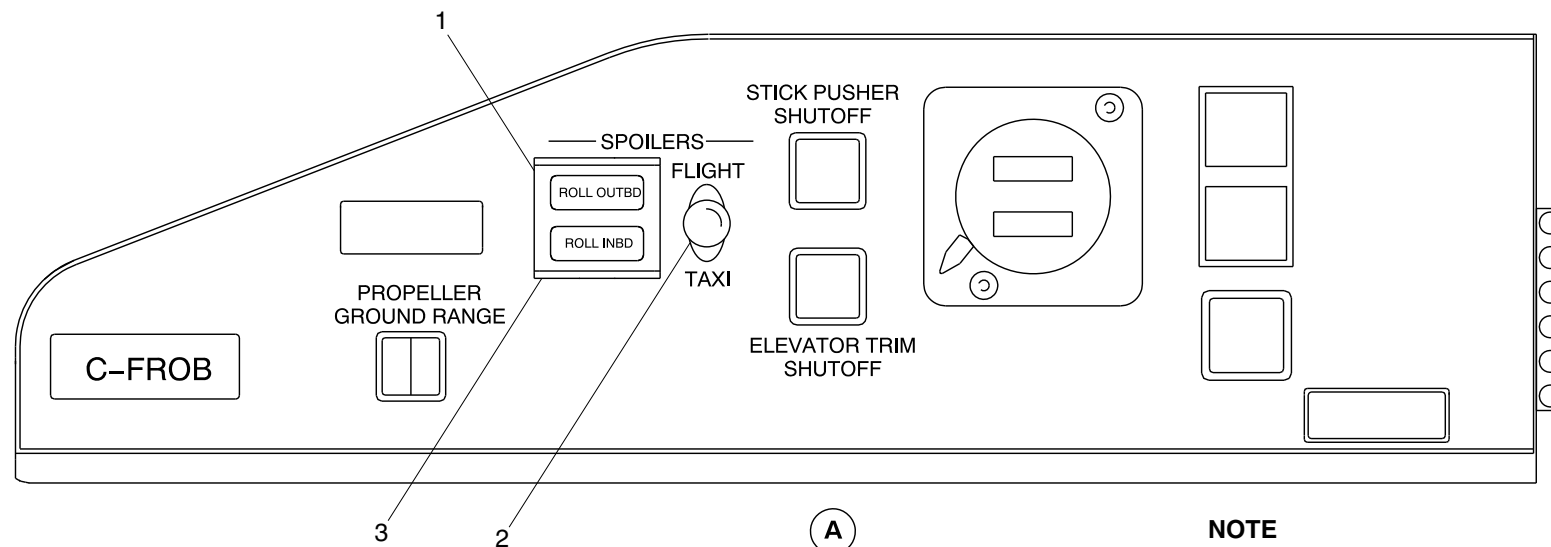
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GLARESHIELD PANEL

LEGEND

1. Roll outboard annunciator light.
2. Flight/taxi switch.
3. Roll inboard annunciator light.



NOTE

Pilot side shown.
Copilot side similar.

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Flight / Taxi Switch
Figure 5

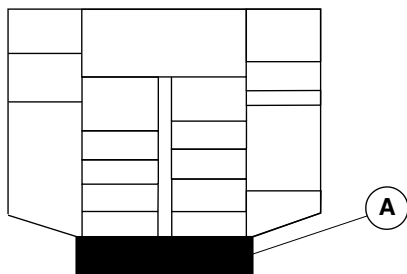
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Config 001

27-61-00

Config 001
Page 10
Sep 05/2021



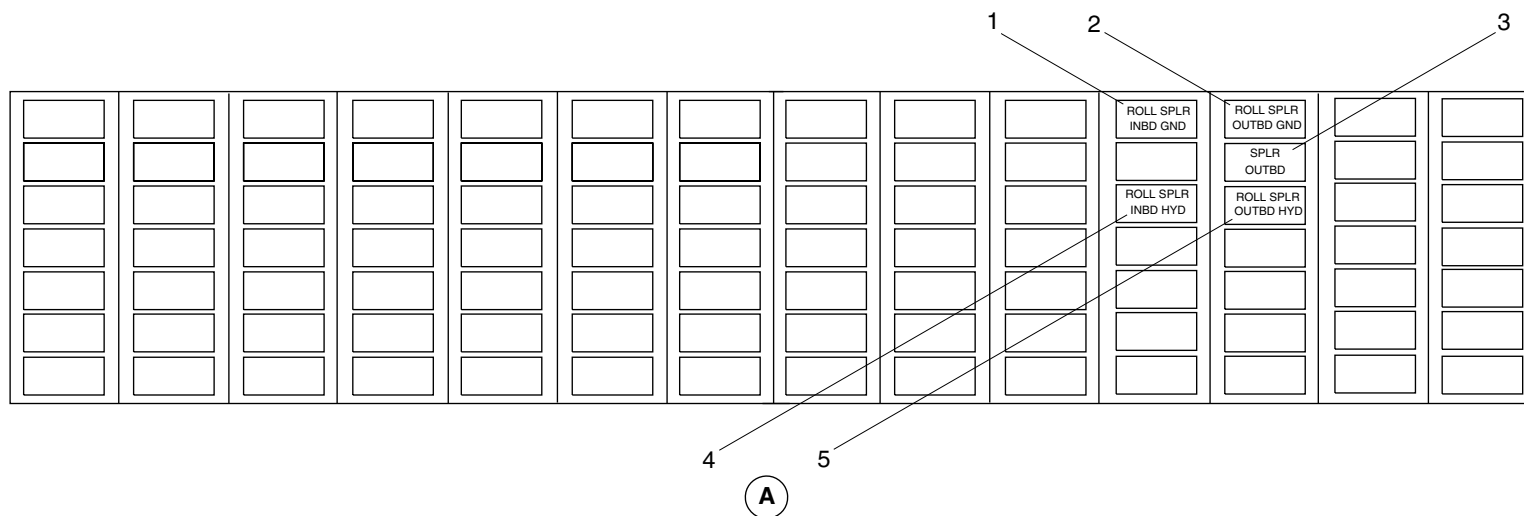
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



OVERHEAD CONSOLE

LEGEND

1. Roll Spoiler Inboard Ground (Amber).
2. Roll Spoiler Outboard Ground (Amber).
3. Spoiler Outboard (Amber).
4. Roll Spoiler Inboard Hydraulic (Amber).
5. Roll Spoiler Outboard Hydraulic (Amber).



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SPOILER CAUTION LIGHTS
Figure 6

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See first effectivity on page 2 of 27-61-00
Config 001

27-61-00

Config 001
Page 11
Sep 05/2021



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**A SPOILER CONTROL SYSTEM
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PSM 1-84-2A
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Config 001

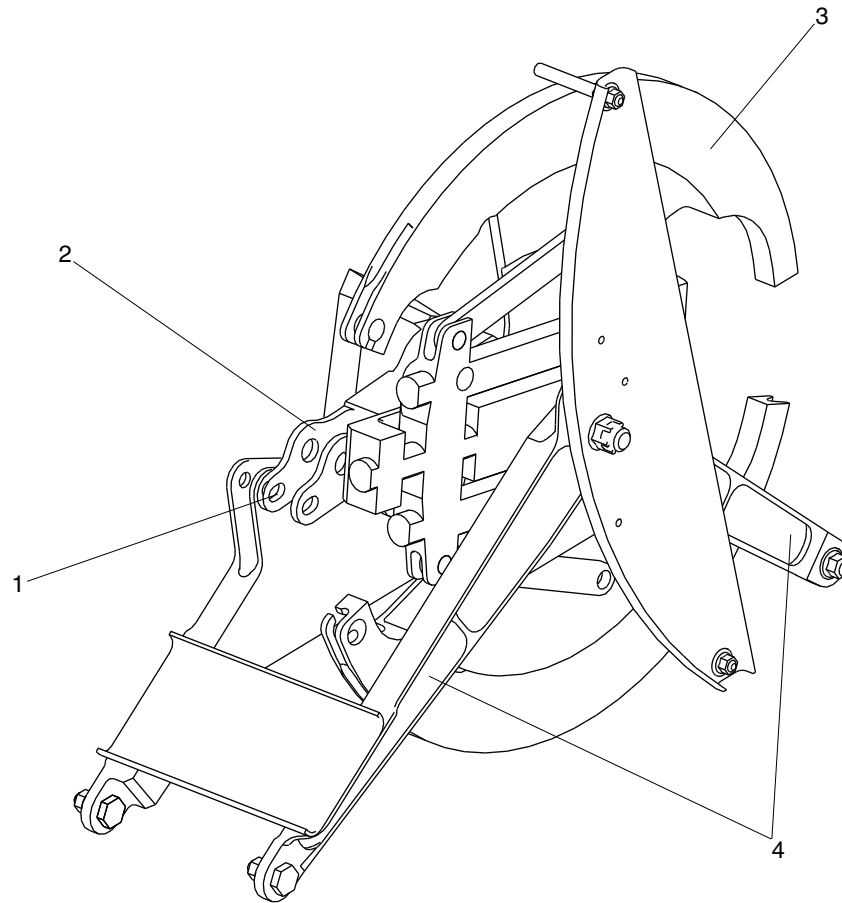
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Config 001
Page 12
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Rig Pin Hole.
- 2. Pushrod Attachment.
- 3. Cable Tension Regulator.
- 4. Mounting Brackets.



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SPOILER QUADRANT PAGE 2
Figure 8

PSM 1-84-2A
EFFECTIVITY:
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Config 001

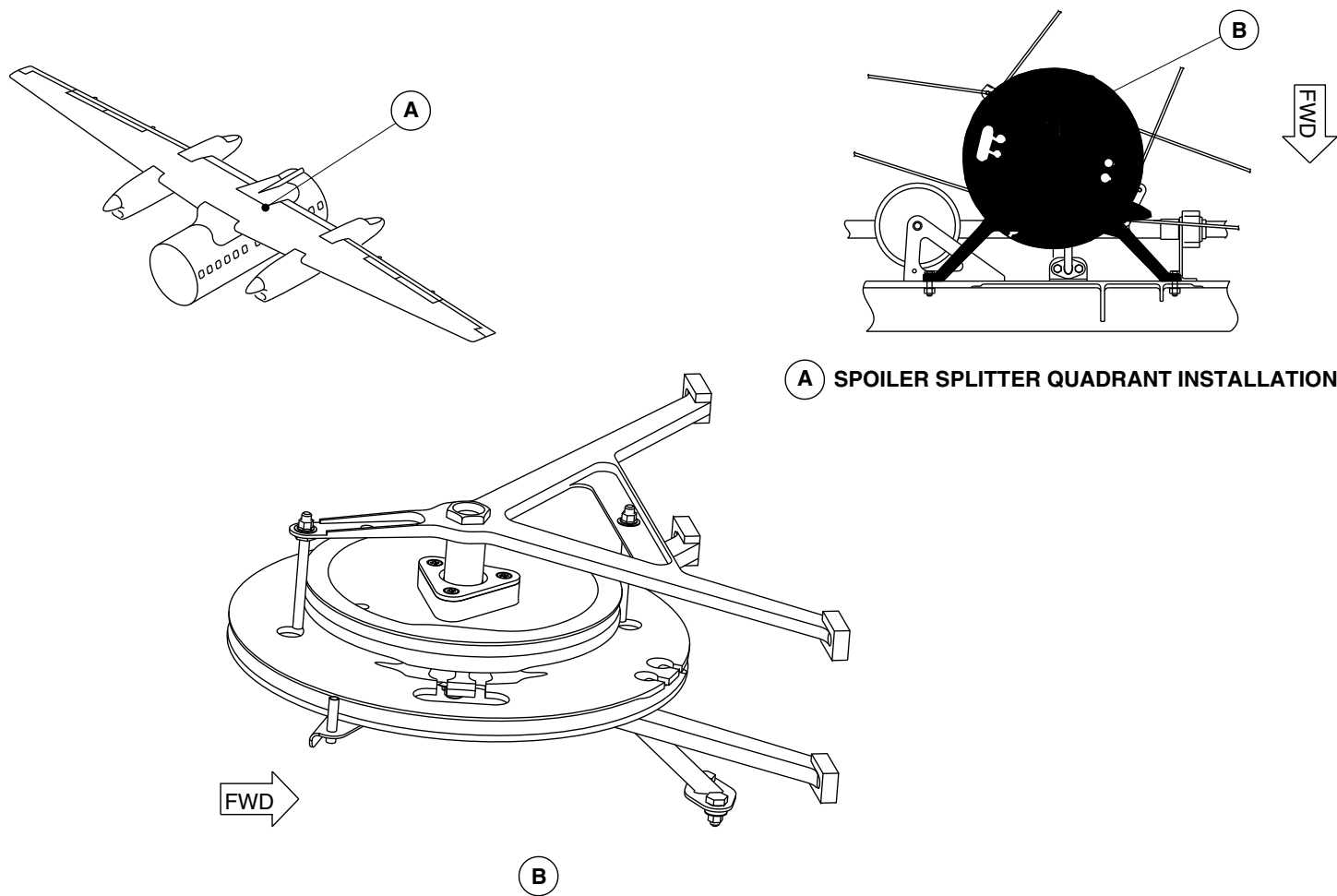
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Config 001
Page 13
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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SPOILER SPLITTER QUADRANT PAGE 1
Figure 9

PSM 1-84-2A
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Config 001

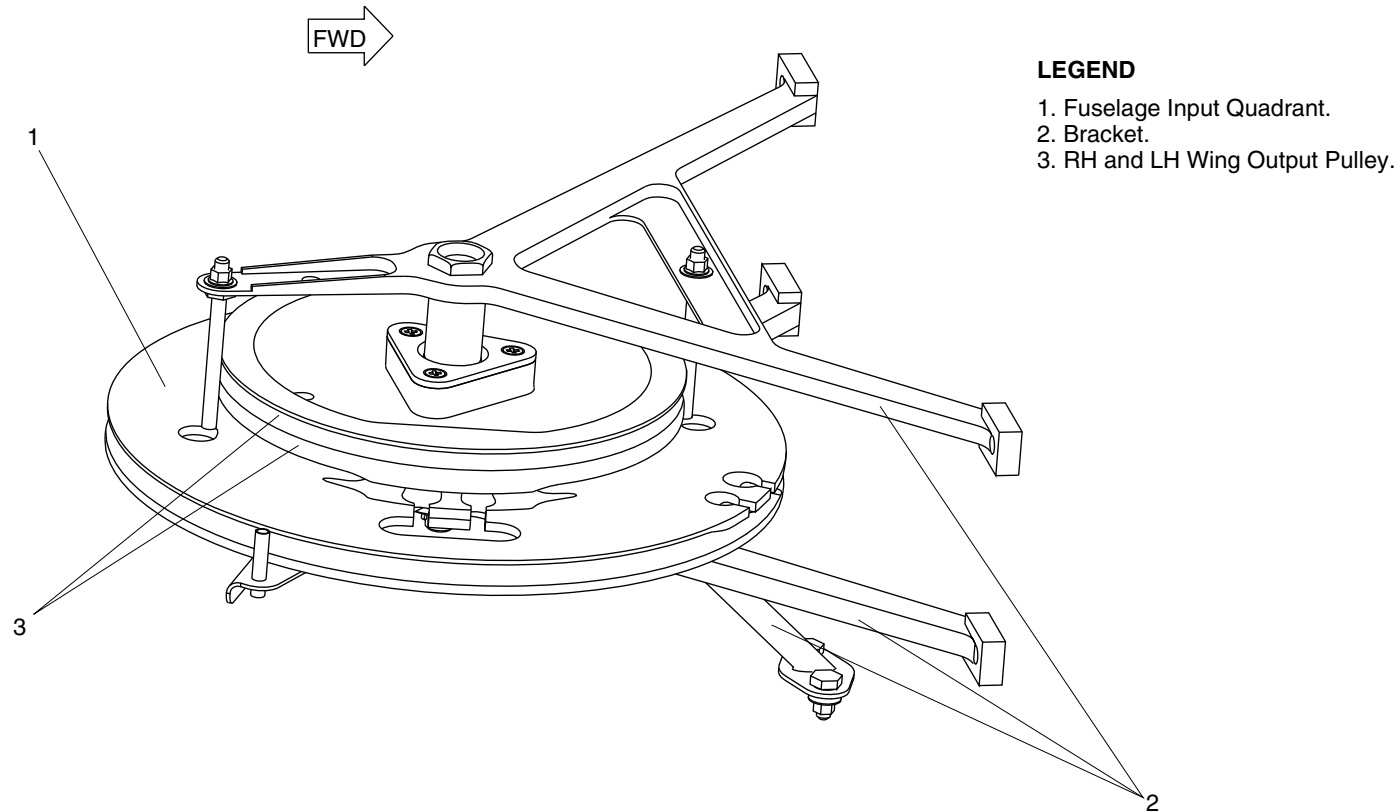
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Config 001
Page 14
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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SPOILER SPLITTER QUADRANT PAGE 2
Figure 10

PSM 1-84-2A
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Config 001

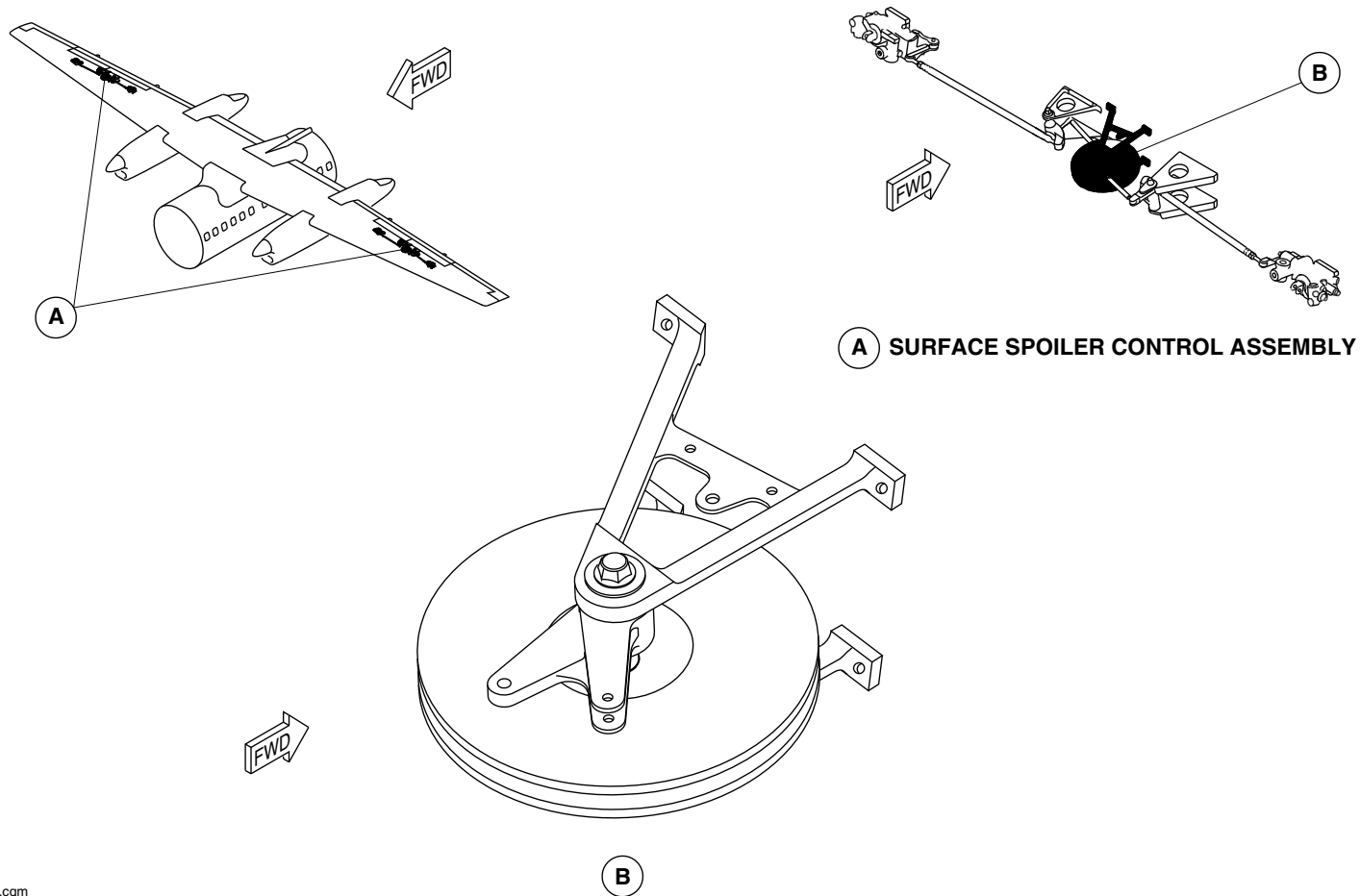
27-61-00

Config 001
Page 15
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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SPOILER CLUTCH UNITS PAGE 1
Figure 11

PSM 1-84-2A
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See first effectivity on page 2 of 27-61-00
Config 001

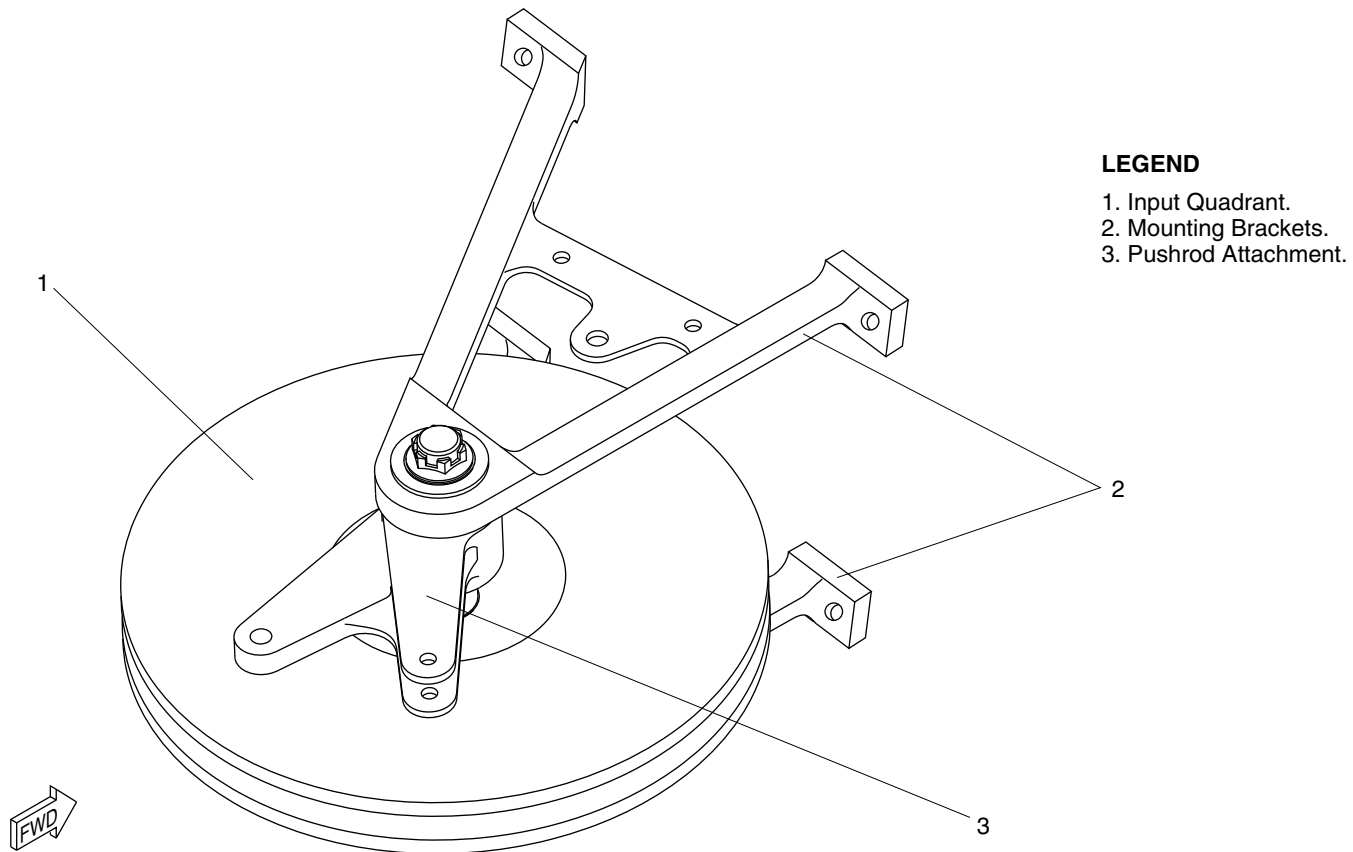
27-61-00

Config 001
Page 16
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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SPOILER CLUTCH UNITS PAGE 2
Figure 12

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-61-00
Config 001

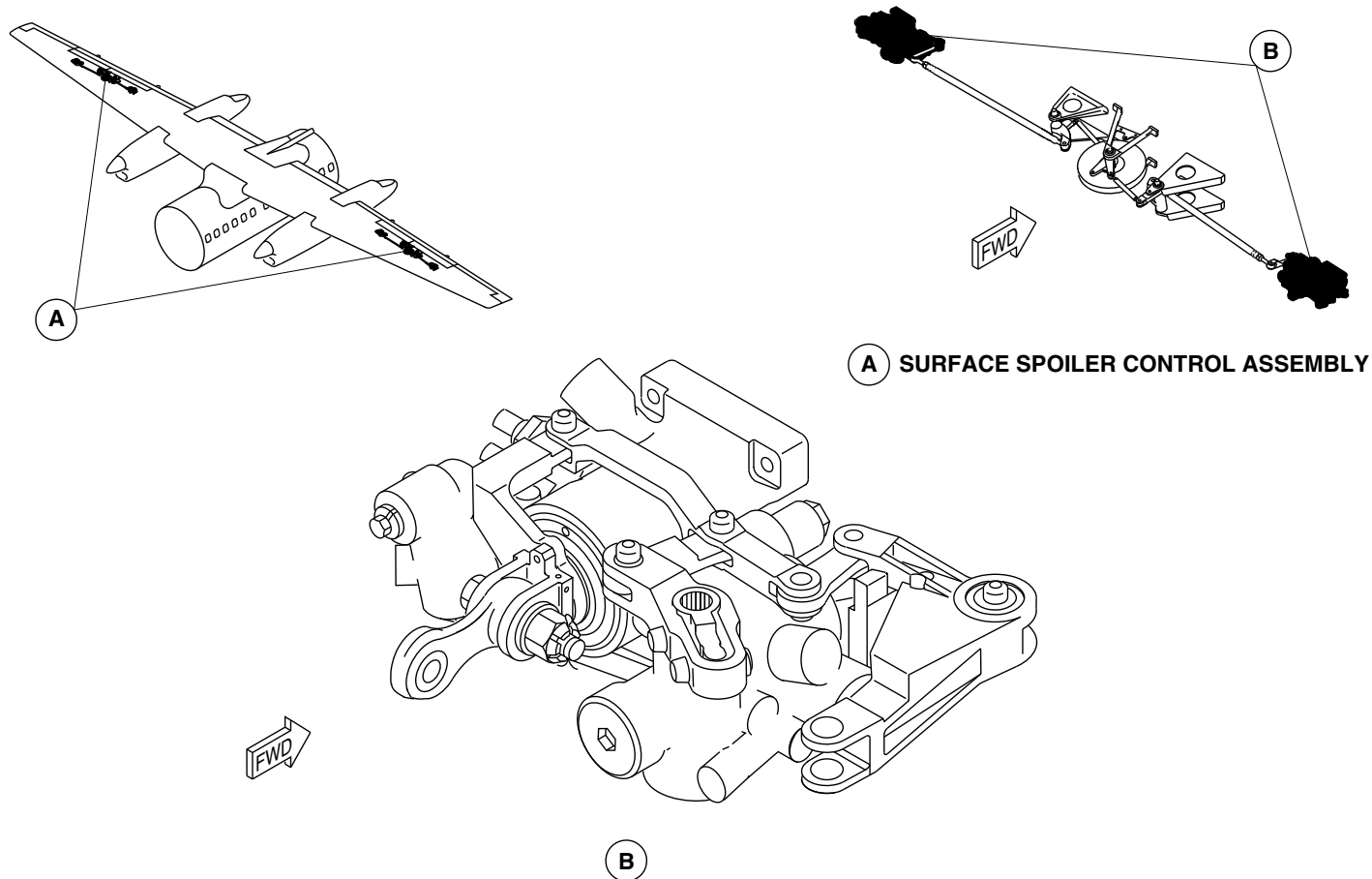
27-61-00

Config 001
Page 17
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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SPOILER POWER CONTROL UNIT (PCU) PAGE 1
Figure 13

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-61-00
Config 001

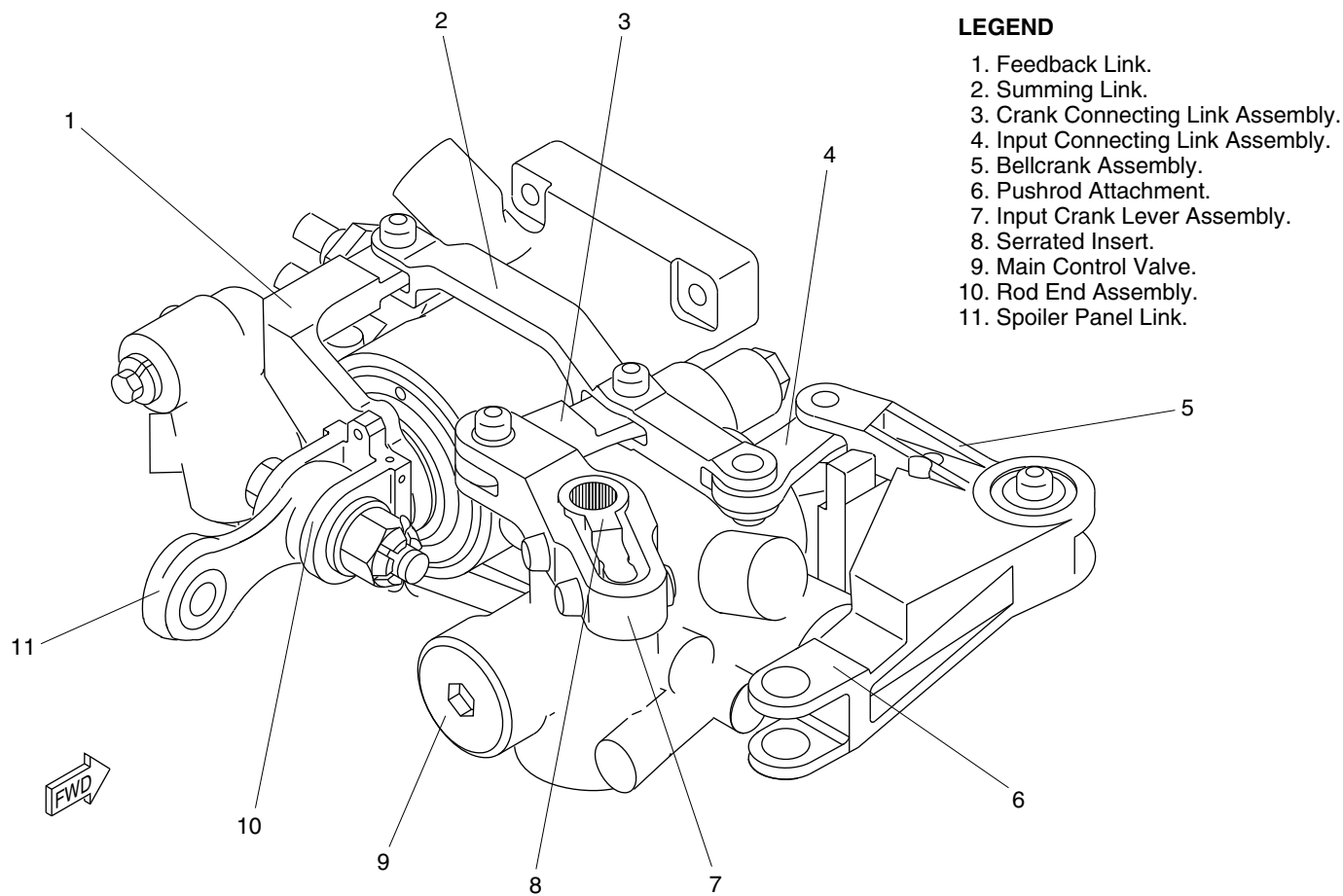
27-61-00

Config 001
Page 18
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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SPOILER POWER CONTROL UNIT (PCU) PAGE 2
Figure 14

PSM 1-84-2A
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See first effectivity on page 2 of 27-61-00
Config 001

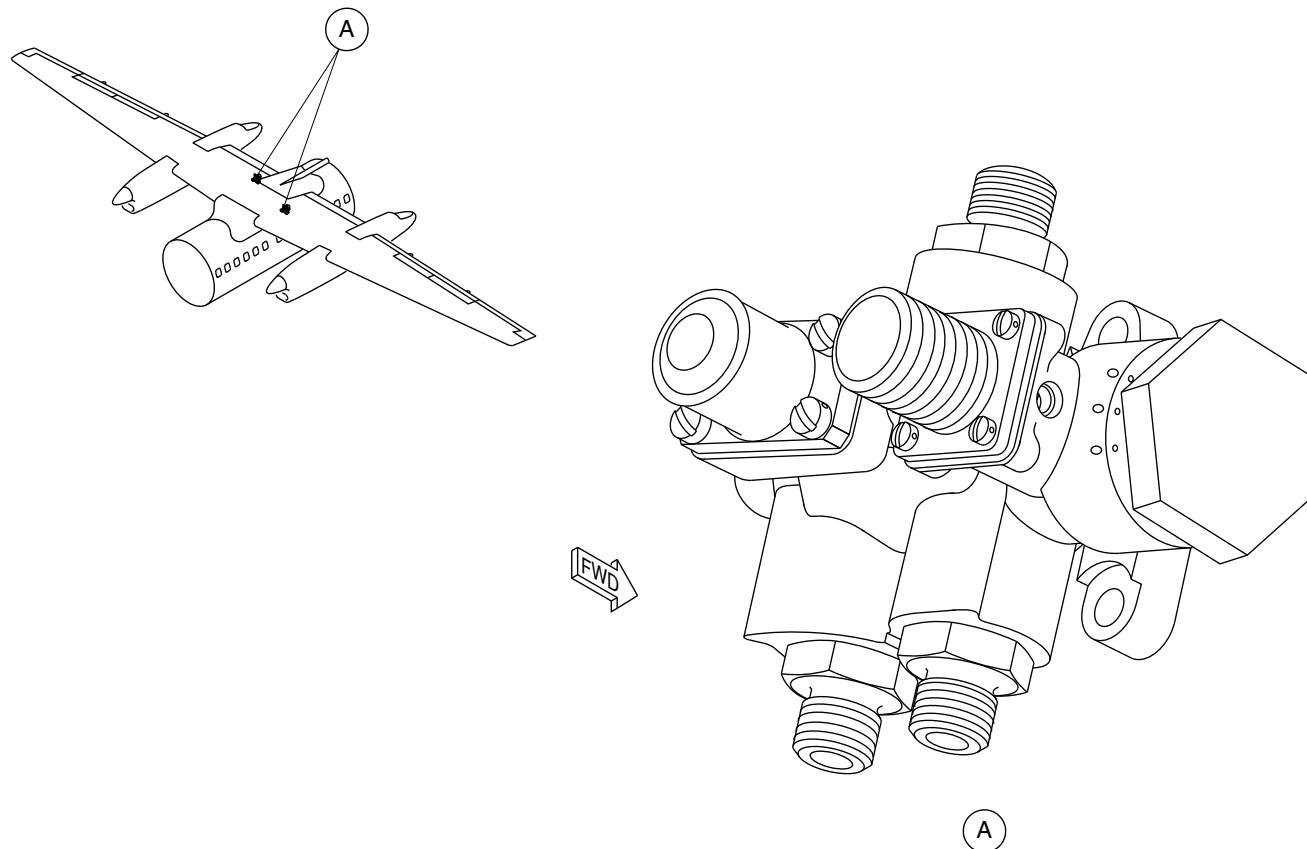
27-61-00

Config 001
Page 19
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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SPOILER UNLOAD VALVE PAGE 1
Figure 15

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-61-00
Config 001

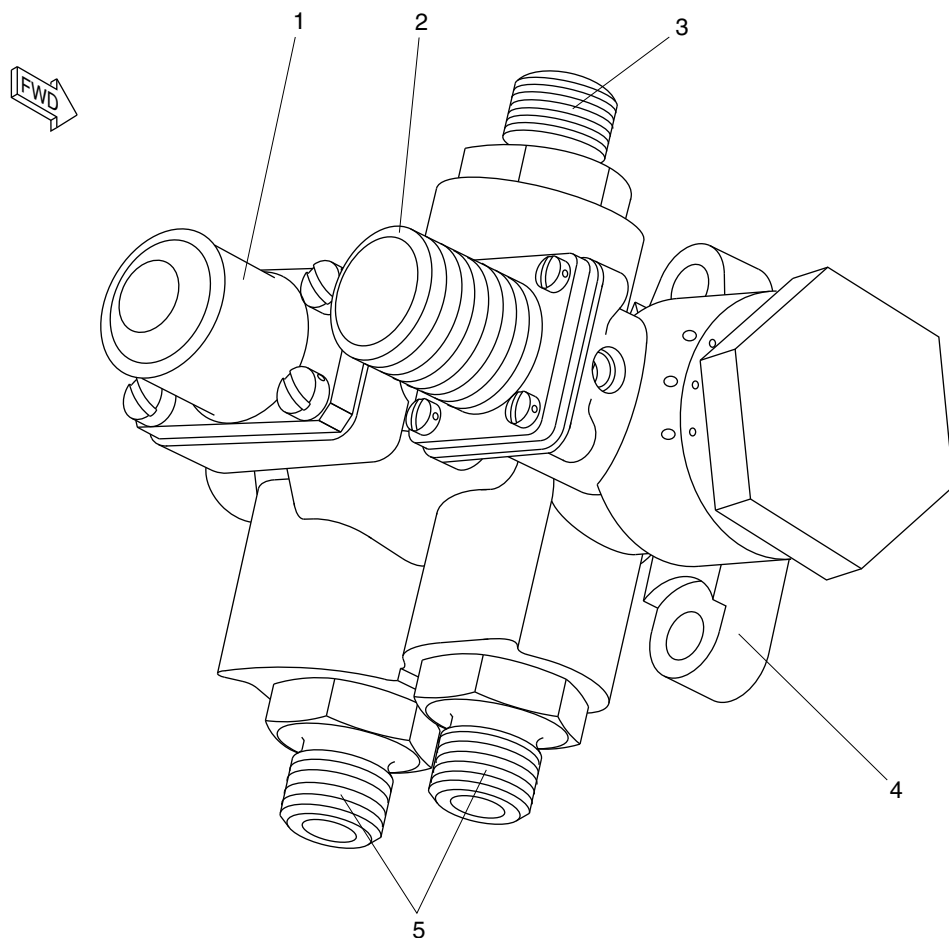
27-61-00

Config 001
Page 20
Sep 05/2021



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OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Solenoid.
- 2. Electrical Connector.
- 3. Hydraulic Port Return.
- 4. Mounting Bracket.
- 5. Hydraulic Pressure Port.

fs099a01.cgm

SPOILER UNLOAD VALVE PAGE 2
Figure 16

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-61-00
Config 001

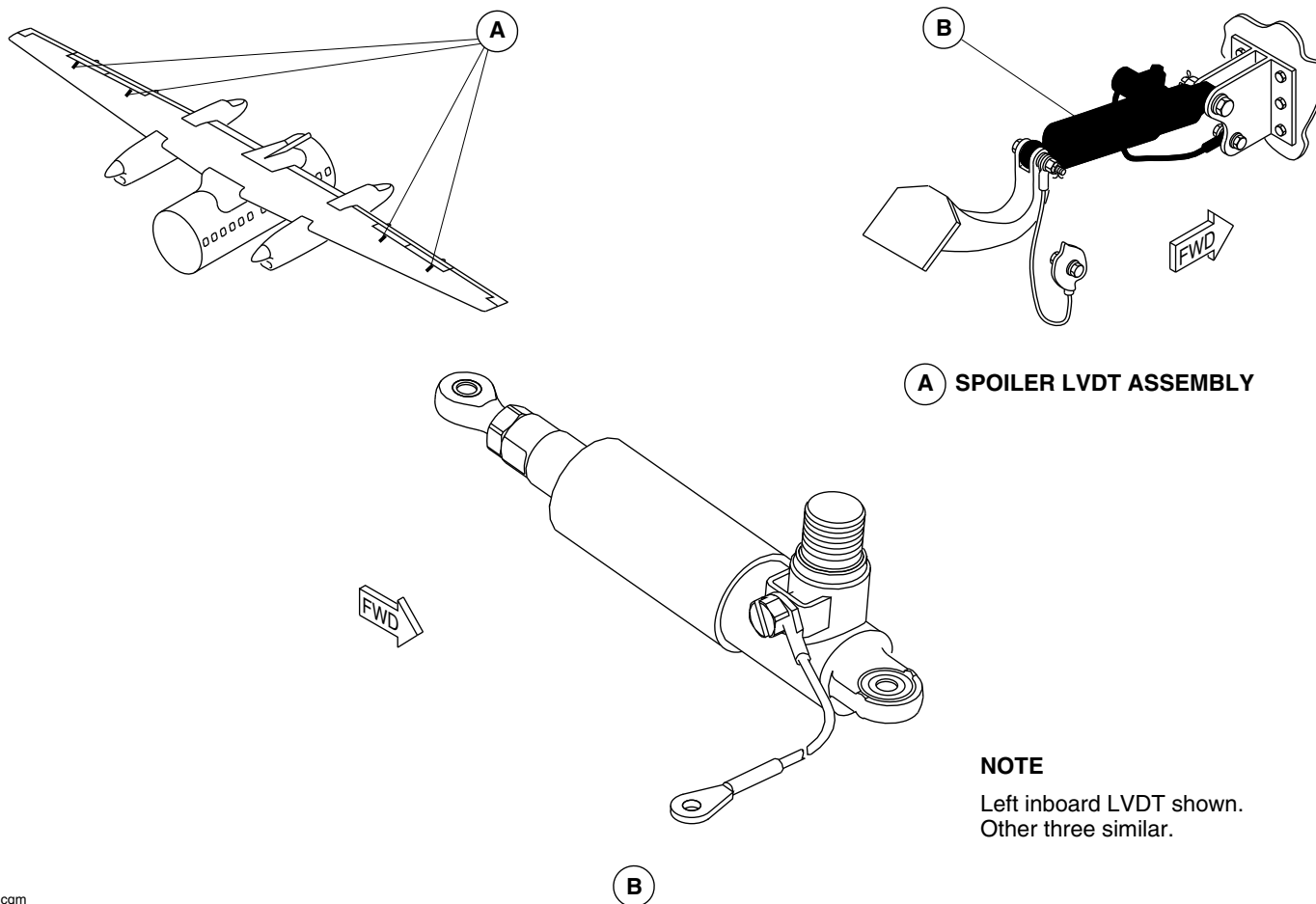
27-61-00

Config 001
Page 21
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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SPOILER LVDT PAGE 1
Figure 17

PSM 1-84-2A
EFFECTIVITY:
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Config 001

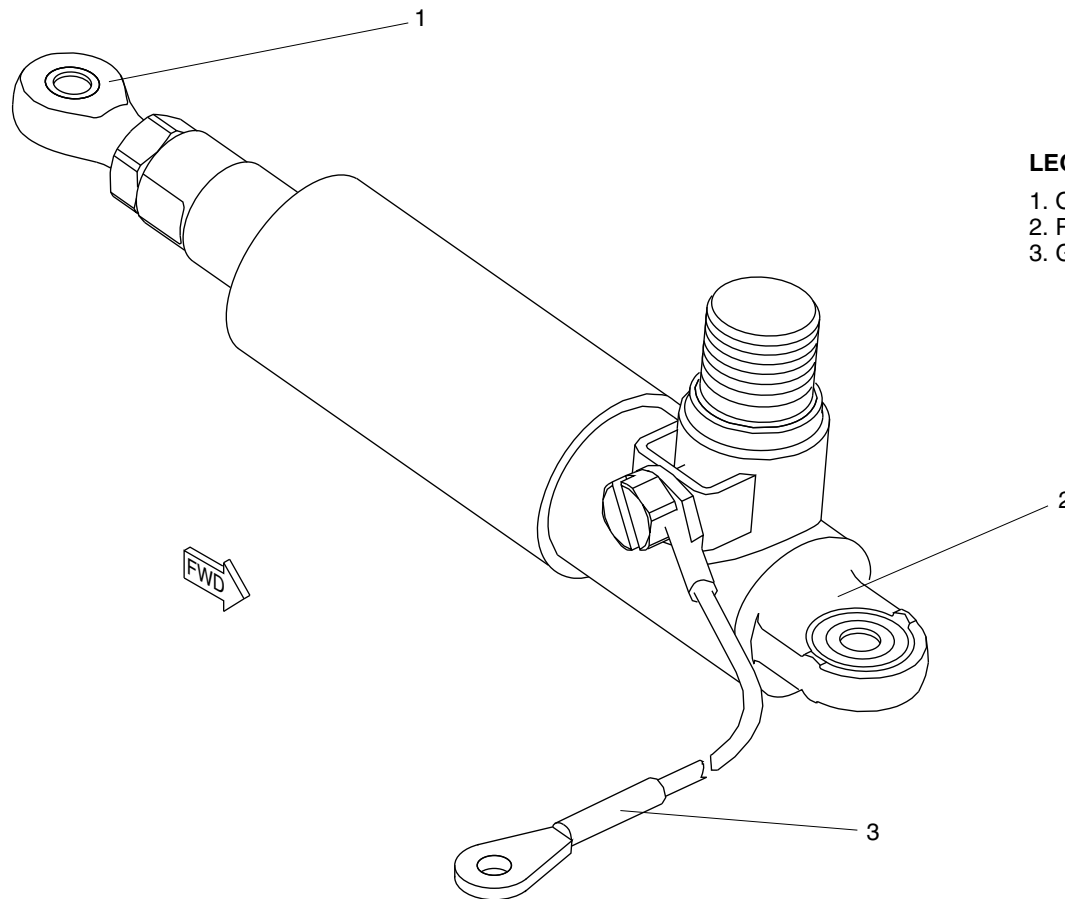
27-61-00

Config 001
Page 22
Sep 05/2021



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OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. Outboard Spoiler Attachment.
- 2. Rear Spar Attachment.
- 3. Ground Strap.

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SPOILER LVDT PAGE 2
Figure 18

PSM 1-84-2A
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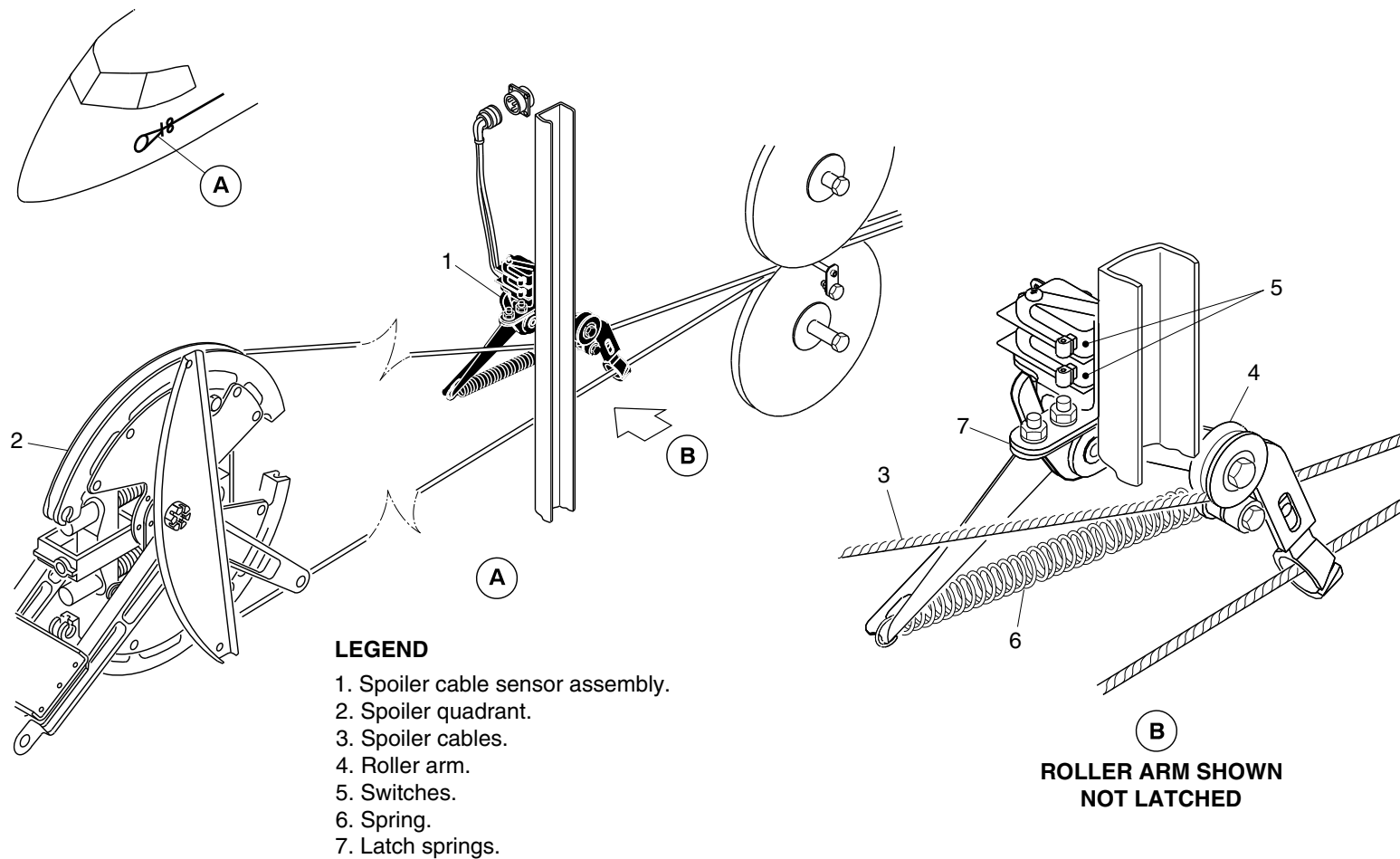
27-61-00

Config 001
Page 23
Sep 05/2021



DE HAVILLAND AIRCRAFT
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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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SPOILER CABLE SENSOR
Figure 19

PSM 1-84-2A
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See first effectivity on page 2 of 27-61-00
Config 001

27-61-00

Config 001
Page 24
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

27-62-00-001

GROUND SPOILERS SYSTEM

Introduction

The spoiler panels are deployed symmetrically to decrease or “dump” the lift made by the wings. This action puts all of the weight of the aircraft on the landing gear, so that the wheel brakes can slow the aircraft more efficiently.

The flight spoilers are symmetrically deployed as ground spoilers, when signalled by the Weight-On-Wheels (WOW) and power lever sensors, if the spoiler FLIGHT/TAXI switch is in the FLIGHT position.

General Description

The roll spoiler system automatically changes from the flight mode to the ground mode after the aircraft has landed. The roll spoilers operate symmetrically on landing (overriding flight mode operation).

Ground mode operation occurs when:

- The Proximity Sensor Electronic Unit (PSEU) gives the Weight-On-Wheels (WOW1 and WOW2 respectively) signal to the Flight Control Electronic Control Unit (FCECU)
- Power levers 1 and 2 are set to less than 12 degrees above flight idle
- The spoiler FLIGHT/TAXI switch is in the FLIGHT position.

When the power levers and WOW inputs are valid, the lift dump solenoid valves are energized to extend all roll spoiler panels. When the spoilers extend the ROLL INBD and ROLL OUTBD advisory lights come on. In the event that a solenoid valve fails to energize, the inboard or outboard ground spoilers will not extend. The applicable ROLL SPLR INBD GND or ROLL SPLR OUTBD GND caution light comes on after a time delay of 5 seconds. If a go-around is necessary, the power levers are advanced by the pilot, removing the power to the lift dump solenoid valves and to the roll advisory lights. All spoiler panels will retract and the roll advisory lights will go off.

Selection of the spoilers' FLIGHT/TAXI switch to TAXI removes hydraulic power from all spoiler lift dump valve solenoids and all spoiler panels retract. The switch is held in the TAXI position by a magnetic latch solenoid. Advancing either power lever above flight idle +12 degrees, or removing electrical power from the aircraft allows the spring loaded switch to return to the FLIGHT position.

The ground spoiler system includes the components that follow:

- Valves, Lift-dump (27-62-01).

Detailed Description

When the aircraft lands, the lift dump valves are energized through electrical signals from the Flight Control Electronic Control Unit (FCECU) and the Proximity Sensor Electronic Unit (PSEU). When signaled, the lift dump valves give hydraulic input commands to each spoiler PCU, which override roll commands from the pilot, and extend the spoiler panels to 75 degrees. When the spoiler panels are extended the wing lift is “dumped”, increasing the weight on the landing gear to give maximum braking efficiency.

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See first effectivity on page 2 of 27-62-00
Config 001

27-62-00

Config 001
Page 2
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

Refer to Figures 1 and 2.

The ROLL INBD and ROLL OUTBD advisory lights come on when the inboard and outboard lift dump valves are operated after landing.

The ROLL SPLR INBD GND or ROLL SPLR OUTBD GND caution lights come on:

- If the Flight Control Electronic Control Unit (FCECU) senses a loss of lift dump valve function
- If the lift dump valves are operated but the spoiler panels fail to extend after landing
- If the spoiler panels extend when the FCECU carries out the lift dump Built In Test (BIT), with the FLIGHT/TAXI switch set to TAXI.

Valves, Lift Dump

Refer to Figures 3 and 4.

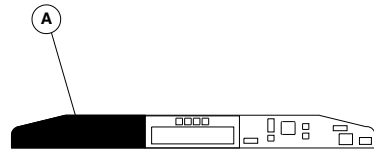
The two lift dump valves for the inboard (No. 1) spoiler system are located on the left side of the aircraft in the wing-to-fuselage attachment area. The two lift dump valves for the outboard (No. 2) spoiler system are located on the right side of the aircraft also in the wing-to-fuselage attachment area.

The two valves in each system are connected hydraulically in series. Both valves must open to let extend pressure through to the related spoiler PCU. This reduces the possibility of in flight symmetric deployment. After landing, with the aircraft weight-on-wheels signal and both power levers set below flight idle +12 degrees, all lift dump solenoid valves are energized. This supplies hydraulic pressure to the extend side of both spoiler actuators, extending all spoiler panels symmetrically.



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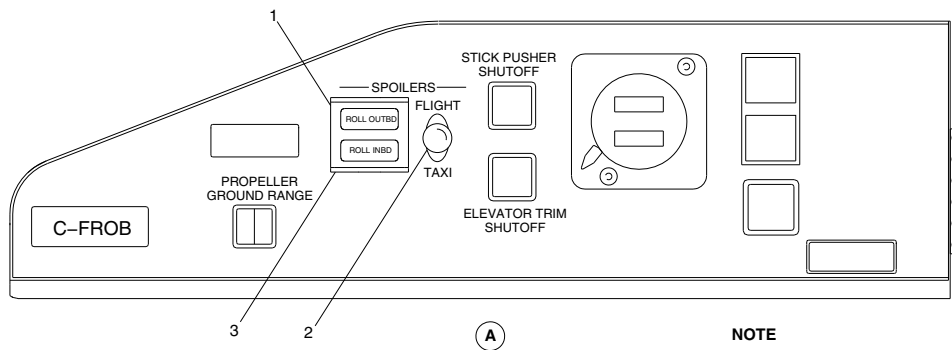
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



GLARESHIELD PANEL

LEGEND

1. Roll Outboard Annunciator Light.
2. Flight/Taxi Switch.
3. Roll Inboard Annunciator Light.



NOTE

Pilot side shown.
Copilot side similar.

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SPOILER ADVISORY LIGHTS
Figure 1

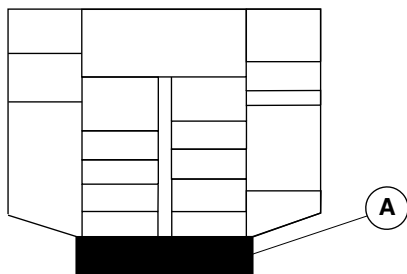
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Config 001

27-62-00

Config 001
Page 4
Sep 05/2021



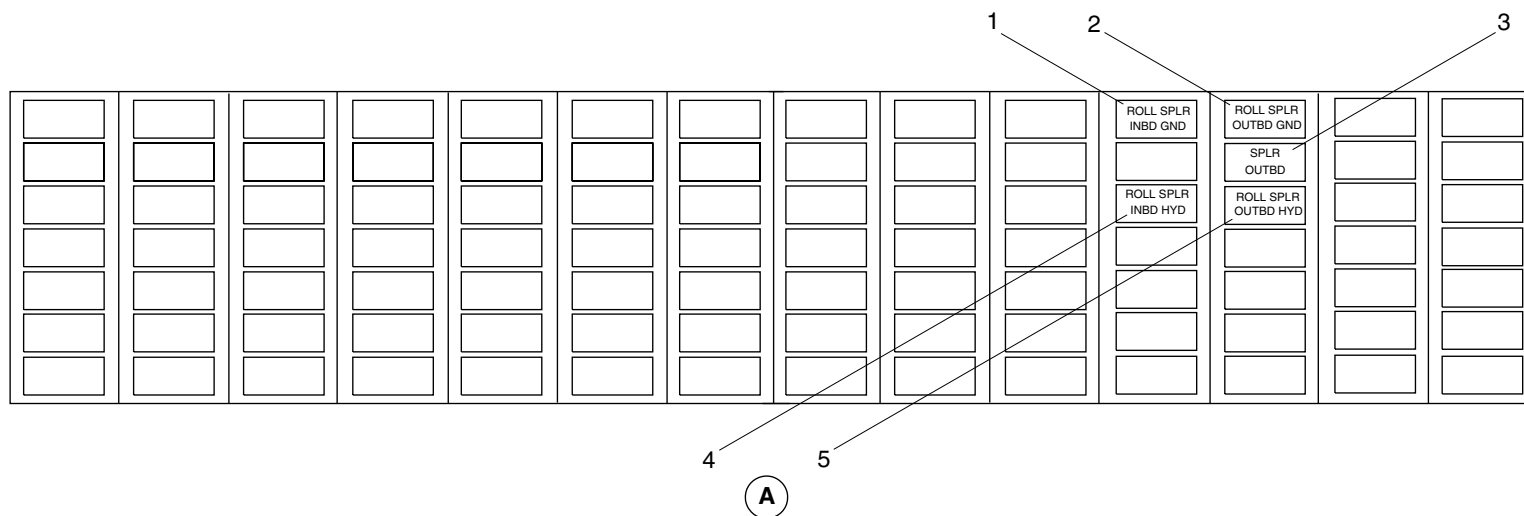
AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



OVERHEAD CONSOLE

LEGEND

1. Roll Spoiler Inboard Ground (Amber).
2. Roll Spoiler Outboard Ground (Amber).
3. Spoiler Outboard (Amber).
4. Roll Spoiler Inboard Hydraulic (Amber).
5. Roll Spoiler Outboard Hydraulic (Amber).



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SPOILER CAUTION LIGHTS
Figure 2

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-62-00
Config 001

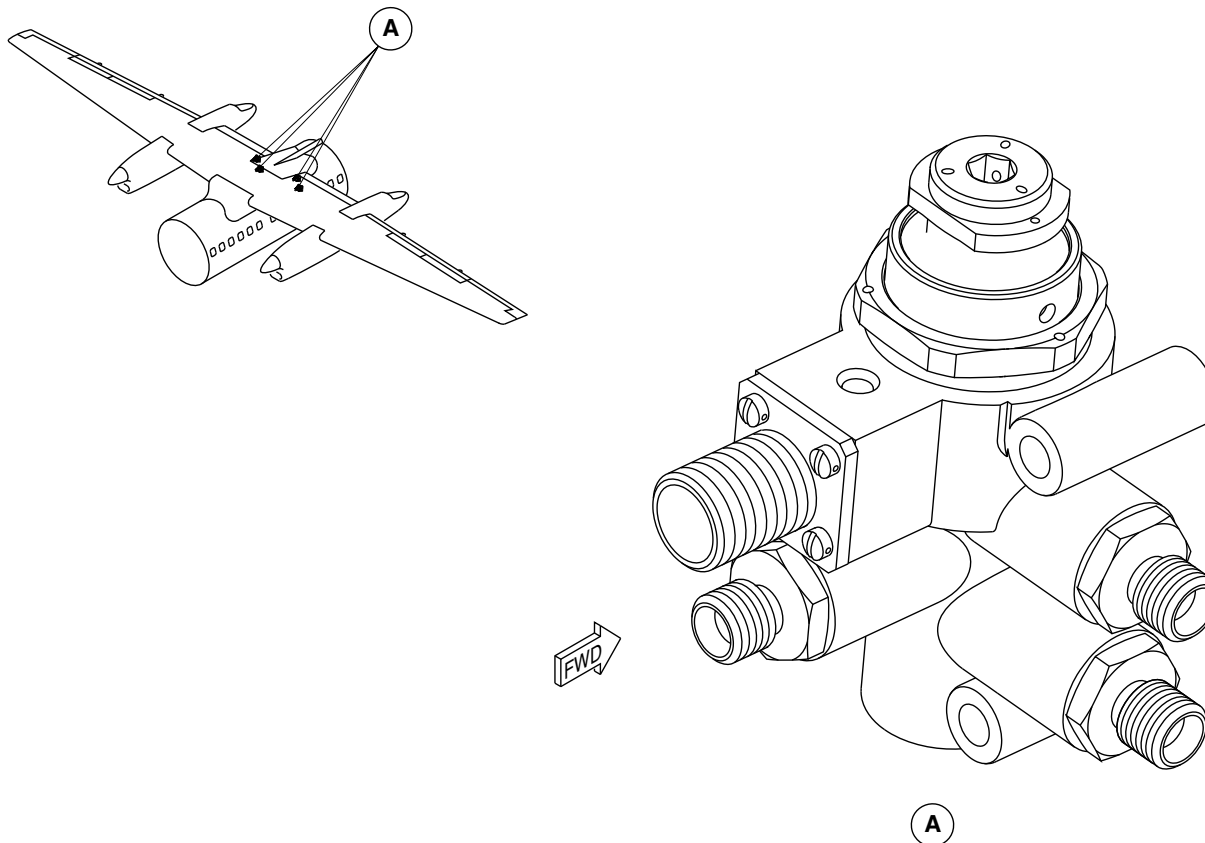
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Config 001
Page 5
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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SPOILER LIFT DUMP VALVE PAGE 1
Figure 3

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-62-00
Config 001

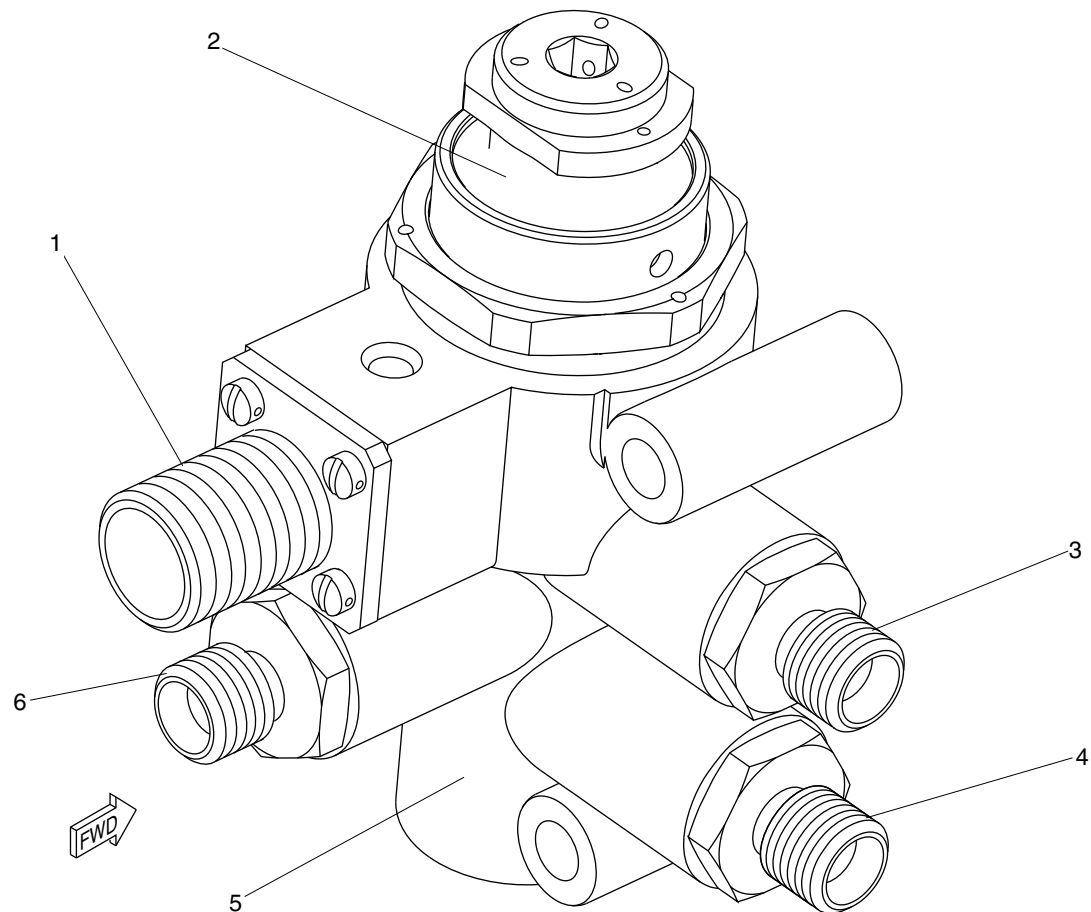
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Config 001
Page 6
Sep 05/2021



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LEGEND

- 1. Electrical Connector.
- 2. Solenoid.
- 3. Return Port.
- 4. Pressure Port.
- 5. Check Valve.
- 6. Cylinder Port.

SPOILER LIFT DUMP VALVE PAGE 2
Figure 4

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-62-00
Config 001

27-62-00

Config 001
Page 7
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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See first effectivity on page 2 of 27-62-00
Config 001

27-62-00

Config 001
Page 8
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

27-63-00-001

SPOILER PANELS

Introduction

The spoiler panels extend, in proportion to upgoing aileron movement, into the airstream flowing over the top surface of the wing. This reduces the amount of lift produced by the wing.

General Description

There are four spoiler panels. Each wing has an inboard and an outboard spoiler panel.

Detailed Description

Four spoiler panels are installed on the upper surface of the wing forward of the outboard flaps. The panels are identified as inboard and outboard and are operated hydromechanically using control wheel inputs in conjunction with the ailerons. The inboard and outboard spoilers are hinged to the upper surface of the wing. The Power Control Units (PCUs) attached to the inboard panels, are supplied with hydraulic pressure from No.1 hydraulic system. The PCUs attached to the outboard panels are supplied by No.2 hydraulic system.

Panels, Spoiler Inboard

[Refer to Figures 1 and 2.](#)

The inboard panels are located on the upper surface of each wing forward of the outboard flaps. They are attached by three hinge fittings on the aft face of the rear wing spar. The Power Control Unit is attached to the center hinge fitting. The spoiler panels fair with the wing top skin when retracted and extend upwards up to a maximum of 75 degrees when actuated.

Panels, Spoiler Outboard

[Refer to Figures 3 and 4.](#)

The outboard panels are located on the upper surface of each wing forward of the outboard flaps. They are attached by three hinge fittings on the aft face of the rear wing spar. The Power Control Unit is attached to the center hinge fitting. The spoiler panels fair with the wing top skin when retracted and extend upwards up to a maximum of 75 degrees when actuated.

PSM 1-84-2A

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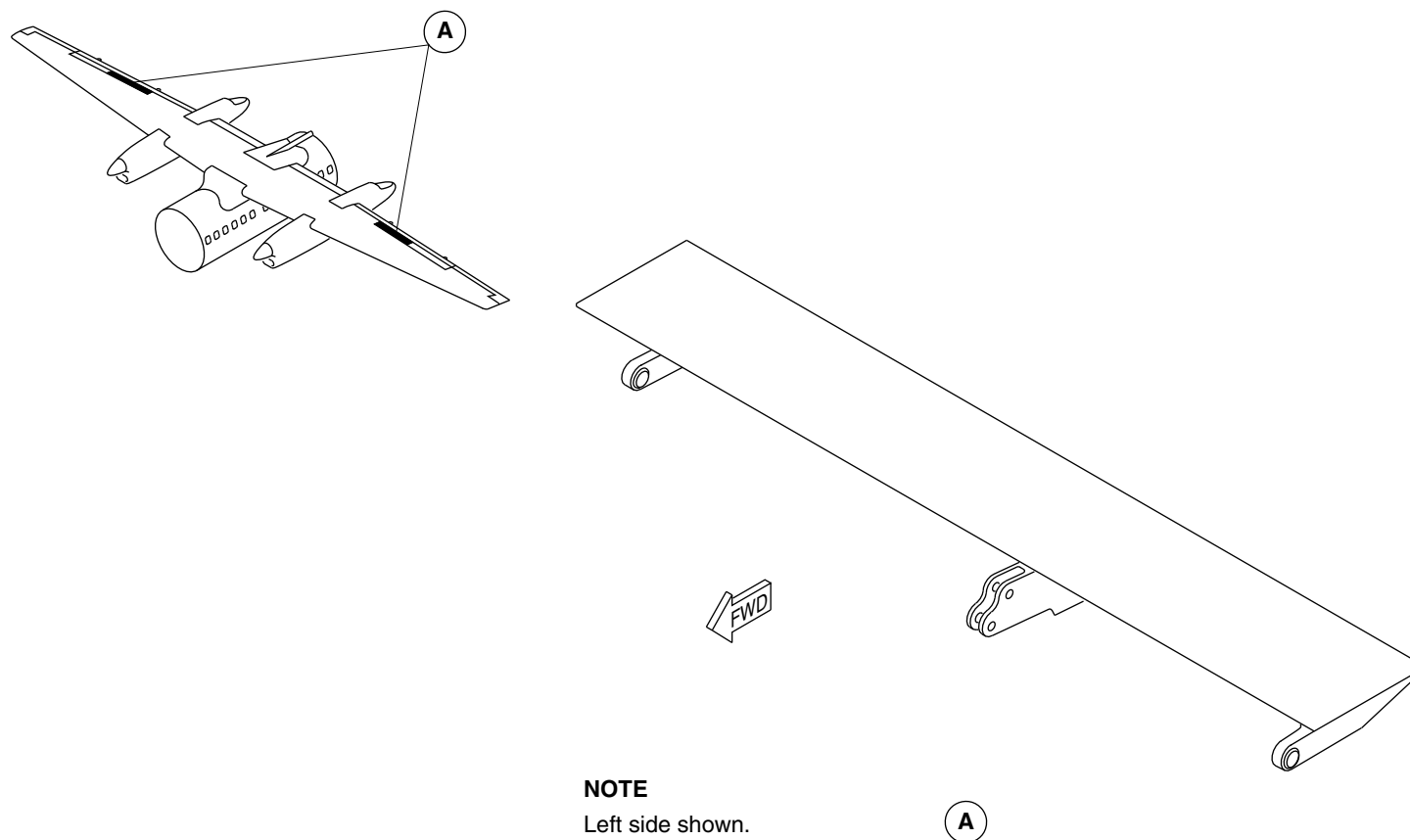
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Config 001
Page 2
Sep 05/2021



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NOTE

Left side shown.
Right side similar.

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INBOARD SPOILER PANEL PAGE 1
Figure 1

PSM 1-84-2A
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See first effectivity on page 2 of 27-63-00
Config 001

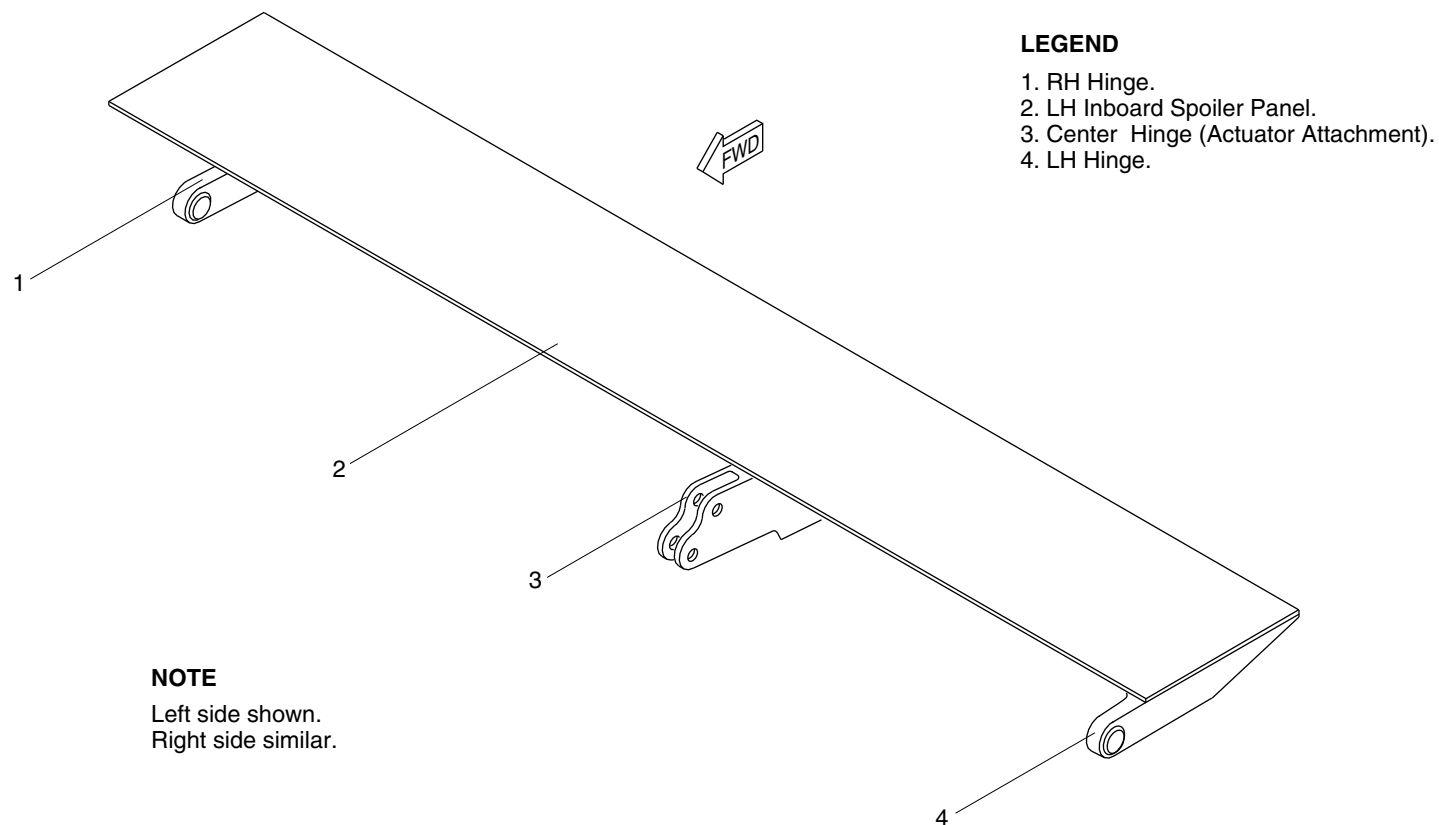
27-63-00

Config 001
Page 3
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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INBOARD SPOILER PANEL PAGE 2
Figure 2

PSM 1-84-2A
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See first effectivity on page 2 of 27-63-00
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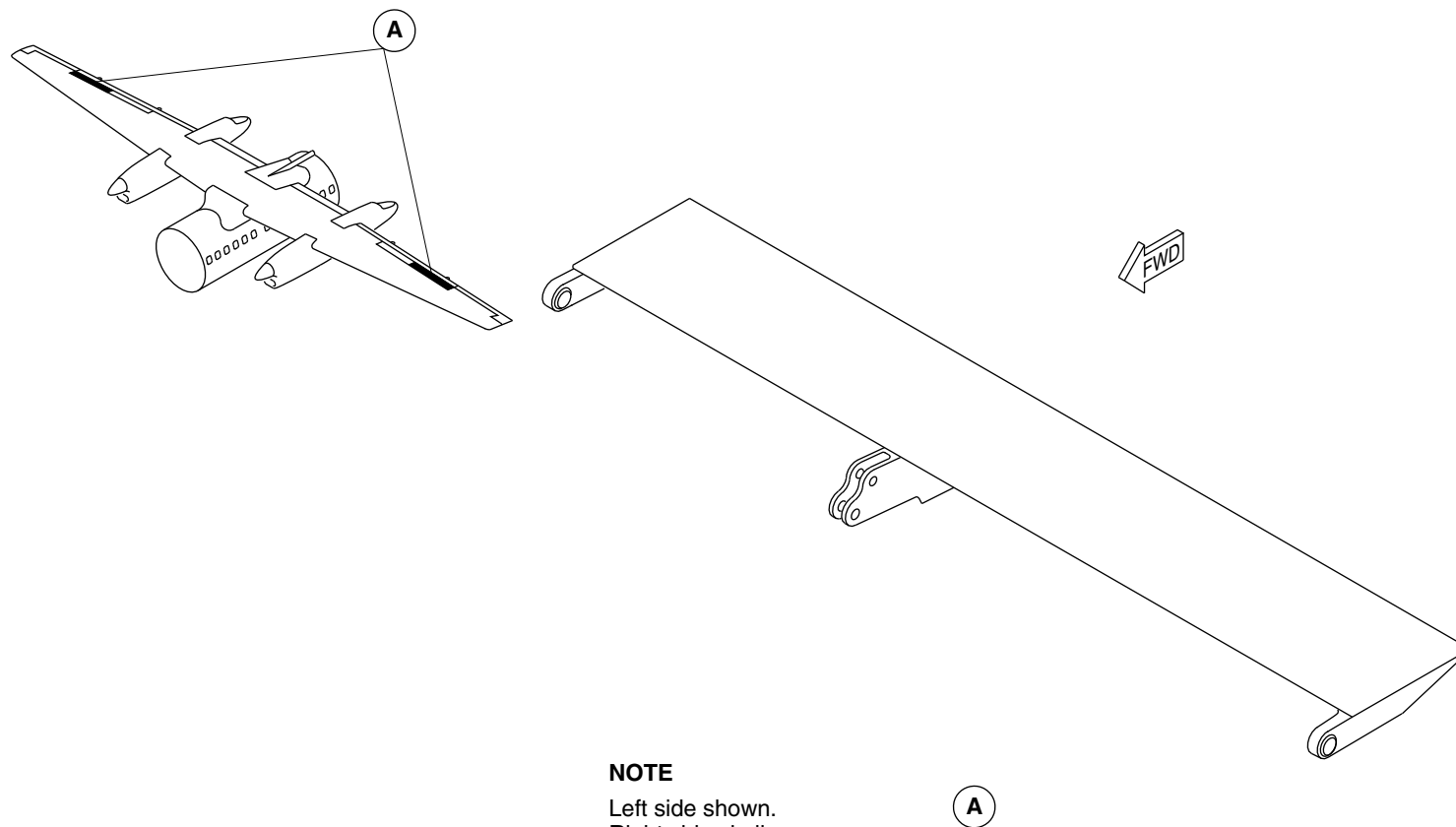
27-63-00

Config 001
Page 4
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



NOTE

Left side shown.
Right side similar.

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OUTBOARD SPOILER PANEL PAGE 1
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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Config 001

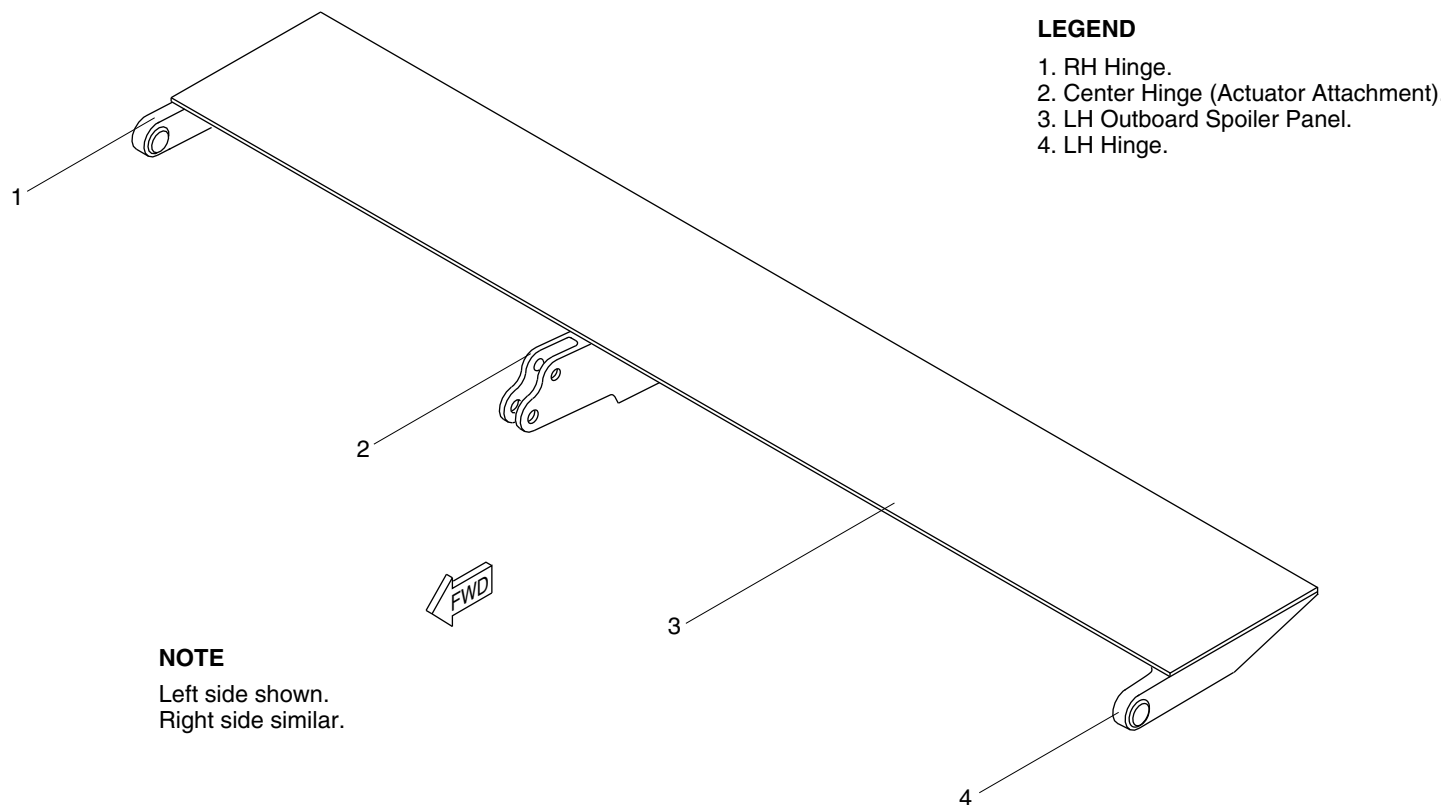
27-63-00

Config 001
Page 5
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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OUTBOARD SPOILER PANEL PAGE 2
Figure 4

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-63-00
Config 001

27-63-00

Config 001
Page 6
Sep 05/2021



AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

**ON A/C ALL

27-70-00-001

GUST LOCK SYSTEMS

Introduction

The gust lock system locks the aileron control surfaces in the neutral position to protect the aileron surfaces from damage.

General Description

The aileron control system is a mechanically operated system. Gust protection for the ailerons is supplied by a gust lock mechanism operated from the flight compartment. When the CONTROL LOCK lever is moved to the ON position, the ailerons are locked in neutral.

The aileron gust lock system has the sub-system that follows:

- Mechanism, Aileron Gust Lock System (27-71-01)

Detailed Description

The aileron gust lock system locks the ailerons in a neutral position when the aircraft is parked or being taxied. This protects the aileron surfaces from damage due to strong gusts.

[Refer to Figures 1 and 2.](#)

The CONTROL LOCK lever is located in the flight compartment, on the power quadrant. When the lever is moved to the ON position, the aileron splitter quadrant is locked in the neutral position and the power levers cannot be advanced. The ailerons are locked in the neutral position.

If a gust lock cable fails, the gust lock mechanism fails safe to the unlocked position.

Mechanism, Aileron Gust Lock System

[Refer to Figures 3 and 4.](#)

The control lock lever is connected by a pushrod and lever/layshaft assembly to the gust lock quadrant located below the flight compartment floor. A cable circuit extends aft from the quadrant, up the forward face of the flight compartment bulkhead, then aft along the cabin roof. The cables then pass through pressure seals in the cabin roof and end at the aileron gust lock mechanism on the wing rear spar.

[Refer to Figures 5 and 6.](#)

The aileron gust lock mechanism is located on the aft face of the wing rear spar, right of the aircraft centreline. The mechanism has a cable-operated gust lock lever, a spring-loaded plunger, two tension springs and a lock plate on the aileron splitter quadrant.

[Refer to Figures 7 and 8.](#)

When the CONTROL LOCK lever release is pulled up and the lever is moved from the 'OFF' to the 'ON' position, the gust lock lever is pivotted by the cable to the locked position. If the ailerons are in the neutral position, the plunger engages in the lock plate slot and prevents the aileron splitter quadrant from rotating. If the ailerons are not in the neutral position, the plunger is compressed against the face of the lock plate. When the ailerons are moved to the neutral position the plunger springs out to the locked position.

Two tension springs pull the gust lock lever towards the OFF position. This gives positive disengagement of the gust lock mechanism and unlocks the ailerons if a cable fails.



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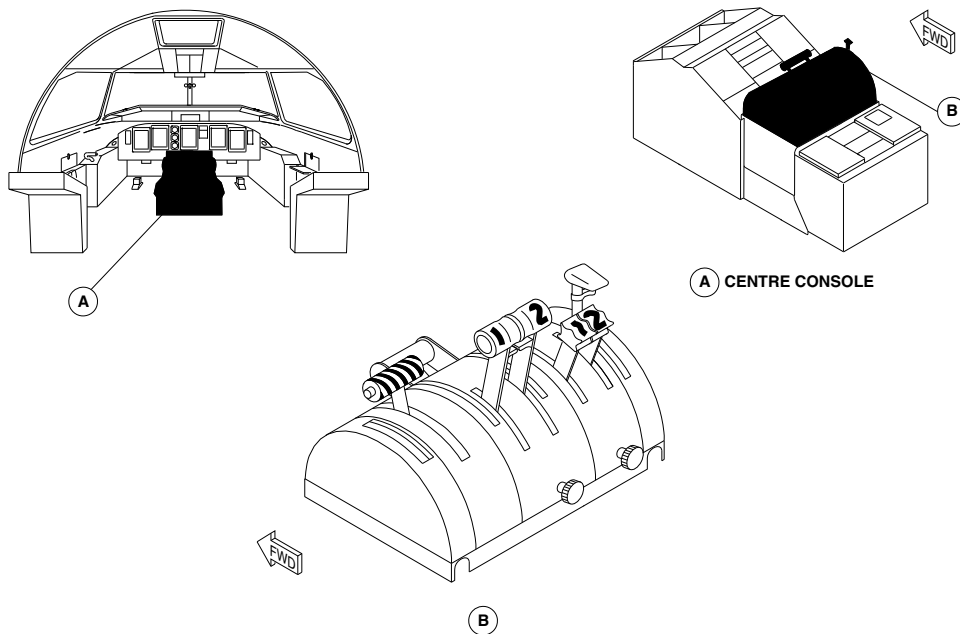
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Config 001
Page 3
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



4761a001 Power Quadrant Locator.iso

POWER QUADRANT
Figure 1

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-70-00
Config 001

27-70-00

Config 001
Page 4
Sep 05/2021



A

CENTRE CONSOLE

LEGEND

1. Control Lock Lever.
2. Control Lock Off Position.
3. Control Lock On Position.

A

AILERON GUST LOCK CONTROL LEVER
Figure 2

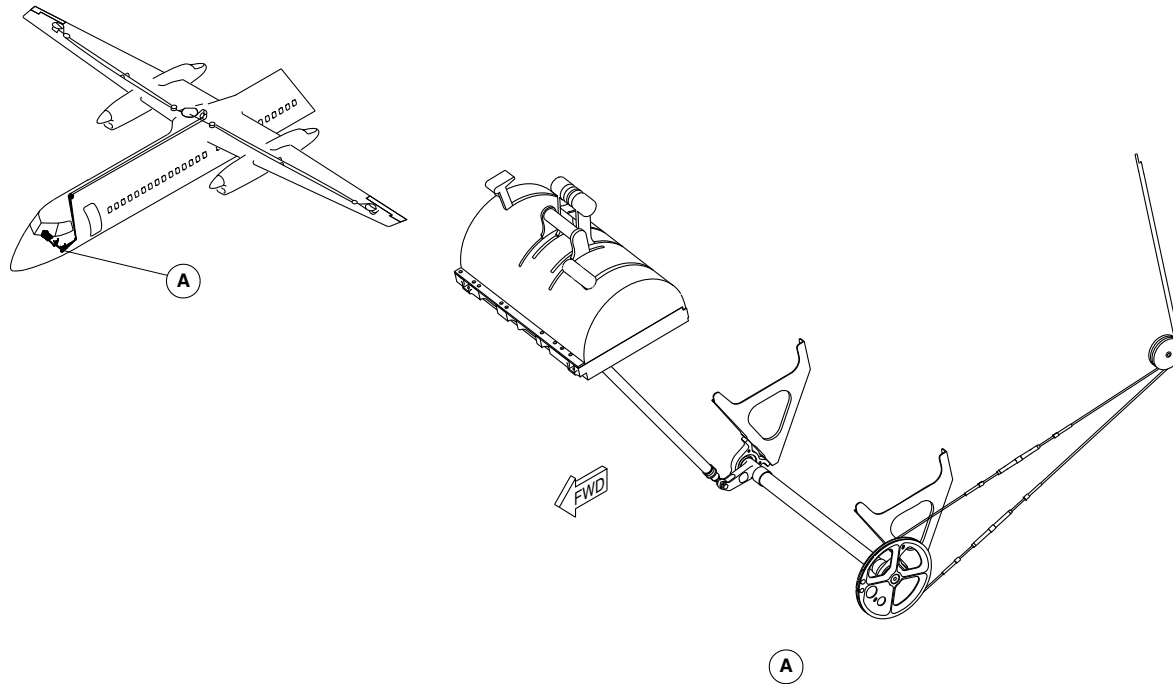
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Config 001
Page 5
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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AILERON GUST LOCK / FLIGHT COMPARTMENT
Figure 3

PSM 1-84-2A
EFFECTIVITY:
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Config 001

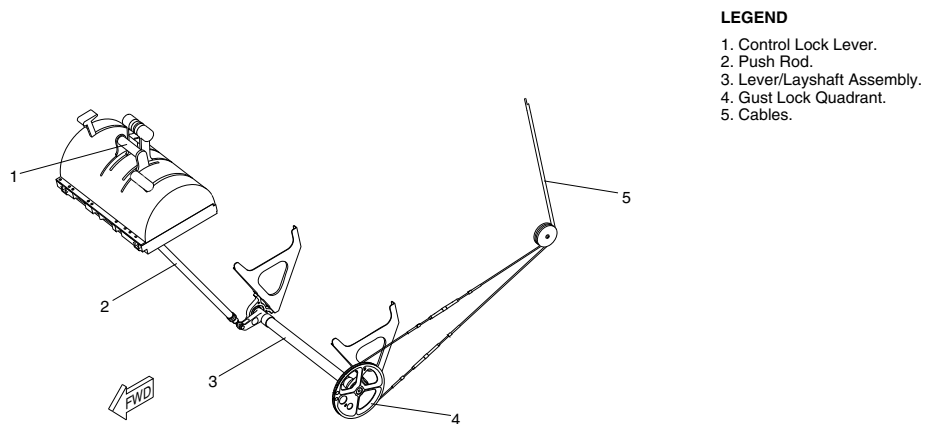
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Config 001
Page 6
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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AILERON GUST LOCK SYSTEM / FLIGHT COMPARTMENT
Figure 4

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-70-00
Config 001

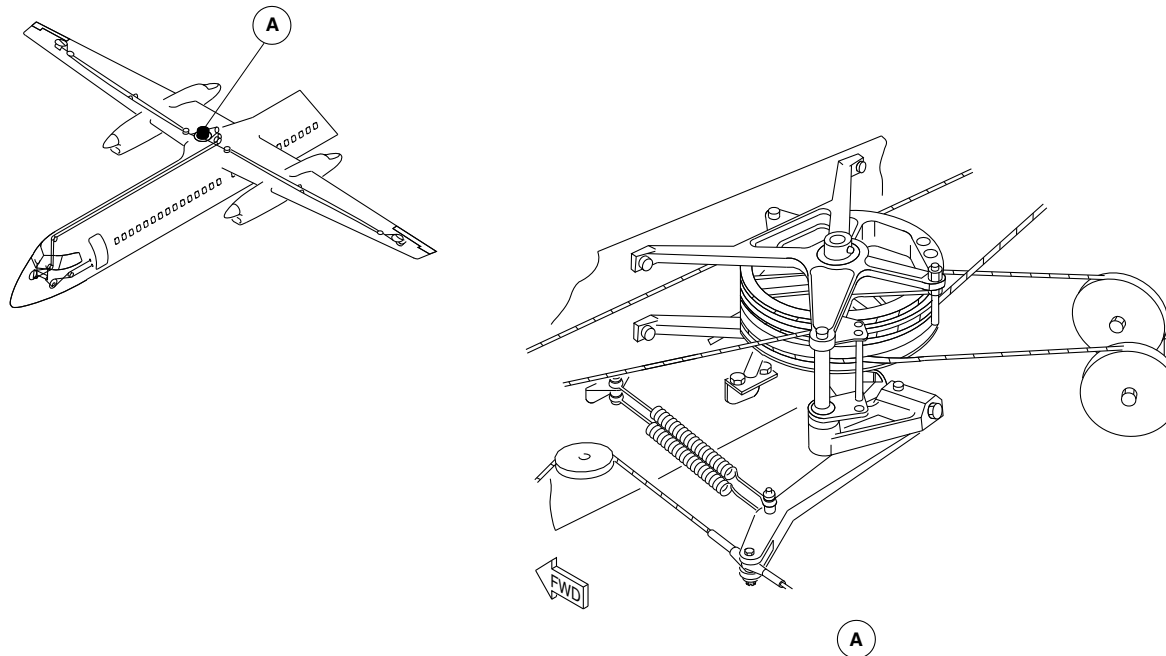
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Config 001
Page 7
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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AILERON GUST LOCK / SPLITTER QUADRANT PAGE 1
Figure 5

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-70-00
Config 001

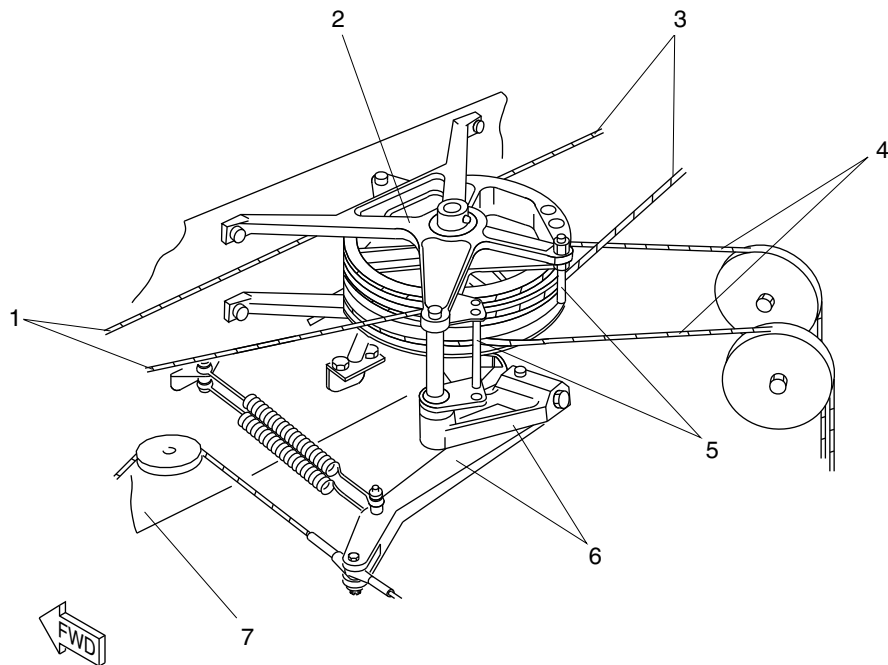
27-70-00

Config 001
Page 8
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

- 1. LH Aileron Control Cables.
- 2. Aileron Splitter Quadrant.
- 3. RH Aileron Control Cables.
- 4. Quadrant Input Cables.
- 5. Cable Guards.
- 6. Gust Lock Mechanism.
- 7. Wing Rear Spar.

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AILERON GUST LOCK / SPLITTER QUADRANT PAGE 2
Figure 6

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-70-00
Config 001

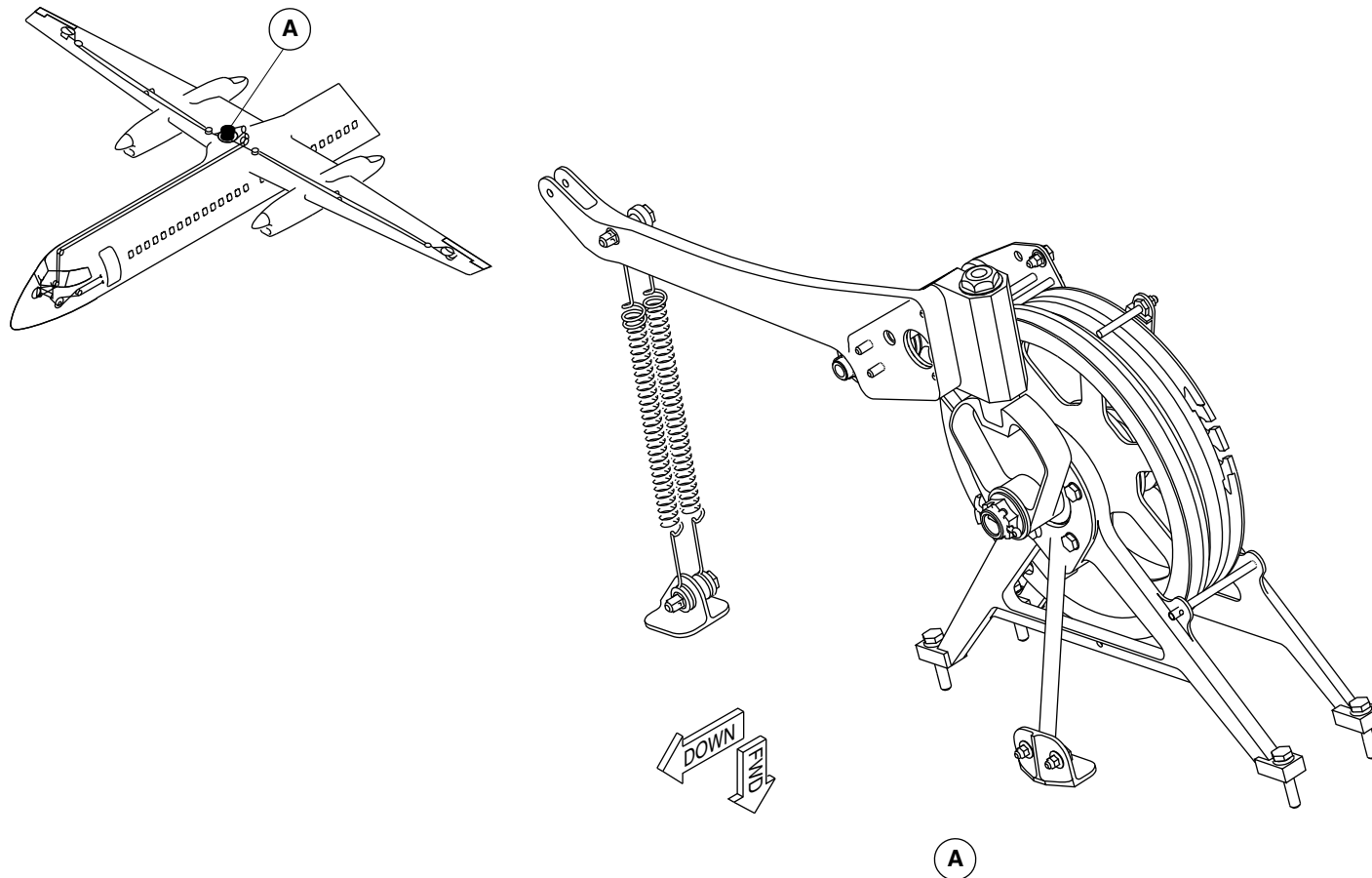
27-70-00

Config 001
Page 9
Sep 05/2021



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AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



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AILERON GUST LOCK MECHANISM PAGE 1
Figure 7

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-70-00
Config 001

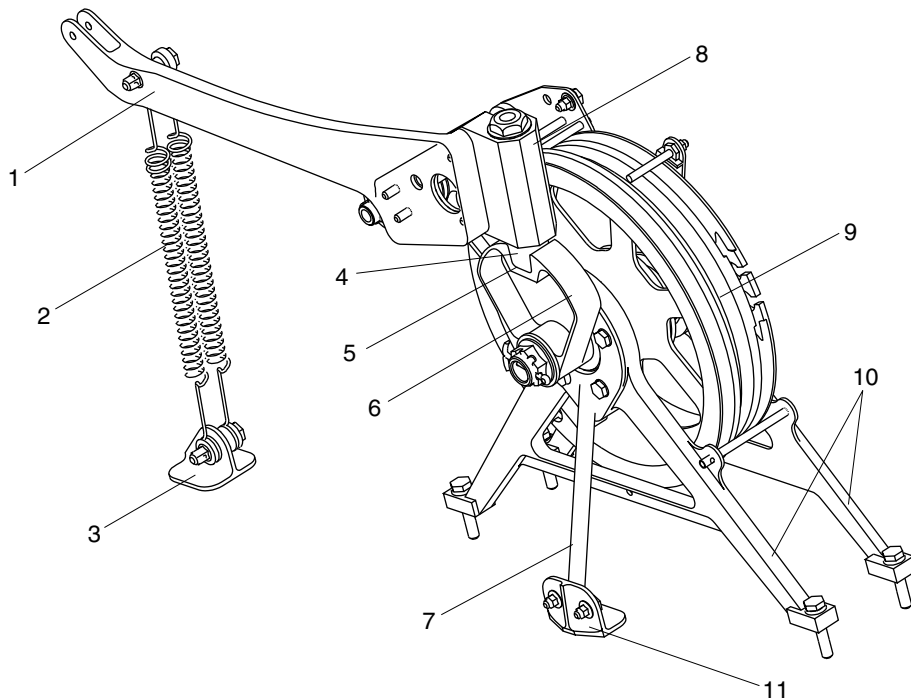
27-70-00

Config 001
Page 10
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION



LEGEND

1. Gust Lock Lever.
2. Tension Springs.
3. Tension Spring Bracket.
4. Plunger.
5. Slot.
6. Lock Plate.
7. Strut.
8. Gust Lock Housing.
9. Aileron Splitter Quadrant.
10. Support Brackets.
11. Strut Bracket.

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AILERON GUST LOCK MECHANISM PAGE 2
Figure 8

PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-70-00
Config 001

27-70-00

Config 001
Page 11
Sep 05/2021



DE HAVILLAND AIRCRAFT
OF CANADA LIMITED

AIRCRAFT MAINTENANCE MANUAL – SYSTEM DESCRIPTION SECTION

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PSM 1-84-2A
EFFECTIVITY:
See first effectivity on page 2 of 27-70-00
Config 001

27-70-00

Config 001
Page 12
Sep 05/2021